

PROLOGUE

THE TRUST STANDARD - VERSION 1.0 (2026)

The Crisis of Confidence in Physical Assets

We live in an unprecedented paradox in the history of global finance. While digital markets reach levels of sophistication once unimaginable, the verification of the physical assets underpinning trillions of dollars in global wealth remains anchored in twentieth-century methodologies.

Every day, Family Offices in Geneva, sovereign wealth funds

in the Middle East, and reinsurers in London make investment decisions based on inspection reports designed for a world without blockchain, without artificial intelligence, without the Internet of Things, and without the reality of tokenized Real World Assets (RWA).

The result is a systemic crisis of confidence.

The Seven Gaps Threatening Trillions in Wealth

For decades, physical asset verification standards have operated under assumptions that are no longer valid:

1. Sectoral Fragmentation: A $50 million yacht requires hiring five different experts (structural, mechanical, legal, financial, environmental) with no integration between their reports.

2. Technological Obsolescence: RICS, IIMS, and Big Four methodologies were designed in the paper era, without contemplating drones, LiDAR, IoT sensors, or digital twins.

3. Point-in-Time Inspection vs. Continuous Reality: An inspection report is a momentary "snapshot." Between audits, the asset may deteriorate, suffer damage, or face undetected risks.

4. Physical-Digital Disconnect: When a physical asset is tokenized, no standard guarantees the cryptographic linkage between the real asset and its digital representation.

5. Absence of Predictive Approach: Current standards describe the present state but do not anticipate future

risks or model scenarios.

6. Insufficient Authenticity Verification: In a world of high-value assets, provenance and authenticity verification protocols are inadequate.

7. Lack of Standard for Integrated Wealth Management: Family Offices need a 360-degree view of their assets, but existing standards are designed for point transactions, not continuous management.

The Response: The Trust Standard

This document is not a best practices manual. It is the world's first programmable trust infrastructure for physical assets.

The Trust Standard is born from the convergence of three decades of experience in real estate, naval, and industrial expert assessment with the most advanced technologies of cryptographic verification, artificial intelligence, and distributed computing.

Three Layers of Innovation

Our standard operates in three layers that, together, eliminate the seven identified gaps:

LAYER 1: Comprehensive Physical Verification
The traditional physical inspection methodology, radically improved with advanced technology (drones, LiDAR, thermography, IoT sensors) and unified under a multi-asset protocol covering real estate, naval, industrial, and infrastructure assets.

LAYER 2: Cryptographic Proof of Existence (CPEP)
Every inspection datum is cryptographically signed at the device of origin (edge signing) by an auditor with decentralized identity (DID), creating an immutable digital chain of custody that enables mathematical verification of authenticity.

LAYER 3: Programmable Compliance (SCCIP)
Verification does not emit a static PDF. It translates into logical rules injected directly into smart contracts (ERC-3643), enabling automatic transfer pauses of tokens when critical risks are detected in the physical asset.

The Bridge Between the Physical and Digital Worlds

The Trust Standard is the missing link in the value chain of Real World Assets.

Until now, the tokenization of physical assets has faced a fundamental barrier: How can an institutional investor trust that the physical asset truly exists, is in the declared state, and will continue to be so?

Our standard responds with mathematical cryptographic proof and programmable compliance. It is not a promise. It is executable code.

For Whom This Standard Is Designed

This standard is designed for:

- Family Offices needing comprehensive and continuous verification of their patrimonial assets

- Investment funds and reinsurers requiring robust due diligence for high-value assets
- RWA tokenization platforms needing an institutional trust layer
- International law firms advising on physical asset transactions
- Inspection bodies seeking top-tier accreditation
- Asset owners wishing to maximize the value and liquidity of their holdings

Governance and Updates

The Trust Standard is a living standard. It is updated quarterly through a review process driven by:

- The Aurema Group Technical Advisory Board
- Feedback from accredited bodies
- Technological and regulatory evolution
- Emerging market use cases

The governance of the standard follows the principles of transparency, impartiality, and consensus that define the most prestigious international standards.

The Purpose of Jire

Behind every inspection, every signed datum, every smart contract, there is a fundamental principle: integrity.

The Trust Standard not only protects wealth. It protects the trust that makes the global economy possible. Every verified asset is a promise fulfilled. Every programmable token is an executable guarantee.

In a world of uncertainty, we build certainty.

Aurelio Tamarit Blay
Technical Director
Aurema Group
July 2026

CHAPTER 1: INTRODUCTION AND SCOPE

1.1 PURPOSE AND OBJECTIVES OF THE STANDARD

1.1.1 Fundamental Purpose

The Prop Trust Verified Standard (hereinafter "the Standard" or "PTVS") establishes the requirements, methodologies, and protocols for comprehensive verification of physical assets in the digital era. Developed by Aurema Group, this standard represents the convergence between traditional physical verification and programmable trust technologies.

The fundamental purpose of the Standard is to eliminate the systemic crisis of confidence that exists between:

- The physical assets of the real world (real estate, naval, industrial, infrastructure)
- The digital representations of those assets (tokens, NFTs, tokenized assets)
- The market participants who rely on that information to make decisions

1.1.2 Strategic Objectives

The Standard pursues five interconnected strategic objectives:

OBJECTIVE 1: Methodological Unification
Provide a unified framework for verification of multiple physical asset typologies under a coherent methodology, eliminating the sectoral fragmentation that forces hiring multiple experts without integration.

OBJECTIVE 2: Advanced Technological Integration
Incorporate cutting-edge technologies (drones, LiDAR, IoT, AI, blockchain) into verification processes, overcoming the technological obsolescence of traditional standards.

OBJECTIVE 3: Continuous Monitoring
Transition from point-in-time inspections to a continuous

monitoring model through IoT sensors and real-time analysis, providing permanent certainty about the asset's state.

OBJECTIVE 4: Physical-Digital Bridge
Establish cryptographic protocols that mathematically link the physical asset with its digital representation, enabling secure tokenization of real-world assets (RWA).

OBJECTIVE 5: Programmable Compliance
Enable injection of compliance rules directly into smart contracts, enabling automatic execution of protections when critical risks are detected.

1.1.3 Guiding Principles

The Standard is based on seven guiding principles that guide all its application:

1. Absolute Integrity: Every verification must reflect the reality of the asset without biases, commercial pressures, or conflicts of interest.

2. Technical Independence: Accredited inspection bodies must operate with technical and financial autonomy from interested parties.

3. Demonstrable Competence: All personnel involved must demonstrate training, experience, and continuous updating in their areas of specialization.

4. Strict Confidentiality: Sensitive information about the asset and client must be protected through cryptographic and legal protocols.

5. Procedural Transparency: Methods, limitations, and results must be communicated clearly, including uncertainties and

degrees of confidence.

6. Responsible Innovation: Adoption of new technologies must prioritize security, privacy, and reliability over novelty.

7. Comprehensive Sustainability: Verifications must consider the environmental, social, and governance (ESG) impact of assets.

1.2 SCOPE AND APPLICABILITY

1.2.1 Covered Asset Typologies

The Standard is applicable to the following categories of physical assets:

CATEGORY A: REAL ESTATE ASSETS

A.1 Residential
• Single-family homes
• Collective residential buildings
• Luxury residences and second homes
• Historical and protected properties

A.2 Commercial and Offices
• Office buildings
• Shopping centers and retail
• Hotels and resorts
• Mixed-use buildings

A.3 Industrial and Logistics
• Industrial warehouses
• Warehouses and logistics centers
• Production plants
• Energy facilities

A.4 Land and Plots
• Urban land
• Industrial land
• Agricultural and forest land
• Land with patrimonial value

CATEGORY B: NAVAL ASSETS

B.1 Recreational Vessels
• Yachts and megayachts (>24m)
• Recreational boats (10-24m)
• Sailboats and catamarans
• Jet skis and light vessels

B.2 Commercial Vessels
• Merchant ships
• Fishing vessels
• Tugboats and barges
• Service vessels

B.3 Specialized Vessels
• Offshore platforms
• Research vessels
• Military vessels (when applicable)
• Submarines and submersible vehicles

CATEGORY C: INDUSTRIAL AND INFRASTRUCTURE ASSETS

C.1 Industrial Machinery
• Production equipment
• Heavy machinery
• Construction equipment
• Industrial vehicle fleets

C.2 Infrastructure
• Bridges and viaducts
• Roads and highways
• Port facilities
• Airports and heliports

C.3 Energy Facilities
• Solar and wind plants
• Electrical substations
• Storage facilities
• Distribution networks

CATEGORY D: PATRIMONIAL AND HIGH-VALUE ASSETS

D.1 Art and Collectibles
• Works of art
• Antiques
• High-value collections
• High-end jewelry and watchmaking

D.2 Collector Vehicles
• Classic automobiles
• Collector motorcycles
• Historical aircraft
• Demilitarized military vehicles

1.2.2 Scope Exclusions

The Standard is NOT applicable to:

1. Purely financial assets: Stocks, bonds, derivatives, native digital crypto-assets

2. Intellectual property: Patents, trademarks, copyrights (except when part of a physical asset)

3. Intangible assets: Goodwill, brand value, commercial relationships

4. Biological assets: Livestock, crops, standing forests (except the underlying land)

5. Assets in active litigation: When there is an ongoing judicial procedure affecting ownership or value of the asset

1.2.3 Jurisdictions of Application

The Standard is designed for global application, with specific jurisdictional adaptations for:

• European Union: Compliance with MiCA, national inspection regulations

• United Kingdom: Alignment with RICS, UKAS, post-Brexit regulations

• United States: Compatibility with USPAP, ASHRAE standards

• Middle East (ADGM, DIFC): Compliance with financial zone regulations

• Asia-Pacific (Singapore, Hong Kong): Adaptation to local regulatory frameworks

[DIAGRAM 1.1: World map with application jurisdictions and their specific adaptations]

1.2.4 Liability Limits

The Standard explicitly recognizes the following limits:

1. Inherent uncertainty: Every verification has a degree of uncertainty that must be quantified and communicated

2. Inaccessible conditions: Areas of the asset not accessible during inspection must be explicitly declared

3. Future events: The Standard does not guarantee future behavior of the asset, only provides predictive analysis based on available data

4. Sophisticated fraud: The Standard significantly reduces fraud risk but cannot guarantee detection of extremely sophisticated fraud

1.3 DOCUMENT STRUCTURE

1.3.1 General Organization

The Standard is structured in five main parts, each with a specific purpose:

PART I: GENERAL FRAMEWORK (Chapters 1-4)
Establishes the conceptual foundations, principles, definitions, and governance of the Standard.

PART II: INTEGRAL VERIFICATION METHODOLOGY (Chapters 5-8)
Details specific verification protocols for physical, documentary, technical, and financial aspects.

PART III: ADVANCED PROTOCOLS (Chapters 9-12)
Defines technological protocols for continuous monitoring, cryptographic proof, predictive analysis, and authenticity verification.

PART IV: REPORTING AND CERTIFICATION (Chapters 13-15)
Establishes report structure, certification levels, and seal usage conditions.

PART V: IMPLEMENTATION AND GOVERNANCE (Chapters 16-18)
Defines requirements for inspection bodies, accreditation process, and continuous improvement mechanisms.

1.3.2 Drafting Conventions

To ensure clarity and consistency, the Standard uses the following conventions:

Normative terms:
- "SHALL": Mandatory requirement
- "SHALL NOT": Prohibition
- "SHOULD": Strong recommendation
- "SHOULD NOT": Discouraged
- "MAY": Permission or possibility
- "CANNOT": Technical or legal impossibility

Cross-references:
- All references to other chapters or sections follow the format: [see Section X.X]
- References to annexes follow the format: [see Annex X]

Notes and examples:
- Explanatory notes are presented in gray boxes
- Examples are presented in blue boxes
- Critical warnings are presented in red boxes

1.3.3 Normative and Informative Annexes

The Standard includes two types of annexes:

Normative Annexes:
Form an integral part of the Standard's requirements. Their compliance is mandatory for accreditation.

• Annex A: Inspection checklists by asset typology
• Annex B: Technical requirements for measurement equipment
• Annex C: Calibration and traceability protocols
• Annex D: Edge cryptographic signature specifications

Informative Annexes:
Provide additional guidance, examples, and best practices. Their compliance is recommended but not mandatory.

• Annex E: Implementation guide for inspection bodies
• Annex F: Detailed case studies
• Annex G: Expanded technical glossary
• Annex H: Bibliographic and normative references

1.4 VERIFICATION LEVELS

1.4.1 Philosophy of Levels

The Standard recognizes that not all assets require the same level of verification. An $50 million yacht destined for tokenization requires a much higher level of rigor than a $300,000 home for a standard transaction.

Therefore, the Standard defines four verification levels, each with specific requirements for depth, technology, and monitoring.

1.4.2 Level 1: Basic Verification

Application:

• Moderate-value assets (<$500,000)
• Standard transactions without special requirements
• Preliminary due diligence

Requirements:
• Complete visual inspection of the asset
• Basic documentary verification (titles, main licenses)
• Standard report with critical findings
• Representative photographs
• Approximate market valuation

Technology:
• Traditional in-person inspection
• Digital photographic documentation
• Basic measurement tools

Deliverable:
• 20-40 page PDF report
• Validity: 6 months
• No continuous monitoring

Certification:
• "Prop Trust Verified - Level 1" seal
• Digital signature of accredited inspector

1.4.3 Level 2: Standard Verification

Application:
• Medium-value assets ($500,000 - $5M)
• Transactions with robust due diligence requirements
• Bank financing

Requirements:
• Detailed physical inspection with advanced technology
• Exhaustive documentary due diligence
• Technical analysis of critical systems

• Preliminary environmental assessment
• Market valuation with multiple methodologies
• Qualitative risk analysis

Technology:
• Drones for roof and facade inspection
• Infrared thermography
• Moisture and materials analysis
• Automated valuation software

Deliverable:
• 50-100 page PDF report
• Detailed technical annexes
• Validity: 12 months
• No continuous monitoring

Certification:
• "Prop Trust Verified - Level 2" seal
• Digital signature of inspector and technical reviewer
• Registration in Aurema Group database

1.4.4 Level 3: Premium Verification

Application:
• High-value assets ($5M - $50M)
• Family Offices and institutional investors
• Asset tokenization (RWA)
• Large-risk insurance

Requirements:
• All Level 2 requirements, plus:
• Inspection with LiDAR and 3D scanning
• Advanced structural analysis
• Complete environmental assessment
• Quantitative risk analysis with AI
• Digital twin of the asset

• Continuous monitoring via IoT
• Integration with smart contracts (SCCIP)

Technology:
• LiDAR for precise 3D modeling
• IoT sensors for continuous monitoring
• Predictive AI for risk analysis
• Blockchain for immutable registration
• Edge cryptographic signature (CPEP)

Deliverable:
• Interactive report on web platform
• Accessible digital twin
• Real-time monitoring dashboard
• Automatic risk alerts
• Validity: 24 months with continuous updates
• Cryptographically verifiable on-chain certificate

Certification:
• "Prop Trust Verified - Level 3 Premium" seal
• Cryptographic signature of inspector with DID
• Report hash registered on blockchain
• Active compliance oracle

1.4.5 Level 4: Institutional Verification

Application:
• Very high-value assets (>$50M)
• Ultra-high-net-worth Family Offices
• Sovereign wealth funds and reinsurers
• Multi-asset portfolios
• Large-scale institutional tokenization

Requirements:
• All Level 3 requirements, plus:
• Dedicated multidisciplinary team

• Advanced forensic analysis (when applicable)
• Exhaustive regulatory compliance audit
• Extreme scenario stress testing
• Cross-asset risk correlation analysis
• Personalized reporting for investment committees
• Dedicated 24/7 support

Technology:
• All Level 3 technologies, plus:
• Zero-Knowledge Proofs for privacy
• Multi-sig for critical decision governance
• Custom APIs for client system integration
• Big data analysis with proprietary models

Deliverable:
• Custom white-label platform
• Monthly executive reporting
• Quarterly review meetings
• Priority alerts with impact analysis
• Validity: 36 months with continuous updates
• Cryptographic certificates for each portfolio asset

Certification:
• "Prop Trust Verified - Level 4 Institutional" seal
• Dedicated Advisory Board
• Annual maintenance audit
• Priority access to Standard updates

1.4.6 Selection of Appropriate Level

The selection of verification level must be based on:

1. Asset value: Quantified according to objective criteria
2. Technical complexity: Assets with complex systems require higher levels
3. Verification purpose: Tokenization, financing, insurance,

etc.
4. Regulatory requirements: Some jurisdictions require minimum levels
5. Client risk appetite: Conservative investors prefer higher levels
6. Cost-benefit: Verification cost must be proportional to protected value

Aurema Group provides an automated decision tool that recommends the optimal level based on these criteria.

1.5 NORMATIVE REFERENCES

1.5.1 International Reference Standards

The Standard aligns with and complements the following international standards:

ISO Standards:
• ISO 17020: Criteria for the operation of inspection bodies
• ISO 9001: Quality management systems
• ISO 14001: Environmental management systems
• ISO 45001: Occupational health and safety management systems
• ISO 27001: Information security management systems
• ISO 31000: Risk management

Sectoral Standards:
• RICS Valuation Standards (Red Book): Real estate valuation
• IVS (International Valuation Standards): International valuation
• IIMS Standards: Marine inspection
• Lloyd's Register Rules: Naval classification
• USPAP: Appraisal practice in the USA

Financial Regulations:
- MiCA (Markets in Crypto-Assets): European regulation of crypto-assets
- AMLD (Anti-Money Laundering Directives): Prevention of money laundering
- GDPR: Data protection in the EU
- SOX (Sarbanes-Oxley): Internal control in the USA

1.5.2 Technological Standards

The Standard incorporates references to key technological standards:

Blockchain and Cryptography:
- ERC-3643: Token standard with identity for real-world assets
- ERC-721 and ERC-1155: Non-fungible token standards
- DID (Decentralized Identifiers): W3C DID Core Specification
- Verifiable Credentials: W3C Verifiable Credentials Data Model

Internet of Things:
- IEEE 802.15.4: Low-rate personal area networks
- MQTT: Telemetry protocol
- OPC UA: Industrial communication

Artificial Intelligence:
- ISO/IEC 23894: AI risk management
- IEEE P2894: Ethics standard in AI

1.5.3 Normative Hierarchy

In case of conflict between references, the following hierarchy applies:

1. Mandatory legal regulations of the applicable jurisdiction

2. Specific requirements of the Standard (PTVS)
3. International ISO standards
4. Sectoral standards (RICS, IIMS, etc.)
5. Industry best practices

1.6 KEY DEFINITIONS

1.6.1 Fundamental Terms

Physical Asset
Tangible good with independent material existence, susceptible to appropriation and economic value. Includes real estate, vessels, machinery, infrastructure, and patrimonial assets.

Verification
Systematic process of examination, analysis, and evaluation of a physical asset to determine its existence, state, value, conformity, and risks, in accordance with Standard requirements.

Inspection Body
Entity accredited by Aurema Group to perform verifications according to the Standard, meeting competence, independence, and impartiality requirements.

Accredited Inspector
Professional with specific training, experience, and certification to perform verifications under the Standard, contractually linked to an accredited inspection body.

Verification Report
Technical document presenting verification results, including findings, analysis, conclusions, and recommendations, according to the structure defined in the Standard.

1.6.2 Technological Terms

Edge Signing
Cryptographic process by which a device (sensor, drone, camera) digitally signs data at the moment of capture, before any transmission or subsequent processing.

Decentralized Identity (DID - Decentralized Identifier)
Unique identifier cryptographically verifiable that does not require a centralized registry, allowing auditors to demonstrate their identity and credentials in a sovereign manner.

Oracle
Mechanism connecting real-world data with smart contracts on blockchain, enabling automatic execution based on externally verified conditions.

Digital Twin
Dynamic digital representation of a physical asset that updates in real-time through IoT sensor data, enabling simulation, analysis, and monitoring.

Zero-Knowledge Proof
Cryptographic protocol that allows demonstrating the validity of a claim without revealing the underlying information, protecting the confidentiality of sensitive data.

1.6.3 Compliance Terms

Programmable Compliance
Ability to inject compliance rules directly into smart contracts, enabling automatic execution of actions (pauses, alerts, restrictions) when specific conditions are detected.

SCCIP (Smart Contract Compliance Integration Protocol)

Protocol defined in the Standard for integration of physical verifications with compliance modules in smart contracts, particularly ERC-3643.

CPEP (Cryptographic Proof of Existence Protocol)
Protocol defined in the Standard for cryptographic verification of existence and state of physical assets through edge signing and digital chain of custody.

CAP (Continuous Assurance Protocol)
Continuous monitoring protocol through IoT sensors that provides real-time updates of asset state and automatic alerts for deviations.

1.6.4 Risk Terms

Critical Risk
Condition that threatens the existence, integrity, or legal value of the asset immediately, requiring urgent corrective action and potentially automatic transfer pause in tokenized assets.

High Risk
Condition that may significantly affect the value or functionality of the asset in the medium term, requiring priority corrective action and manual approval for transfers in tokenized assets.

Medium Risk
Condition that may affect the value or functionality of the asset in the long term, requiring monitoring and planning of corrective actions.

Low Risk
Minor condition that does not significantly affect the value or functionality of the asset, but must be documented for

owner awareness.

CHAPTER 2: FUNDAMENTAL PRINCIPLES

2.1 INTRODUCTION TO GUIDING PRINCIPLES

2.1.1 Philosophy of the Principles

The fundamental principles of the Prop Trust Verified Standard are not mere statements of intent. They are mandatory requirements governing every aspect of the Standard's application, from inspection planning to the issuance of a cryptographic certificate.
These principles constitute the ethical and professional framework that distinguishes the Standard from traditional methodologies and guarantees market participants' confidence.

2.1.2 Hierarchy of Principles

The seven principles have equal normative hierarchy. In case of conflict between principles, the following order of priority applies:

1. Absolute Integrity (maximum priority)
2. Technical Independence
3. Strict Confidentiality
4. Demonstrable Competence
5. Procedural Transparency

6.Responsible Innovation
7.Comprehensive Sustainability

2.1.3 Compliance Mechanisms

Compliance with the principles is verified through:
• Periodic audits of accredited inspection bodies • Confidential reporting systems for non-compliance reports • Escalating sanctions ranging from warnings to accreditation revocation • Public transparency of serious sanctions in the Aurema Group registry

2.2 PRINCIPLE 1: ABSOLUTE INTEGRITY

2.2.1 Definition

Absolute integrity requires that every verification reflect the reality of the asset without biases, commercial pressures, conflicts of interest, or external influences.
This principle is the cornerstone of the Standard. Without integrity, all other principles lose their meaning.

2.2.2 Specific Requirements

REQUIREMENT 1.1: Absence of Biases
The inspection body MUST ensure that:
• Inspectors have no direct or indirect financial interest in the verified asset • No commercial relationships exist that could compromise objectivity • Analysis methods are free from systematic cognitive biases • Conclusions are based exclusively on verifiable evidence
REQUIREMENT 1.2: Resistance to Pressures
The inspection body MUST:
• Reject any request to modify conclusions by the client • Document and report attempts at undue influence •

Protect inspectors from retaliation for maintaining objective conclusions • Refuse to issue favorable reports in exchange for additional compensation
REQUIREMENT 1.3: Documentary Truthfulness
The inspection body MUST:
• Present accurate and complete data, without relevant omissions • Clearly distinguish between verified facts and professional opinions • Recognize limitations and degrees of uncertainty in conclusions • Correct material errors immediately upon detection

2.2.3 Practical Application

CASE 1: Financial Conflict of Interest
Situation: An inspector discovers that their spouse owns shares in the company requesting verification of an industrial asset.
Required action: • The inspector MUST declare the conflict immediately • MUST be reassigned to another case • The body MUST document the conflict and reassignment • If the conflict is not declared and is later discovered, the body faces serious sanctions
CASE 2: Client Pressure
Situation: A client offers to double fees if the report concludes that a building has no structural problems, when the inspection revealed deficiencies.
Required action: • The inspector MUST reject the offer • MUST document the influence attempt • MUST issue the report with objective conclusions • The body MUST evaluate whether to continue the commercial relationship with that client

2.2.4 Verification Mechanisms

• Annual declaration of conflicts of interest from all personnel • Mandatory rotation of inspectors on high-value assets • Independent peer review of randomly selected reports • Forensic audit upon reports of integrity breaches

2.2.5 Sanctions for Non-Compliance

• Minor (minor omission): Formal warning + mandatory training • Moderate (unintentional bias): 3-6 month suspension + intensive audit • Serious (undeclared conflict): 12-month suspension + review of all issued reports • Critical (intentional fraud): Permanent accreditation revocation + legal action

2.3 PRINCIPLE 2: TECHNICAL INDEPENDENCE

2.3.1 Definition

Technical independence requires that inspection bodies operate with technical and financial autonomy from interested parties, ensuring that professional decisions are not influenced by commercial, regulatory, or other pressures.

2.3.2 Specific Requirements

REQUIREMENT 2.1: Structural Autonomy
The inspection body MUST:
• Have its own legal personality differentiated from other commercial activities • Have a governance structure that protects technical independence • Have sufficient financial resources to operate without dependence on specific clients • Not be subject to control by entities that could compromise independence
REQUIREMENT 2.2: Financial Independence
The inspection body MUST:
• Not obtain more than 30% of its income from a single client or group of related clients • Not accept commissions, gifts, or benefits from service providers related to

inspections • Establish transparent pricing policies based on work performed, not results • Disclose any funding source that could be perceived as conflicting
REQUIREMENT 2.3: Personnel Independence
The inspection body MUST:
• Contractually protect inspectors against dismissals or retaliation for maintaining objective conclusions • Avoid compensation systems based on inspection results • Ensure inspectors do not participate in business development activities that could compromise independence • Provide continuous training in professional ethics and resistance to pressures

2.3.3 Practical Application

CASE 1: Financial Dependence
Situation: An inspection body obtains 45% of its income from a single investment fund commissioning verification of multiple assets.
Required action: • The body MUST diversify its client portfolio • MUST implement additional safeguards for inspections of that client • MUST disclose the dependence in its transparency reports • If dependence persists, Aurema Group may require a correction plan
CASE 2: Regulatory Pressure
Situation: A regulatory authority informally requests that an inspection body issue favorable conclusions about an asset of public interest.
Required action: • The body MUST maintain its technical independence • MUST document the request and its response • MUST issue the report based exclusively on technical evidence • MAY consult with Aurema Group on how to handle the situation

2.3.4 Verification Mechanisms

• Annual analysis of income concentration • Review of governance structures every 3 years • Confidential

surveys to staff on pressures received • Independent financial audit of income sources

2.3.5 Sanctions for Non-Compliance

• Minor (moderate dependence): Correction plan in 6 months • Moderate (failure to disclose): 3-month suspension + financial audit • Serious (critical dependence): 12-month suspension + mandatory restructuring • Critical (total loss of independence): Accreditation revocation

2.4 PRINCIPLE 3: DEMONSTRABLE COMPETENCE

2.4.1 Definition

Demonstrable competence requires that all personnel involved in verifications demonstrate training, experience, and continuous updating in their areas of specialization, ensuring that inspections are performed by qualified professionals.

2.4.2 Specific Requirements

REQUIREMENT 3.1: Personnel Qualifications
The inspection body MUST:
• Hire only inspectors with recognized academic qualifications in relevant disciplines • Minimum 5 years of experience in inspection of similar assets • Current professional certifications (RICS, IIMS, engineering associations, etc.) • Demonstrated knowledge of advanced technologies required by the Standard (drones, IoT, blockchain)
REQUIREMENT 3.2: Continuous Training

The inspection body MUST:
• Provide a minimum of 40 hours of continuous training annually per inspector • Include training on Standard updates, new technologies, and ethical aspects • Maintain records of all training completed • Evaluate training effectiveness through tests or practical assessments
REQUIREMENT 3.3: Competence Evaluation
The inspection body MUST:
• Conduct periodic performance evaluations of inspectors • Include report reviews, inspection observations, and client feedback • Identify areas for improvement and provide specific training • Not assign complex inspections to inspectors without demonstrated competence

2.4.3 Competence Matrix by Asset Typology

Real Estate Residential: Architecture, Civil Engineering - 5 years minimum - RICS, Professional Associations
Real Estate Commercial: Architecture, Engineering - 7 years minimum - RICS, CCIM, LEED AP
Naval (Yachts): Naval Engineering - 7 years minimum - IIMS, Yacht Design
Naval (Merchant): Naval Engineering, Nautical Science - 10 years minimum - IIMS, Lloyd's Register
Industrial: Industrial Engineering, Mechanical Engineering - 7 years minimum - Engineering Associations
Infrastructure: Civil Engineering - 10 years minimum - Civil Engineering Associations
Art and Collectibles: Art History, Restoration - 5 years minimum - Specific certifications

2.4.4 Practical Application

CASE 1: Inspector Without Sufficient Competence
Situation: A body assigns an inspector with experience in residential housing to verify a $20M yacht.
Required action: • The body MUST NOT make the assignment • MUST hire or subcontract an inspector with

specific competence in naval assets • If the assignment was already made, MUST suspend it and reassign
CASE 2: Competence Obsolescence
Situation: An inspector with 20 years of experience has not trained in drone technologies, IoT, or blockchain.
Required action: • The body MUST provide intensive training in required technologies • While not completing training, MUST NOT assign Level 3 or 4 inspections • MUST evaluate whether the inspector can adapt to Standard requirements

2.4.5 Verification Mechanisms

• Audit of personal files of all inspectors • Verification of qualifications and certifications with issuing entities • Periodic technical competence tests • Review of reports by senior inspectors

2.4.6 Sanctions for Non-Compliance

• Minor (lack of continuous training): Mandatory training plan in 3 months • Moderate (inspector without competence): Suspension of inspections until accreditation • Serious (credential falsification): Inspector accreditation revocation + body sanction • Critical (fraud in competencies): Body accreditation revocation + legal action

2.5 PRINCIPLE 4: STRICT

CONFIDENTIALITY

2.5.1 Definition

Strict confidentiality requires that all sensitive information about the asset and client be protected through

cryptographic and legal protocols, ensuring that only authorized parties have access to the data.

2.5.2 Specific Requirements

REQUIREMENT 4.1: Client Data Protection
The inspection body MUST:
• Identify all sensitive information of the client and asset • Implement technical and organizational measures to protect that information • Limit access to strictly necessary personnel • Destroy or return information when the contractual relationship ends
REQUIREMENT 4.2: Cryptographic Security
The inspection body MUST:
• Encrypt all sensitive information in transit and at rest • Use internationally recognized cryptographic algorithms (AES-256, RSA-2048) • Implement role-based access controls (RBAC) • Conduct periodic security audits
REQUIREMENT 4.3: Confidentiality Agreements
The inspection body MUST:
• Sign confidentiality agreements (NDA) with all clients • Require NDAs from all personnel and subcontractors • Establish confidentiality clauses in all contracts • Define penalties for confidentiality breaches

2.5.3 Practical Application

CASE 1: Unauthorized Access
Situation: An administrative employee accesses verification reports of Family Office assets without operational need.
Required action: • The body MUST immediately investigate the access • MUST revoke the employee's permissions • MUST evaluate whether there was a confidentiality breach • MUST notify the client if there was access to sensitive information • MUST apply disciplinary sanctions to the employee
CASE 2: Security Breach

Situation: A cyberattack compromises the body's database of reports.
Required action: • The body MUST contain the breach immediately • MUST notify affected clients within a maximum of 72 hours • MUST notify regulatory authorities if applicable (GDPR) • MUST hire computer forensics experts to investigate • MUST implement corrective measures to prevent future breaches

2.5.4 Verification Mechanisms

• Annual information security audits • Penetration testing by independent third parties • Quarterly review of access controls • Compliance surveys to personnel

2.5.5 Sanctions for Non-Compliance

• Minor (accidental access without damage): Mandatory training + reinforcement of controls • Moderate (violation by employee): Employee dismissal + security audit • Serious (security breach): 6-month suspension + remediation plan • Critical (intentional violation): Accreditation revocation + legal action

2.6 PRINCIPLE 5: PROCEDURAL TRANSPARENCY

2.6.1 Definition

Procedural transparency requires that methods, limitations, and results of verifications be communicated clearly, including uncertainties and degrees of confidence, allowing stakeholders to understand and evaluate the quality of the work performed.

2.6.2 Specific Requirements

REQUIREMENT 5.1: Methodological Clarity
The inspection body MUST:
• Clearly document the methods used in each inspection • Explain the limitations of those methods • Justify the technical decisions made • Provide references to applicable standards and best practices
REQUIREMENT 5.2: Communication of Uncertainties
The inspection body MUST:
• Quantify degrees of uncertainty in conclusions • Distinguish between verified facts and estimates • Explain the factors contributing to uncertainty • Provide confidence intervals when applicable
REQUIREMENT 5.3: Disclosure of Conflicts
The inspection body MUST:
• Disclose any real or potential conflict of interest • Inform about commercial relationships that could be perceived as conflicting • Publish information about governance and ownership structure • Maintain a public record of accreditations and sanctions

2.6.3 Practical Application

CASE 1: Limitation in Inspection
Situation: During inspection of a building, 20% of the homes could not be accessed due to owners' refusal.
Required action: • The report MUST explicitly declare this limitation • MUST explain the impact on conclusions • MUST recommend additional inspections to cover non-accessible areas • MUST NOT issue conclusions about uninspected areas
CASE 2: Uncertainty in Valuation
Situation: The valuation of a historical asset has a high degree of uncertainty due to lack of comparables.
Required action: • The report MUST quantify the degree of uncertainty (e.g., ±25%) • MUST explain the factors

contributing to uncertainty • MUST provide a range of values instead of a single value • MUST recommend additional methodologies to reduce uncertainty

2.6.4 Verification Mechanisms

• Peer review of reports by independent parties • Client satisfaction surveys on clarity of communication • Audit of conflict disclosures • Publication of annual transparency reports

2.6.5 Sanctions for Non-Compliance

• Minor (minor omission in disclosure): Report correction + training • Moderate (concealment of limitations): 3-month suspension + report review • Serious (systematic lack of transparency): 12-month suspension + intensive audit • Critical (fraud in disclosures): Accreditation revocation + legal action

2.7 PRINCIPLE 6: RESPONSIBLE INNOVATION

2.7.1 Definition

Responsible innovation requires that the adoption of new technologies prioritize security, privacy, and reliability over novelty, ensuring that the tools used improve the quality of verifications without introducing unacceptable risks.

2.7.2 Specific Requirements

REQUIREMENT 6.1: Evaluation of Technological Risks
The inspection body MUST:
• Evaluate the security, privacy, and reliability risks of any new technology • Implement technologies only after

validating their effectiveness and security • Maintain contingency plans for technological failures • Document known limitations of the technologies used
REQUIREMENT 6.2: Privacy by Design
The inspection body MUST:
• Design data collection systems with privacy from the outset • Minimize collection of unnecessary personal data • Implement anonymization and pseudonymization techniques • Use Zero-Knowledge Proofs when possible to protect confidentiality
REQUIREMENT 6.3: Scientific Validation
The inspection body MUST:
• Use only scientifically validated technologies • Document the scientific basis of the methods used • Stay updated on advances in technology validation • Reject technologies that lack sufficient validation

2.7.3 Practical Application

CASE 1: Unvalidated Technology
Situation: A provider offers a new AI algorithm for detection of structural defects, but it has not been validated by independent third parties.
Required action: • The body MUST NOT use the algorithm in official inspections • MUST request independent validation of the algorithm • MAY use the algorithm in experimental mode with explicit disclosure to the client • MUST document the results of validation tests
CASE 2: Privacy Risk
Situation: A drone with a high-resolution camera accidentally captures images of neighboring properties during an inspection.
Required action: • The body MUST implement protocols to avoid unauthorized image capture • MUST immediately delete non-relevant images • MUST notify affected parties if there was capture of sensitive information • MUST review and improve privacy protocols

2.7.4 Verification Mechanisms

• Internal technology evaluation committee • Security and privacy audits of technological systems • Review of scientific validation of methods used • Monitoring of technological advances and updating of procedures

2.7.5 Sanctions for Non-Compliance

• Minor (use of non-optimal technology): Training + updating of procedures • Moderate (lack of validation): Suspension of technology use + review • Serious (privacy violation): 6-month suspension + remediation plan • Critical (technological fraud): Accreditation revocation + legal action

2.8 PRINCIPLE 7: COMPREHENSIVE SUSTAINABILITY

2.8.1 Definition

Comprehensive sustainability requires that verifications consider the environmental, social, and governance (ESG) impact of assets, promoting responsible practices and contributing to sustainable development.

2.8.2 Specific Requirements

REQUIREMENT 7.1: Environmental Assessment
The inspection body MUST:
• Evaluate the environmental impact of the asset (carbon footprint, energy consumption, waste management) • Identify environmental risks (contamination, energy efficiency, regulatory compliance) • Recommend

sustainability improvements when applicable • Consider ESG criteria in asset valuation
REQUIREMENT 7.2: Social Impact
The inspection body MUST:
• Evaluate the social impact of the asset (accessibility, community integration, working conditions) • Identify social risks (displacement, inequality, community conflicts) • Consider social responsibility criteria in recommendations • Promote inclusive and equitable practices
REQUIREMENT 7.3: Corporate Governance
The inspection body MUST:
• Evaluate the governance of the asset owner (transparency, ethics, compliance) • Identify governance risks (corruption, conflicts of interest, lack of transparency) • Promote good governance practices in recommendations • Consider governance criteria in risk evaluation

2.8.3 Practical Application

CASE 1: Negative Environmental Impact
Situation: Inspection of a building reveals energy consumption well above average and lack of efficiency measures.
Required action: • The report MUST quantify the environmental impact • MUST recommend improvement measures (insulation, renewable energy, etc.) • MUST consider the impact on asset valuation • MUST inform about compliance with environmental regulations
CASE 2: Social Risk
Situation: Verification of a real estate project reveals it may cause displacement of local communities.
Required action: • The report MUST identify the social risk • MUST recommend mitigation measures (compensation, community integration) • MUST consider the risk in the overall evaluation • MUST inform about compliance with social regulations

2.8.4 Verification Mechanisms

• ESG evaluation checklists in all inspections • Sustainability training for all personnel • ESG impact audits of verified assets • Publication of annual sustainability reports

2.8.5 Sanctions for Non-Compliance

• Minor (omission of ESG evaluation): Training + updating of procedures • Moderate (lack of ESG consideration): Report review + improvement plan • Serious (ignorance of critical ESG risks): 3-month suspension + ESG audit • Critical (complicity in unsustainable practices): Accreditation revocation + legal action

2.9 INTEGRATION OF PRINCIPLES

2.9.1 Holistic Approach

The seven principles do not operate in isolation. Their effective application requires a holistic approach that considers the interrelationships between them.
Example of integration:
An inspection of a $50M yacht must:
• Integrity: Objectively report all defects found • Independence: Resist pressures from the owner to minimize defects • Competence: Have qualified naval inspectors • Confidentiality: Protect the owner's information • Transparency: Clearly communicate inspection limitations • Innovation: Safely use drones and IoT sensors • Sustainability: Evaluate the yacht's environmental impact

2.9.2 Conflict Resolution

When principles conflict, the hierarchy established in Section 2.1.2 applies.
Example of conflict:
Situation: A client demands absolute confidentiality about a critical defect affecting the asset's safety.
Conflict: Confidentiality (Principle 4) vs. Integrity (Principle 1)
Resolution: • Integrity has priority over confidentiality • The body MUST report the critical defect • MAY protect the client's identity if it does not affect public safety • MUST communicate to the client that confidentiality has limits when safety is affected

CHAPTER 3: DEFINITIONS AND TERMINOLOGY

3.1 PURPOSE OF THE GLOSSARY

3.1.1 Need for Terminological Precision

In a standard that integrates traditional disciplines (real estate, naval, and industrial expert assessment) with emerging technologies (blockchain, IoT, AI), terminological precision is critical.
A misunderstood term can lead to: • Errors in the application of protocols • Legal disputes over the scope of verifications • Misunderstanding of report limitations • Failures in physical-digital integration

3.1.2 Structure of the Glossary

The glossary is organized into seven thematic categories:

1. General Terms of the Standard
2. Physical Asset Terms
3. Verification and Inspection Terms
4. Advanced Technological Terms
5. Risk and Compliance Terms
6. Financial and Valuation Terms
7. Governance and Accreditation Terms

3.1.3 Definition Conventions

Each definition includes: • The term in English • A precise and unambiguous definition • Explanatory notes when necessary to clarify scope or limitations • Cross-references to other related terms

3.2 GENERAL TERMS OF THE STANDARD

3.2.1 Fundamental Terms

The Trust Standard (or PTVS) The set of requirements, methodologies, and protocols established by Aurema Group for the comprehensive verification of physical assets in the digital era. Also known as the "Prop Trust Verified Standard."
Verification The systematic process of examining, analyzing, and evaluating a physical asset to determine its existence, state, value, conformity, and risks, in accordance with the Standard's requirements. Note: Verification does not guarantee the total absence of defects or risks, but provides a defined level of certainty according to the applied verification level.

Certification The formal act by which Aurema Group or an accredited body attests that an asset or an inspection body complies with the Standard's requirements.
Accreditation The formal recognition by Aurema Group that an inspection body meets the competence, independence, and impartiality requirements established in the Standard.
The Trust Verified Seal (or PTVS Seal) The certification mark awarded to verified assets in accordance with the Standard, indicating the verification level achieved and the validity date.

3.2.2 Scope Terms

Asset A tangible or intangible good with economic value susceptible to appropriation. In the context of this Standard, it refers exclusively to physical assets.
Physical Asset A tangible good with independent material existence, susceptible to appropriation and economic value. Includes real estate, vessels, machinery, infrastructure, and patrimonial assets.
Tokenized Asset The digital representation of a physical asset through a token on a blockchain, according to standards such as ERC-3643, ERC-721, or ERC-1155.
Real World Asset (RWA) An asset from the real world that physically exists and can be digitally represented through tokenization. Includes real estate, vessels, works of art, raw materials, etc.
Asset Portfolio A set of physical assets managed by the same owner or entity, which can be verified comprehensively according to Level 4 of the Standard.

3.3 PHYSICAL ASSET TERMS

3.3.1 Real Estate Assets

Real Estate Asset A physical asset consisting of land, buildings, or structures fixed to the ground, with independent economic value.
Residential Property A building intended wholly or partially for human housing, including single-family homes, collective residential buildings, luxury residences, and second homes.
Commercial Property A building intended for commercial, service, or office activities, including shopping centers, hotels, and office buildings.
Industrial Property A building intended for industrial, logistics, or production activities, including industrial warehouses, storage facilities, and production plants.
Urban Land Land classified as urban according to current regulations, with access to services and suitable for construction.
Heritage Property A property with recognized historical, artistic, or cultural value, subject to specific legal protections that limit permitted modifications.

3.3.2 Naval Assets

Naval Asset (Marine Asset) A physical asset consisting of vessels, ships, or floating structures with independent economic value.
Yacht A recreational vessel with a length greater than 24 meters, intended for private use or luxury charter.
Megayacht A recreational vessel with a length greater than 50 meters, with professional crew and luxury services.
Commercial Vessel A ship intended for the transport of goods, passengers, or commercial activities, including merchant ships, ferries, and fishing vessels.

Recreational Boat A vessel between 10 and 24 meters in length intended for private or recreational use.
Offshore Platform A fixed or floating structure intended for the exploration or exploitation of marine resources, primarily hydrocarbons.

3.3.3 Industrial and Infrastructure Assets

Industrial Asset A physical asset consisting of machinery, equipment, or facilities intended for industrial or production processes.
Industrial Machinery Mechanical or electromechanical equipment used in production, transformation, or manufacturing processes.
Infrastructure A large-scale physical asset that provides essential services to society, including roads, bridges, ports, airports, and utility networks.
Energy Facility A physical asset intended for the generation, transformation, storage, or distribution of energy, including solar and wind plants, electrical substations, etc.

3.3.4 Patrimonial Assets

Patrimonial Asset (Heritage Asset) A physical asset with historical, artistic, cultural, or collector value, including works of art, antiques, classic vehicles, and high-value jewelry.
Artwork An original artistic creation with economic value, including paintings, sculptures, installations, and digital art with physical representation.
Collector Vehicle A land, air, or sea vehicle with historical or collector value, generally over 25 years old or of limited production.

3.4 VERIFICATION AND INSPECTION TERMS

3.4.1 Verification Process

Inspection The direct physical examination of an asset through observation, measurement, and analysis, performed by qualified personnel according to the Standard's protocols.
Due Diligence The exhaustive and systematic investigation of an asset, including physical, legal, financial, environmental, and regulatory aspects, to evaluate risks and opportunities.
Documentary Due Diligence The exhaustive review of all relevant documentation of the asset, including property titles, licenses, permits, contracts, and regulatory records.
Physical Due Diligence The detailed physical inspection of the asset to verify its state, conformity, and risks, using advanced methodologies and technologies.
Audit The systematic and independent evaluation of an inspection body or a verified asset, to determine compliance with the Standard's requirements.

3.4.2 Personnel and Bodies

Inspection Body A legally constituted entity accredited by Aurema Group to perform verifications according to the Standard, meeting the competence, independence, and impartiality requirements established in Part V.
Accredited Inspector A professional with specific training, experience, and certification to perform verifications under the Standard, contractually linked to an accredited inspection body.

Technical Director The professional with technical responsibility for the verifications performed by an inspection body, ensuring compliance with the Standard's requirements.
Technical Reviewer An independent professional who reviews verification reports before issuance, ensuring technical quality and protocol compliance.
Technical Advisory Board A group of independent experts who advise Aurema Group on the evolution of the Standard, validation of new technologies, and resolution of technical disputes.

3.4.3 Documents and Reports

Verification Report The technical document presenting the verification results, including findings, analysis, conclusions, and recommendations, according to the structure defined in Chapter 13.
Verification Certificate The official document attesting that an asset has been verified according to the Standard, indicating the verification level, issuance date, and validity period.
Digital Twin The dynamic digital representation of a physical asset that updates in real-time through IoT sensor data, enabling simulation, analysis, and continuous monitoring.
Monitoring Dashboard The visual interface presenting the real-time state of the verified asset, including key metrics, alerts, and trends, accessible to the owner and authorized parties.

3.5 ADVANCED TECHNOLOGICAL TERMS

3.5.1 Blockchain and Cryptography

Blockchain A distributed and immutable ledger of transactions and data, organized in cryptographically linked blocks, operating without the need for a central authority.
Smart Contract A computer program that executes automatically on a blockchain when predefined conditions are met, without the need for intermediaries.
Token A unit of digital value registered on a blockchain, which can represent physical assets, property rights, or access to services.
Non-Fungible Token (NFT) A unique and non-replicable token representing the ownership of a digital or physical asset, according to standards such as ERC-721 or ERC-1155.
ERC-3643 A token standard on Ethereum designed specifically for real-world assets (RWA), incorporating compliance modules and identity management.
Decentralized Identifier (DID) A unique identifier cryptographically verifiable that does not require a centralized registry, allowing persons or organizations to demonstrate their identity and credentials in a sovereign manner.
Verifiable Credential A digitally signed cryptographic assertion that can be independently verified, used to demonstrate qualifications, certifications, or attributes.

3.5.2 Signature and Cryptographic Verification

Edge Signing The cryptographic process by which a device (sensor, drone, camera) digitally signs data at the moment of capture, before any transmission or subsequent processing, guaranteeing the authenticity and integrity of the data from the origin.
Digital Chain of Custody An immutable and chronological record of all manipulations, transfers, and accesses to a digital data or asset, from its creation to its final destination, guaranteeing complete traceability.
Cryptographic Hash A mathematical function that converts data of arbitrary size into a fixed-length string, such that any modification of the original data produces a completely different hash, allowing verification of information integrity.
Zero-Knowledge Proof (ZKP) A cryptographic protocol that allows one party to prove to another that a statement is true without revealing any information other than the validity of the statement itself.
Oracle A mechanism or service connecting real-world data with smart contracts on a blockchain, enabling automatic execution based on externally verified conditions.
Compliance Oracle A specialized oracle that injects compliance rules into smart contracts, enabling automatic execution of actions (pauses, alerts, restrictions) when specific conditions are detected in the verified physical asset.

3.5.3 Inspection Technologies

LiDAR (Light Detection and Ranging) A remote sensing technology that uses laser pulses to measure distances and create precise three-dimensional models of objects and terrains, used in asset inspection to generate point clouds and 3D models.

Infrared Thermography A non-destructive inspection technique using infrared cameras to detect and visualize temperature distribution on an asset's surface, allowing identification of defects, leaks, or insulation problems.
Drone (Unmanned Aerial Vehicle - UAV) An uncrewed aircraft equipped with sensors (cameras, LiDAR, thermal) used for remote inspection of assets, especially in hard-to-reach or high-risk areas.
IoT Sensor An electronic device that measures physical variables (temperature, humidity, vibration, pressure, etc.) and transmits the data via the internet for real-time monitoring and analysis.

3.5.4 Standard Protocols

SCCIP (Smart Contract Compliance Integration Protocol) The protocol defined in the Standard (Chapter 10) for integrating physical verifications with compliance modules in smart contracts, particularly ERC-3643, enabling automatic execution of actions when critical risks are detected in the physical asset.
CPEP (Cryptographic Proof of Existence Protocol) The protocol defined in the Standard (Chapter 11) for cryptographic verification of the existence and state of physical assets through edge signing and digital chain of custody, guaranteeing the authenticity and integrity of inspection data.
CAP (Continuous Assurance Protocol) The continuous monitoring protocol via IoT sensors that provides real-time updates of the asset's state and automatic alerts for deviations, defined in the Standard (Chapter 9).
PRA (Predictive Risk Assessment) The risk analysis methodology based on artificial intelligence and predictive models that anticipates future risks and models scenarios, defined in the Standard (Chapter 12).
APV (Authenticity & Provenance Verification) The protocol for verifying the authenticity and provenance of assets,

especially relevant for patrimonial and high-value assets, defined in the Standard (Chapter 12).
PDBP (Physical-Digital Bridge Protocol) The protocol establishing the cryptographic link between the physical asset and its digital representation (token), guaranteeing the correspondence between both worlds, defined in the Standard (Chapter 10).

3.6 RISK AND COMPLIANCE TERMS

3.6.1 Risk Classification

Risk The possibility of an event or condition occurring that negatively affects the existence, integrity, value, or conformity of a physical asset.
Critical Risk A condition that threatens the existence, integrity, or legal value of the asset immediately, requiring urgent corrective action and potentially automatic transfer pause in tokenized assets.
High Risk A condition that may significantly affect the value or functionality of the asset in the medium term, requiring priority corrective action and manual approval for transfers in tokenized assets.
Medium Risk A condition that may affect the value or functionality of the asset in the long term, requiring monitoring and planning of corrective actions.
Low Risk A minor condition that does not significantly affect the value or functionality of the asset, but must be documented for owner awareness.

3.6.2 Compliance Terms

Compliance The conformity of an asset, body, or process with applicable legal, regulatory, contractual, and Standard requirements.
Programmable Compliance The ability to inject compliance rules directly into smart contracts, enabling

automatic execution of actions (pauses, alerts, restrictions) when specific conditions are detected.
KYC (Know Your Customer) The process of verifying the identity of clients and users, mandatory in the financial sector to prevent money laundering and terrorist financing.
AML (Anti-Money Laundering) The set of norms, regulations, and procedures aimed at preventing and detecting money laundering.
GDPR (General Data Protection Regulation) The European data protection regulation establishing strict requirements for the processing of personal data of individuals in the European Union.
MiCA (Markets in Crypto-Assets Regulation) The European regulation establishing a regulatory framework for crypto-assets, including tokens representing physical assets (RWA).

3.6.3 Governance Terms

Governance The system of rules, processes, and structures by which an inspection body or a verified asset is directed and controlled.
Conflict of Interest A situation in which the personal, financial, or commercial interests of an inspector or inspection body could compromise the objectivity and independence of the verification.
Impartiality The absence of biases, prejudices, or undue influences in the performance of verifications, ensuring that conclusions are based exclusively on technical evidence.
Independence The structural, financial, and technical autonomy of an inspection body from interested parties, ensuring that professional decisions are not influenced by external pressures.

3.7 FINANCIAL AND VALUATION TERMS

3.7.1 Valuation Terms

Valuation The process of estimating the economic value of a physical asset through recognized methodologies and verifiable market data.
Market Value The estimated price for which an asset would be exchanged in a transaction between willing, knowledgeable, and uncoerced parties, after adequate marketing, under normal market conditions.
Liquidation Value The estimated value of an asset in the event of a forced or urgent sale, generally lower than market value due to time pressure.
Replacement Value The cost of acquiring or constructing an equivalent asset with the same characteristics and functionality, under current market conditions.
Residual Value The estimated value of an asset at the end of its economic useful life or at the end of a determined investment period.

3.7.2 Investment Terms

ROI (Return on Investment) The ratio measuring the profitability of an investment, calculated as net profit divided by investment cost.
Cap Rate (Capitalization Rate) The ratio used in real estate valuation, calculated as net operating income divided by the market value of the asset.
IRR (Internal Rate of Return) The discount rate that makes the net present value of an investment's cash flows equal to zero, used to evaluate project profitability.

Financial Due Diligence The exhaustive analysis of the financial situation of an asset or company, including income, expenses, debts, and future projections.
Stress Testing The analysis of extreme scenarios to evaluate an asset's or portfolio's resilience under adverse conditions (economic crises, natural disasters, regulatory changes).

3.8 GOVERNANCE AND ACCREDITATION TERMS

3.8.1 Accreditation Terms

Accreditation The formal recognition by Aurema Group that an inspection body meets the competence, independence, and impartiality requirements established in the Standard.
Accreditation Audit The systematic and independent evaluation of an inspection body to determine compliance with the Standard's requirements before granting or renewing accreditation.
Surveillance Audit The periodic audit performed on accredited inspection bodies to verify continuous compliance with the Standard's requirements.
Sanction The disciplinary measure imposed on an accredited inspection body or inspector for non-compliance with the Standard's requirements, which may include warnings, suspensions, or accreditation revocation.
Revocation The withdrawal of accreditation from an inspection body or inspector for serious or critical non-compliance with the Standard's requirements.

3.8.2 Quality Terms

Quality Management System (QMS) The set of processes, procedures, and responsibilities established by an inspection body to ensure consistent quality of verifications performed.
Continuous Improvement The systematic process of evaluating and optimizing the processes, methods, and results of an inspection body to increase the quality and efficiency of verifications.
Non-Conformity The non-fulfillment of a specific requirement of the Standard, classified as minor, major, or critical according to its severity and impact.
Corrective Action The action taken to eliminate the cause of an identified non-conformity and prevent its recurrence.
Preventive Action The action taken to eliminate the cause of a potential non-conformity and prevent its occurrence.

3.9 ABBREVIATIONS AND ACRONYMS

3.9.1 Standard Abbreviations

PTVS: Prop Trust Verified Standard SCCIP: Smart Contract Compliance Integration Protocol CPEP: Cryptographic Proof of Existence Protocol CAP: Continuous Assurance Protocol PRA: Predictive Risk Assessment APV: Authenticity & Provenance Verification PDBP: Physical-Digital Bridge Protocol DID: Decentralized Identifier RWA: Real World Asset

3.9.2 Technological Abbreviations

IoT: Internet of Things AI: Artificial Intelligence ML: Machine Learning LiDAR: Light Detection and Ranging

UAV: Unmanned Aerial Vehicle NFT: Non-Fungible Token ZKP: Zero-Knowledge Proof RBAC: Role-Based Access Control

3.9.3 Normative and Regulatory Abbreviations

ISO: International Organization for Standardization RICS: Royal Institution of Chartered Surveyors IIMS: International Institute of Marine Surveying MiCA: Markets in Crypto-Assets Regulation GDPR: General Data Protection Regulation AML: Anti-Money Laundering KYC: Know Your Customer ESG: Environmental, Social, and Governance

3.9.4 Financial Abbreviations

ROI: Return on Investment IRR: Internal Rate of Return NPV: Net Present Value EBITDA: Earnings Before Interest, Taxes, Depreciation, and Amortization HVAC: Heating, Ventilation, and Air Conditioning

CHAPTER 4: STANDARD GOVERNANCE

4.1 PURPOSE OF GOVERNANCE

4.1.1 Importance of Robust Governance

The credibility of the Prop Trust Verified Standard (PTVS) depends fundamentally on the robustness of its governance structure. A technically brilliant standard that is poorly governed quickly loses market confidence, especially in the institutional realm where Family Offices, sovereign wealth funds, and reinsurers demand guarantees of independence, transparency, and rigor. The governance of the Standard must guarantee: • Independence of technical decisions from commercial interests • Transparency in update processes and dispute resolution • Scientific rigor in the validation of new methodologies and technologies • Representativeness of the diverse stakeholders in the ecosystem • Agility to adapt to rapid technological evolution • Integrity in protecting the value and meaning of the Trust Verified Seal

4.1.2 Governance Principles

The governance of the Standard is governed by the following principles:

1. Separation of Powers: The functions of technical development, accreditation, auditing, and dispute resolution are separated to avoid conflicts of interest.

2.Multistakeholder Participation: The Technical Advisory Board includes representatives from all relevant sectors (real estate, naval, industrial, technological, financial, legal).
3.Radical Transparency: Processes, decisions, and conflicts are documented and made public, except when confidentiality requires otherwise.
4.Scientific Basis: Every update to the Standard must be backed by empirical evidence, technical validation, and expert consensus.
5.Protection of Public Interest: The Standard prioritizes the protection of investors, owners, and society over the commercial interests of Aurema Group.

4.1.3 Scope of Governance

The governance of the Standard encompasses: • Development and updating of technical content • Accreditation of inspection bodies • Supervision of compliance by accredited bodies • Resolution of technical and commercial disputes • Management of the brand and the Trust Verified Seal • Relations with external normative bodies (ISO, RICS, regulators) • Protection of the intellectual property of the Standard

4.2 ORGANIZATIONAL STRUCTURE

4.2.1 Governance Architecture

The governance of the Standard is structured into four levels with clearly differentiated responsibilities:
LEVEL 1: STRATEGIC GOVERNANCE • Standard Governing Board • Aurema Group Executive Management • Responsible for strategic vision and critical decisions

LEVEL 2: TECHNICAL SUPERVISION • Technical Advisory Board (TAB) • Specialized Technical Committees • Responsible for scientific and technical validity
LEVEL 3: OPERATIONS AND COMPLIANCE • Accreditation Department • Audit and Surveillance Department • Dispute Resolution Department • Responsible for operational execution
LEVEL 4: ECOSYSTEM PARTICIPATION • Accredited inspection bodies • Standard users (Family Offices, funds, RWA platforms) • External observers and collaborators • Responsible for feedback and continuous improvement

4.2.2 Separation of Functions

To guarantee independence and impartiality, the following functions are strictly separated:

Function	Responsible	Required Independence
Standard Development	Technical Advisory Board	Technical independence
Body Accreditation	Accreditation Department	Commercial independence
Body Auditing	Audit Department	Total independence
Dispute Resolution	Dispute Committee	Judicial independence
Commercial Management	Aurema Group	Separated from technical functions

CRITICAL PROHIBITION: No person or organization may simultaneously participate in: • Standard development and body accreditation • Accreditation and auditing of the same body • Provision of verification services and auditing of that verification

4.2.3 Legal Structure

The Standard is legally protected through: • Copyright of the content (registered in the Intellectual Property

Registry) • Trademark registration of "Prop Trust Verified Standard" (EUIPO, USPTO, WIPO) • Patents for specific technical processes (SCCIP, CPEP) where applicable • Usage licenses for accredited bodies and users • Confidentiality agreements with all participants
Aurema Group Holdings OÜ (Estonia) is the entity holding the rights to the Standard, with Aurema Group LLC (Delaware) acting as the protector of intellectual property in common law jurisdictions.

4.3 TECHNICAL ADVISORY BOARD (TAB)

4.3.1 Composition of the TAB

The Technical Advisory Board is the supreme body for technical supervision of the Standard. It is composed of 12 members with the following distribution:
PERMANENT MEMBERS (4):

1.Technical Director of Aurema Group (TAB President) Responsibility: General coordination and institutional representation Profile: 20+ years of experience in expert assessment and asset verification
2.Director of Technological Innovation Responsibility: Supervision of technological protocols (SCCIP, CPEP, CAP) Profile: Expert in blockchain, IoT, AI, and cryptography
3.Director of Quality and Accreditation Responsibility: Supervision of accreditation and audit processes Profile: Expert in quality management systems (ISO 17020)
4.Legal and Compliance Director Responsibility: Supervision of regulatory and legal aspects Profile:

Expert in real estate, naval, MiCA, and financial regulation law

INDEPENDENT MEMBERS (8): 5-6. Real Estate Sector Representatives Profiles: RICS Fellow, internationally renowned architect Designation: Proposed by professional associations, appointed by TAB

7-8. Naval Sector Representatives Profiles: IIMS Fellow, naval engineer with megayacht experience Designation: Proposed by professional associations, appointed by TAB

9.Industrial and Infrastructure Sector Representative Profile: Industrial engineer with experience in large assets Designation: Proposed by engineering associations

10.Financial and Family Office Sector Representative Profile: Family Office Director or ultra-high-net-worth wealth manager Designation: Proposed by Family Office associations (e.g., FOX, Campden Wealth)

11.Technological and Blockchain Sector Representative Profile: Expert in RWA tokenization, smart contracts, oracles Designation: Proposed by blockchain consortia (e.g., Token Alliance)

12.Academic and Scientific Sector Representative Profile: University professor in engineering, architecture, or applied sciences Designation: Proposed by prestigious universities

4.3.2 Designation Process

For independent members:

1.Public call for candidacies every 3 years

2.Evaluation by an independent selection committee

3.Verification of credentials and absence of conflicts of interest

4.Approval by qualified majority (2/3) of the existing TAB

5.Formal appointment by the Aurema Group Governing Board

6.Publication in the public registry of the Standard Mandate: • Duration: 3 years, renewable once • Staggered rotation: 4 members are renewed every 18 months • Maximum limit: 2 consecutive terms (6 years)

4.3.3 Functions of the TAB

MANDATORY FUNCTIONS:

1.Approval of Standard updates • Review and approval of all technical modifications • Validation of new protocols and methodologies • Approval by qualified majority (2/3)

2.Supervision of accreditation • Review of complex or controversial cases • Approval of accreditation criteria • Resolution of appeals from rejected bodies

3.Technological validation • Evaluation of new technologies for integration into the Standard • Scientific validation of inspection methods • Approval of reference technical standards

4.Resolution of technical disputes • Arbitration in disputes between inspection bodies • Resolution of conflicts over Standard interpretation • Issuance of binding technical opinions

5.Quality supervision • Review of audit reports • Analysis of trends and patterns of non-conformities • Recommendation of systemic corrective actions

ADVISORY FUNCTIONS: 6. Strategic advice • Recommendations on market evolution • Identification of opportunities and threats • Proposals for expansion into new sectors

7.Institutional representation • Participation in international forums (ISO, RICS, IIMS) • Representation of the Standard at conferences and

events • Relations with regulators and normative bodies

4.3.4 Operation of the TAB

Meetings: • Ordinary: Quarterly (in-person or virtual) • Extraordinary: When requested by the President or 4 members • Minimum quorum: 8 members (including at least 2 independents)
Decision-making: • Ordinary decisions: Simple majority • Critical decisions (Standard updates, serious sanctions): Qualified majority (2/3) • Strategic decisions (governance changes, expansions): Consensus or supermajority (3/4)
Confidentiality: • TAB members sign strict confidentiality agreements • Deliberations are confidential; decisions are public • Conflicts of interest must be declared before each vote
Compensation: • Independent members receive compensation for meeting attendance • Compensation is fixed, independent of the Standard's business volume • Travel and accommodation expenses are reimbursed

4.3.5 Specialized Technical Committees

The TAB may create Specialized Technical Committees to address specific areas:
REAL ESTATE ASSETS COMMITTEE • Composition: 5 real estate experts • Function: Development of specific protocols for real estate • Reports to: TAB quarterly
NAVAL ASSETS COMMITTEE • Composition: 5 naval asset experts • Function: Development of specific protocols for naval assets • Reports to: TAB quarterly
ADVANCED TECHNOLOGIES COMMITTEE • Composition: 5 experts in blockchain, IoT, AI • Function: Development of SCCIP, CPEP, CAP protocols • Reports to: TAB monthly
RISK AND COMPLIANCE COMMITTEE • Composition: 5 experts in risk, compliance, legal • Function: Development

of risk matrices and compliance protocols • Reports to: TAB quarterly
SUSTAINABILITY AND ESG COMMITTEE • Composition: 5 experts in sustainability, environment, ESG • Function: Integration of ESG criteria into the Standard • Reports to: TAB quarterly

4.4 STANDARD UPDATE PROCESS

4.4.1 Philosophy of Updates

The Prop Trust Verified Standard is a living standard that must evolve with: • Technological advances (new capabilities in AI, IoT, blockchain) • Regulatory evolution (MiCA, national regulations, ISO standards) • Market needs (new asset types, new use cases) • Stakeholder feedback (accredited bodies, users, regulators) • Lessons learned (non-conformities, disputes, best practices)

4.4.2 Types of Updates

MAJOR UPDATES (Versions X.0) • Structural changes to the Standard • New fundamental protocols or methodologies • Modifications to guiding principles • Frequency: Every 3-5 years • Process: Complete review with public consultation • Example: PTVS 2.0, PTVS 3.0
MINOR UPDATES (Versions X.Y) • Refinements of existing protocols • New asset typologies • Updates to normative references • Frequency: Annual • Process: TAB review with limited consultation • Example: PTVS 1.1, PTVS 1.2
TECHNICAL UPDATES (Versions X.Y.Z) • Error corrections • Clarification of ambiguities • Updates to technological references • Frequency: Quarterly • Process: Approval by Technical Director • Example: PTVS 1.1.1, PTVS 1.1.2

4.4.3 Major Update Process

PHASE 1: IDENTIFICATION OF NEEDS (Months 1-3)

1.Collection of feedback from all stakeholders • Surveys to accredited bodies • Consultations with Standard users • Analysis of non-conformities and disputes • Review of technological and regulatory advances
2.Preliminary analysis by the Technical Director • Identification of critical areas for improvement • Evaluation of technical and commercial feasibility • Prioritization of proposals
3.Update proposal to the TAB • Scope and objectives document • Impact analysis on accredited bodies • Resource and timeline estimation

PHASE 2: TECHNICAL DEVELOPMENT (Months 4-9) 4. Creation of a specific working group • 5-7 experts from the TAB and external collaborators • Representation from all affected sectors • Coordination by a TAB member 5. Drafting of technical proposals • Development of new protocols or refinements • Scientific and technical validation • Impact and compatibility analysis 6. Internal review by the TAB • Detailed technical evaluation • Identification of conflicts or inconsistencies • Adjustments and refinements

PHASE 3: PUBLIC CONSULTATION (Months 10-12) 7. Publication of the draft for public consultation • Available on the Standard's portal • Consultation period: 60 days • Open access to all stakeholders 8. Collection of comments • Structured feedback reception system • Categorization by type (technical, commercial, regulatory) • Quantitative and qualitative analysis 9. Final review and adjustments • Integration of valid comments • Justification of rejected comments • Final version for approval

PHASE 4: APPROVAL AND LAUNCH (Months 13-15) 10. Approval by the TAB • Voting with qualified majority (2/3) • Detailed minutes of the vote • Publication of the decision 11. Launch preparation • Final layout and design • Translation into key languages (English, Spanish, French, German, Chinese) • Communication and training materials 12. Official launch • Publication on the Standard's portal • Communication to all stakeholders • Transition period for accredited bodies (6-12 months)

4.4.4 Transition Period

When a major update is published: • Transition period: 12 months from publication • During this period: • Accredited bodies may operate under the previous or new version • Aurema Group provides training and resources for the transition • Transition audits are conducted to verify compliance • End of period: • All bodies must comply with the new version • Certificates issued under the previous version maintain validity until expiration • New verifications must follow the new version

4.4.5 Version Control

All versions of the Standard are kept available on the public portal: • Current version: The version in force and mandatory for compliance • Previous versions: Archived with read-only access • Drafts in consultation: Available during the public consultation period • Change log: Detailed document of all modifications between versions Each version includes: • Version number (X.Y.Z) • Publication date • Summary of main changes • List of authors and contributors • TAB approval

4.5 DISPUTE RESOLUTION MECHANISMS

4.5.1 Typology of Disputes

The Standard recognizes four categories of disputes: TECHNICAL DISPUTES • Disagreements over the interpretation of technical protocols • Conflicts over the application of methodologies • Disputes over the validity of technical conclusions • Example: Two inspectors reach different conclusions about an asset's state ACCREDITATION DISPUTES • Appeals from bodies rejected in the accreditation process • Conflicts over sanctions or revocations of accreditation • Disputes over the scope of accreditation • Example: A body considers the suspension of its accreditation unfair COMMERCIAL DISPUTES • Conflicts between inspection bodies and clients • Disputes over fees or scope of services • Claims for alleged professional negligence • Example: A client considers the report does not reflect the asset's reality BRAND AND SEAL DISPUTES • Unauthorized use of the Trust Verified Seal • Disputes over the validity of certificates • Conflicts over the reputation of the Standard • Example: An entity uses the seal without being accredited

4.5.2 Committee for Dispute Resolution (CDR)

Composition: • 5 independent members appointed by the TAB • Profiles: Legal, technical, commercial, ethical, sectoral • Mandate: 3 years, not consecutively renewable • Total independence from Aurema Group and the disputing parties

Functions: • Resolution of disputes of any category • Issuance of binding opinions or recommendations • Imposition of administrative sanctions • Mediation between conflicting parties

4.5.3 Resolution Process

PHASE 1: SUBMISSION OF DISPUTE (Days 1-5)

1.Formal submission by the affected party • Standardized form available on the portal • Detailed description of the dispute • Relevant documentary evidence • Identification of the parties involved
2.Admission to proceedings by the CDR • Verification of CDR competence • Confirmation that internal channels have been exhausted • Notification to the other party
3.Precautionary measure (if applicable) • Precautionary suspension of certificates or accreditations • Protection of critical evidence • Measures to prevent irreparable damage

PHASE 2: INVESTIGATION (Days 6-30) 4. Collection of evidence • Request for documentation from the parties • Interviews with relevant witnesses • Independent technical analysis (if applicable) • Third-party expert opinions 5. Preliminary analysis • Identification of key facts • Evaluation of the applicability of the Standard • Determination of whether there is a breach of principles 6. Hearing of the parties (if necessary) • In-person or virtual session • Presentation of arguments by both parties • Questions from the CDR

PHASE 3: RESOLUTION (Days 31-45) 7. Deliberation of the CDR • Collective analysis of the evidence • Application of the Standard's principles • Voting by simple majority 8. Issuance of the opinion • Motivated and detailed resolution • Determination of responsibilities • Imposition of sanctions or corrective measures • Deadline for compliance 9. Notification to the parties • Formal communication of the

resolution • Explanation of available appeals • Anonymized publication in the public registry
PHASE 4: COMPLIANCE AND APPEALS (Days 46-60)
10. Compliance with the resolution • Parties must comply within the established deadline • Verification of compliance by the CDR • Sanctions for non-compliance 11. Appeals (if applicable) • Appeal to the TAB in exceptional cases • Deadline: 15 days from notification • Only for procedural errors or incorrect interpretation of the Standard • TAB decision is final

4.5.4 Sanctions

The CDR may impose the following sanctions:
ADMINISTRATIVE SANCTIONS: • Private reprimand • Public reprimand • Temporary suspension of accreditation (3-12 months) • Permanent revocation of accreditation • Prohibition of use of the Trust Verified Seal
ECONOMIC SANCTIONS: • Fines from €5,000 to €100,000 • Compensation to affected parties • Loss of deposits or guarantees
REPUTATIONAL SANCTIONS: • Publication of the resolution in the public registry • Notification to relevant regulatory bodies • Communication to professional associations

4.5.5 Public Registry of Disputes

All CDR resolutions are published in a public registry that includes: • Case reference number • Resolution date • Category of the dispute • Summary of facts (anonymized) • Resolution and reasoning • Sanctions imposed
Exceptions: • Cases involving confidential client information • Cases where parties request anonymity for legitimate reasons • Cases where publication could cause disproportionate harm

4.6 TRANSPARENCY AND PUBLIC REPORTING

4.6.1 Philosophy of Transparency

Transparency is a fundamental principle of the Standard (see Chapter 2, Section 2.6). The governance of the Standard must be exemplary in this principle, providing complete and accessible information about: • Strategic and technical decisions • Composition and operation of governance bodies • Results of accreditations and audits • Resolution of disputes and sanctions • Evolution of the Standard and updates • Impact and reach of the Standard in the market

4.6.2 Mandatory Public Reports

ANNUAL GOVERNANCE REPORT Published every year in March, it includes: • Executive summary of the year • Activities of the TAB and technical committees • Standard updates carried out and planned • Accreditation statistics: • Number of accredited bodies • New accreditations granted • Suspended or revoked accreditations • Geographical and sectoral distribution • Audit statistics: • Number of audits performed • Non-conformities identified • Corrective actions required • Trends and patterns • Dispute resolution: • Number and type of disputes • Resolutions issued • Sanctions imposed • Financial evolution: • Income from accreditations and licenses • Investment in Standard development • Compensation of TAB members • Perspectives for the following year QUARTERLY ACTIVITY REPORT Published every quarter, it includes: • Summary of quarterly activities • New

accreditations granted • Technical updates published • Disputes resolved • Relevant events and conferences
PUBLIC REGISTRY OF ACCREDITED BODIES Updated in real-time, it includes: • Complete list of accredited bodies • Scope of accreditation (asset typologies, levels) • Accreditation date and expiration • Status (active, suspended, revoked) • History of audits and non-conformities (summarized)
PUBLIC REGISTRY OF CERTIFICATES Updated in real-time, it includes: • Verification of certificate validity • Verification level • Issuance and expiration dates • Inspection body that issued the certificate • Cryptographic hash for on-chain verification (Levels 3 and 4)

4.6.3 Access to Information

PUBLIC INFORMATION (Free access): • Full text of the Standard (current and previous versions) • Annual and quarterly reports • Registries of accredited bodies and certificates • Dispute resolutions (anonymized) • Summarized TAB minutes (without confidential information) • Training materials and guides
RESTRICTED INFORMATION (Registration required): • Drafts in public consultation • Detailed statistical data • Specific technical information for accredited bodies • Access to exclusive webinars and events
CONFIDENTIAL INFORMATION (Limited access): • Information on clients of inspection bodies • Details of unpublished audits • Internal TAB deliberations • Sensitive commercial information of Aurema Group

4.6.4 Feedback Mechanisms

The Standard provides multiple channels for stakeholder feedback:
CHANNEL 1: Web Portal • General contact form • Ticketing system for technical queries • Moderated discussion forum • Periodic surveys

CHANNEL 2: Stakeholder Meetings • Annual meetings with accredited bodies • Public consultations for major updates • Focus groups with Standard users • Sectoral conferences and events
CHANNEL 3: Confidential Reporting • Anonymous system for reporting non-compliance • Whistleblower protection • Independent investigation of reports • Communication of results to the whistleblower

4.7 RELATIONSHIP WITH EXTERNAL BODIES

4.7.1 International Normative Bodies

ISO (International Organization for Standardization) The Standard aligns with and complements ISO norms, particularly: • ISO 17020: Criteria for the operation of inspection bodies • ISO 9001: Quality management systems • ISO 14001: Environmental management systems • ISO 45001: Occupational health and safety management systems • ISO 27001: Information security management systems
Strategy of relationship: • Active participation in relevant ISO technical committees • Contribution to the development of new ISO norms • Mutual recognition of accreditations • Possible future proposal of PTVS as an ISO norm (via PAS in BSI/AENOR)
RICS (Royal Institution of Chartered Surveyors) The Standard complements the RICS Valuation Standards (Red Book): • PTVS focuses on comprehensive verification, not just valuation • PTVS incorporates advanced technologies not covered by RICS • PTVS provides the physical-digital bridge for tokenization

Strategy of relationship: • Mutual recognition of competencies • Participation in RICS events and conferences • Possible mutual recognition agreements • Collaboration in the development of sectoral standards
IIMS (International Institute of Marine Surveying) The Standard complements IIMS standards for marine inspection: • PTVS provides a unified multi-asset framework • PTVS incorporates advanced technological protocols • PTVS enables the tokenization of naval assets
Strategy of relationship: • Recognition of IIMS certifications as valid for PTVS • Joint participation in the development of naval standards • Collaboration in the training of naval inspectors

4.7.2 Regulatory Bodies

Financial Market Authorities: • ESMA (Europe): Market regulation, MiCA • FCA (UK): Post-Brexit financial regulation • SEC (USA): Securities regulation • ADGM/DIFC (Middle East): Special financial zones
Strategy of relationship: • Proactive compliance with applicable regulations • Participation in regulatory public consultations • Contribution to the development of frameworks for tokenized RWAs • Recognition of PTVS as a verification standard for RWAs
National Accreditation Bodies: • ENAC (Spain): Accreditation of inspection bodies • UKAS (UK): British accreditation • COFRAC (France): French accreditation • DAkkS (Germany): German accreditation
Strategy of relationship: • Alignment of PTVS accreditation requirements with ISO 17020 • Possible mutual recognition of accreditations • Participation in accreditation body forums (ILAC, EA)

4.7.3 Professional and Sectoral Associations

Family Office Associations: • Family Office Exchange (FOX) • Campden Wealth • Family Office Club

Strategy: Position PTVS as the reference standard for patrimonial asset verification.
Tokenization and Blockchain Associations: • Token Alliance • Global Digital Finance (GDF) • RWA working groups of various blockchains
Strategy: Position PTVS as the institutional trust layer for tokenized RWAs.
Insurance and Reinsurance Associations: • Lloyd's of London • International Underwriting Association (IUA) • Association of British Insurers (ABI)
Strategy: Position PTVS as the standard for verifying high-value insured assets.

4.7.4 Mutual Recognition Agreements (MRA)

The Standard may establish Mutual Recognition Agreements with other standards or bodies, under the following conditions:
REQUIREMENTS FOR MRA:

1. Technical equivalence: The other standard must have equivalent or superior technical requirements
2. Verified independence: The other body must demonstrate independence and impartiality
3. Transparency: The other body must have comparable transparency practices
4. Regulatory recognition: The other standard must be recognized by relevant regulators
5. Reciprocity: The recognition must be mutual and equitable

EFFECTS OF MRA: • Bodies accredited by the other standard may obtain simplified PTVS accreditation • Certificates issued under the other standard may be recognized as equivalent • Joint participation in standard development • Joint representation in international forums

4.8 PROTECTION OF STANDARD INTEGRITY

4.8.1 Protection of the Brand and Seal

Trademark "Prop Trust Verified Standard": • Registered in EUIPO (Europe), USPTO (USA), WIPO (international) • Classes: 35 (consulting), 36 (financial services), 42 (technical services), 45 (legal services) • Legal protection against unauthorized use
Trust Verified Seal: • Graphic design registered as a trademark • Exclusive use by accredited bodies • Strict technical specifications for use • Active monitoring of unauthorized uses

4.8.2 Market Monitoring

Monitoring Program:

1. Digital monitoring: • Periodic searches on the internet and social media • Automatic alerts for use of protected terms • Analysis of tokenization platforms and marketplaces
2. Sectoral monitoring: • Participation in conferences and events • Relations with professional associations • Feedback from accredited bodies and users
3. Legal monitoring: • Review of trademark and patent registries • Monitoring of technical publications • Analysis of competing standards

4.8.3 Action Against Unauthorized Use

UNAUTHORIZED USE OF THE SEAL:

1. Detection: Identification of unauthorized use

2.Investigation: Verification of the facts
3.Notification: Formal communication to the infringer
4.Demand: Requirement of immediate cessation
5.Legal action: If there is no compliance, legal action for trademark infringement

INCORRECT USE BY ACCREDITED BODIES:

1.Detection: Identification of incorrect use
2.Notification: Communication to the body
3.Correction: Requirement of correction within a specified deadline
4.Sanction: If there is no correction, administrative sanction
5.Revocation: In serious cases, revocation of accreditation

IMITATION OR COPYING OF THE STANDARD:

1.Detection: Identification of standards copying PTVS
2.Analysis: Evaluation of the degree of similarity
3.Legal action: If there is copyright or patent infringement, legal action
4.Differentiation: Communication to the market about the differences
5.Evolution: Acceleration of innovation to maintain leadership

4.8.4 Protection of Intellectual Property

Copyright of Content: • Registration in the Intellectual Property Registry (Spain) • Registration in the Copyright Office (USA) • Automatic protection in 181 countries (Berne Convention) • Usage licenses for accredited bodies and users

Patents for Technical Processes: • SCCIP (Smart Contract Compliance Integration Protocol) • CPEP (Cryptographic Proof of Existence Protocol) • Other technically novel

processes • Protection in key jurisdictions (Europe, USA, Asia)
Trade Secrets: • Proprietary AI predictive algorithms • Model training data • Unpublished internal methodologies • Protection through NDAs and technical measures

4.8.5 Reputation Defense

Reputation Protection Program:

1. Reputation monitoring: • Sentiment analysis in media and social networks • Identification of criticisms or controversies • Detection of smear campaigns
2. Rapid response: • Protocol for responding to reputational crises • Transparent and proactive communication • Collaboration with media and stakeholders
3. Legal action: • Against defamation or slander • Against unfair competition • Against improper use of the brand

CHAPTER 5: PHASE 1 - DOCUMENTARY DUE DILIGENCE

5.1 PURPOSE AND SCOPE OF PHASE 1

5.1.1 Critical Importance of Documentary Verification

Phase 1: Documentary Due Diligence constitutes the fundamental foundation upon which the entire verification process is built. A physically perfect asset with documentary defects represents an unacceptable risk for any institutional investor.
Experience demonstrates that 67% of critical problems in high-value asset transactions originate in documentary deficiencies, not physical defects. Therefore, the Prop Trust Verified Standard grants this phase the same importance as physical inspection.

5.1.2 Objectives of Phase 1

Phase 1 pursues five interconnected objectives:

OBJECTIVE 1: Verification of Ownership Confirm that the seller or owner holds valid and complete legal title to the asset, free of disputes or third-party claims.
OBJECTIVE 2: Identification of Charges and Encumbrances Detect any limitation on property rights, including mortgages, embargoes, easements, usufructs, or contractual restrictions.
OBJECTIVE 3: Validation of Regulatory Compliance Verify that the asset complies with all applicable regulations (urban planning, environmental, safety, tax).
OBJECTIVE 4: Analysis of Legal Risks Identify potential legal risks that may affect the value or transferability of the asset.
OBJECTIVE 5: Preparation for Physical Verification Provide the necessary documentary basis to plan and execute the physical inspection efficiently.

5.1.3 Scope by Asset Typology

Phase 1 applies to all asset typologies covered by the Standard, with specific adaptations:
• Real Estate Residential: Deed, cadastre, licenses (Medium complexity) • Real Estate Commercial: Deed, activity licenses, lease contracts (High complexity) • Real Estate Industrial: Deed, environmental licenses, operation permits (Very High complexity) • Yacht/Megayacht: Naval registry, flag certificate, insurance (High complexity) • Commercial Vessel: Registry, class certificates, international conventions (Very High complexity) • Industrial Machinery: Invoices, conformity certificates, maintenance logs (Medium complexity) • Artwork: Provenance, authenticity certificates, export permits (Very High complexity)

5.2 GENERAL PROTOCOL FOR DOCUMENTARY VERIFICATION

5.2.1 Guiding Principles

PRINCIPLE 1: COMPLETENESS The inspection body SHALL obtain and verify ALL relevant documentation, without omissions. The lack of a critical document SHALL be reported as a non-conformity.
PRINCIPLE 2: AUTHENTICITY All documentation SHALL be verified as authentic through comparison with official registries, verification of signatures and seals, internal consistency analysis, and cryptographic validation (when applicable).
PRINCIPLE 3: VALIDITY All documentation SHALL be valid on the date of verification. Expired or pending renewal documents SHALL be identified and reported.
PRINCIPLE 4: CONSISTENCY All documentation SHALL be consistent with each other and with the physical reality of the asset. Inconsistencies SHALL be investigated and resolved.
PRINCIPLE 5: TRACEABILITY All documentation SHALL have a clear chain of custody from its origin to its presentation. Documents without verifiable traceability SHALL be treated with caution.

5.2.2 Verification Methodology

Documentary verification follows a four-level depth process:
LEVEL 1: BASIC VERIFICATION • Visual review of presented documents • Verification of dates and signatures • Basic consistency check • Application: Level 1 of the Standard

LEVEL 2: STANDARD VERIFICATION • All of the above, plus: • Consultation with official registries (property registry, cadastre, naval registries) • Verification of licenses with issuing authorities • Analysis of charges and encumbrances in registries • Application: Level 2 of the Standard
LEVEL 3: PREMIUM VERIFICATION • All of the above, plus: • Exhaustive legal analysis by specialized lawyers • Complete regulatory compliance verification • Quantitative legal risk analysis • Cryptographic verification of digital documents • Application: Level 3 of the Standard
LEVEL 4: INSTITUTIONAL VERIFICATION • All of the above, plus: • Complete forensic legal audit • Historical provenance verification (especially for patrimonial assets) • Multi-jurisdictional legal risk analysis • Validation by a multidisciplinary legal team • Application: Level 4 of the Standard

5.2.3 Technological Tools

The Standard incorporates advanced technological tools for documentary verification:
BLOCKCHAIN FOR DOCUMENT VERIFICATION • Hash of original documents registered on blockchain • Integrity verification through hash comparison • Cryptographic timestamp of verification • Application: Especially useful for Levels 3 and 4
ARTIFICIAL INTELLIGENCE FOR DOCUMENT ANALYSIS • Advanced OCR (Optical Character Recognition) for scanned documents • Automatic consistency analysis between documents • Detection of anomalies and suspicious patterns • Automatic risk classification • Application: All levels
INTEGRATED LEGAL DATABASES • Access to official registries of multiple jurisdictions • Automated search for charges and encumbrances • Verification of licenses and

permits • Monitoring of regulatory changes • Application: Levels 2, 3, and 4

5.3 LEGAL AND REGISTRY

VERIFICATION

5.3.1 Verification of Ownership

REQUIRED DOCUMENTS:
For Real Estate Assets: • Public deed of sale or constitution • Updated registry note (maximum 30 days old) • Cadastral ownership certificate • Latest property tax receipt • Certificate of being up to date with the homeowners' association
For Naval Assets: • Naval registry certificate (matriculation) • Flag certificate • Certificate of ownership (Bill of Sale) • Certificate of mortgages and charges (if applicable) • Class certificate (for commercial vessels)
For Industrial Assets: • Purchase invoice or financial leasing contract • Manufacturer's certificate of ownership • Industrial property registry (if applicable) • Current maintenance contracts
For Patrimonial Assets: • Certificate of authenticity • Provenance document • Original purchase invoice • Export/import certificates • Current insurance policies
VERIFICATION PROTOCOL:

1.Obtaining original documents • The client MUST provide original documents or certified copies • The inspection body SHALL NOT accept simple copies without additional verification • For Levels 3 and 4, it is recommended to obtain documents directly from official registries
2.Authenticity verification • Verification of official seals and signatures • Verification of registry

numbers and cross-references • Analysis of paper, inks, and watermarks (for physical documents) • Cryptographic validation (for digitally signed documents)
3.Consultation with official registries • Property Registry (real estate) • Movable Property Registry (naval and industrial assets) • Cadastre (real estate) • Naval registries (vessels) • Intellectual property registries (works of art, designs)
4.Ownership analysis • Verification that the registry owner matches the seller • Identification of co-owners or holders of real rights • Verification of the legal capacity of the owner (individuals/legal entities) • Analysis of historical ownership chain (especially for patrimonial assets)

5.3.2 Analysis of Property Titles

CRITICAL ELEMENTS TO VERIFY:
For Real Estate Assets: • Asset description: Match with physical reality (High risk if deficient) • Surface area: Match with cadastre and reality (Medium risk if deficient) • Boundaries: Clarity and match with reality (High risk if deficient) • Charges: Complete identification (Critical risk if deficient) • Easements: Identification and impact (Medium-High risk if deficient) • Special conditions: Use restrictions, protections (Variable risk)
ANALYSIS PROTOCOL:

1.Documentary-physical comparison • The registry description MUST match physical reality • Surface discrepancies greater than 10% MUST be investigated • Boundary discrepancies MUST be resolved before certification
2.Charge analysis • Identification of ALL registered charges • Evaluation of impact on value and transferability • Verification of validity and amount • Analysis of registry priority

3.Easement analysis • Identification of active and passive easements • Evaluation of impact on use and value • Verification of ownership and scope • Analysis of possible modifications or extinctions
4.Verification of special conditions • Urban planning restrictions (historical protection, green zones) • Previous sale conditions (right of first refusal, right of withdrawal) • Maintenance or conservation obligations • Use or activity limitations

5.3.3 Registry Verification

REGISTRIES TO CONSULT:
Real Estate Registries: • Property Registry • Real Estate Cadastre • Intellectual Property Registry (for protected elements) • Special administrative registries (BIC, protected environments)
Movable Property Registries: • Movable Property Registry (ships and aircraft section) • Movable Property Registry (industrial machinery section) • Intellectual and industrial property registries • Cultural property registries
Administrative Registries: • Municipal registries (licenses, permits) • Regional registries (environment, heritage) • State registries (customs, taxes)
VERIFICATION PROTOCOL:

1.Identification of applicable registries • According to asset typology and location • According to applicable jurisdiction and regulations • According to specific asset characteristics
2.Request for certificates • Updated certificates (maximum 30 days for real estate) • Complete certificates (not extracts) • Certificates with charge and owner information
3.Analysis of results • Match with presented documentation • Identification of discrepancies • Verification of pending updates

4.Documentation of findings • Record of all consultations performed • Copy of certificates obtained • Analysis of discrepancies and resolutions

5.4 REVIEW OF LICENSES AND PERMITS

5.4.1 Real Estate Licenses

REQUIRED LICENSES BY TYPOLOGY:
Residential Real Estate: • First occupancy license or habitability certificate • Building license (if recent construction or renovation) • Energy efficiency certificate • Final work certificate (if applicable)
Commercial Real Estate: • Commercial activity license • Building license (if applicable) • Energy efficiency certificate • Opening license • Exterior advertising permits (if applicable)
Industrial Real Estate: • Industrial activity license • Integrated environmental authorization (if applicable) • Building license • Permits for storage of dangerous products (if applicable) • Industrial safety certificates
VERIFICATION PROTOCOL:

1.Inventory of required licenses • According to asset typology • According to activity carried out • According to applicable jurisdiction
2.Verification of existence • Obtaining copies of licenses • Consultation with issuing authorities • Verification of validity
3.Compliance analysis • Match between license and actual activity • Compliance with license conditions • Identification of possible infractions

4.Risk evaluation • Risk of closure or sanction • Cost of regularization (if applicable) • Impact on asset value

5.4.2 Naval Certificates

REQUIRED CERTIFICATES BY TYPOLOGY:
Yachts and Megayachts: • Naval registry certificate • Flag certificate • Class certificate (Lloyd's, RINA, ABS, etc.) • Safety certificate (according to international conventions) • Pollution prevention certificate (MARPOL) • Tonnage certificate • P&I (Protection & Indemnity) insurance certificate • Navigation license
Commercial Vessels: • All of the above, plus: • Cargo safety certificate • Passenger safety certificate (if applicable) • Compliance certificates with international conventions (SOLAS, MARPOL, STCW) • Flag state inspection certificates • Labor safety certificates
VERIFICATION PROTOCOL:

1.Inventory of required certificates • According to vessel typology • According to flag and jurisdiction • According to navigation area
2.Validity verification • Issuance and expiration dates • Periodic inspections performed • Pending renewals
3.Compliance analysis • Compliance with class conditions • Compliance with international conventions • Identification of pending conditions or recommendations
4.Risk evaluation • Risk of port detention • Cost of regularization • Impact on insurance and operability

5.4.3 Industrial and Environmental Permits

REQUIRED PERMITS:
Industrial Facilities: • Activity license • Integrated environmental authorization (IEA) • Discharge permit •

Atmospheric emissions permit • Waste producer registry • Chemical storage permit
Energy Facilities: • Administrative construction authorization • Exploitation authorization • Grid connection permit • Nuclear safety certificates (if applicable) • Environmental impact assessment
VERIFICATION PROTOCOL:

1.Identification of applicable permits • According to type of activity • According to sectoral regulations • According to location and characteristics
2.Verification of obtaining • Existence of permits • Validity and conditions • Compliance with requirements
3.Compliance analysis • Compliance with permit conditions • Identification of infractions • Evaluation of sanction risks
4.Evaluation of regularization costs • Cost of obtaining missing permits • Cost of adaptation to conditions • Potential fines and sanctions

5.5 ANALYSIS OF CHARGES AND ENCUMBRANCES

5.5.1 Typology of Charges

REAL CHARGES:
Mortgages: • Voluntary mortgages • Legal mortgages • Reverse mortgages • Analysis: holder, amount, term, conditions, partial cancellations
Embargoes: • Preventive embargoes • Executive embargoes • Analysis: origin, amount, court, procedure status

Easements: • Right of way easements • Light and view easements • Aqueduct easements • Administrative easements • Analysis: ownership, scope, impact on value
Usufructs and Use Rights: • Vital or temporary usufructs • Use and habitation rights • Analysis: holder, duration, conditions
PERSONAL CHARGES:
Leases: • Current lease contracts • Analysis: duration, rent, conditions, extension rights
Purchase Options: • Registered purchase options • Analysis: holder, term, price, conditions
Right of First Refusal and Withdrawal: • Administration right of first refusal • Neighboring right of withdrawal • Analysis: holder, term, conditions

5.5.2 Analysis Methodology

STEP 1: COMPLETE IDENTIFICATION The inspection body SHALL identify ALL charges registered in the registry, including: • Main charges (mortgages, embargoes) • Accessory charges (interest, costs) • Extinguished but uncancelled charges • Conditional or future charges
STEP 2: IMPACT ANALYSIS For each identified charge, it SHALL evaluate: • Financial impact: Total amount (principal + interest + costs) • Legal impact: Restrictions on transferability • Impact on value: Reduction of market value • Execution risk: Probability of forced execution
STEP 3: RISK PRIORITIZATION Charges are classified according to their risk:
• Critical Risk: Executive embargoes, mortgages in default. Action: Resolution before certification. • High Risk: Current mortgages, restrictive easements. Action: Detailed evaluation and disclosure. • Medium Risk: Leases, purchase options. Action: Impact analysis and disclosure. • Low Risk: Extinguished uncancelled charges. Action: Recommendation for cancellation.

STEP 4: RECOMMENDATIONS The report SHALL include specific recommendations: • Charges that must be cancelled before the transaction • Charges that may be maintained with buyer consent • Charges requiring negotiation or renegotiation • Estimated costs of cancellation or regularization

5.5.3 Verification of Cancellations

REQUIRED DOCUMENTS:
For supposedly cancelled charges: • Mortgage cancellation deed • Registry cancellation certificate • Full payment receipt • Creditor's payment letter
VERIFICATION PROTOCOL:

1. Registry verification • Consultation with the registry to confirm inscribed cancellation • If not inscribed, require immediate inscription
2. Payment verification • Full payment receipt from the creditor • Verification that no amounts are pending
3. Effect analysis • Confirmation that the cancellation completely frees the asset • Verification that no subordinate charges move to first place

5.6 REGULATORY COMPLIANCE VERIFICATION

5.6.1 Urban Planning Compliance (Real Estate)

ASPECTS TO VERIFY:
Urban Planning: • Land classification (urban, developable, rural) • Urban qualification (permitted use, buildability, heights) • Special reform interior plan (if applicable) • Heritage protection (BIC, protected environments)

Legality of Constructions: • Match between built and permitted • Existence of unlicensed works • Existence of works in disagreement with license • Prescription of urban planning infractions
VERIFICATION PROTOCOL:

1.Planning consultation • Obtaining applicable urban planning regulations • Verification of classification and qualification • Analysis of urban planning parameters
2.Physical-legal comparison • Verification that what is built matches licenses • Identification of unlicensed works • Identification of works in disagreement
3.Risk analysis • Risk of demolition order • Risk of economic sanction • Possibility of regularization
4.Impact evaluation • Impact on asset value • Cost of regularization • Estimated deadlines

5.6.2 Environmental Compliance

ASPECTS TO VERIFY:
Soil Contamination: • History of land uses • Existence of potentially polluting activities • Contaminated land situation declarations • Soil quality studies
Emissions and Discharges: • Atmospheric emission permits • Water discharge permits • Waste producer registry • Compliance with legal limits
Nature Protection: • Location in protected spaces (Natura 2000 Network) • Impact on protected species • Environmental impact assessment
VERIFICATION PROTOCOL:

1.Historical analysis • Review of previous land uses • Identification of potentially polluting activities • Consultation of contaminated land registries
2.Permit verification • Inventory of required environmental permits • Verification of obtaining and validity • Analysis of compliance with conditions

3.Risk evaluation • Contamination risk • Risk of sanction for non-compliance • Remediation cost (if applicable)
4.Recommendations • Additional environmental studies • Remediation measures • Environmental management plan

5.6.3 Tax Compliance

ASPECTS TO VERIFY:
Local Taxes: • Property tax (IBI) up to date • Economic activities tax (IAE) if applicable • Pending municipal capital gains tax
State Taxes: • Wealth tax if applicable • Inheritance and gift tax (if recent acquisition) • VAT or property transfer tax (depending on case)
International Taxes: • Withholding taxes for non-residents • Applicable double taxation treaties • International tax reporting (FATCA, CRS)
VERIFICATION PROTOCOL:

1.Payment verification • Certificates of being up to date with the Tax Agency • Receipts of paid local taxes • Verification of tax debts
2.Analysis of future obligations • Estimation of pending taxes • Municipal capital gains tax in case of sale • Applicable withholdings
3.Risk evaluation • Risk of complementary settlements • Risk of sanctions for non-compliance • Impact on investment profitability

5.6.4 Specific Sectoral Compliance

For Naval Assets: • Compliance with international conventions (SOLAS, MARPOL, STCW) • Compliance with flag regulations • Compliance with port regulations • Compliance with marine environmental regulations
For Industrial Assets: • Compliance with industrial safety regulations • Compliance with occupational risk prevention

regulations • Compliance with chemical product regulations (REACH, CLP) • Compliance with machinery regulations
For Patrimonial Assets: • Compliance with heritage protection regulations • Export restrictions • Conservation and public access obligations • Special tax regime

5.7 DOCUMENTARY RISK MATRIX

5.7.1 Risk Classification

The documentary risk matrix classifies findings into four levels:
CRITICAL RISK (RED): • Defects that prevent legal transfer of the asset • Examples: Disputed ownership, executive embargo, total lack of documentation • Action: Mandatory resolution before certification • SCCIP Impact: Automatic pause of token transfers
HIGH RISK (ORANGE): • Defects that significantly affect value or transferability • Examples: Current mortgage, restrictive easement, expired license • Action: Detailed evaluation and mandatory disclosure • SCCIP Impact: Requires manual approval for transfers
MEDIUM RISK (YELLOW): • Defects that may affect value in the medium term • Examples: Current lease, purchase option, uncancelled charge • Action: Monitoring and planning of corrective actions • SCCIP Impact: Alert to holders but allows trading
LOW RISK (GREEN): • Minor defects without significant impact • Examples: Outdated documentation, uncancelled extinguished charge • Action: Documentation and recommendation for improvement • SCCIP Impact: Normal operation

5.7.2 Risk Matrix by Typology

Real Estate: • Typical Critical Risk: Embargo, lack of deed • Typical High Risk: Mortgage, easement • Typical Medium Risk: Lease • Typical Low Risk: Uncancelled charge
Naval: • Typical Critical Risk: Lack of registry, embargo • Typical High Risk: Naval mortgage, expired class • Typical Medium Risk: Current charter • Typical Low Risk: Certificate expiring soon
Industrial: • Typical Critical Risk: Lack of activity license • Typical High Risk: Pending IEA • Typical Medium Risk: Maintenance contract • Typical Low Risk: Incomplete technical documentation
Patrimonial: • Typical Critical Risk: Forgery, theft • Typical High Risk: Lack of provenance • Typical Medium Risk: Restricted export • Typical Low Risk: Insurance expiring soon

5.7.3 Documentary Risk Score (SRD)

The Standard calculates a Documentary Risk Score (SRD) from 0 to 100:
FORMULA: SRD = 100 - (Σ Critical Risks × 25) - (Σ High Risks × 10) - (Σ Medium Risks × 3) - (Σ Low Risks × 1)
INTERPRETATION: • 90-100: Excellent documentation, no significant risks • 75-89: Good documentation, manageable minor risks • 60-74: Acceptable documentation, medium risks require attention • 40-59: Deficient documentation, high risks require resolution • 0-39: Critical documentation, unacceptable risks
CERTIFICATION THRESHOLD: • Level 1: Minimum SRD 60 • Level 2: Minimum SRD 70 • Level 3: Minimum SRD 80 • Level 4: Minimum SRD 90

5.8 INTEGRATION WITH SUBSEQUENT PHASES

5.8.1 Handoff to Phase 2: Physical Inspection

Phase 1 provides Phase 2 with:
CRITICAL INFORMATION: • Plans and technical documents for physical-documentary comparison • Licenses and permits to verify compliance • Charges and restrictions affecting asset use • Identified risks requiring specific inspection
INSPECTION PLANNING: • Critical areas requiring detailed inspection • Specific technologies needed (e.g., LiDAR to verify surfaces) • Required specialized experts (e.g., urban planning lawyer) • Additional time needed to resolve findings

5.8.2 Integration with Technological Protocols

CPEP (Cryptographic Proof of Existence Protocol): • Hash of all verified documents registered on blockchain • Cryptographic signature of the inspector who performed the verification • Timestamp of documentary verification • Chain of custody of digital documents
SCCIP (Smart Contract Compliance Integration Protocol): • Critical documentary risks injected as rules in smart contracts • Automatic pauses if new embargoes or charges are detected • Real-time update of documentary state • Integration with official registries (when available)

5.9 DOCUMENTATION AND REPORTING

5.9.1 Structure of the Phase 1 Report

The Phase 1 report SHALL include:
SECTION 1: EXECUTIVE SUMMARY • Critical findings and main recommendations • Documentary Risk Score (SRD) • Level of documentary certification achieved • Recommendation to proceed or not to Phase 2
SECTION 2: OWNERSHIP VERIFICATION • Legal ownership analysis • Historical ownership chain • Findings and recommendations
SECTION 3: ANALYSIS OF CHARGES AND ENCUMBRANCES • Complete inventory of charges • Impact analysis per charge • Cancellation or acceptance recommendations
SECTION 4: REVIEW OF LICENSES AND PERMITS • Inventory of required licenses • Status of each license • Risks of non-compliance
SECTION 5: REGULATORY COMPLIANCE • Urban planning analysis (if applicable) • Environmental analysis • Tax analysis • Specific sectoral compliance
SECTION 6: RISK MATRIX • Classification of all findings by risk level • Detailed Documentary Risk Score • Required corrective actions
ANNEXES: • Copy of all verified documents • Obtained registry certificates • Correspondence with authorities • Photographs of physical documents

5.9.2 Phase 1 Certification

Upon completing Phase 1, the inspection body issues:

DOCUMENTARY VERIFICATION CERTIFICATE • Level of verification achieved (1, 2, 3, or 4) • Documentary Risk Score (SRD) • Date of issuance and validity (6-12 months according to level) • Digital signature of inspector and technical reviewer • Cryptographic hash registered on blockchain (Levels 3 and 4)
CONDITIONS OF CERTIFICATION: • The certification is valid only for the date of issuance • Changes in the legal situation require new verification • The certification does not substitute independent legal advice • The client must report any relevant change

CHAPTER 6: PHASE 2 - PHYSICAL INSPECTION

6.1 PURPOSE AND SCOPE OF PHASE 2

6.1.1 Importance of Physical Inspection

Phase 2: Physical Inspection constitutes the technical core of the Prop Trust Verified Standard (PTVS). It is in this phase that documentary theory confronts the material

reality of the asset, where hidden defects are detected, where documentary claims are validated (or invalidated), and where the data that will feed the advanced technological protocols are generated.
Physical inspection under the PTVS is not a mere "technical visit." It is a structured forensic process that combines traditional expert assessment methodologies validated by decades of professional practice, cutting-edge technologies (drones, LiDAR, thermography, IoT), cryptographic edge signing protocols (CPEP), and predictive analysis based on artificial intelligence.

6.1.2 Objectives of Phase 2

Phase 2 pursues six fundamental objectives:
OBJECTIVE 1: Verification of Physical Existence Confirm that the asset physically exists, in the declared location, with the documentary characteristics verified in Phase 1.
OBJECTIVE 2: Evaluation of Current State Determine with precision the state of conservation, functionality, and conformity of the asset at the time of inspection.
OBJECTIVE 3: Identification of Defects and Non-Conformities Detect structural, functional, environmental, or safety defects, classifying them according to their severity and impact.
OBJECTIVE 4: Documentary-Physical Validation Contrast physical reality with the documentation verified in Phase 1, identifying critical discrepancies.
OBJECTIVE 5: Generation of Data for Digital Twin Capture structured data that allows the construction of a digital twin of the asset for Levels 3 and 4.
OBJECTIVE 6: Cryptographic Signing of Evidence Apply the CPEP protocol to cryptographically sign all generated evidence, guaranteeing its authenticity and integrity.

6.1.3 Principles of Physical Inspection

PRINCIPLE 1: SYSTEMATIC APPROACH The inspection SHALL follow structured and reproducible protocols, not depending on the intuition or individual experience of the inspector.
PRINCIPLE 2: COMPLETENESS The inspection SHALL cover ALL accessible areas and systems of the asset, without unjustified omissions. Non-accessible areas SHALL be explicitly declared.
PRINCIPLE 3: OBJECTIVITY Conclusions SHALL be based on measurable and verifiable evidence, not on subjective impressions. Professional opinions SHALL be clearly distinguished from facts.
PRINCIPLE 4: APPROPRIATE TECHNOLOGY The inspection body SHALL use the most appropriate technology for each type of inspection, prioritizing precision, safety, and efficiency.
PRINCIPLE 5: SAFETY The inspection SHALL be carried out guaranteeing the safety of the inspection personnel, the inspected asset, and third parties.
PRINCIPLE 6: CRYPTOGRAPHIC TRACEABILITY All generated evidence (photographs, measurements, samples) SHALL be cryptographically signed at the edge according to the CPEP protocol.

6.2 GENERAL METHODOLOGY OF INSPECTION

6.2.1 Phases of Physical Inspection

Physical inspection is structured into four sequential sub-phases:

SUB-PHASE 1: PREPARATION (Days -7 to 0) • Review of Phase 1 documentation • Logistical and access planning • Calibration and verification of equipment • Briefing of the inspection team • Coordination with the owner/client • Verification of access permits and insurance
SUB-PHASE 2: IN SITU INSPECTION (Days 1 to N) • Systematic visual inspection • Measurements and data collection • Photographic and video capture • Use of advanced technologies (drones, LiDAR, thermography) • Material sampling (when applicable) • Cryptographic edge signing of all evidence
SUB-PHASE 3: LABORATORY ANALYSIS (Days N+1 to N+14) • Analysis of material samples • Processing of drone and LiDAR data • Detailed thermographic analysis • Construction of the digital twin (Levels 3 and 4) • Predictive analysis with AI
SUB-PHASE 4: CONSOLIDATION (Days N+15 to N+21) • Integration of all findings • Risk classification • Generation of the technical report • Peer review • Issuance of Phase 2 certificate

6.2.2 Minimum Required Equipment

BASIC EQUIPMENT (All levels): • Professional digital camera (minimum 24MP) • Laser distance meter • Digital hygrometer and thermometer • Moisture detector • Personal protective equipment (PPE) • Tablet with inspection software • Edge cryptographic signing system
ADVANCED EQUIPMENT (Level 2 and above): • Drone with 4K and thermal camera • Professional thermal imaging camera • Industrial endoscope • Air quality analyzer • Gas detector • Vibration meter
PREMIUM EQUIPMENT (Level 3 and above): • Terrestrial LiDAR scanner • Drone with LiDAR • Ground Penetrating Radar (GPR) • Ultrasonic structural analyzer • Portable weather station • Temporary IoT monitoring system

6.2.3 Levels of Inspection Depth

Level 1: Visual complete coverage | Basic photography | 4-8 hours | 1 inspector Level 2: Detailed with technology | Drones, thermography | 1-3 days | 2-3 inspectors Level 3: Forensic with 3D modeling | LiDAR, IoT, AI | 3-7 days | 3-5 specialists Level 4: Integral audit | All technologies | 1-4 weeks | Multidisciplinary team

6.3 SPECIFIC PROTOCOLS BY ASSET TYPOLOGY

6.3.1 Residential Real Estate

CRITICAL AREAS OF INSPECTION:
A. Structure and Foundation • Foundation: settlements, cracks, dampness • Load-bearing structure: columns, beams, slabs • Load-bearing walls: cracks, deformations • Roof: state, waterproofing, insulation • Facades: cracks, detachments, dampness
B. Installations • Electrical: panel, wiring, outlets, differential protection • Plumbing: pipes, valves, water heaters, pressure • HVAC: heating, cooling, ventilation systems • Gas: installation, detectors, ventilation • Telecommunications: wiring, connection points
C. Finishes and Elements • Floors: state, leveling, dampness • Walls and ceilings: cracks, dampness, finishes • Carpentry: windows, doors, blinds • Sanitary ware: state, leaks, ventilation • Kitchen: state, hood, outlets
D. Exterior and Common Areas • Gardens and terraces • Garages and storage rooms • Swimming pools (if applicable) • Common areas (in collective buildings)

6.3.2 Commercial Real Estate

ADDITIONAL CRITICAL AREAS:
A. Specific Regulatory Compliance • Accessibility (disabled access) • Fire protection • Ventilation and indoor air quality • Emergency exits • Signage
B. Advanced Technical Systems • Centralized HVAC • Access control systems • CCTV and security • Building Management Systems (BMS/GTC) • High voltage installations
C. Energy Efficiency • Real vs. declared energy certification • Efficient lighting systems • Thermal insulation • Renewable energy systems • Smart consumption management

6.3.3 Industrial Real Estate

ADDITIONAL CRITICAL AREAS:
A. Special Structures • Large-span warehouses • Overhead cranes and hoists • Industrial flooring • Special roofs (translucent, sheet metal)
B. Industrial Installations • Power electrical installations • Fluid networks (water, air, gas, steam) • Extraction and filtration systems • Chemical product storage • Special extinguishing systems
C. Environment and Safety • Spill containment • Forced ventilation systems • Explosion protection (ATEX) • Waste management • Atmospheric emissions

6.3.4 Yachts and Megayachts

CRITICAL AREAS OF INSPECTION:
A. Hull and Structure • Hull: osmosis, deformations, impact damage • Keel and appendages: state, fastenings • Deck: waterproofing, drains • Superstructure: cracks, deformations • Bulkheads and watertight compartments

B. Propulsion and Machinery • Main engines: hours, maintenance, compression • Generators: state, capacity • Transmissions: shafts, propellers, steering • Fuel systems • Cooling systems
C. Naval Systems • Bilge and pumping systems • Fresh water and purification systems • Air conditioning systems • Electrical systems (AC/DC) • Navigation and communication systems
D. Interiors and Finishes • Luxury carpentry and joinery • Upholstery and textiles • Appliances and equipment • Entertainment systems • High-end finishes
E. Safety and Equipment • Life-saving equipment (rafts, life jackets) • Firefighting systems • Navigation equipment (radar, GPS, AIS) • Communication equipment (VHF, satellite) • Anchoring and mooring equipment

6.3.5 Commercial Vessels

ADDITIONAL CRITICAL AREAS:
A. Compliance with International Conventions • SOLAS (Safety of Life at Sea) • MARPOL (Marine Pollution) • STCW (Standards of Training, Certification and Watchkeeping) • MLC (Maritime Labour Convention) • Applicable IMO conventions
B. Class Certifications • Valid class certificate • History of repairs and modifications • Pending class conditions • Recommendations from the classification society
C. Operational Systems • Cargo loading and unloading systems • Ballast systems • Inerting systems (tankers) • Tank cleaning systems • Cargo handling equipment

6.3.6 Industrial Machinery

CRITICAL AREAS OF INSPECTION:
A. General State • Identification and traceability • Operating hours • Maintenance history • State of conservation • Modifications and updates

B. Mechanical Systems • Main components (engine, transmission, etc.) • Hydraulic and pneumatic systems • Bearings and shafts • Lubrication systems • Safety elements
C. Control Systems • Automation and PLC • Safety systems • Instrumentation • Operator interfaces • Connectivity and telemetry
D. Performance and Efficiency • Performance tests • Energy consumption • Emissions (if applicable) • Noise and vibrations • Precision and tolerances

6.3.7 Infrastructure

CRITICAL AREAS OF INSPECTION:
A. Linear Structures (Roads, Railways) • Alignment and geometry • Pavement and roadbed • Drainage • Signage • Structures (bridges, tunnels, viaducts)
B. Singular Structures (Bridges, Dams) • Foundations and supports • Main structural elements • Joints and bearings • Drainage systems • Monitoring instrumentation
C. Port and Airport Facilities • Quays and docks • Special pavements • Mooring systems • Navigational aids • Service installations

6.3.8 Patrimonial Assets

CRITICAL AREAS OF INSPECTION:
A. Authenticity and Provenance • Analysis of materials and techniques • Comparison with works by the author • Verification of marks and signatures • Scientific analysis (if applicable) • Aging study
B. State of Conservation • Structural damage • Material degradation • Previous interventions • Environmental conditions • Deterioration risks
C. Historical and Artistic Valuation • Historical context • Artistic importance • Singularity and rarity • Market state • Valuation trends

6.4 ADVANCED INSPECTION TECHNOLOGIES

6.4.1 Drones and UAVs

APPLICATIONS: • Inspection of hard-to-reach roofs and facades • Georeferenced aerial photography • Aerial thermography • 3D modeling of exteriors • Inspection of elevated structures
OPERATIONAL REQUIREMENTS: • Pilot certified according to local regulations (EASA, FAA, etc.) • Specific civil liability insurance • Approved flight plan (when required) • Prior weather verification • Safety and emergency protocols

6.4.2 LiDAR and 3D Scanning

APPLICATIONS: • Millimeter-precision 3D modeling • Verification of surfaces and volumes • Detection of structural deformations • Heritage documentation • Integration with BIM
TECHNICAL SPECIFICATIONS: • Point density: minimum 100 points/m² • Precision: ±2mm at distances up to 50m • Range: up to 350m for terrestrial equipment • Capture speed: minimum 500,000 points/second

6.4.3 Infrared Thermography

APPLICATIONS: • Detection of hidden dampness • Thermal insulation analysis • Identification of critical electrical points • Leak detection in installations • Analysis of operating machinery
TECHNICAL SPECIFICATIONS: • Spectral range: 7.5-14 μm (long wave) • Thermal resolution: minimum 40 mK

(0.04°C) • Spatial resolution: minimum 320×240 pixels • Temperature range: -20°C to +650°C

6.4.4 IoT Sensors for Monitoring

APPLICATIONS: • Continuous monitoring of critical parameters • Early detection of anomalies • Validation of environmental conditions • Support for the CAP protocol
TYPES OF SENSORS: • Accelerometer: Vibrations (Machinery, structures) • Extensometer: Deformations (Structures, bridges) • Hygrometer: Humidity (Buildings, warehouses) • Thermocouple: Temperature (Machinery, processes) • Strain gauge: Stresses (Critical structures) • Piezometer: Water pressure (Dams, foundations) • Inclinometer: Inclinations (Structures, slopes) • Gas sensor: Gas concentration (Industrial, parking)

6.4.5 Materials Analysis

NON-DESTRUCTIVE TESTING (NDT): • Ultrasound: detection of internal defects • Industrial radiography: analysis of welds • Liquid penetrants: detection of surface cracks • Magnetic particles: detection of defects in ferromagnetic materials • Eddy currents: detection of defects in conductive materials
DESTRUCTIVE ANALYSIS (when permitted): • Chemical composition analysis • Mechanical tests (tension, hardness) • Metallographic analysis • Accelerated aging analysis

6.5 INTEGRATION WITH CPEP PROTOCOL

6.5.1 Edge Signing of Evidence

The CPEP protocol guarantees that all evidence generated during physical inspection is cryptographically authentic and immutable.
DATA TO BE SIGNED:
Photographs and Videos: • SHA-256 hash of the original file • EXIF metadata (date, time, GPS, device) • Inspector identification (DID) • Cryptographic timestamp
Measurements: • Measured value with units • Measurement uncertainty • Identification of the measuring equipment • Environmental conditions during measurement • Device signature
Sensor Data: • Sensor reading • Capture timestamp • Sensor identification • Geolocation • Device signature
Samples: • Unique sample identification • Sampling point (geolocation) • Date and time • Responsible inspector • Documented chain of custody

6.5.2 Cryptographic Signing Flow

STEP 1: CAPTURE AT DEVICE • Device (camera, sensor, drone) captures the data • Generates SHA-256 hash of the original data • Adds metadata (timestamp, GPS, device DID)
STEP 2: DEVICE SIGNING • Device signs the hash with its private key • Generates a signature certificate • Original data remains on the device
STEP 3: TRANSFER TO INSPECTOR • Signed data is transferred to the inspector's device • Inspector verifies the device signature • Adds their own signature with their DID

STEP 4: REGISTRATION ON BLOCKCHAIN • Hash of the signed data is registered on the blockchain • Immutable timestamp is generated • Linked to the asset's digital twin
STEP 5: VERIFICATION BY THIRD PARTIES • Any third party can verify authenticity • Chain of signatures is checked • Blockchain registration is verified • Data integrity is confirmed

6.5.3 Benefits of CPEP

FOR THE OWNER: • Mathematical proof of the asset's existence • Guarantee of evidence authenticity • Complete traceability of the inspection • Protection against manipulation
FOR THE INVESTOR: • Verifiable confidence in the data • Capacity for independent audit • Reduction of fraud risk • Support for investment decisions
FOR THE TOKENIZATION PLATFORM: • Solid foundation for RWA tokenization • Native integration with smart contracts • Regulatory compliance (MiCA) • Competitive differentiation

6.6 PHYSICAL RISK MATRIX

6.6.1 Classification of Defects

CRITICAL DEFECTS (RED): Those that compromise the safety, structural integrity, or legality of the asset. • Examples: Severe structural damage in a load-bearing element, severe soil contamination, electrical installation with fire risk, hull damage compromising buoyancy, failure in a critical safety system. • Required action: Immediate resolution or before certification. Automatic pause of transfers in tokenized assets (SCCIP). Notification to authorities if applicable.

HIGH DEFECTS (ORANGE): Those that significantly affect the value, functionality, or useful life of the asset. • Examples: Significant dampness in structural elements, failure in central HVAC system, advanced wear of critical machinery, deficiencies in thermal insulation, damage to high-value finishes. • Required action: Detailed evaluation and repair budgeting. Manual approval for transfers (SCCIP). Priority repair plan.
MEDIUM DEFECTS (YELLOW): Those that may affect the value in the medium term or require planned maintenance. • Examples: Normal wear of non-critical elements, minor deficiencies in installations, opportunities for efficiency improvement, recommended preventive maintenance, recommended technological updates. • Required action: Monitoring and planning. Alert to holders but allows trading (SCCIP). Inclusion in maintenance plan.
LOW DEFECTS (GREEN): Minor ones without significant impact on value or functionality. • Examples: Cosmetic defects, normal wear of low-value elements, minor optimization opportunities, aesthetic improvement recommendations. • Required action: Documentation for awareness. Normal operation (SCCIP). Consideration in future renovations.

6.6.2 Physical Risk Score (SRF)

The Standard calculates a Physical Risk Score (SRF) from 0 to 100:
FORMULA: SRF = 100 - (Σ Critical Defects × 30) - (Σ High Defects × 12) - (Σ Medium Defects × 4) - (Σ Low Defects × 1)
INTERPRETATION: • 90-100: Excellent state, no significant defects • 75-89: Good state, manageable minor defects • 60-74: Acceptable state, medium defects require attention • 40-59: Deficient state, high defects require resolution • 0-39: Critical state, unacceptable defects

CERTIFICATION THRESHOLD: • Level 1: Minimum SRF 60 • Level 2: Minimum SRF 70 • Level 3: Minimum SRF 80 • Level 4: Minimum SRF 90

6.6.3 Risk Matrix by Typology

Residential Real Estate: • Typical Critical Risk: Structural damage, electrical risk • Typical High Risk: Severe dampness, deteriorated roof • Typical Medium Risk: Installation wear, deficient insulation • Typical Low Risk: Cosmetic defects
Commercial Real Estate: • Typical Critical Risk: Fire system failure, accessibility issues • Typical High Risk: Deficient HVAC, low efficiency • Typical Medium Risk: Worn finishes, obsolete technology • Typical Low Risk: Aesthetic improvements
Industrial Real Estate: • Typical Critical Risk: Structural, ATEX, contamination • Typical High Risk: Deteriorated critical machinery • Typical Medium Risk: Pending preventive maintenance • Typical Low Risk: Minor updates
Yacht/Megayacht: • Typical Critical Risk: Hull damage, engine failure • Typical High Risk: Osmosis, deficient systems • Typical Medium Risk: Finish wear, old electronics • Typical Low Risk: Cosmetic improvements
Commercial Vessel: • Typical Critical Risk: Structural failure, safety issues • Typical High Risk: Critical machinery, certificates • Typical Medium Risk: General wear, updates • Typical Low Risk: Aesthetic improvements
Industrial Machinery: • Typical Critical Risk: Catastrophic failure, safety • Typical High Risk: Critical wear, low efficiency • Typical Medium Risk: Pending maintenance, updates • Typical Low Risk: Cosmetic improvements
Infrastructure: • Typical Critical Risk: Collapse, user safety • Typical High Risk: Deformations, deficient drainage • Typical Medium Risk: Normal wear, signage • Typical Low Risk: Aesthetic improvements

Patrimonial Asset: • Typical Critical Risk: Forgery, irreversible damage • Typical High Risk: Active degradation, previous interventions • Typical Medium Risk: Preventive conservation, environment • Typical Low Risk: Cleaning, minor touch-ups

6.7 HANDOFF TO PHASE 3:

EVALUATION OF RISKS

6.7.1 Information Transferred

Phase 2 provides Phase 3 with:
TECHNICAL DATA: • Detailed state of all systems and elements • Measurements and analysis results • Laboratory analysis (when applicable) • 3D models and digital twin (Levels 3 and 4)
DOCUMENTARY EVIDENCE: • Cryptographically signed photographs and videos • Sensor and monitoring data • Reports from specialized analyses • Equipment calibration certificates
FINDINGS AND DEFECTS: • Complete inventory of identified defects • Classification by risk level • Preliminary repair cost estimation • Immediate action recommendations
DATA FOR PREDICTIVE ANALYSIS: • Maintenance history (if available) • Real operating environmental conditions • Usage and wear patterns • Data for AI predictive models

6.7.2 Integration with Technological Protocols

CAP (Continuous Assurance Protocol): • Installation of IoT sensors for continuous monitoring (Levels 3 and 4) • Configuration of alert thresholds • Integration with monitoring platform • Feeding the digital twin

SCCIP (Smart Contract Compliance Integration Protocol): • Injection of critical defects as rules in smart contracts • Configuration of automatic pauses • Definition of risk thresholds • Integration with compliance oracles
PRA (Predictive Risk Assessment): • Feeding AI models with inspection data • Training of predictive algorithms • Generation of future scenarios • Estimation of remaining useful life

6.8 REPORTING AND CERTIFICATION OF PHASE 2

6.8.1 Structure of the Phase 2 Report

The Phase 2 report SHALL include:
SECTION 1: EXECUTIVE SUMMARY • Critical findings and main recommendations • Physical Risk Score (SRF) • Level of certification achieved • Recommendation to proceed or not to Phase 3
SECTION 2: INSPECTION METHODOLOGY • Description of the inspection process • Equipment and technologies used • Participating personnel and qualifications • Limitations and non-accessible areas
SECTION 3: DETAILED FINDINGS • Complete inventory of defects • Classification by risk level • Photographs and evidence • Technical analysis of each finding
SECTION 4: TECHNOLOGICAL ANALYSIS • Thermography results • LiDAR and 3D modeling results • Materials analysis results • IoT sensor data
SECTION 5: RISK MATRIX • Classification of all findings • Detailed Physical Risk Score • Required corrective actions • Cost estimation

SECTION 6: DIGITAL TWIN (Levels 3 and 4) • Description of the 3D model • Integration of inspection data • Access to monitoring platform • Simulation capabilities
ANNEXES: • Complete photographs of the inspection • Laboratory analysis reports • Drone and LiDAR data • Equipment calibration certificates • Cryptographic evidence (CPEP)

6.8.2 Phase 2 Certification

Upon completing Phase 2, the inspection body issues:
PHYSICAL INSPECTION CERTIFICATE • Level of verification achieved (1, 2, 3, or 4) • Physical Risk Score (SRF) • Date of issuance and validity (6-24 months according to level) • Digital signature of inspector and technical reviewer • Cryptographic hash registered on blockchain (Levels 3 and 4) • Access to digital twin (Levels 3 and 4)
CONDITIONS OF CERTIFICATION: • The certification is valid only for the date of issuance • Changes in the state of the asset require new inspection • The certification does not guarantee the total absence of defects • The client must report any relevant change • Continuous monitoring (Levels 3 and 4) updates the state

CHAPTER 7: PHASE 3 - TECHNICAL AND RISK ASSESSMENT

7.1 PURPOSE AND SCOPE OF PHASE 3

7.1.1 Transition from Inspection to Analysis

Phase 3: Technical and Risk Assessment represents the qualitative leap of the Prop Trust Verified Standard (PTVS) compared to traditional standards. While the previous phases (documentary and physical) are common in conventional expert assessment methodologies, Phase 3 introduces predictive analysis, multidisciplinary integration, and prospective evaluation that distinguish the PTVS as a next-generation standard.

In this phase, the data collected in Phases 1 and 2 are transformed into actionable intelligence: patterns are identified, evolutions are predicted, future risks are quantified, and strategic recommendations are generated that go beyond a mere diagnosis of the current state.

7.1.2 Objectives of Phase 3

OBJECTIVE 1: Advanced Structural Analysis Evaluate the structural integrity of the asset using advanced engineering methodologies, including Finite Element Method (FEM) modeling, dynamic analysis, and estimation of remaining useful life.
OBJECTIVE 2: Integral Systems Evaluation Analyze the operation, efficiency, and state of all technical systems of the asset (electrical, mechanical, HVAC, hydraulic, control), identifying optimization opportunities and failure risks.
OBJECTIVE 3: Environmental Analysis Evaluate the environmental impact of the asset, its vulnerability to climate change, and its compliance with ESG standards and current and future environmental regulations.
OBJECTIVE 4: Energy Efficiency Evaluation Determine the actual energy efficiency of the asset, identify improvement opportunities, and project future energy behavior under different scenarios.
OBJECTIVE 5: Natural and Climate Risks Analysis Evaluate the asset's exposure to natural risks (seismic, hydrological, meteorological) and climate risks, quantifying probabilities and potential impacts.
OBJECTIVE 6: Integration with PRA Feed the Predictive Risk Assessment (PRA) protocol with data from previous phases to generate AI-based predictive analyses, modeling future scenarios and estimating remaining useful lives.

7.1.3 Principles of Technical Assessment

PRINCIPLE 1: MULTIDISCIPLINARITY The assessment SHALL integrate perspectives from multiple technical disciplines (structural, mechanical, electrical, environmental, energy) to provide a holistic view of the asset.

PRINCIPLE 2: QUANTIFICATION Risks and opportunities SHALL be quantified whenever possible, using objective metrics, calculated probabilities, and estimated economic impacts.
PRINCIPLE 3: PROSPECTIVITY The assessment SHALL NOT be limited to the current state, but SHALL project future scenarios, considering technological evolution, regulatory changes, and market trends.
PRINCIPLE 4: SCIENTIFIC BASIS Every conclusion SHALL be supported by scientific evidence, validated models, or internationally recognized methodologies.
PRINCIPLE 5: EXPLICIT UNCERTAINTY Degrees of uncertainty SHALL be quantified and clearly communicated, distinguishing between verified facts, high-confidence estimates, and speculative projections.

7.2 ADVANCED STRUCTURAL

ANALYSIS

7.2.1 Structural Analysis Methodologies

LINEAR STATIC ANALYSIS • Evaluation of gravitational and use loads • Verification of stresses and deformations • Comparison with applicable codes (Eurocodes, ACI, AISC) • Application: Conventional structures under normal conditions
NON-LINEAR STATIC ANALYSIS • Evaluation of post-elastic behavior • Seismic capacity analysis (pushover analysis) • Identification of collapse mechanisms • Application: Structures in seismic zones or with visible damage
DYNAMIC ANALYSIS • Modal analysis (natural frequencies, vibration modes) • Spectral analysis for

seismic loads • Dynamic wind analysis • Application: Slender structures, bridges, tall buildings
FINITE ELEMENT ANALYSIS (FEM) • 3D modeling of complex structural elements • Analysis of stress concentrations • Simulation of extreme load scenarios • Application: Singular elements, complex joints, patrimonial structures

7.2.2 Evaluation of Structural Materials

CONCRETE: • Compressive strength: Rebound hammer, ultrasonic testing • Carbonation depth: Phenolphthalein solution • Chloride content: Chemical analysis • Cover thickness: Pachometer • Reinforcement corrosion: Corrosion potential
STEEL: • Element thickness: Ultrasonic testing • Mechanical strength: Hardness testing (conversion) • Welds: Ultrasonic testing, magnetic particles • Corrosion: Loss measurement • Deformations: Topography, laser scanning
WOOD: • Strength: Resistograph, stress wave • Moisture: Hygrometer • Biological attack: Visual inspection, endoscopy • Deformations: Topography, laser scanning • Joints: Visual inspection, extraction

7.2.3 Foundation Evaluation

METHODOLOGIES: • Visual inspection: Cracks in basement walls, differential settlements, rising dampness • Geotechnical tests: SPT, plate load tests, piezometers for water table, soil laboratory analysis • Geophysical techniques: Ground Penetrating Radar (GPR) for void detection, electrical resistivity tomography • Monitoring: Inclinometers for foundation rotation, settlement gauges, piezometers for water pressure, crack gauges for crack evolution

7.2.4 Remaining Useful Life Estimation

METHODOLOGY:

1.Current state analysis: Quantification of observed deterioration, identification of degradation mechanisms, evaluation of deterioration rate.
2.Degradation modeling: Selection of appropriate degradation model, calibration with historical data, projection of future evolution.
3.Definition of limit states: Identification of relevant limit states, definition of acceptance thresholds, consideration of safety factors.
4.Useful life estimation: Calculation of time until limit states are reached, uncertainty analysis (Monte Carlo), generation of confidence intervals.

DEGRADATION MODELS: • Concrete: Fick's Law (carbonation) • Steel: Linear section loss • Wood: Exponential degradation • Roofs: Degradation by UV and thermal cycles

7.2.5 Integration with PRA

The structural analysis feeds the PRA protocol with: • Current state of structural elements • Measured degradation rates • Environmental and load conditions • Maintenance and repair history
PRA outputs: • Modeling of deterioration evolution • Prediction of structural failures • Estimation of future maintenance costs • Optimization of intervention plans

7.3 EVALUATION OF TECHNICAL SYSTEMS

7.3.1 Electrical System

AREAS OF EVALUATION: • Supply and main panel: Contracted vs. installed power, supply state, panel protection and organization. • Distribution and wiring: Conductor sections vs. loads, insulation state, circuit organization. • Protections: Circuit breakers and RCDs, grounding, surge protection, protection selectivity. • Lighting: Illuminance levels per use, energy efficiency of luminaires, control and regulation systems, emergency lighting.
METHODOLOGIES: • Electrical thermography: Detection of hot spots in connections, identification of overloads, analysis of phase imbalances. • Power quality analysis: Measurement of voltages and frequencies, harmonic analysis, detection of voltage sags, transient analysis. • Insulation tests: Conductor insulation resistance, grounding resistance, RCD operation, protective conductor continuity.

7.3.2 Mechanical System

AREAS OF EVALUATION: • Motors and machinery: General state and operating hours, anomalous vibrations and noises, operating temperatures, energy consumption. • Transmissions: Shafts, couplings, gearboxes, belts, chains, gears, lubrication and component state, alignment and balancing. • Hydraulic and pneumatic systems: Pumps, compressors, valves, pipes, connections, actuators, leaks and pressure losses, fluid state. • Lifting and transport systems: Elevators, freight lifts, overhead

cranes, hoists, safety systems, certifications and maintenance.
METHODOLOGIES: • Vibration analysis: Measurement at critical points, spectral analysis for fault identification, comparison with reference values (ISO 10816). • Oil analysis: Lubrication oil sampling, particle analysis (ferrography), particle counting (ISO 4406), physicochemical analysis. • Mechanical thermography: Detection of hot spots in bearings, identification of excessive friction, analysis of associated electrical systems. • Ultrasound analysis: Detection of compressed air leaks, identification of electrical discharges, bearing analysis at high frequency, cavitation detection in pumps.

7.3.3 HVAC System

AREAS OF EVALUATION: • Generation: Boilers, chillers, heat pumps, actual vs. nominal energy efficiency, state of burners, compressors, exchangers. • Distribution: Pipe and duct networks, pumps, fans, valves, thermal insulation, pressure losses. • Emission: Air handling units (AHU), fan-coils, splits, VRV/VRF, diffusers, grilles, extraction systems. • Regulation and control: Building management systems (BMS/GTC), sensors and actuators, control strategies, integration with other systems.
METHODOLOGIES: • HVAC energy audit: Measurement of energy consumption, analysis of efficiencies (COP, EER), evaluation of operating hours, comparison with sector benchmarks. • Indoor air quality analysis: Measurement of CO2, CO, VOC, analysis of particles (PM2.5, PM10), measurement of temperature and humidity, evaluation of ventilation rates. • Tightness tests: Duct pressure testing (EN 12599), leak detection with anemometer, insulation thermography, analysis of energy losses. • Comfort analysis: Measurement of PMV/PPD (Predicted Mean Vote), analysis of thermal stratification, evaluation of air currents, acoustic analysis of equipment.

7.3.4 Specialized Systems

FIRE PROTECTION SYSTEMS: • Detection and alarm, extinguishing (sprinklers, fire hose reels, extinguishers), smoke control, signage and evacuation, evaluation according to local building codes.
SECURITY SYSTEMS: • Access control, CCTV and video surveillance, intrusion detection, integration with emergency systems.
TELECOMMUNICATION SYSTEMS: • Structured cabling, data networks, telephony systems, connectivity and bandwidth.
VERTICAL TRANSPORT SYSTEMS: • Elevators and freight lifts, escalators, certifications and maintenance, accessibility.

7.4 ENVIRONMENTAL ANALYSIS

7.4.1 Environmental Impact Evaluation

CARBON FOOTPRINT: • Scope 1 (Direct emissions): Combustion in boilers and furnaces, own vehicles, refrigerant gas leaks. • Scope 2 (Indirect energy emissions): Grid electricity consumption, grid heat/cold consumption. • Scope 3 (Other indirect emissions): Employee transport, business travel, supply chain, generated waste.
METHODOLOGY: • Inventory of emission sources, applicable emission factors (GHG Protocol, ISO 14064), calculation of emissions in tCO2eq, third-party verification (when applicable).
WASTE MANAGEMENT: • Hazardous waste: Used oils, solvents and chemicals, sanitary waste, batteries and accumulators, electronic equipment (WEEE). • Non-hazardous waste: Construction and demolition waste, packaging waste, municipal waste, production process

waste. • Methodology: Inventory of generated waste, classification according to European Waste List (EWL), evaluation of authorized managers, analysis of recycling rates, identification of reduction opportunities.

7.4.2 Climate Change Vulnerability

PHYSICAL RISKS: • Acute Risks (Extreme events): Floods and surges, storms and extreme winds, heat and cold waves, forest fires, extreme droughts. • Chronic Risks (Long-term trends): Increase in average temperature, sea level rise, changes in precipitation patterns, ocean acidification (naval assets), permafrost degradation (arctic infrastructure).
METHODOLOGY:

1. Hazard identification: Historical analysis of extreme events, climate projections (IPCC, local meteorological agencies), risk maps (floodability, fires), climate scenarios (RCP 4.5, RCP 8.5).
2. Exposure evaluation: Geographical location of the asset, physical characteristics, dependence on external services, interconnections with other assets.
3. Sensitivity analysis: Structural vulnerability to extreme events, sensitivity of technical systems, impact on operability, dependence on natural resources.
4. Impact evaluation: Potential physical damage, estimated economic losses, activity interruption, supply chain impacts.
5. Adaptation strategies: Structural measures (protections, reinforcements), non-structural measures (insurance, emergency plans), risk diversification, relocation (when necessary).

7.4.3 ESG Compliance

ENVIRONMENTAL CRITERIA (E): • Carbon footprint and reduction targets, energy efficiency and use of renewables, waste management and circular economy, water consumption and water management, biodiversity and land use.
SOCIAL CRITERIA (S): • Working conditions and safety, relations with local communities, human rights in the supply chain, diversity and inclusion, social impact of the asset.
GOVERNANCE CRITERIA (G): • Transparency and reporting, business ethics, risk management, board composition, executive remuneration.
REPORTING ALIGNMENT: • Alignment with GRI Standards, compliance with SASB Standards, preparation for TCFD (Task Force on Climate-related Financial Disclosures), compliance with CSRD (Corporate Sustainability Reporting Directive).

7.5 ENERGY EFFICIENCY

EVALUATION

7.5.1 Energy Audit

PHASES:

1.Data collection: Energy bills (minimum 24 months), plans and technical documentation, inventories of consuming equipment, operating schedules, production or activity data.
2.Preliminary analysis: Global energy balance, breakdown by uses and equipment, sector benchmarks, identification of improvement areas.

3.In situ inspection: Consumption measurements, load profile analysis, evaluation of consuming equipment, detection of leaks and losses.
4.Detailed analysis: Energy simulation (if applicable), analysis of improvement opportunities, technical and economic evaluation, prioritization of measures.
5.Action plan: Catalog of efficiency measures, profitability analysis (NPV, IRR, payback), implementation plan, monitoring and verification system.

7.5.2 Improvement Opportunities

BUILDING ENVELOPE: • Insulation of facades and roofs, window replacement, solar protection, sealing of infiltrations.
HVAC SYSTEMS: • Replacement of obsolete equipment, installation of variable frequency drives, heat recovery, improvement of controls and regulation.
LIGHTING: • Replacement with LED, control systems (presence detectors, regulation), use of natural light, maintenance and cleaning.
EQUIPMENT AND PROCESSES: • Replacement with efficient motors, optimization of compressed air systems, process heat recovery, automation and control.
RENEWABLE ENERGIES: • Photovoltaic installation, solar thermal, geothermal, biomass.
MANAGEMENT MEASURES: • Energy management systems (ISO 50001), monitoring and targeting, training and awareness, preventive maintenance, efficient energy contracting.

7.5.3 Energy Certification

BUILDINGS: • LEED (Leadership in Energy and Environmental Design) • BREEAM (Building Research Establishment Environmental Assessment Method) •

WELL Building Standard (focused on health) • Passivhaus (nearly zero consumption standard)
VESSELS: • EEDI (Energy Efficiency Design Index) - IMO • EEXI (Energy Efficiency Existing Ship Index) - IMO • CII (Carbon Intensity Indicator) - IMO
INDUSTRY: • ISO 50001 (Energy Management Systems) • Motor Systems Optimization • Compressed Air Challenge • Process Heating Assessment

7.5.4 Future Consumption Projection

SCENARIOS: • Base Scenario (Business as Usual): Maintenance of current conditions, natural degradation of equipment, evolution of energy prices, foreseeable regulatory changes. • Improvement Scenario (With Measures): Implementation of efficiency measures, investments in equipment renewal, integration of renewable energies, operation optimization. • Decarbonization Scenario (Net Zero): Complete electrification of consumption, 100% renewable energy, compensation of residual emissions, alignment with climate objectives.
METHODOLOGY: • Dynamic energy modeling (EnergyPlus, DesignBuilder), time series analysis, Monte Carlo simulation for uncertainty, sensitivity analysis to key variables.

7.6 ANALYSIS OF NATURAL AND CLIMATE RISKS

7.6.1 Seismic Risk

METHODOLOGY: • Seismic hazard: Seismic zoning, basic seismic acceleration (ab), return period (475 years for no-collapse limit state), soil type (I, II, III, IV). •

Structural vulnerability: Structural typology (masonry, concrete, steel, wood), year of construction and applicable code, regularity in plan and height, state of conservation, existence of vulnerable non-structural elements. • Exposure: Replacement value of the asset, contents and interior goods, loss of use (business interruption), civil liability.
CALCULATION: Risk = Hazard × Vulnerability × Exposure
ADVANCED METHODOLOGIES: • Pushover analysis for evaluation of seismic capacity • Non-linear dynamic analysis (time-history) • Fragility curves for different damage states • Analysis of expected economic losses (PML, AAL)

7.6.2 Flood Risk

TYPES OF FLOODING: • Fluvial flooding: Overflow of rivers and streams, floodable zones according to management plans, return periods (10, 100, 500 years), expected depths and velocities. • Coastal flooding: Maritime storms, sea level rise, tsunamis (in risk zones), saltwater intrusion. • Runoff flooding: Intense rainfall on impermeable surfaces, saturation of drainage networks, topographically depressed zones, flash floods.
METHODOLOGY: • Analysis of floodability maps, hydrological-hydraulic study (if applicable), evaluation of existing protection measures, analysis of climate change scenarios, estimation of potential damages.

7.6.3 Wind Risk

EVALUATION OF WIND LOADS: • Basic wind speed: Wind zoning, exposure category, return period (50 years for ordinary buildings). • Exposure coefficients: Terrain category (0, I, II, III, IV), roughness and orography, building height, surrounding effects. • Pressure coefficients: External pressure on facades and roofs,

internal pressure according to permeability, global and local forces, dynamic effects (vortices, flutter).
EVALUATION OF POTENTIAL DAMAGES: • Damage to enclosures and roofs, damage to facade elements, damage to light structures, falling objects and trees, activity interruption.

7.6.4 Forest Fire Risk

EXPOSURE EVALUATION: • Hazard indices: Meteorological danger index (FWI), fire danger index (FFDI), historical fires in the area, climate change projections. • Environmental characteristics: Distance to forest mass, type of vegetation and fuel load, topography and slopes, accessibility for extinguishing means. • Asset vulnerability: Construction materials (combustible/non-combustible), design of defensible perimeter, protection systems (sprinklers, shutters), vegetation management around the asset.
METHODOLOGY: • Analysis of urban-forest interfaces, modeling of fire spread (FARSITE, Prometheus), evaluation of damage by radiation and convection, analysis of vulnerability of critical elements.

7.7 INTEGRATION WITH PRA

PROTOCOL

7.7.1 PRA Architecture

The PRA protocol constitutes the analytical brain of the PTVS, integrating data from all previous phases to generate predictive analyses based on artificial intelligence.
LAYERS:

1.Data Ingestion: Data from Phase 1 (documentary), Phase 2 (physical inspection), Phase 3 (technical assessment), continuous monitoring data (CAP), external data (market, climate, regulatory).
2.Processing and Analysis: Data cleaning and normalization, feature engineering, exploratory analysis, detection of patterns and anomalies.
3.Predictive Modeling: Component degradation models, failure probability models, maintenance cost models, future asset value models.
4.Insight Generation: Predictive maintenance recommendations, early risk alerts, investment plan optimization, asset evolution scenarios.

7.7.2 Predictive Models

DEGRADATION MODELS: • Physics-based: Degradation laws of materials, fatigue and fracture models, corrosion models, accelerated aging models. • Data-driven: Regression for historical trends, time series (ARIMA, Prophet), Random Forest for state classification, neural networks for complex patterns. • Hybrid: Physics-Informed Neural Networks (PINN), grey-box models, transfer learning between similar assets.
FAILURE PROBABILITY MODELS: • Reliability analysis: Weibull distributions for failure times, Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis (FTA), common cause analysis. • Machine learning for prediction: Classifiers for imminent failure prediction, anomaly detection in sensor data, Remaining Useful Life (RUL) prediction, maintenance interval optimization.

7.7.3 PRA Applications

PREDICTIVE MAINTENANCE: • Objective: Optimize maintenance intervals based on the actual state of the asset, not on fixed schedules. • Benefits: Reduction of

maintenance costs (15-30%), increase in asset availability (5-10%), prevention of catastrophic failures, optimization of spare parts stocks.
PREDICTIVE RISK MANAGEMENT: • Objective: Anticipate risks before they materialize, allowing preventive actions. • Applications: Prediction of structural failures, anticipation of environmental problems, early detection of regulatory risks, alerts for security vulnerabilities.
INVESTMENT OPTIMIZATION: • Objective: Maximize the return on investments in maintenance and asset improvement. • Analysis: Life Cycle Cost (LCC), profitability analysis of measures, optimization of investment plans, prioritization of interventions.

7.7.4 Integration with SCCIP and CAP

SCCIP (Smart Contract Compliance Integration Protocol): The PRA feeds the SCCIP with: • Dynamic risk thresholds: Adjusted according to predictions. • Evolving compliance rules: That adapt to the predicted state. • Preventive alerts: Before thresholds are reached. • Rating updates: Based on predictive trends.
CAP (Continuous Assurance Protocol): The PRA integrates with the CAP through: • Real-time analysis: Of IoT sensor data. • Deviation detection: With respect to predictive models. • Model adjustment: Based on real monitoring data. • Continuous validation: Of previous predictions.

7.8 INTEGRATED RISK MATRIX OF PHASE 3

7.8.1 Technical Risk Score (SRT)

The Standard calculates a Technical Risk Score (SRT) from 0 to 100, integrating all Phase 3 analyses:
FORMULA: SRT = (0.25 × SRE_structural) + (0.25 × SRE_systems) + (0.20 × SRE_environmental) + (0.15 × SRE_energy) + (0.15 × SRE_natural)
Where: • SRE_structural = Structural Risk Score • SRE_systems = Systems Risk Score • SRE_environmental = Environmental Risk Score • SRE_energy = Energy Risk Score • SRE_natural = Natural Risk Score
INTERPRETATION: • 90-100: Technical excellence, no significant risks • 75-89: Good technical level, manageable minor risks • 60-74: Acceptable level, medium risks require attention • 40-59: Deficient level, high risks require resolution • 0-39: Critical level, unacceptable risks
CERTIFICATION THRESHOLD: • Level 1: Minimum SRT 60 • Level 2: Minimum SRT 70 • Level 3: Minimum SRT 80 • Level 4: Minimum SRT 90

7.9 REPORTING AND CERTIFICATION OF PHASE 3

7.9.1 Structure of the Phase 3 Report

The Phase 3 report SHALL include:
SECTION 1: EXECUTIVE SUMMARY • Main findings and strategic recommendations • Technical Risk Score (SRT) •

Level of certification achieved • Main predictive risks identified
SECTION 2: STRUCTURAL ANALYSIS • Results of advanced analyses • Remaining useful life evaluation • Intervention recommendations • Integration with PRA
SECTION 3: TECHNICAL SYSTEMS EVALUATION • State of electrical, mechanical, HVAC systems • Efficiency and improvement opportunities • Optimal maintenance plan • Failure predictions
SECTION 4: ENVIRONMENTAL ANALYSIS • Carbon footprint and waste management • Vulnerability to climate change • ESG compliance • Adaptation strategies
SECTION 5: ENERGY EFFICIENCY • Energy audit results • Identified improvement opportunities • Profitability analysis of measures • Future consumption projection
SECTION 6: NATURAL AND CLIMATE RISKS • Evaluation of seismic, flood, wind, fire risks • Climate change scenarios • Recommended mitigation measures • Analysis of insurance and coverages
SECTION 7: PREDICTIVE ANALYSIS (PRA) • Applied predictive models • Degradation and failure predictions • Predictive maintenance recommendations • Asset evolution scenarios
SECTION 8: STRATEGIC PLAN • Prioritized intervention roadmap • Life cycle cost analysis • Investment optimization • Tracking KPIs
ANNEXES: • Structural calculation models (FEM) • Systems analysis reports • Energy certificates • Natural risk maps • Predictive dashboards

7.9.2 Phase 3 Certification

Upon completing Phase 3, the inspection body issues: TECHNICAL AND RISK ASSESSMENT CERTIFICATE • Level of verification achieved (1, 2, 3, or 4) • Technical Risk Score (SRT) • Date of issuance and validity (12-36 months according to level) • Digital signature of the

multidisciplinary team • Cryptographic hash registered on blockchain (Levels 3 and 4) • Access to the predictive dashboard (Levels 3 and 4)
CONDITIONS OF CERTIFICATION: • The certification is valid for the date of issuance • Continuous monitoring (Levels 3 and 4) updates predictions • Extraordinary events require re-evaluation • The client must report significant changes • Predictions have explicit confidence intervals

CHAPTER 8: PHASE 4 - FINANCIAL AND VALUE VERIFICATION

8.1 PURPOSE AND SCOPE OF PHASE 4

8.1.1 The Financial Dimension of Verification

Phase 4: Financial and Value Verification represents the culmination of the Prop Trust Verified Standard (PTVS) integral verification process. While the previous phases have established the documentary, physical, and technical reality of the asset, Phase 4 translates all that information into economic and financial terms, providing investors,

owners, and tokenization platforms with the quantitative basis for decision-making.
This phase is particularly critical in the context of Real World Assets (RWA) tokenization, where the value of the token must accurately reflect the underlying physical asset's value, adjusted for all risks identified in the previous phases.

8.1.2 Objectives of Phase 4

OBJECTIVE 1: Life Cycle Cost Analysis Determine the total costs of ownership (TCO), including maintenance, operation, repairs, and eventual dismantling or final disposal.
OBJECTIVE 2: Remaining Useful Life Estimation Project the remaining economic life of the asset, considering technical factors, obsolescence, regulatory changes, and market trends.
OBJECTIVE 3: Rigorous Market Valuation Determine the market value of the asset using internationally recognized methodologies, adjusted for the findings of the previous phases.
OBJECTIVE 4: Profitability and ROI Analysis Evaluate the current and projected profitability of the asset, including income, expenses, and return on investment under different scenarios.
OBJECTIVE 5: Financial Stress Testing Model the behavior of the asset under adverse scenarios (economic crises, regulatory changes, natural disasters) to evaluate its financial resilience.
OBJECTIVE 6: Preparation for Tokenization Provide the solid financial basis for the tokenization of the asset, including value per token, dividend distribution, and governance mechanisms.

8.1.3 Principles of Financial Verification

PRINCIPLE 1: METHODOLOGICAL TRANSPARENCY Every valuation MUST be based on internationally recognized methodologies (IVS, RICS Red Book, USPAP), with explicit and justifiable assumptions.
PRINCIPLE 2: PRUDENT CONSERVATISM Valuations MUST apply a conservative approach, avoiding overvaluations that could harm investors or token holders.
PRINCIPLE 3: MULTIDIMENSIONAL INTEGRATION The valuation MUST integrate all findings from the previous phases (documentary, physical, technical) as adjustments to the base value.
PRINCIPLE 4: MULTIPLE SCENARIOS Every valuation MUST present multiple scenarios (optimistic, base, conservative, stress) with associated probabilities.
PRINCIPLE 5: DYNAMIC UPDATING For Levels 3 and 4, the valuation MUST be updated continuously through the CAP protocol, reflecting changes in the asset's state and the market.

8.2 LIFE CYCLE COST ANALYSIS

8.2.1 TCO Structure (Total Cost of Ownership)

The life cycle cost analysis identifies and quantifies all costs associated with the asset during its remaining useful life, structured in the following categories:
ACQUISITION COSTS (Already incurred - Context): • Purchase price of the asset • Transaction costs (notary, registries, taxes) • Financing costs (interest, commissions) • Initial adaptation costs (renovations, installations)
OPERATING COSTS (OPEX - Operating Expenses): • Preventive Maintenance: Inspections, lubrication, adjustments. Frequency: Monthly/Quarterly. Variability: Low. • Corrective Maintenance: Repairs, replacements.

Frequency: Unexpected. Variability: High. • Energy and Utilities: Electricity, water, gas, fuels. Frequency: Monthly. Variability: Medium. • Insurance: Annual premium, deductibles. Frequency: Annual. Variability: Low. • Taxes: Property tax, rates, patrimonial taxes. Frequency: Annual. Variability: Low. • Personnel: Operators, technicians, managers. Frequency: Monthly. Variability: Low. • Supplies: Consumables, spare parts, materials. Frequency: Variable. Variability: Medium. • Technology: Software, licenses, updates. Frequency: Annual. Variability: Medium.
CAPITAL COSTS (CAPEX - Capital Expenditures): • Improvements and Renovations: Updates, expansions. Horizon: 5-15 years. Uncertainty: Medium. • Major Replacements: Equipment at end of useful life. Horizon: 10-25 years. Uncertainty: High. • Regulatory Adaptations: Compliance with new regulations. Horizon: Unpredictable. Uncertainty: Very High. • Structural Repairs: Major interventions. Horizon: Unpredictable. Uncertainty: Very High.
DISPOSAL COSTS: • Dismantling or demolition • Waste management (especially hazardous) • Site restoration (if applicable) • Environmental costs (contamination, remediation)

8.2.2 Cost Estimation Methodology

STEP 1: COMPONENT INVENTORY Breakdown of the asset into its main components, with estimated useful life and replacement cost for each.
STEP 2: OPERATING COST PROJECTION Based on historical data, sector benchmarks, and current asset conditions. • Preventive maintenance: 1-3% of replacement value annually. • Corrective maintenance: 0.5-2% of replacement value annually (depending on state). • Energy: According to energy audit (Phase 3) and price projection. • Insurance: According to valuation and

identified risks. • Taxes: According to current regulations and projections.
STEP 3: FUTURE CAPEX ANALYSIS Identification of necessary investments to maintain operability and asset value. • Value maintenance: Investments necessary to avoid accelerated depreciation. • Competitive improvements: Investments to maintain market competitiveness. • Regulatory adaptations: Investments to comply with foreseeable future regulations. • Scheduled replacements: According to component inventory.
STEP 4: FINANCIAL DISCOUNTING All future costs are discounted to present value using an appropriate discount rate.
Formula: PV = Σ [Cost_t / (1 + r)^t] Where: PV = Present Value, Cost_t = Cost in year t, r = Discount rate (WACC or required return), t = Future year.
Recommended Discount Rates: • Real estate assets: 6-8% • Naval assets: 8-10% • Industrial assets: 10-12% • Patrimonial assets: 5-7% (lower obsolescence risk)

8.2.3 Equivalent Annual Cost (EAC)

The Equivalent Annual Cost expresses the total life cycle cost as a constant annual payment, facilitating comparison between alternatives.
Formula: EAC = PV_total × [r(1+r)^n] / [(1+r)^n - 1]
Application: Comparison of investment alternatives, evaluation of net profitability (Income - EAC), repair vs. replace decision, long-term budgeting.

8.2.4 Integration with PRA and CAP

PRA (Predictive Risk Assessment): • Prediction of corrective maintenance costs based on asset state. • Estimation of probability of costly failures. • Optimization of maintenance plans to minimize EAC. • Projection of regulatory adaptation costs.

CAP (Continuous Assurance Protocol): • Real-time monitoring of energy consumption. • Early detection of anomalies that increase costs. • Dynamic update of cost projections. • Alerts for budget deviations.

8.3 REMAINING USEFUL LIFE AND DEPRECIATION

8.3.1 Fundamental Concepts

TOTAL USEFUL LIFE (TUL): Period during which the asset can function economically and safely, from its installation to its final disposal.
REMAINING USEFUL LIFE (RUL): Remaining period from the valuation date to the end of the total useful life.
ECONOMIC LIFE: Period during which the asset generates positive net economic benefits (may be shorter than the technical useful life due to obsolescence).
OBSOLESCENCE: Loss of value due to factors external to physical deterioration. • Technological obsolescence: New technologies make the asset less competitive. • Functional obsolescence: Design inadequate for current uses. • Economic obsolescence: Market changes reduce profitability. • Regulatory obsolescence: New regulations make the asset non-compliant.

8.3.2 RUL Estimation Methodologies

METHOD 1: PHYSICAL STATE ANALYSIS Based on physical inspection (Phase 2) and technical evaluation (Phase 3). • Percentage of useful life consumed according to physical state. • Identification of critical components. • Observed degradation rate. • Degradation projection (linear, exponential, S-curve). • Failure or replacement threshold. • Time until threshold is reached. • Confidence

interval (uncertainty analysis). • Scenarios (optimistic, base, conservative).
METHOD 2: ACTUARIAL ANALYSIS Based on statistical data of similar assets. • Sources: Sectorial body useful life studies, insurer data, academic studies, industry benchmarks. • Application: Survival tables by asset typology, Weibull distributions for failure times, cohort analysis.
METHOD 3: ECONOMIC ANALYSIS Based on the future profitability of the asset. • Criteria for end of economic life: Maintenance cost > Annualized replacement cost, Generated income < Operating costs, Residual value < Disposal cost, Superior alternatives available in the market.
METHOD 4: REGULATORY ANALYSIS Based on future regulatory compliance. • Factors: Environmental regulations (emissions, efficiency), Safety regulations (seismic, fire), Accessibility regulations, Future use restrictions.

8.3.3 Depreciation and Amortization

ACCOUNTING DEPRECIATION METHODS: • Linear Depreciation: Annual depreciation = (Value - Residual Value) / Useful Life. Application: Assets with uniform wear. • Accelerated Depreciation: Sum of years' digits, 200% declining balance. Application: Assets with higher initial wear. • Units of Production Depreciation: Depreciation per unit = (Value - Residual Value) / Total expected units. Application: Industrial machinery, vehicles.
ECONOMIC DEPRECIATION (VS. ACCOUNTING): Economic depreciation reflects the real loss of market value, which may differ significantly from accounting depreciation. • Factors: Real physical deterioration, technological obsolescence, changes in market preferences, location factors, general economic

conditions. • Calculation: Economic Depreciation = Current Market Value - Projected Future Market Value.

8.3.4 Residual and Disposal Value

RESIDUAL VALUE: Estimated value of the asset at the end of its economic useful life or at the end of the investment period.
Estimation Methods: • Market Method: Analysis of sales of similar assets at end of life. Application: Assets with active secondary market. • Adjusted Replacement Cost Method: Replacement cost new minus accumulated depreciation. Application: Specialized assets without secondary market. • Income Capitalization Method: Present value of future income during the remaining period. Application: Income-generating assets.
DISPOSAL VALUE: Net value obtained after the final disposal of the asset. Formula: Disposal Value = Sale Value (or Scrap) - Disposal Costs. Typical disposal costs: Dismantling or demolition, transport, waste management, site restoration, environmental costs, administrative and legal costs.

8.3.5 Integration with PRA

The PRA protocol feeds the RUL estimation with: • Continuous monitoring data: Real degradation rate vs. theoretical models, detection of incipient failures, impact of operating conditions on useful life. • Predictive models: RUL prediction based on sensor data, trend analysis of critical parameters, simulation of maintenance scenarios. • Dynamic updating: Adjustment of RUL estimates according to real data, alerts for accelerated reduction of useful life, recommendations for interventions to extend useful life.

8.4 MARKET VALUATION

8.4.1 Valuation Principles

DEFINITION OF MARKET VALUE (IVS - International Valuation Standards): "The estimated amount for which an asset or liability should exchange on the valuation date between a willing buyer and a willing seller in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion."
FUNDAMENTAL ASSUMPTIONS: • Motivated but not desperate buyer and seller. • Adequate marketing period. • Information available to both parties. • Transaction between independent parties. • Price expressed under normal market conditions.

8.4.2 Valuation Methodologies

APPROACH 1: MARKET COMPARISON METHOD Fundament: The value of an asset is determined by comparing it with similar assets recently sold in the market. Application: Residential and commercial real estate (with abundant comparables), vehicles and vessels (with active secondary markets), standard machinery. Process: Identification of comparables, analysis of unit prices, adjustments for differences, determination of value. APPROACH 2: COST METHOD Fundament: The value of an asset is equal to the replacement or reproduction cost of an equivalent asset, minus accumulated depreciation. Application: Specialized assets without active secondary market, singular properties (churches, museums, monuments), infrastructure and public works, patrimonial assets. Process: Estimation of new replacement cost, calculation of depreciation (physical, functional, economic), determination of value.

APPROACH 3: INCOME METHOD (CAPITALIZATION) Fundament: The value of an asset is the present value of the future income it generates. Application: Income-generating assets (rents, concessions), operating businesses, assets with predictable cash flows. Process: Estimation of future income, estimation of operating expenses, calculation of net income, capitalization. Capitalization Rate (Cap Rate): Reflects the return required by the market. Typically: 4-8% for real estate, 8-12% for industrial, 10-15% for naval.
APPROACH 4: RWA TOKENIZATION VALUATION METHOD Fundament: Specific valuation for assets destined for tokenization, considering token liquidity, governance, and value distribution mechanisms. Application: Tokenized assets or destined for tokenization, fractions of high-value assets, assets with multiple holders. Process: Valuation of underlying asset, adjustment for token liquidity, adjustment for governance, adjustment for programmable compliance (SCCIP), determination of value per token.

8.4.3 Adjustments for Previous Phase Findings

The final valuation MUST adjust the base value according to the findings of the previous phases:
ADJUSTMENTS FOR PHASE 1 (DOCUMENTARY): • Financial charges: Reduction (-100% of the charge). • Restrictive easements: Reduction (-5% to -20%). • Pending licenses: Reduction (-10% to -30%). • Registry discrepancies: Reduction (-5% to -15%). • Deficient regulatory compliance: Reduction (-10% to -40%).
ADJUSTMENTS FOR PHASE 2 (PHYSICAL): • Critical defects: Reduction (-20% to -50%). • High defects: Reduction (-5% to -20%). • Medium defects: Reduction (-2% to -10%). • Exceptional state: Increment (+5% to

+15%). • Undocumented improvements: Increment (+3% to +10%).
ADJUSTMENTS FOR PHASE 3 (TECHNICAL): • Energy inefficiency: Reduction (-5% to -15%). • Environmental risk: Reduction (-10% to -30%). • Climate vulnerability: Reduction (-5% to -20%). • Obsolete systems: Reduction (-5% to -15%). • ESG certifications: Increment (+5% to +15%). • Superior energy efficiency: Increment (+3% to +10%).
FORMULA OF ADJUSTED VALUE: Final Value = Base Value × (1 - Σ Negative Adjustments + Σ Positive Adjustments)

8.4.4 Valuation Scenarios

Every valuation MUST present multiple scenarios:
OPTIMISTIC SCENARIO: Favorable market conditions, no materialization of identified risks, successful execution of improvement plans. Probability: 20-25%.
BASE SCENARIO: Normal market conditions, partial materialization of risks, standard execution of plans. Probability: 50-60%.
CONSERVATIVE SCENARIO: Unfavorable market conditions, materialization of main risks, delays in improvement plans. Probability: 15-20%.
STRESS SCENARIO: Economic or sectoral crisis, materialization of critical risks, extreme events. Probability: 5-10%.
EXPECTED VALUE: Expected Value = Σ (Value_scenario × Probability_scenario)

8.4.5 Integration with SCCIP

The SCCIP protocol uses the valuation for: • Determination of value per token: Initial token value based on valuation, update according to continuous monitoring (CAP), adjustment for critical events (automatic pauses). • Distribution mechanisms: Distribution of income based on

relative value, adjustments for capital contributions, dilution by issuance of new tokens. • Governance: Voting rights weighted by value, approval thresholds according to impact on value, protection of minorities.

8.5 PROFITABILITY AND ROI ANALYSIS

8.5.1 Profitability Metrics

CURRENT YIELD: Current Yield = Annual Net Income / Market Value.
TOTAL RETURN: Total Return = (Net Income + Capital Appreciation) / Initial Value.
RETURN ON INVESTMENT (ROI): ROI = (Net Profit / Investment Cost) × 100.
INTERNAL RATE OF RETURN (IRR): The discount rate that makes the Net Present Value (NPV) of the cash flows equal to zero. Interpretation: IRR > Cost of capital: Profitable investment. IRR < Cost of capital: Unprofitable investment. IRR = Cost of capital: Break-even point.
NET PRESENT VALUE (NPV): NPV = Σ [FCF_t / (1+r)^t] - Initial Investment. Interpretation: NPV > 0: Investment creates value. NPV < 0: Investment destroys value. NPV = 0: Investment neither creates nor destroys value.
PAYBACK PERIOD: Period of time necessary to recover the initial investment.

8.5.2 Cash Flow Analysis

CASH FLOW STRUCTURE: • Inflows: Rental or sales income, operating income, subsidies and grants, residual value at the end of the period. • Outflows: Operating costs (OPEX), capital investments (CAPEX), taxes, debt service (if applicable), final disposal costs. • Free Cash Flow

(FCF): FCF = Income - Operating Expenses - Taxes - CAPEX.
CASH FLOWS BY ASSET TYPOLOGY: • Real Estate: Inflows (rents, sales), Outflows (maintenance, taxes, insurance, management), CAPEX (improvements, major repairs), Horizon: 10-30 years. • Naval: Inflows (charter, freight, sales), Outflows (crew, fuel, maintenance, port), CAPEX (repairs, dry-docking, updates), Horizon: 5-25 years. • Industrial: Inflows (production, sales), Outflows (raw materials, energy, personnel, maintenance), CAPEX (machinery, process improvements), Horizon: 5-15 years. • Patrimonial: Inflows (appreciation, sales if realized), Outflows (insurance, storage, conservation), CAPEX (restoration, conservation), Horizon: Indefinite (long term).

8.5.3 Sensitivity Analysis

OBJECTIVE: Evaluate how the profitability of the asset changes in the face of variations in key parameters.
KEY PARAMETERS TO ANALYZE: • For real estate: Occupancy rate, rent level, capitalization rate, maintenance costs, discount rate. • For naval: Freight rates, fuel costs, operating days vs. port days, maintenance costs, residual value. • For industrial: Production volume, selling prices, raw material costs, production efficiency, energy costs.
METHODOLOGY:

1. Identification of critical variables (Tornado analysis).
2. Definition of variation ranges (±10%, ±20%, ±30%).
3. Calculation of impact (Recalculation of IRR, NPV, profitability).
4. Identification of critical points (Thresholds of negative profitability).

VISUALIZATION: Tornado chart, Spider chart.

8.5.4 Profitability Benchmarking

COMPARISON WITH MARKET BENCHMARKS: • Real Estate: Cap rates by typology and location, REIT returns, corporate bond yields (spread). • Naval: Freight rates by vessel typology, market indices (Baltic Exchange), naval investment fund returns. • Industrial: ROIC (Return on Invested Capital) by sector, EBITDA margins, cost of capital by sector. • Patrimonial: Art price indices (Artprice, Sotheby's), art fund returns, comparison with other safe-haven assets (gold, bonds).

8.6 STRESS TESTING AND SCENARIO ANALYSIS

8.6.1 Purpose of Stress Testing

Stress testing evaluates the financial resilience of the asset under adverse scenarios, identifying vulnerabilities and quantifying potential losses. It is particularly critical for Family Offices with wealth concentrations, investment funds with risk mandates, tokenization platforms that must protect investors, and insurers and reinsurers covering the asset.

8.6.2 Typology of Stress Scenarios

MACROECONOMIC SCENARIOS: • Economic Recession: GDP drop (-3% to -5%), unemployment increase, asset price drop (-15% to -30%), interest rate increase, rent reduction, vacancy rate increase. • Financial Crisis: Stock market drop (-30% to -50%), credit liquidity crisis, bond spread increase, currency devaluation, systemic contagion. • Stagflation: High inflation (+5% to +8%), low or negative economic growth, increase in

operating costs, difficulty in passing costs to clients, margin compression.
SECTORAL SCENARIOS: • Real Estate: Real estate sector crisis, adverse regulatory changes (rent control), structural changes (telework, e-commerce), oversupply in specific segments. • Naval: Drop in freight rates (-30% to -50%), increase in fuel costs (+50% to +100%), geopolitical crisis (route blockades), environmental regulatory changes (IMO 2030, 2050). • Industrial: Supply chain disruption, disruptive technological changes, loss of main clients, increase in raw material costs.
IDIOSYNCRATIC SCENARIOS: • Specific events of the asset: Severe physical damage (fire, flood), environmental contamination, important litigation, loss of licenses or permits, accelerated technological obsolescence.

8.6.3 Stress Testing Methodology

STEP 1: DEFINITION OF SCENARIOS Identification of relevant scenarios according to asset typology, geographical location, economic sector, and specific characteristics.
STEP 2: QUANTIFICATION OF IMPACTS For each scenario, estimate: Impact on income (percentage reduction), impact on costs (percentage increase), impact on asset value (depreciation), duration of the scenario (temporary vs. permanent).
STEP 3: FINANCIAL MODELING Recalculation of financial metrics under each scenario: Projected cash flows, NPV under stress scenario, IRR under stress scenario, payback period, debt coverage ratio (if applicable).
STEP 4: ANALYSIS OF RESULTS Identification of: Maximum potential losses, break-even points, most sensitive variables, mitigation measures.
STEP 5: RECOMMENDATIONS Strategies to improve resilience: Income diversification, reduction of fixed costs,

improvement of operational efficiency, adequate insurance coverages, emergency exit strategies.

8.6.4 Specific Stress Scenarios

SCENARIO 1: SEVERE RECESSION Assumptions: GDP drop -4%, unemployment increase +4%, real estate price drop -25%, interest rate increase +2.5%, rent reduction -15%, vacancy increase +8%. Impact on real estate asset: Income reduction -20%, expense increase (defaults) +5%, value depreciation -25%, NPV reduces by -40%, IRR drops from 8% to 3%. Mitigation measures: Renegotiation of lease contracts, reduction of operating costs, debt refinancing (if possible), sale of non-strategic assets.
SCENARIO 2: ENERGY CRISIS Assumptions: Energy price increase +100%, fuel cost increase +80%, general inflation +6%, mild economic recession -1% GDP. Impact on industrial asset: Energy cost increase +50%, margin reduction -30%, need for efficiency investment: €500,000, NPV reduces by -35%, IRR drops from 10% to 5%. Mitigation measures: Urgent investment in energy efficiency, long-term supply contracts, passing costs to clients (if possible), diversification of energy sources.
SCENARIO 3: ADVERSE REGULATORY CHANGE Assumptions: New environmental regulation (e.g., prohibition of fossil fuels), adaptation cost: €2,000,000, compliance deadline: 5 years, fines for non-compliance: €500,000/year. Impact on industrial asset: Mandatory CAPEX: €2,000,000, operating cost increase +15%, income reduction (if not adapted) -30%, NPV reduces by -45%, IRR drops from 12% to 4%. Mitigation measures: Gradual adaptation plan, search for subsidies and grants, negotiation of deadlines with regulators, business reorientation.
SCENARIO 4: NATURAL DISASTER Assumptions: Extreme event (earthquake, flood, hurricane), physical damage: 40% of asset value, activity interruption: 6

months, repair cost: €1,500,000, income loss during interruption: €500,000. Impact on asset: Immediate depreciation -40%, income loss €500,000, repair cost €1,500,000, NPV reduces by -60%, IRR drops from 9% to -2% (value destruction). Mitigation measures: Adequate insurance (full coverage), business continuity plan, emergency fund, geographical diversification of assets.

8.6.5 Reversibility Analysis

TEMPORARY SCENARIOS: • V-shaped recovery: Rapid recovery after shock. • U-shaped recovery: Slow but complete recovery. • L-shaped recovery: Prolonged stagnation.
PERMANENT SCENARIOS: • Irreversible structural changes. • Definitive technological obsolescence. • Permanent regulatory changes. • Irreversible environmental deterioration.
REAL OPTIONS ANALYSIS: • Option to wait (delay). • Option to expand (expand). • Option to contract (contract). • Option to abandon (abandon). • Option to change use (switch).

8.6.6 Integration with SCCIP

The SCCIP protocol uses stress testing results for: • Definition of risk thresholds: Dynamic thresholds according to scenario, automatic pauses in severe scenarios, preventive alerts in moderate scenarios. • Protection mechanisms: Capital reserve for adverse scenarios, automatic parametric insurance, emergency governance mechanisms. • Communication to investors: Transparency about identified risks, public stress scenarios, documented mitigation strategies.

8.7 INTEGRATION WITH TOKENIZED ASSET MARKETS

8.7.1 Valuation for Tokenization

The tokenization of physical assets requires a specific valuation that considers:
FRACTIONING OF THE ASSET: • Total number of tokens to be issued. • Nominal value per token. • Rights associated with each token. • Structure of token classes (if applicable).
TOKEN LIQUIDITY: • Expected secondary market. • Market depth. • Buy-sell spreads. • Transaction costs.
GOVERNANCE: • Voting rights per token. • Decision mechanisms. • Protection of minorities. • Information transparency.
VALUE DISTRIBUTION: • Dividends or income distributions. • Appreciation of token value. • Buyback mechanisms. • Liquidity events (redemption).

8.7.2 Capital Structure for Tokens

EQUITY TOKENS (Capital): • Represent fractional ownership of the asset. • Rights to income and appreciation. • Voting rights on important decisions. • Similar to shares of a company.
DEBT TOKENS: • Represent loans to the asset owner. • Right to fixed interest. • Priority in liquidation. • Similar to bonds.
UTILITY TOKENS: • Give access to services or benefits. • Do not represent ownership or debt. • Value based on utility. • Similar to coupons or vouchers.

HYBRID TOKENS: • Combine characteristics of equity and debt. • Complex distribution structures. • Conversion mechanisms. • Flexibility in rights.

8.7.3 Value Distribution Mechanisms

INCOME DISTRIBUTION: • Dividend model: Periodic distribution of net income. Proportional to number of tokens. Frequency: monthly, quarterly, annual. Mechanism: Smart contract executes automatically. • Reinvestment model: Reversal of income for asset improvements. Increases underlying asset value. Appreciation reflected in token value. Decision by governance. • Mixed model: Part of income distributed, part reinvested. Ratio determined by governance. Flexibility according to needs.
APPRECIATION MECHANISMS: • Automatic revaluation: Update of token value according to valuation. Frequency: quarterly, semi-annual, annual. Based on valuation by Aurema Group (PTVS). Reflected in market price. • Realization events: Partial sale of the asset, refinancing with equity extraction, extraordinary distribution. Decision by governance. • Token buyback: Use of income to buy back tokens. Reduces supply, increases remaining value. Price support mechanism. Executed by smart contract.

8.7.4 Dynamic Value Update

INTEGRATION WITH CAP (Continuous Assurance Protocol): The CAP protocol provides real-time data that feeds the update of the token's value. • Monitoring data: Physical state of the asset (IoT sensors), operating conditions, compliance with maintenance, detection of anomalies. • Impact on valuation: Physical deterioration → Value reduction. Implemented improvements → Value increase. Detected anomalies → Risk adjustment. Ongoing maintenance → Value preservation. • Update

frequency: Level 3: Quarterly update. Level 4: Monthly update. Critical events: Immediate update.
INTEGRATION WITH SCCIP (Smart Contract Compliance Integration Protocol): The SCCIP protocol automatically executes value adjustments based on predefined rules. • Adjustment rules: If SRF < 70 → Token value reduction 10%. If SRD < 80 → Pause transfers. If SRT < 75 → Alert to holders. If external valuation confirms → Value adjustment. • Protection mechanisms: Automatic pause for critical risks, manual release after resolution, automatic notification to holders, immutable record on blockchain.

8.7.5 Reporting for Token Holders

MANDATORY INFORMATION: • Financial information: Updated valuation of underlying asset, value per token, generated and distributed income, incurred costs, profitability metrics (IRR, NPV, Cap Rate). • Technical information: Current physical state of the asset (executive summary), risk scores (SRF, SRT, SRD, SRE), critical findings and corrective actions, maintenance and CAPEX plan. • Risk information: Updated risk matrix, stress testing results, adverse scenarios and probabilities, mitigation strategies. • Governance information: Decisions made by holders, proposals under vote, voting results, execution of decisions.
REPORTING FREQUENCY: • Level 1: Annual. Basic summary. • Level 2: Semi-annual. Complete report. • Level 3: Quarterly. Detailed report + dashboard. • Level 4: Monthly. Complete report + dashboard + meetings.
REPORTING FORMAT: • Interactive dashboard: Key metrics in real time, evolution charts, alerts and notifications, access from web platform. • PDF reports: Executive report (2-5 pages), complete technical report (20-50 pages), annexes with detail of findings, certificates and validations. • Meetings with holders: Level 3: Annual.

Level 4: Quarterly. Presentation of results, Q&A session, live voting.

8.7.6 Regulatory Compliance (MiCA and others)

MiCA (Markets in Crypto-Assets Regulation - EU): Requirements applicable to tokenized RWAs: Transparent information to the investor, assessment of suitability and appropriateness, management of conflicts of interest, regulatory reporting, investor protection. Issuer obligations: White paper with complete information of the asset, independent and verifiable valuation, clear governance mechanisms, business continuity plan, risk management policies. Investor rights: Clear and non-misleading information, right of withdrawal (14 days), complaint mechanisms, protection against fraud, cost transparency.
OTHER JURISDICTIONS: • United States (SEC): Possible classification as security, registration or exemption requirements, periodic reporting (10-K, 10-Q), compliance with anti-fraud norms. • Switzerland (FINMA): Classification according to token type, proportional requirements to risk, AML/KYC obligations, reporting to regulator. • Singapore (MAS): Payment Token vs. Security Token, licensing requirements, AML/CFT compliance, investor protection measures.
INTEGRATION WITH PTVS: The Prop Trust Verified Standard provides the compliance basis for MiCA and other regulations. • Independent verification: Valuation by accredited body, audit of technical findings, certification of compliance, continuous updating. • Transparency: Public valuation methodology, documented inspection findings, identified and quantified risks, stress scenarios. • Investor protection: Objective risk scores, automatic alerts (SCCIP), governance mechanisms, periodic reporting.

8.8 FINANCIAL RISK MATRIX

8.8.1 Classification of Financial Risks

MARKET RISKS: • Price risk: Fluctuations in asset value, changes in market prices, volatility of comparables. Impact: High. • Interest rate risk: Increase in interest rates, impact on financing cost, impact on asset values. Impact: Medium-High. • Exchange rate risk: Currency fluctuations (international assets), impact on value in local currency, impact on income/costs. Impact: Variable. • Liquidity risk: Difficulty in selling the asset, wide buy-sell spreads, shallow markets. Impact: Medium.
CREDIT RISKS: • Default risk: Tenants or clients not paying, counterparties breaching contracts, loss of income. Impact: High. • Concentration risk: Dependence on few clients/tenants, lack of diversification, vulnerability to specific losses. Impact: High. • Counterparty risk: Risk that counterparties default, suppliers, tenants, partners, impact on operations. Impact: Medium.
OPERATIONAL RISKS: • Cost risk: Increase in operating costs, inflation of maintenance costs, increase in taxes and insurance. Impact: Medium. • Income risk: Reduction of income, vacancies, defaults, drops in demand, changes in market preferences. Impact: High. • Management risk: Poor asset management, incorrect investment decisions, lack of adequate maintenance. Impact: Medium-High.
LEGAL AND REGULATORY RISKS: • Regulatory risk: Changes in applicable regulations, new compliance requirements, adaptation costs. Impact: High. • Legal risk: Litigation with third parties, liability claims, contractual disputes. Impact: Variable. • Tax risk: Changes in tax regulations, tax increases, disputes with tax authorities. Impact: Medium.

8.8.2 Financial Risk Score (SRE)

The Standard calculates a Financial Risk Score (SRE) from 0 to 100:
FORMULA: SRE = 100 - (Σ Critical Risks × 20) - (Σ High Risks × 10) - (Σ Medium Risks × 3) - (Σ Low Risks × 1)
INTERPRETATION: • 90-100: Excellent financial profile, minimal risks. • 75-89: Good financial profile, manageable minor risks. • 60-74: Acceptable financial profile, medium risks require attention. • 40-59: Deficient financial profile, high risks require resolution. • 0-39: Critical financial profile, unacceptable risks.
CERTIFICATION THRESHOLD: • Level 1: Minimum SRE 60. • Level 2: Minimum SRE 70. • Level 3: Minimum SRE 80. • Level 4: Minimum SRE 90.

8.8.3 Risk Matrix by Typology

Real Estate Residential: Market (Medium), Credit (Medium), Operational (Low), Legal (Low). Typical SRE: 78. Real Estate Commercial: Market (High), Credit (High), Operational (Medium), Legal (Medium). Typical SRE: 68. Real Estate Industrial: Market (High), Credit (Medium), Operational (High), Legal (High). Typical SRE: 65. Yacht/Megayacht: Market (High), Credit (Low), Operational (High), Legal (Medium). Typical SRE: 62. Commercial Vessel: Market (Very High), Credit (Medium), Operational (High), Legal (High). Typical SRE: 58. Industrial Machinery: Market (Medium), Credit (Medium), Operational (High), Legal (Low). Typical SRE: 70. Infrastructure: Market (Low), Credit (Low), Operational (Medium), Legal (High). Typical SRE: 75. Patrimonial Asset: Market (High), Credit (Low), Operational (Low), Legal (Medium). Typical SRE: 72.

8.8.4 Mitigation Strategies

DIVERSIFICATION: Geographical diversification, diversification by asset typology, diversification of tenants/clients, diversification of currencies.
HEDGING: Adequate insurance (property, liability, loss of income), financial hedging (interest rates, currencies), contracts with protection clauses, reserve funds.
OPTIMIZATION: Optimal capital structure, active cost management, continuous improvement of efficiency, efficient tax planning.
MONITORING: Real-time financial KPIs, early alerts for deviations, periodic reporting to stakeholders, review and adjustment of strategies.

8.9 REPORTING AND

CERTIFICATION OF PHASE 4

8.9.1 Structure of the Phase 4 Report

The Phase 4 report SHALL include:
SECTION 1: EXECUTIVE SUMMARY • Final valuation of the asset. • Key profitability metrics (IRR, NPV, Cap Rate). • Financial Risk Score (SRE). • Main findings and recommendations. • Stress testing scenarios.
SECTION 2: LIFE CYCLE COST ANALYSIS • Cost structure (OPEX, CAPEX). • Projection of future costs. • Equivalent Annual Cost (EAC). • Optimization opportunities.
SECTION 3: REMAINING USEFUL LIFE AND DEPRECIATION • Estimation of remaining useful life. • Depreciation analysis. • Projected residual value. • Obsolescence factors.

SECTION 4: MARKET VALUATION • Applied valuation methodology. • Valuation by comparison, cost, and income methods. • Adjustments for findings of previous phases. • Valuation scenarios (optimistic, base, conservative, stress). • Specific valuation for tokenization (if applicable).
SECTION 5: PROFITABILITY ANALYSIS • Projected cash flows. • Profitability metrics (IRR, NPV, ROI, Payback). • Sensitivity analysis. • Market benchmarking.
SECTION 6: STRESS TESTING • Defined stress scenarios. • Impact on financial metrics. • Resilience analysis. • Mitigation strategies.
SECTION 7: INTEGRATION WITH TOKENIZATION • Capital structure for tokens. • Value distribution mechanisms. • Dynamic value update. • Reporting for holders.
SECTION 8: FINANCIAL RISK MATRIX • Classification of financial risks. • Financial Risk Score (SRE). • Mitigation strategies. • Monitoring and control.
SECTION 9: STRATEGIC RECOMMENDATIONS • Investment and divestment strategy. • Capital structure optimization. • Value creation plan. • Tracking KPIs.
ANNEXES: • Detailed financial models. • Analysis of comparables. • Market studies. • Detailed stress testing scenarios. • Valuation certificates.

8.9.2 Phase 4 Certification

Upon completing Phase 4, the inspection body issues:
FINANCIAL AND VALUE VERIFICATION CERTIFICATE • Level of verification achieved (1, 2, 3, or 4). • Final valuation of the asset (with confidence intervals). • Main profitability metrics. • Financial Risk Score (SRE). • Date of issuance and validity (3-12 months according to level). • Digital signature of the valuation team. • Cryptographic hash registered on blockchain (Levels 3 and 4). • Access to the interactive financial model (Levels 3 and 4).

CONDITIONS OF CERTIFICATION: • The valuation is valid for the date of issuance. • Significant market changes require revaluation. • Critical events activate immediate revaluation (SCCIP). • Continuous monitoring (Levels 3 and 4) updates valuations. • The certificate does not constitute investment advice.

CHAPTER 9: CONTINUOUS ASSURANCE PROTOCOL (CAP)

9.1 PURPOSE AND SCOPE OF THE PROTOCOL

9.1.1 From Point-in-Time Inspection to Continuous Assurance

The Continuous Assurance Protocol (CAP) represents the most radical transformation introduced by The Trust Standard compared to traditional verification

methodologies. While conventional standards operate under a paradigm of a "momentary snapshot"—where a report reflects the asset's state at a specific instant and loses validity over time—CAP establishes a paradigm of "perpetual monitoring" that maintains certainty about the asset continuously.
This transformation is particularly critical in the context of tokenized Real World Assets (RWA), where institutional investors and token holders require continuous guarantees regarding the physical state of the underlying asset. A $50 million yacht or a $100 million commercial building cannot rely on an annual inspection to validate its value; it requires permanent, verifiable monitoring.

9.1.2 Objectives of CAP

OBJECTIVE 1: Continuous Physical State Monitoring Maintain permanent knowledge of the asset's state through IoT sensor networks, detecting deviations, anomalies, and deterioration trends in real-time.
OBJECTIVE 2: Early Risk Detection Identify emerging risks before they materialize into significant damage, allowing preventive interventions that preserve the asset's value.
OBJECTIVE 3: Dynamic Digital Twin Updating Continuously feed the asset's digital twin with verified data, maintaining a precise and updated virtual replica of the physical asset.
OBJECTIVE 4: Support for Programmable Compliance (SCCIP) Provide the real-time data necessary for the SCCIP protocol to automatically execute compliance rules in smart contracts.
OBJECTIVE 5: Feeding Predictive Analysis (PRA) Supply continuous data streams to the PRA protocol to improve the accuracy of predictive models and validate previous predictions.

OBJECTIVE 6: Generation of Continuous Cryptographic Evidence Produce an immutable and verifiable record of the asset's evolution over time, cryptographically signed according to the CPEP protocol.

9.1.3 Application by Verification Level

Level 1: Not applicable. Level 2: Optional, limited to critical points. Level 3: Mandatory. Complete sensor network with real-time monitoring. Level 4: Mandatory. Redundant sensor network with advanced edge computing and 24/7 institutional support.

9.2 CAP SYSTEM ARCHITECTURE

9.2.1 Five-Layer Architecture

The CAP system is structured into five technological layers that operate in a coordinated manner:
LAYER 1: PERCEPTION (Edge Layer) Physical devices that capture data from the real world: specialized IoT sensors, embedded measurement devices, autonomous drones, computer vision systems, and remote inspection equipment.
LAYER 2: CONNECTIVITY (Network Layer) Data transmission infrastructure: LPWAN networks (LoRaWAN, NB-IoT, Sigfox), cellular connectivity (4G/5G), satellite (for naval and remote assets), WiFi/Mesh for local environments, and industrial protocols (Modbus, OPC-UA, BACnet).
LAYER 3: EDGE COMPUTING (Processing Layer) Decentralized processing near the origin: intelligent gateways with processing capacity, data filtering and aggregation, local anomaly detection, cryptographic edge signing, and temporary storage (buffer).
LAYER 4: CLOUD PLATFORM (Data Layer) Central management infrastructure: data lakes for massive

storage, AI and analysis engines, alert management systems, APIs for external integration, dashboards, and visualization tools.
LAYER 5: APPLICATION (Integration Layer) Application and connection layer with other protocols: asset digital twin, SCCIP protocol (smart contracts), PRA protocol (predictive analysis), CPEP protocol (cryptographic proof), and client systems (ERP, CMMS, BIM).

9.2.2 End-to-End Data Flow

1.Capture (Milliseconds): Sensor detects physical phenomenon, converts to digital, generates SHA-256 hash, adds precise timestamp and GPS metadata.
2.Edge Signing (Milliseconds): Device signs the hash with its private key, generates signature certificate, verifies local integrity.
3.Transmission (Milliseconds to seconds): Encrypted packet transmitted via secure protocol, gateway confirms receipt, automatic retries on failure.
4.Edge Processing (Seconds): Gateway verifies device signature, aggregates and filters data, detects local anomalies, stores temporarily if cloud connection is lost.
5.Cloud Processing (Seconds to minutes): Cloud verifies cryptographic signature, stores in data lake, registers hash on blockchain (CPEP), runs AI analysis (PRA), updates digital twin, evaluates SCCIP rules.
6.Action (Minutes to hours): Generates alerts if thresholds are exceeded, notifies stakeholders according to severity, executes automatic SCCIP actions, updates dashboards and reports.

9.2.3 Performance Requirements

Availability: Level 3 requires 99.5% uptime; Level 4 requires 99.99% uptime. Latency: Capture to gateway <1 second; gateway to cloud <5 seconds; cloud processing <30 seconds; critical alert to stakeholder <5 minutes. Security: End-to-end encryption (AES-256), mutual device-server authentication, automatic key rotation (every 90 days), complete access auditing. Scalability: Support for millions of sensors per asset (large infrastructures), processing terabytes of daily data, horizontally scalable distributed architecture.

9.3 TYPES OF SENSORS AND DEVICES

9.3.1 Classification by Parameter

STRUCTURAL SENSORS: • Triaxial accelerometers: Vibrations and acceleration. Application: Machinery, structures. • Extensometers: Deformations and cracks. Application: Bridges, buildings. • Strain gauges: Stresses and deformations. Application: Critical structures. • Inclinometers: Inclinations and tilts. Application: Structures, slopes. • Crack meters: Crack opening. Application: Structural elements.
ENVIRONMENTAL SENSORS: • Thermocouples/Thermistors: Temperature. Application: General, machinery. • Hygrometers: Relative humidity. Application: Buildings, warehouses. • Anemometers: Wind speed. Application: Structures, naval assets. • Gas sensors: CO, CO2, VOC, CH4. Application: Industrial, parking.

ELECTRICAL AND ENERGY SENSORS: • Network analyzers: Voltage, current, power, THD. Application: Electrical installations. • Energy meters: Active/reactive energy. Application: Energy audits. • Pyranometers: Solar radiation. Application: Solar installations.
NAVAL SPECIALIZED SENSORS: • Bilge level sensors: Water level in bilges. Application: Naval safety. • Gyrocompass/IMU: Orientation, pitch, roll. Application: Naval stability. • Tachometers: Engine/propulsion RPM. Application: Propulsion. • Oil analyzers: Particles, viscosity, water. Application: Predictive maintenance.

9.3.2 Selection Criteria

Technical Suitability: Measurement range appropriate to the parameter, precision adequate to the asset's risk, response time compatible with the phenomenon's dynamics. Environmental Robustness: Ingress protection (minimum IP67 for outdoors, IP68 for immersion), operating temperature range, resistance to vibrations, shocks, and corrosion, ATEX certifications for explosive zones. Energy Autonomy: Power consumption compatible with required autonomy, power options (battery, solar energy, energy harvesting), battery life (minimum 2 years for Level 3, 5 years for Level 4). Connectivity: Protocol compatible with available infrastructure, appropriate transmission range, sufficient bandwidth for data volume, roaming support (mobile assets). Regulatory Compliance: Metrological certifications (ISO 17025 for calibration), electrical safety certifications, EMC certifications, specific sectoral certifications (marine, ATEX).

9.4 MONITORING BY ASSET TYPOLOGY

9.4.1 High-Value Residential Real Estate

Critical Parameters: • Structure: Cracks in structural elements, differential settlements, dampness in foundation and walls, reinforcement corrosion. • Installations: Water leaks, ambient temperature and humidity, indoor air quality, energy consumption by circuit. • Security: Intrusion, fire, flooding in basements, gas leaks. Recommended Density: 15-25 sensors for a single-family home; 50-100 sensors for a luxury villa. Sampling frequency: 1-15 minutes depending on the parameter.

9.4.2 Commercial and Office Real Estate

Additional Critical Parameters: • HVAC Systems: Supply and return temperatures, duct pressures, filter status, energy consumption of consumers, air quality by zone. • Vertical Transport: Cabin vibrations, cycle times, energy consumption, door and mechanism status. • Space Management: Occupancy by zones, meeting room usage, temperature by zone, illuminance at workstations. Recommended Density: 200-500 sensors for a medium office building; 1000+ sensors for emblematic buildings. Frequency: 1-5 minutes for HVAC, 15 minutes for structural.

9.4.3 Industrial Real Estate

Additional Critical Parameters: • Production Processes: Process variables (pressure, temperature, flow), status of critical machinery, product quality, energy efficiency by line. • Environment: Atmospheric emissions, discharges,

ambient noise, air quality in work areas. • Industrial Safety: Detection of hazardous gases, ATEX zones, access to restricted areas, status of safety equipment. Recommended Density: 500-2000 sensors for a medium industrial plant; 5000+ sensors for refineries or chemical plants. Frequency: 1 second to 1 minute for critical processes.

9.4.4 Yachts and Megayachts

Specific Critical Parameters: • Naval Safety: Bilge level in all compartments (critical), watertight door status, pumping systems, fire detection by zone, gas detection in enclosed spaces. • Machinery and Propulsion: Engine temperatures (exhaust, cooling, oil), fuel and oil pressures, engine and generator RPM, shaft line vibrations, continuous oil analysis. • Hull and Structure: Stresses at critical points, hull deformations, corrosion, humidity in spaces. • Naval Systems: Tank levels (fuel, water, black water), system pressures, battery status, electrical power quality. Recommended Density: 200-500 sensors for a 30-40m yacht; 500-1500 sensors for a 60m+ megayacht. Frequency: 1 second for safety, 1 minute for comfort.

9.5 THRESHOLDS, ALERTS, AND ESCALATION

9.5.1 Hierarchy of Thresholds

CAP establishes four levels of thresholds for each monitored parameter:
NORMAL THRESHOLD (GREEN): Values within expected ranges. Operation without intervention. Data recorded for trend analysis. Asset state: Optimal.

ATTENTION THRESHOLD (YELLOW): Moderate deviation from reference values. Notification to the maintenance team. Trend analysis to detect patterns. Asset state: Requires increased monitoring.
ALERT THRESHOLD (ORANGE): Significant deviation that may indicate an emerging problem. Immediate notification to the technical manager. Activation of additional verification protocols. Asset state: Requires planned intervention. SCCIP Impact: Alert to holders, allows trading.
CRITICAL THRESHOLD (RED): Value that compromises safety, integrity, or asset value. Immediate multi-channel notification (phone, SMS, email). Activation of emergency protocols. Asset state: Requires immediate intervention. SCCIP Impact: Automatic pause of transfers.

9.5.2 Multi-Channel Alert System

Email: Attention and higher level alerts. Daily summary of events. Weekly trend reports. SMS: Alert and Critical level alerts. Concise messages with required action. Mandatory receipt confirmation. Push Notifications (App): All alerts in real-time. Rich interface with graphs and context. Ability to acknowledge and comment. Automated Phone Call: Critical alerts only. Voice message with essential information. Requires confirmation by code. Escalation if not answered in 5 minutes. Real-Time Dashboard: Visualization of all parameters. Asset heat maps. Alert history. Public access (limited) for token holders.

9.5.3 False Alarm Management

False alarms erode trust in the system and generate "alert fatigue," where operators ignore legitimate notifications. Mitigation Strategies: • Multi-Sensor Confirmation: An alert requires confirmation from multiple related sensors. • Trend Analysis: Not just instantaneous value, but rate of change. Avoids alerts for transient peaks. • Temporal

Contextualization: Consideration of hourly, daily, and seasonal patterns. Adaptive thresholds based on context. • Machine Learning: AI models that learn from previous false alarms. Automatic threshold adjustment based on history. • Quality Metrics: False alarm rate (target: <5%), real detection rate (target: >95%), mean confirmation time.

9.6 INTEGRATION WITH DIGITAL TWIN

9.6.1 Concept of Digital Twin in PTVS

The digital twin is a dynamic virtual replica of the physical asset that updates in real-time through CAP data. It is not a static model (like a traditional BIM), but a living organism that faithfully reflects the current state of the asset and allows simulation, analysis, and prediction.
Fundamental Characteristics: • Bidirectional Synchronization: The twin receives data from the physical (sensors) and can send commands to the physical (actuators). • Multi-Scale Fidelity: Macro level (general state), meso level (systems and subsystems), micro level (individual components). • Predictive Capacity: Simulation of future scenarios, what-if analysis, prediction of failures and degradation. • Historical Memory: Complete record of temporal evolution, long-term trend analysis, support for strategic decisions.

9.6.2 Real-Time Updating

Update Flow:

1. Sensor captures data → Edge signing (CPEP).
2. Gateway processes → Aggregates and filters.
3. Cloud receives → Verifies and stores.

4. Twin engine → Updates component state.
5. Visualization → Reflects change in interface.
6. Analysis → PRA evaluates implications.
7. SCCIP → Executes rules if applicable.

Latency Objective: Level 3 <5 minutes from capture to visualization; Level 4 <30 seconds.

9.6.3 Use Cases of the Digital Twin

Remote Inspection: Inspector accesses the twin from any location, visualizes current state of all components, identifies areas requiring physical inspection, reduces travel costs and time. Scenario Simulation: Simulate impact of an earthquake on a building, model effect of climate changes on a yacht, evaluate consequences of structural modifications, optimize maintenance plans. Operator Training: Training in a safe virtual environment, simulation of emergency situations, familiarization with complex systems, risk reduction during training. Maintenance Planning: Visualization of components requiring intervention, optimization of maintenance routes, simulation of interventions before executing them, management of spare parts and resources. Stakeholder Communication: Visual presentation of asset state, transparency for token holders, executive reporting with impactful visualizations, support for investment decisions.

9.7 CYBERSECURITY AND PRIVACY

9.7.1 Threats to the CAP System

Device Threats: Physical manipulation of sensors, device spoofing, extraction of cryptographic keys, local Denial of Service (DoS). Communication Threats: Data interception (eavesdropping), data manipulation in transit (MITM), injection of false data, replay attacks. Platform Threats: Unauthorized access to data, manipulation of business

logic, exfiltration of sensitive data, ransomware and cryptolockers. Integration Threats: Compromise of connected systems, lateral propagation from IoT, manipulation of smart contracts (SCCIP), oracle attacks.

9.7.2 Defense in Depth Strategy

Physical Security: Tamper detection on devices, inviolable security seals, protected location of gateways, access control to facilities. Device Security: Secure boot with chain of trust, secure storage of keys (TPM, secure enclave), digitally signed updates, cryptographic capabilities in hardware. Communication Security: TLS 1.3 for cloud communications, end-to-end encryption (AES-256-GCM), mutual authentication (X.509 certificates), perfect forward secrecy. Platform Security: Zero-trust architecture, network segmentation, SIEM (Security Information and Event Management), anomaly detection with AI. Data Security: Encryption at rest (AES-256), key management (HSM, KMS), masking of sensitive data, encrypted and geo-distributed backups.

9.7.3 Privacy by Design

The system MUST implement privacy from its architecture, minimizing the capture of personal data and applying anonymization and Zero-Knowledge Proofs techniques when necessary.
Compliance: GDPR and data protection, MiCA and financial regulations, ISO 27001 and information security, specific sectoral standards.
Identity and Access Management: Role-Based Access Control (RBAC). Owner (Total access), Technical Manager (Operational access), Inspector (Read access), Token Holder (Limited access, Level 3-4), Maintenance (Operational access), Auditor (Read access + logs). Multi-Factor Authentication (MFA) mandatory for all remote accesses. Complete access auditing.

9.8 DATA GOVERNANCE AND QUALITY

9.8.1 Data Quality Dimensions

Accuracy: Data correctly reflects physical reality. Regular sensor calibration. Cross-verification with physical inspections. Target: <1% error for critical parameters. Completeness: Absence of gaps in time series. Coverage of all required parameters. Target: >99.5% data available. Consistency: Coherence between related sensors. Absence of contradictions. Precise temporal synchronization. Target: 100% logical consistency. Timeliness: Data available when needed. Latency within defined SLA. Target: <5 min for Level 3, <30 sec for Level 4. Validity: Data within expected ranges. Correct format and structure. Target: 100% valid data.

9.8.2 Quality Assurance Processes

Edge Validation: Acceptable value range, maximum rate of change, consistency with related sensors, rejection of impossible values. Gateway Validation: Statistical aggregation, outlier detection, sequence verification, compression with controlled loss. Cloud Validation: Historical trend analysis, sensor drift detection, cross-validation with models, correlation analysis. Periodic Validation: Calibration according to manufacturer (typically annual), comparison with reference instruments, data quality audit, metrological certification (ISO 17025).

9.8.3 Retention and Archiving

Critical Data (Safety): 5 years online, 25 years archive, permanent final destination. Operational Data: 2 years

online, 10 years archive, deletion. Environmental Data: 1 year online, 5 years archive, deletion. Comfort Data: 6 months online, 2 years archive, deletion. System Logs: 1 year online, 3 years archive, deletion.
File Formats: Raw data in open formats (Parquet, CSV), metadata in JSON-LD, XML, 3D models in IFC, glTF, documents in PDF/A for long-term.

9.9 INTEGRATION WITH PRA AND SCCIP

9.9.1 Feeding the PRA (Predictive Risk Assessment)

Data provided by CAP to PRA: • Time series of parameters: Historical evolution of all monitored parameters, seasonal and cyclical patterns, long-term trends, anomalous events and their causes. • Asset states: Current state of each component, historical states, transitions between states, time in each state. • Events and alerts: Complete record of generated alerts, response time, effectiveness of interventions, associated costs. • Operating conditions: Real loads and solicitations, environmental conditions, usage patterns, maintenance performed.
Applications fed by CAP: Failure prediction (RUL models, detection of pre-failure patterns, estimation of failure probability, intervention recommendations), maintenance optimization (condition-based maintenance, interval optimization, spare parts needs prediction, resource planning), degradation analysis (calibrated degradation models, identification of accelerating factors, remaining useful life prediction, operation recommendations),

anomaly detection (identification of unusual behaviors, early detection of problems, classification of anomaly types, response prioritization).

9.9.2 Support to SCCIP (Smart Contract Compliance Integration Protocol)

Data provided by CAP to SCCIP: • Updated scores: SRF (Physical Risk Score) in real-time, SRT (Technical Risk Score) updated, SRE (Structural Risk Score) dynamic, SRS (Systems Risk Score) continuous. • Risk states: Current risk level (Normal, Attention, Alert, Critical), duration of current state, trend (improving, stable, worsening), history of state changes. • Critical events: Detection of events requiring pause, verification of event resolution, confirmation of return to normality, immutable record of events. • Compliance conditions: Verification of up-to-date maintenance, compliance with operating conditions, respect for environmental limits, conformity with regulations.
Integration mechanisms: Compliance oracle (CAP acts as data source for SCCIP oracle, cryptographically signed data (CPEP), secure transmission to smart contracts, verification by multiple nodes), dynamic rules (thresholds adjustable according to context, rules that evolve with the asset, complex conditional logic, governance mechanisms for changes), automatic execution (automatic transfer pause for critical risks, manual release after resolution, automatic notification to holders, immutable record of actions on blockchain).

9.10 CAP REPORTING AND CERTIFICATION

9.10.1 Types of Reports

Real-Time Report (Dashboard): Current state of all parameters, asset heat map, active alerts, recent trends (last 24h). Access: 24/7 for authorized stakeholders. Daily Report (Automatic): Summary of events of the day, system operation statistics, alerts generated and resolved, detected anomalies. Access: Automatic email to technical team. Weekly Report: Weekly trend analysis, comparison with previous week, maintenance KPIs, intervention recommendations. Access: Asset manager, technical team. Monthly Report: Detailed evolution analysis, asset health state, maintenance compliance, avoided cost analysis. Access: Owner, holders (Level 3-4). Quarterly Report: Strategic analysis of the period, PRA prediction validation, valuation update (with PRA), strategic recommendations. Access: Owner, board of directors, holders. Annual Report: Complete balance of the year, historical evolution of the asset, CAP system ROI analysis, strategic plan for the following year. Access: All stakeholders.

9.10.2 CAP Certification

For assets with active CAP, Aurema Group issues a specific Continuous Monitoring Certificate:
Information included: Identification of monitored asset (AUI), monitoring level (3 or 4), number and types of installed sensors, certification period, current system state (operational, degraded, down), system uptime (last 12

months), data quality score, cryptographic hash of current state.
Validity: Level 3: 12 months, with quarterly updates. Level 4: 24 months, with monthly updates. Certification is suspended if the system is down >24h without justification.
Conditions: Certificate reflects state at time of issuance, significant changes require update, system must remain continuously operational, annual quality audits of the system.

9.10.3 Annual Audit of the CAP System

Scope: Calibration of all sensors, verification of cryptographic integrity, review of security policies, analysis of data quality, evaluation of alert effectiveness.
Acceptance Criteria: >95% of sensors within tolerances, 100% cryptographic integrity verified, 0 critical security vulnerabilities, >99% data quality, <5% false alarm rate.
Corrective Actions: Maximum 30 days for corrections, re-audit of corrected elements, possible suspension of certificate if not corrected.

CHAPTER 10: PHYSICAL-DIGITAL BRIDGE PROTOCOL (PDBP) AND SMART CONTRACT COMPLIANCE INTEGRATION PROTOCOL (SCCIP)

10.1 PURPOSE AND SCOPE OF THE PROTOCOLS

10.1.1 The Fundamental Challenge of Tokenized Real World Assets

The tokenization of Real World Assets (RWA) faces an existential challenge that no previous standard has comprehensively solved: **How can an institutional investor mathematically verify that the digital token they hold is backed by a real physical asset, in the declared state, and that it will continue to be so in the future?**
Existing standards (ERC-721, ERC-1155, ERC-3643) provide technical frameworks for the digital representation of assets, but none address the continuous verification of the underlying physical asset. This creates a trust gap that has limited the institutional adoption of tokenized RWAs.
The **Physical-Digital Bridge Protocol (PDBP)** and the **Smart Contract Compliance Integration Protocol (SCCIP)** constitute The Trust Standard's response to this challenge. Together, they form the institutional bridge between the physical and digital worlds.

10.1.2 Differentiation between PDBP and SCCIP

While both protocols work in coordination, they have clearly differentiated responsibilities:
PDBP (Physical-Digital Bridge Protocol): • Function: Cryptographically link the physical asset with its digital representation. • Responsibility: Guarantee that the token faithfully represents the physical asset. • Focus: Authenticity, integrity, physical-digital correspondence. • Key Technology: Oracles, cryptographic signing, digital twin.
SCCIP (Smart Contract Compliance Integration Protocol): • Function: Inject compliance rules into smart contracts based on verified physical data. • Responsibility: Automatically execute protections when the physical asset changes state. • Focus: Programmable compliance,

automatic execution, governance. • Key Technology: Smart contracts, compliance oracles, multi-sig.

10.1.3 Joint Objectives

OBJECTIVE 1: Verifiable Institutional Trust Provide institutional investors with mathematical certainty that the token is backed by a real, continuously verified physical asset.
OBJECTIVE 2: Automatic Programmable Compliance Enable the automatic execution of compliance rules in smart contracts when changes are detected in the physical asset's state, without manual intervention.
OBJECTIVE 3: Dynamic Value Protection Protect the value of tokens through automatic transfer pauses when critical risks are detected in the underlying physical asset.
OBJECTIVE 4: Radical Transparency Provide all stakeholders with verifiable access to the asset's state and its correspondence with the token via real-time dashboards and immutable records.
OBJECTIVE 5: Multi-Level Governance Establish governance mechanisms that balance automation with human control for critical decisions, using multi-sig signatures and supervisory committees.
OBJECTIVE 6: Standard Interoperability Ensure compatibility with major tokenization standards (ERC-3643, ERC-721, ERC-1155) and oracle networks (Chainlink, Band Protocol, API3).

10.1.4 Application by Verification Level

Level 1: Not applicable. Level 2: Not applicable. Level 3: Basic linking, basic automatic pauses, 2-of-3 multi-sig governance. Level 4: Advanced linking with Zero-Knowledge Proofs (ZKP), complete programmable compliance, 3-of-5 multi-sig with committee.

10.2 ARCHITECTURE OF THE PHYSICAL-DIGITAL BRIDGE PROTOCOL (PDBP)

10.2.1 Fundamental Concept: The Cryptographic Bridge

The PDBP establishes a bidirectional cryptographic bridge between the physical asset and its digital representation (token). This bridge is not a simple reference or metadata; it is a mathematically verifiable link that guarantees:

1. Existence: The physical asset truly exists.
2. Correspondence: The token specifically represents that asset.
3. State: The current state of the asset is as declared.
4. Continuity: The link is maintained over time.
5. Integrity: No part has been manipulated.

10.2.2 Four-Layer Architecture

LAYER 1: PHYSICAL LAYER The real physical asset, the IoT sensor network (CAP), periodic physical inspections (Phase 2), and the continuously updated digital twin.
LAYER 2: VERIFICATION LAYER Accredited inspection bodies, cryptographic edge signing (CPEP), verification certificates with hash on blockchain, and periodic compliance audits.
LAYER 3: ORACULIZATION LAYER Oracles that extract data from the physical world, decentralized oracle networks, data aggregation and validation, and feeding of smart contracts.

LAYER 4: DIGITAL LAYER The representative token (ERC-3643, ERC-721, etc.), compliance smart contracts (SCCIP), immutable registry on blockchain, and stakeholder interfaces.

10.2.3 End-to-End Data Flow

1.Capture: IoT sensor captures physical data, cryptographic edge signing (CPEP), precise timestamp and geolocation, secure transmission to gateway.
2.Verification: Gateway processes and aggregates data, cloud platform validates integrity, analysis engine (PRA) evaluates state, generation of state certificate.
3.Oraculization: Oracle extracts verified data, signs with its private key, transmits to oracle network, consensus among multiple oracles, publication of data on blockchain.
4.Execution: Smart contract receives data from oracle, evaluates compliance rules (SCCIP), executes automatic actions (if applicable), updates token state, notifies stakeholders.

10.2.4 Linking Models

DIRECT LINKING (1:1): One token represents one complete physical asset. Application: High-value assets (yachts, singular buildings). Advantage: Simplicity, clarity of ownership.
FRACTIONAL LINKING (1:N): Multiple tokens represent fractions of an asset. Application: Very high-value assets (megayachts, commercial buildings). Advantage: High liquidity, accessibility.
AGGREGATED LINKING (N:1): One token represents a portfolio of assets. Application: Diversified asset funds. Advantage: Diversification, professional management.

DYNAMIC LINKING (N:M): Tokens that can represent different assets based on conditions. Application: Complex structured products. Advantage: Flexibility, optimization.

10.2.5 Asset Unique Identifier (AUI)

Every physical asset verified under PTVS receives an Asset Unique Identifier (AUI) that unequivocally links it to its digital representation.
Structure: AUI-[TYPE]-[JURISDICTION]-[REGISTRY]-[VERSION]-[HASH] Example: AUI-NAV-ES-VLC001-v3-0x7f8e9d...
Properties: Globally unique, immutable (only changes with new verification), verifiable on blockchain, human and machine-readable, compatible with international standards.

10.3 CRYPTOGRAPHIC ASSET-

TOKEN LINKING

10.3.1 Linking Mechanism

The link between physical asset and token is established through a system of chained hashes that creates a mathematical proof of correspondence:

1. Generation of Physical Hash: SHA3-256(Sensor Data + Verification Certificate + Digital Twin + Timestamp + Inspector DID).
2. Generation of Digital Hash: SHA3-256(Token Metadata + Smart Contract + Compliance Rules + Timestamp + Issuer DID).
3. Bidirectional Linking: Physical Hash ↔ Digital Hash. Registry on Blockchain.

4.Verification: Any third party can verify the link by obtaining the Physical Hash, obtaining the Digital Hash, verifying they are linked in the registry, and confirming the signature is valid.

10.3.2 Updating the Link

When the physical asset's state changes (improvement, deterioration, maintenance), the link MUST be updated:

1.Detection: IoT sensors detect significant variation, physical inspection identifies modification, maintenance registered in system.
2.New Verification: Accredited body verifies new state, generates new certificate with updated Physical Hash, cryptographic signature.
3.Link Update: New Physical Hash registered on blockchain, previous link marked as historical, new link established, timestamp and version updated.
4.Notification: Token holders notified, dashboard updated, SCCIP evaluates impact on compliance rules.

10.3.3 Proofs of Existence and State

The PDBP provides three types of cryptographic proofs:
PROOF OF EXISTENCE: Demonstrates that the physical asset exists at a specific moment. Includes AUI, Timestamp, Sensor Data, Device Signature, Inspector Signature, Blockchain Hash.
PROOF OF STATE: Demonstrates the current state of the asset according to specific metrics. Includes AUI, Scores (SRF, SRT, SRD, SRE), Timestamp, Body Signature, Blockchain Hash.
PROOF OF COMPLIANCE: Demonstrates that the asset complies with specific compliance rules. Includes AUI, Rules, Compliance Status (True/False), Exceptions, Auditor Signature, Blockchain Hash.

10.3.4 Zero-Knowledge Proofs for Privacy

In many cases, token holders need to verify compliance without accessing sensitive asset data. The PDBP implements Zero-Knowledge Proofs (ZKP) to resolve this tension:
Use Case: A holder wants to verify that a tokenized yacht complies with safety requirements, but does not need to know the specific technical details. Solution: The smart contract verifies a ZKP. It confirms the statement is true (e.g., "SRF >= 80 AND SRT >= 80") without obtaining access to the specific values. The holder sees: TRUE/FALSE, without knowing the exact scores.

10.4 ORACLE NETWORKS AND CHAINLINK INTEGRATION

10.4.1 The Role of Oracles in PTVS

Oracles are services that connect real-world data with smart contracts on the blockchain. In the PTVS context, oracles are critical because they extract verified data from the physical asset (via CAP), transform it into a format usable by smart contracts, transmit it securely to the blockchain, and guarantee the integrity and authenticity of the data. Without reliable oracles, the physical-digital bridge breaks.

10.4.2 Types of Oracles in PTVS

STATE ORACLE: Reports the current state of the asset (scores, metrics). Frequency: According to verification level. Source: Accredited inspection body. Application: Token value update, alerts.

EVENT ORACLE: Reports specific events (maintenance, damage, changes). Frequency: On demand. Source: Monitoring system (CAP) or inspector. Application: Triggers for automatic actions.
COMPLIANCE ORACLE: Reports compliance with specific rules. Frequency: Continuous or on demand. Source: Analysis engine (PRA) + accredited body. Application: SCCIP, automatic pauses.
VALUATION ORACLE: Reports the updated valuation of the asset. Frequency: According to level. Source: Accredited valuation team. Application: Token price adjustment, distributions.

10.4.3 Decentralized Oracle Architecture

To avoid single points of failure and manipulation, PTVS uses decentralized oracle networks (e.g., Chainlink): Process:

1. Request: Smart contract requests data, specifies what data and freshness required.
2. Collection: Multiple oracle nodes collect data from independent sources, cryptographically sign each node's data.
3. Aggregation: Data aggregated using robust algorithms (median, weighted average), outliers detected and rejected.
4. Reporting: Aggregated data sent to smart contract, oracle contract signature, registry on blockchain.
5. Verification: Smart contract verifies oracle signature, accepts or rejects data, executes business logic.

10.4.4 Oracle Security

Threats: Data manipulation (malicious node), Denial of Service, excessive latency, collusion (multiple nodes coordinating to manipulate).

Mitigations: Multiple independent nodes, robust aggregation algorithms (median, not mean), outlier detection, staking tokens as guarantee (slashing), geographic distribution of nodes, independent audits.

10.5 SMART CONTRACT COMPLIANCE INTEGRATION PROTOCOL (SCCIP)

10.5.1 Fundamental Concept: Programmable Compliance

SCCIP transforms regulatory and contractual compliance from a manual, reactive process into an automatic, proactive system. Instead of relying on periodic audits and human actions, SCCIP: • Monitors the physical asset's state continuously (via oracles). • Automatically evaluates compliance with predefined rules. • Executes protective actions when non-compliance is detected. • Notifies all stakeholders transparently. • Records all actions immutably on the blockchain.

10.5.2 SCCIP Architecture

RULES LAYER: Definition of compliance rules, thresholds and conditions, actions associated with each rule, governance of rule changes.
DATA LAYER: Reception of data from oracles, integrity validation, historical storage, fast access cache.
EVALUATION LAYER: Rule evaluation engine, business logic, calculation of compliance states, generation of events.

EXECUTION LAYER: Execution of automatic actions, transfer pauses, stakeholder notifications, state updates. GOVERNANCE LAYER: Human supervision mechanisms, multi-sig signatures for critical decisions, appeal processes, audit and transparency.

10.5.3 Integration with ERC-3643

ERC-3643 is the token standard with identity for real-world assets. SCCIP integrates with ERC-3643 by extending the compliance module:
Architecture: ERC-3643 Token Contract includes Identity Registry (KYC/AML), Compliance Module (SCCIP), and Token Controller (Transfer Restrictions, Pause Mechanisms).
Transfer Restrictions: The token contract overrides transfer functions to verify that the asset is not paused and the risk level is not critical. If the risk level is ALERT, manual approval from the Compliance Officer is required.

10.5.4 Types of Compliance Rules

RISK RULES: • SRF < 60 → PAUSE. • SRF < 70 → ALERT. • SRT < 60 → PAUSE. • SRT < 75 → ESCALATE.
DOCUMENTARY RULES: • SRD < 80 (Level 3) → PAUSE.
COMPOUND RULES: • SRF < 60 OR SRT < 60 → PAUSE.
TEMPORAL RULES: • SRF < 70 for more than 24 hours → PAUSE.

10.5.5 Automatic Pause Mechanisms

SCCIP implements three levels of pause according to severity:
SOFT PAUSE: Trigger: SRF < 70 or SRT < 70 (ATTENTION level). Action: Notification to all holders.

Restriction: None, transfers allowed. Duration: Until state improves.
MODERATE PAUSE: Trigger: SRF < 60, SRT < 60, or SRD < 80 (ALERT level). Action: Notification + requirement for manual approval. Restriction: Transfers require Compliance Officer approval. Duration: Until problem resolution.
FULL PAUSE: Trigger: SRF < 50, SRT < 50, or SRD < 60 (CRITICAL level). Action: Immediate pause of all transfers. Restriction: No transfers allowed. Duration: Until complete resolution. Requires Emergency Committee (3-of-5 multi-sig) to unlock.

10.5.6 Multi-Sig Governance

For critical decisions, SCCIP implements multi-signature mechanisms:
Emergency Committee: 5 independent members (Legal, Technical, Financial, Sectoral, Independent). 2-year mandate. Functions: Unlock assets in full pause, approve changes in critical compliance rules, resolve disputes, authorize emergency contract updates.
Mechanism: Transaction submitted, members confirm, when required confirmations (e.g., 3 of 5) are reached, transaction executes.

10.5.7 Audit and Transparency

Immutable Record: All SCCIP actions are recorded on blockchain (ComplianceAction event).
Audit Dashboard: Web interface showing complete history of rule evaluations, executed actions, current state, compliance metrics.
Query API: Public API to consult compliance history, current state, and rule configurations.

10.6 PRACTICAL TOKENIZATION CASES

10.6.1 Case 1: Tokenized Megayacht

Scenario: 80m megayacht, €50M value, tokenized in 10,000 ERC-3643 tokens. Owned by 150 institutional investors.
Architecture: 500 IoT sensors → CAP Platform → 3 Chainlink Nodes (Aurema, Swiss Re, Lloyd's) → Aggregation → PTVS State Feed → ERC-3643 Smart Contract → SCCIP Logic → Token State.
Compliance Rules: SRF < 50 → PAUSE. SRT < 70 → ALERT. Class certificate expired → PAUSE. Insurance lapsed → PAUSE.
Crisis Scenario: Sensor detects water ingress in engine room. CAP detects anomaly, CRITICAL alert. Oracle reports SRF = 45. SCCIP evaluates rule, executes PAUSE. Token automatically paused. 150 holders notified. Crew initiates emergency protocol. Problem resolved in 2 hours, SRF returns to 75. Inspector verifies. Oracle reports SRF = 75. Emergency Committee (3-of-5) approves unlock. Token unpaused.
Operational Cost: ~€10,000/month (0.22% of asset value annually). Benefits: Problem detection in minutes vs. months, automatic value protection, total transparency, reduced insurance premiums (15-20%), higher token liquidity.

10.6.2 Case 2: Tokenized Commercial Building

Scenario: Office building in Madrid, €30M value, tokenized in 30,000 tokens. Rental income €2M/year.

Specific Rules: Occupancy < 70% → ALERT. Energy efficiency below C rating → ALERT. Structural risk < 75 → ESCALATE. Net income > 0 AND distribution date reached → DISTRIBUTE.
Automatic Dividend Distribution: Smart contract calculates net income since last distribution, verifies asset is not paused and risk is not critical, distributes dividends proportionally to all holders automatically via smart contract.

10.6.3 Case 3: Tokenized Asset Portfolio

Scenario: Diversified portfolio: 5 real estate, 2 yachts, 1 art collection. Total value €100M. Tokenized as a fund.
Portfolio Rules: Single asset > 30% of portfolio → ALERT. Average SRF < 75 → ESCALATE. More than 2 assets in critical state → PAUSE.
Result: The fund token allows institutional investors to gain exposure to a diversified basket of high-value physical assets, with continuous, automated, and transparent risk management.

10.7 REPORTING AND

CERTIFICATION

10.7.1 PDBP-SCCIP Certification

For assets with active PDBP and SCCIP, Aurema Group issues a specific Physical-Digital Bridge Certificate:
Information included: Asset identification (AUI), Token identification (contract address), Integration level (3 or 4), Oracle architecture used, Active compliance rules, Current system state, Date of certification and validity, Cryptographic hash of the certificate.

Validity: Level 3: 12 months. Level 4: 24 months. Requires annual compliance audit.

10.7.2 Reporting to Stakeholders

Monthly Report (Level 3): Compliance state, evaluated rules, executed actions, oracle metrics (uptime, latency), improvement recommendations.
Quarterly Report (Level 3 and 4): Detailed compliance analysis, score evolution (SRF, SRT, SRD, SRE), incidents and resolutions, operational cost of the system, tokenization ROI, perspectives and recommendations.
Annual Report (All levels): Complete balance of the year, independent audit of the system, PDBP-SCCIP integration validation, market analysis and comparables, strategy for the following year.

CHAPTER 11:

CRYPTOGRAPHIC PROOF

OF EXISTENCE

PROTOCOL (CPEP)

11.1 PURPOSE AND SCOPE OF THE PROTOCOL

11.1.1 The Fundamental Problem of Trust in Physical Data

In a world where physical assets are digitally represented via tokens, a critical question arises that no previous standard has satisfactorily resolved: **How can an institutional investor mathematically verify that the data about a physical asset is authentic, has not been manipulated, and truly comes from a legitimate inspection performed by an accredited auditor?** Traditional certification systems rely on **institutional trust** in the issuing entity. A PDF report with an auditing firm's seal is ultimately only as trustworthy as the firm's reputation. This reliance on human trust creates systemic

vulnerabilities: • Document fraud: Forged or altered reports. • Data manipulation: Intercepted and modified sensor data. • Identity spoofing: Unaccredited individuals signing as auditors. • Broken chain of custody: Data losing traceability from the source. • Lack of independent verifiability: Third parties cannot verify without accessing the issuer.

The **Cryptographic Proof of Existence Protocol (CPEP)** resolves these issues through **mathematical proofs** rather than institutional trust. CPEP establishes a cryptographic framework that guarantees:

1. Authenticity of origin: Data truly comes from the device claiming to have captured it.
2. Immutable integrity: Data has not been altered since capture.
3. Verifiable identity: The auditor who performed the inspection is cryptographically identified.
4. Complete chain of custody: Every transfer or processing of data is recorded.
5. Independent verification: Any third party can verify authenticity without intermediaries.

11.1.2 Objectives of CPEP

OBJECTIVE 1: Decentralized Auditor Identity Establish a Decentralized Identifier (DID) system for all accredited auditors, enabling cryptographic verification of their identity and credentials without relying on centralized authorities.

OBJECTIVE 2: Edge Cryptographic Signing Implement cryptographic signing of all data at the exact moment and place of capture (edge computing), before any transmission or processing, guaranteeing authenticity from the source.

OBJECTIVE 3: Immutable Digital Custody Chain Create an immutable record of all manipulations, transfers, and

accesses to data from creation to final destination, using blockchain technology.
OBJECTIVE 4: Independent Third-Party Verification Enable any third party (investors, regulators, independent auditors) to verify the authenticity and integrity of data without needing to trust the issuing body.
OBJECTIVE 5: Privacy via Zero-Knowledge Proofs Allow verification of claims about data (e.g., "the asset complies with ISO 17020") without revealing the underlying sensitive data, protecting client confidentiality.
OBJECTIVE 6: Interoperability with Open Standards Ensure compatibility with W3C standards (DID, Verifiable Credentials), ISO/IEC 27001, and emerging sovereign identity frameworks.

11.1.3 Guiding Principles of CPEP

PRINCIPLE 1: DATA SOVEREIGNTY Data belongs to its subject (the physical asset and its owner). No entity can access, modify, or distribute data without cryptographically verifiable authorization.
PRINCIPLE 2: MINIMIZATION OF TRUST The system MUST minimize the need to trust any central entity. Trust MUST emerge from verifiable mathematical proofs, not institutional reputation.
PRINCIPLE 3: VERIFIABLE TRANSPARENCY All cryptographic processes MUST be auditable and verifiable by independent third parties. The cryptography used MUST be open-source and standardized.
PRINCIPLE 4: CENSORSHIP RESISTANCE The system MUST continue functioning (in degraded mode) even if individual entities are compromised or cease to operate. Decentralization MUST guarantee resilience.
PRINCIPLE 5: PRIVACY BY DESIGN Data protection MUST be integrated into the system's architecture from the start, minimizing personal data capture and applying

anonymization and Zero-Knowledge Proofs where necessary.
PRINCIPLE 6: FORWARD-COMPATIBILITY The system MUST be compatible with future cryptographic advances, including quantum computing. Algorithms MUST be updatable without breaking the validity of historical proofs.

11.1.4 Application by Verification Level

Level 1: Not applicable. Level 2: Basic. Simple DID, basic edge signing for key photographs, partial custody chain. Level 3: Complete. Full DID + Verifiable Credentials, edge signing for all data, complete custody chain, basic ZKP. Level 4: Advanced. DID + VC + Multi-sig, real-time edge signing, complete custody chain + advanced ZKP.

11.2 DECENTRALIZED IDENTITY (DID) FOR AUDITORS

11.2.1 Concept of Decentralized Identity

A **Decentralized Identifier (DID)** is a new type of global identifier that enables cryptographic verification of an entity's identity (person, organization, device) without relying on a centralized registry.
Fundamental Characteristics: • Sovereign Control: The identity subject controls their credentials. • Cryptographic Verification: Verified via public/private key pairs. • Portability: Not tied to a specific provider. • Persistence: Remains valid even if the underlying registry changes.

11.2.2 DID Structure for PTVS Auditors

FORMAT (W3C Standard):

`did:ptvs:es:auditor:0x7f8e9d2c3b4a5f6e7d8c9b0a1f2e3d4c5b6a7f8e`

Components: • `did`: Standard W3C prefix. • `ptvs`: PTVS-specific method. • `es`: ISO country code (optional). • `auditor`: Entity type. • `0x7f8e...`: Unique identifier based on public key.

DID DOCUMENT (JSON): Contains verification methods (public keys), authentication rules, service endpoints, and metadata. It is stored on a decentralized ledger or IPFS.

11.2.3 Verifiable Credentials (VC)

Verifiable Credentials (VC) are digitally signed cryptographic statements accrediting attributes of the subject.
Types of Credentials for Auditors:

1. PTVS Accreditation: Issued by Aurema Group. Contains accreditation level, specializations, jurisdictions.
2. Specific Technical Certification: Issued by RICS, IIMS, etc. Contains qualifications, CPD hours.
3. Continuous Training: Issued by PTVS Academy. Contains course details, competencies acquired.

11.2.4 Auditor Registry on Blockchain

To guarantee immutability and public verifiability, auditor DIDs are registered on a smart contract:
Contract Functions: • `registerAuditor`: Admin registers a new auditor (DID hash, Ethereum address, level, validity). • `updateAuditor`: Admin updates level or validity.

• `revokeAuditor`: Admin revokes accreditation with a stated reason. • `verifyAuditor`: Public function to check if a DID is valid, active, and not expired.

11.2.5 Auditor Wallet

Each accredited auditor receives a cryptographic wallet managing their identity and signatures: • Secure Key Storage: Master private key in HSM/Secure Enclave. • Credential Management: Storage of VCs, selective disclosure, automatic renewal. • Data Signing: Signing captured data, reports, and transactions. • User Interface: Mobile app for on-site inspections, web dashboard for management.

11.2.6 Real-Time Identity Verification

When an auditor presents credentials, the system verifies:

1. DID Resolution: Fetch DID Document, check registry contract for validity and expiration.
2. Credential Verification: Check VC signatures, verify trusted issuer, check revocation status, check expiration.
3. Competency Verification: Ensure the auditor holds the required PTVS level and specific specializations for the asset type and jurisdiction.

11.3 EDGE SIGNING

11.3.1 Concept of Edge Signing

Edge signing is the cryptographic process by which a device (sensor, camera, drone, meter) digitally signs data **at the moment and place of capture**, before any transmission, processing, or intermediate storage.

In traditional systems, data flows: Sensor → Gateway → Cloud. The data is vulnerable between the sensor and the gateway. With edge signing: Sensor [SIGNS HERE] → Gateway → Cloud. Any modification after signing invalidates the signature.
Benefits: Guaranteed authenticity from origin, verifiable integrity, non-repudiation, complete traceability, infrastructure independence.

11.3.2 Edge Signing Architecture

Hardware Components: • Secure Element (SE) or Trusted Platform Module (TPM): Dedicated cryptographic chip storing private keys securely (non-extractable) and performing internal crypto operations. Resistant to physical tampering.
Software Components: • Signing Firmware: Captures data, calculates SHA-256 hash, gets precise NTP timestamp and GPS coordinates, retrieves device ID, signs the hash using the SE, packages the signed data. • Integration SDK: APIs for commercial sensors, support for MQTT/HTTP/CoAP, automatic certificate management, secure OTA updates.

11.3.3 Step-by-Step Signing Process

1. Data Capture: Sensor captures physical phenomenon, digitizes, calibrates, validates range. Generates automatic metadata (NTP timestamp, GPS, device ID, firmware version, battery level).
2. Hash Generation: Calculates SHA-256 hash of the canonical JSON payload (data + metadata).
3. Cryptographic Signing: The Secure Element signs the hash using Ed25519 (or similar) with the device's private key.
4. Packaging: Creates a JSON packet containing the original data, metadata, hash, and signature.

5.Secure Transmission: Encrypts the packet (AES-256-GCM), transmits via TLS 1.3 to the gateway, stores locally for redundancy.

11.3.4 Types of Data to Sign

• IoT Sensor Data: Temperature, humidity, vibration, pressure, levels. • Inspection Data: Georeferenced photos, inspection videos, laser/ultrasound measurements, NDT results. • Drone Data: LiDAR point clouds, orthophotos, 3D models, aerial videos. • Analysis Data: Lab results, test reports, calibration certificates. • Documentary Data: Scanned documents, handwritten signatures, seals.

11.3.5 Signature Verification

Any third party can verify the authenticity:

1.Extract components (data, metadata, hash, signature).
2.Recalculate hash of the data and metadata. Verify it matches the signed hash.
3.Verify the cryptographic signature using the device's public key.
4.Verify the device's public key is registered in the PTVS device registry.
5.Verify the timestamp is within an acceptable drift (e.g., <5 minutes).
6.Verify the geolocation matches the expected location of the asset.

11.3.6 Device Key Management

• Generation: Generated internally in the SE. Never leaves the device. • Registry: Public key registered on blockchain, linked to device DID. • Rotation: Annual rotation or upon compromise suspicion. Old key marked as "revoked but valid for historical data". • Revocation: Immediate if device is lost/stolen. Revocation list published on blockchain. •

Destruction: Securely erased at end of life. Destruction certificate registered.

11.4 DIGITAL CUSTODY CHAIN

11.4.1 Concept of Digital Custody Chain

The **digital custody chain** is an immutable, chronological record of all manipulations, transfers, and accesses to a digital data or asset, from creation to final destination.
It guarantees:

1. Every manipulation is recorded (who, when, what).
2. Every transfer is signed.
3. Every transformation is documented.
4. The history is immutable.
5. Verification is independent.

11.4.2 Structure of the Custody Chain

Modeled as linked blocks. Each event is a "block" referencing the previous one.
Event Structure (JSON): • `blockNumber`: Sequential number. • `timestamp`: Unix timestamp of the event. • `event`: Object containing type, actor, action, dataHash, metadata. • `previousBlockHash`: Hash of the previous event (null for the first). • `hash`: SHA-256 of the current event. • `signature`: Ed25519 signature of the hash by the entity recording the event.

11.4.3 Blockchain Registry

To guarantee immutability, custody chain hashes are registered on a smart contract: • `registerEvent`:

Registers a new event, verifies the previous event exists and matches the dataHash. • `getCustodyChain`: Retrieves the full chain of events for a specific dataHash by traversing backwards. • `verifyCustodyChain`: Public function that cryptographically verifies the integrity of the entire chain.

11.4.4 Types of Events

• CREATION: Data captured by a device. • TRANSFER: Data transmitted to a gateway. • RECEIPT: Data received and verified by a gateway. • PROCESSING: Data aggregated or transformed. • STORAGE: Data stored in permanent archive (with location and encryption details). • ACCESS: Data accessed by an auditor or system. • REGISTRATION: Data hash registered on blockchain. • REVOCATION: Data revoked due to a security incident.

11.4.5 Custody Chain Verification

To verify a chain:

1. Retrieve all event hashes from the smart contract.
2. For each event, verify the hash matches the calculated hash of its contents.
3. Verify the signature of each event by its actor.
4. Verify the continuity (previousBlockHash matches the actual previous event's hash).
5. Verify the temporal order (timestamps are strictly increasing).
6. Verify the first event is a CREATION event.

11.4.6 Use Cases

• Naval Inspection Audit: An investor verifies that hull photos are authentic and unmanipulated by checking the full custody chain from the drone's camera to the final

report. • Legal Dispute: In a lawsuit, a party claims inspection evidence was manipulated. The custody chain provides court-admissible cryptographic proof of integrity. • Regulatory Compliance (MiCA): A regulator requires proof that all data for a tokenized asset complies with MiCA. The custody chain proves 100% of data points have valid, unbroken cryptographic provenance.

11.5 ZERO-KNOWLEDGE PROOFS (ZKP) FOR PRIVACY

11.5.1 The Privacy-Transparency Dilemma

In tokenized asset verification, there is a fundamental tension: • Investors need transparency about the asset's state. • Regulators require access for supervision. • Auditors must verify data authenticity. BUT: • Owners demand confidentiality for sensitive data. • GDPR imposes personal data protection. • Commercial competition requires protecting strategic information.
Zero-Knowledge Proofs (ZKP) resolve this by allowing verification of claims about data without revealing the data itself.

11.5.2 ZKP Concept

A ZKP is a cryptographic protocol where a prover can demonstrate to a verifier that a statement is true without revealing any information other than the validity of the statement.
Properties: • Completeness: If true, an honest prover can convince an honest verifier. • Soundness: If false, a cheating prover cannot convince an honest verifier (except with negligible probability). • Zero-Knowledge: The verifier learns nothing beyond the validity of the statement.

11.5.3 Applicable ZKP Types

ZK-SNARKs: Very compact proofs, extremely fast verification, non-interactive. Requires a trusted setup. Ideal for compliance thresholds and existence proofs. ZK-STARKs: Larger proofs, slower verification, no trusted setup (transparent), quantum-resistant. Ideal for auditing large data volumes. Bulletproofs: No trusted setup, logarithmic proof size, efficient for range proofs. Ideal for proving a value is within a range (e.g., SRF > 80).

11.5.4 ZKP Use Cases in PTVS

1. Compliance Verification without Revealing Scores: A holder verifies the asset meets Level 3 requirements (SRF ≥ 80, SRT ≥ 80) without seeing the exact scores (e.g., SRF=85, SRT=88).
2. Proof of Existence without Revealing Location: An owner proves an asset is in a specific jurisdiction (e.g., Spain) without revealing the exact GPS coordinates.
3. Proof of Solvency without Revealing Balance: An owner proves they have sufficient funds to cover a liability without revealing their total balance.
4. Identity Verification without Revealing Personal Data: An auditor proves they are accredited for a specific inspection without revealing their name, ID number, or other credentials.

11.5.5 Smart Contract Integration

ZKPs can be verified directly on-chain using verifier contracts (e.g., Groth16 verifier for SNARKs). The smart contract receives the proof and public inputs, calls the verifier, and executes logic (like allowing a token transfer) based on the boolean result, without ever seeing the private data.

11.5.6 Performance Considerations

• SNARKs: Generation 1-10s, Verification <100ms, Size ~200 bytes. • STARKs: Generation 10-60s, Verification 1-5s, Size ~10-100KB. • Bulletproofs: Generation 1-5s, Verification 0.5-2s, Size ~700 bytes. Optimizations include batching multiple proofs, precomputing circuits, and using hardware acceleration (GPUs/FPGAs).

11.5.7 Differential Privacy for Aggregated Data

For statistical reports (e.g., market trends), CPEP implements differential privacy by adding calibrated mathematical noise (Laplace noise) to aggregated data. This allows the release of useful statistical insights without compromising the privacy of individual assets.

11.6 INTEGRATION WITH ORACLE NETWORKS

11.6.1 The Role of CPEP in Oraculization

CPEP provides the **foundation of trust** for oracles feeding smart contracts with physical data. Without CPEP, oracles are just "data sources" relying on operator trust. With CPEP, oracles can cryptographically prove that data comes from verified sources, is unmanipulated, is current, and is relevant.

11.6.2 Integration Architecture

Flow: Sensor IoT → Edge Signing (CPEP) → CAP Platform → PTVS Oracle → Aggregation (Multi-Node) → Blockchain → Smart Contract (SCCIP).
CPEP Components in the Flow:

1. Edge Signing Layer: Every sensor/inspector signs their data.
2. Custody Chain Layer: Immutable record of all data manipulation.
3. Identity Layer: DID of all participants.
4. Privacy Layer: ZKP for selective verification.

11.6.3 PTVS Oracle with CPEP Verification

A specialized smart contract (`PTVSOracleWithCPEP`) verifies data before accepting it from an oracle:

1. Verify cryptographic signature of the data.
2. Verify the custody chain hash on the registry contract.
3. Verify the timestamp is fresh (e.g., < 1 hour old).
4. Verify the signer's DID is accredited and not revoked. Only if all checks pass, the data is accepted and forwarded to the SCCIP logic.

11.6.4 Multi-Node Aggregation with CPEP

When multiple oracle nodes provide data, the aggregator verifies the CPEP proof of each node. It filters out nodes with invalid CPEP proofs or broken custody chains. It requires a minimum quorum (e.g., 2 out of 3) of verified nodes to accept the aggregated data, ensuring high reliability and resistance to single-node compromise.

11.6.5 Use Cases

• Valuation Oracle: Proves the valuation is based on verified, CPEP-signed data from the asset, without exposing the raw data. • Compliance Oracle: Proves the asset complies with a regulation based on verified sensor data, generating a ZKP of compliance for the smart contract.

11.7 REPORTING AND CERTIFICATION

11.7.1 CPEP Authenticity Certificate

For each verified asset, Aurema Group issues a CPEP Authenticity Certificate summarizing the cryptographic integrity of all data:
Includes: Certificate ID, Asset AUI, Issue/Expiry Dates, Issuer DID. Summary Statistics: Total data points signed, signature success rate, custody chain integrity status, blockchain anchored events, unique devices, unique auditors. Cryptographic Proofs: Merkle root of all data, blockchain transaction hash, block number. Verification Instructions: Online verification URL, offline verification public key, blockchain explorer link. Signature: Ed25519 signature of the certificate by Aurema Group.

11.7.2 Public Verification Dashboard

Aurema Group maintains a public dashboard where anyone can verify CPEP data authenticity: • Search by Hash: Verify signature, view custody chain, view blockchain registry. • Search by AUI: View all associated data, verify global integrity, download certificate. • Search by Auditor: View all inspections, verify credentials, view signing history. • Batch Verification: Upload multiple hashes for bulk verification.

11.7.3 CPEP Verification API

RESTful API for third-party integration: • `GET /api/v1/cpep/verify/{dataHash}`: Verify authenticity of specific data. • `GET /api/v1/cpep/custody-chain/`

`{dataHash}`: Get full custody chain. • `GET /api/v1/cpep/auditor/{did}`: Verify auditor identity and credentials. • `GET /api/v1/cpep/asset/{aui}`: Get CPEP summary for an asset. • `POST /api/v1/cpep/batch-verify`: Verify multiple data hashes in batch.

11.7.4 Annual Audit of the CPEP System

An independent auditor annually reviews the CPEP system to guarantee: • Cryptographic Integrity: Review of algorithms, key management, penetration testing of secure elements. • Custody Chain: Sampling of chains, end-to-end integrity verification, security incident review. • Identities and Credentials: Review of DID issuance, access controls, revocations. • Privacy and ZKP: Audit of ZK circuits, verification of zero-knowledge properties, data leakage analysis. • Regulatory Compliance: GDPR, MiCA, ISO 27001, sector-specific standards.

CHAPTER 12: PREDICTIVE RISK ASSESSMENT (PRA)

12.1 PURPOSE AND SCOPE OF THE PROTOCOL

12.1.1 From Reaction to Anticipation

The **Predictive Risk Assessment (PRA)** protocol represents the intellectual culmination of The Trust Standard. While previous protocols (CAP, PDBP, SCCIP, CPEP) have established the foundations for capturing, verifying, and executing data on physical assets, the PRA transforms that data into **anticipatory intelligence**: the ability to predict failures before they occur, optimize investments before they are needed, and preserve value before it erodes.

In traditional verification standards, risk analysis is inherently reactive: a problem is identified when it already exists, its impact is evaluated when it has already materialized, and a solution is recommended when the damage is already done. The PRA inverts this paradigm through a proactive and predictive approach that allows: • Anticipating failures weeks or months in advance. • Optimizing maintenance based on actual condition, not calendars. • Maximizing useful life through interventions at the optimal time. • Minimizing total life cycle costs. • Preserving value through data-informed decisions.

12.1.2 Objectives of PRA

OBJECTIVE 1: Component Degradation Modeling Develop precise mathematical models describing how each component of an asset degrades over time under different operational and environmental conditions.
OBJECTIVE 2: Failure Prediction with Sufficient Lead Time Identify pre-failure patterns in sensor and maintenance data, allowing interventions with enough lead time to avoid unplanned downtime and collateral damage.
OBJECTIVE 3: Remaining Useful Life (RUL) Estimation Calculate, with confidence intervals, the remaining useful life of components and systems, providing a quantitative basis for investment, replacement, and disposal decisions.
OBJECTIVE 4: Maintenance Strategy Optimization Transform maintenance from a scheduled activity to a condition-based activity, maximizing availability and minimizing total costs.
OBJECTIVE 5: Investment Decision Support Provide quantitative analyses that inform CAPEX decisions, including optimal timing of investments, prioritization of interventions, and profitability analysis.
OBJECTIVE 6: Continuous Validation of Predictions Establish feedback mechanisms that compare predictions with actual outcomes, enabling continuous improvement of predictive models.

12.1.3 Principles of PRA

PRINCIPLE 1: SCIENTIFIC BASIS Every prediction MUST be based on scientifically validated models, with explicit assumptions and quantified confidence levels.
PRINCIPLE 2: EXPLICIT UNCERTAINTY Every prediction MUST include confidence intervals and sensitivity analysis, avoiding the false precision of point predictions.

PRINCIPLE 3: CONTINUOUS LEARNING Models MUST be continuously updated with real data, improving their accuracy through supervised and unsupervised learning.
PRINCIPLE 4: INTERPRETABILITY Models MUST be interpretable by humans, avoiding "black boxes" that prevent the understanding of predictions and their foundations.
PRINCIPLE 5: RIGOROUS VALIDATION Every prediction MUST be validated against historical data before operational deployment, with clear metrics for precision and recall.
PRINCIPLE 6: ETHICAL GOVERNANCE The use of AI and predictive models MUST respect ethical principles, including transparency, fairness, and accountability.

12.1.4 Application by Verification Level

Level 1: Not applicable. Level 2: Basic. Linear degradation models. Annual updates. Level 3: Intermediate. Weibull + basic Machine Learning. Quarterly updates. Basic SCCIP integration. Level 4: Advanced. Deep Learning + hybrid physics-informed models. Monthly updates. Advanced SCCIP integration.

12.2 PRA SYSTEM ARCHITECTURE

12.2.1 Six-Layer Architecture

The PRA system is structured into six technological layers operating in coordination:
LAYER 1: DATA INGESTION Data from CAP (real-time IoT sensors), physical inspections (Phase 2), technical evaluations (Phase 3), financial data (Phase 4), external data (market, climate, regulatory), and historical maintenance data (CMMS).
LAYER 2: PREPROCESSING Data cleaning and normalization, outlier detection and treatment, missing

data imputation, feature engineering, dimensionality reduction, and temporal synchronization of series.
LAYER 3: MODELING Physics-based models, data-driven models (ML/DL), hybrid models (physics-informed ML), degradation models (Weibull, Markov), reliability models (FMEA, FTA), and optimization models.
LAYER 4: ANALYSIS Anomaly detection, state classification, RUL prediction, trend analysis, scenario simulation, and sensitivity analysis.
LAYER 5: INTEGRATION Feeding SCCIP (dynamic thresholds), updating the digital twin, generating CAP alerts, supporting investment decisions, reporting to stakeholders, and APIs for external systems.
LAYER 6: GOVERNANCE Model validation, drift monitoring, prediction auditing, model version management, explainability (XAI), and ethical and regulatory compliance.

12.2.2 Technology Stack

Storage: Time-series databases (InfluxDB, TimescaleDB), Document databases (MongoDB), Relational (PostgreSQL), Data Lake (S3, Azure Data Lake).
Processing: Batch (Apache Spark), Streaming (Kafka), ML Ops (MLflow, Kubeflow), Orchestration (Airflow). Models: Classical ML (scikit-learn, XGBoost), Deep Learning (TensorFlow, PyTorch), Physics-Informed (DeepXDE, NVIDIA Modulus), Time Series (Prophet, ARIMA, LSTM).
Visualization: Dashboards (Grafana, Superset), 3D (Three.js, Unity for digital twin), Reporting (Jupyter, Plotly).

12.3 COMPONENT DEGRADATION MODELS

12.3.1 Fundamentals of Degradation Modeling

Component degradation is a stochastic process that can be described through mathematical models. The PRA uses three families of models:
PHYSICS-BASED MODELS: Based on physical and chemical laws governing degradation mechanisms. Advantages: High interpretability, reliable extrapolation, requires less training data. Disadvantages: Complex implementation, requires expert knowledge of the mechanism. Application: Components with well-understood degradation mechanisms (fatigue, corrosion, wear).
DATA-DRIVEN MODELS: Based on patterns extracted from historical data using ML techniques. Advantages: Captures complex non-linear patterns, does not require explicit mechanism knowledge, scales well with large data volumes. Disadvantages: Requires large amounts of data, low interpretability (black box), unreliable extrapolation. Application: Complex systems with multiple interconnected failure modes.
HYBRID MODELS (PHYSICS-INFORMED ML): Combines physical knowledge with machine learning. Advantages: Interpretability + ability to capture complex patterns, requires less data than purely data-driven, more reliable extrapolation. Disadvantages: Complex implementation, requires both physical knowledge and data. Application: Critical components with limited data but solid physical knowledge.

12.3.2 Specific Degradation Models

MODEL 1: LINEAR DEGRADATION Applicable to components with uniform and constant wear. Equation: $D(t) = D_0 + \beta \cdot t + \varepsilon(t)$ Where $D(t)$ is degradation at time t, D_0 is initial degradation, β is degradation rate, and $\varepsilon(t)$ is stochastic noise.
MODEL 2: WEIBULL DISTRIBUTION The most used model in reliability analysis, applicable to components with variable failure rates. Parameters: Shape parameter (β) indicates failure type (infant mortality, random, wear-out). Scale parameter (η) indicates characteristic life.
MODEL 3: MARKOV PROCESSES Applicable to components with multiple degradation states and transitions between them. Uses a transition probability matrix to model the evolution of the component's state over discrete time steps.
MODEL 4: GAUSSIAN PROCESSES Applicable to degradation with quantifiable uncertainty and limited data. Uses covariance functions (kernels) to model the degradation trend and its confidence intervals.

12.3.3 Model Selection by Component

Typology

• Bearings: Weibull + Vibrations (Fatigue failure, predictive vibration data). • Electric Motor: Markov + Temperature (Multiple degradation states). • Concrete Structure: Physics-based (Fick's Law for carbonation). • Battery: Gaussian Process + ML (Non-linear degradation, limited data). • Pipes: Linear + Corrosion (Uniform wear + physical model). • HVAC System: Random Forest (Multiple interdependent variables). • Roof: Weibull + Climate (Influence of environmental factors).

12.4 MACHINE LEARNING FOR FAILURE PREDICTION

12.4.1 Anomaly Detection

Identifies unusual behaviors that may indicate incipient failures.
ISOLATION FOREST: Effective for multivariate anomaly detection in sensor data. Isolates anomalies by randomly selecting a feature and then randomly selecting a split value, requiring fewer splits to isolate anomalies than normal points.
AUTOENCODERS: Neural networks that learn to reconstruct normal data; reconstruction failures indicate anomalies. Trained exclusively on normal operating data, they produce high reconstruction errors when fed anomalous data.

12.4.2 Remaining Useful Life (RUL) Prediction

APPROACH 1: DIRECT REGRESSION Predicts the RUL directly using algorithms like Random Forest or Gradient Boosting. Features are extracted from time windows of sensor data (statistical moments, trends, frequency domain features).
APPROACH 2: LONG SHORT-TERM MEMORY (LSTM) Recurrent neural networks specialized in time series. They capture long-term dependencies in sequential sensor data, making them highly effective for predicting the degradation trajectory and the exact point of failure.

12.4.3 Failure Mode Classification

Uses algorithms like Random Forest or Support Vector Machines (SVM) to classify the specific type of impending

failure (e.g., unbalance, misalignment, bearing wear, electrical fault) based on the signature of the sensor data. This allows for targeted maintenance interventions.

12.4.4 Real-Time Anomaly Detection System

Implements a continuous processing pipeline:

1. Ingests real-time sensor data.
2. Passes data through the anomaly detection model.
3. If an anomaly is detected, triggers the failure mode classifier.
4. Calculates the RUL based on the classified mode.
5. Generates an alert with severity, predicted failure mode, RUL, and recommended action.

12.5 REMAINING USEFUL LIFE (RUL)

12.5.1 Concept and Importance

The Remaining Useful Life (RUL) is the estimated time remaining before a component or system reaches a failure state or requires maintenance. It is the fundamental metric for predictive maintenance, spare parts management, investment decisions, asset valuation, and insurance.

12.5.2 RUL Estimation Methodologies

PHYSICS-BASED METHOD: Uses physical degradation models. Calculates the current degradation rate and extrapolates the time until the failure threshold is reached. Includes uncertainty propagation via Monte Carlo simulations.

DATA-DRIVEN METHOD: Uses ML models trained on historical run-to-failure data. Predicts RUL directly from current sensor readings and historical trends. Uncertainty is estimated using ensemble methods or bootstrapping. HYBRID METHOD: Combines physics-based and data-driven predictions using Bayesian Model Averaging. Weights the predictions based on the historical accuracy of each model for the specific component.

12.5.3 Model Validation Metrics

To ensure reliability, RUL models are evaluated using: • Mean Absolute Error (MAE): Average absolute difference between predicted and actual RUL. • Root Mean Square Error (RMSE): Penalizes larger errors more heavily. • Prognostic Horizon: The time index at which predictions remain within a specified accuracy bound until the end of life. • Relative Accuracy: Percentage of predictions falling within an acceptable error margin (e.g., ±10% of true RUL).

12.6 PREDICTIVE MAINTENANCE

12.6.1 Evolution of Maintenance Strategies

• Corrective: Repair after failure. Low initial cost, high unplanned downtime. • Preventive: Scheduled maintenance. Reduces failures, but leads to unnecessary interventions. • Predictive: Condition-based maintenance. Optimal balance of cost and availability. • Prescriptive: Automated recommendations and execution. Continuous optimization.

12.6.2 Maintenance Plan Optimization

Uses mathematical optimization (e.g., Linear Programming, Mixed-Integer Linear Programming) to

schedule maintenance activities. The objective function minimizes total cost (maintenance cost + expected failure cost) subject to constraints (resource availability, operational windows, safety regulations).

12.6.3 Automatic Work Order Generation

Integrates with Computerized Maintenance Management Systems (CMMS). When the PRA predicts a failure within a specific timeframe, it automatically generates a work order detailing: • Component and predicted failure mode. • Urgency and scheduled date. • Required spare parts and tools. • Required skills and safety protocols. • Estimated duration and cost.

12.7 INVESTMENT OPTIMIZATION

12.7.1 Life Cycle Cost (LCC) Analysis with PRA

Calculates the total cost of ownership over the asset's life, incorporating predictive maintenance costs, expected failure costs, energy costs (adjusted for efficiency degradation), and disposal costs. Discounted to present value using the Weighted Average Cost of Capital (WACC).

12.7.2 ROI of Predictive Maintenance

Calculates the Return on Investment for implementing the PRA system. ROI = (Baseline Maintenance Cost - Predictive Maintenance Cost) / Investment in PRA. Includes secondary benefits: reduced downtime, extended asset life, lower insurance premiums, and higher resale value.

12.7.3 Multi-Criteria Investment Prioritization

When multiple assets or components require investment, PRA provides a scoring model based on: • Safety impact (weight: 30%). • Reliability improvement (weight: 25%). • Cost savings (weight: 20%). • Regulatory compliance (weight: 15%). • Strategic alignment (weight: 10%). This ensures capital is allocated to the interventions that yield the highest overall value.

12.8 MODEL VALIDATION AND GOVERNANCE

12.8.1 Validation Framework

Before deployment, every model must pass a rigorous validation process:

1. Historical backtesting on unseen data.
2. Stress testing with synthetic anomaly data.
3. Peer review by domain experts.
4. Approval by the Technical Advisory Board (for critical models).

12.8.2 Drift Monitoring

Models degrade over time as the physical asset and its environment change. The PRA continuously monitors for: • Data Drift: Changes in the distribution of input sensor data (detected using Kolmogorov-Smirnov tests). • Concept Drift: Changes in the relationship between input data and the target variable (detected by monitoring prediction accuracy against actual outcomes). When drift exceeds thresholds, the model is flagged for retraining.

12.8.3 AI Governance Framework

Ensures ethical and compliant use of AI: • Transparency: Models are documented and explainable (using SHAP or LIME for feature importance). • Fairness: Models are tested for bias across different asset types or operating conditions. • Accountability: Clear ownership of model performance and decisions. • Privacy: Sensitive operational data is protected using differential privacy or federated learning where applicable.

12.9 INTEGRATION WITH CAP AND SCCIP

12.9.1 Dynamic Threshold Feeding to SCCIP

Instead of static thresholds, the PRA provides SCCIP with dynamic, context-aware thresholds. For example, a vibration threshold might be higher during startup than during steady-state operation. If the PRA predicts a rapid degradation, it can instruct SCCIP to tighten the thresholds automatically, triggering an earlier pause or alert.

12.9.2 Digital Twin Updating

The PRA feeds the digital twin with predicted future states. This allows stakeholders to visualize not just the current state of the asset, but its probable future trajectory under different maintenance and operational scenarios, facilitating "what-if" analysis.

12.10 PRACTICAL CASES

12.10.1 Case 1: Failure Prediction in a Yacht Fleet

Scenario: A fleet of 15 megayachts with 500 sensors each. Implementation: Weibull models for mechanical components, LSTM for complex systems. Results after 12 months: Unplanned failures reduced by 67%, maintenance costs reduced by 23%, fleet availability increased by 12%. ROI of implementation: 340% in 3 years.

12.10.2 Case 2: Maintenance Optimization in a Commercial Building

Scenario: 50,000 m² office building with 2,000 IoT sensors. Implementation: Markov models for HVAC, Random Forest for energy systems. Results after 24 months: Energy consumption reduced by 18%, comfort complaints reduced by 45%, operational costs reduced by 27%. Equipment useful life extended by an average of 3 years.

12.10.3 Case 3: Structural Degradation Prediction in a Bridge

Scenario: 500m bridge with continuous structural monitoring. Implementation: Physics-based concrete degradation model combined with ML for crack detection. Results: Optimal maintenance plan generated for 50 years. Total LCC reduced by 35% compared to reactive maintenance. Risk of unexpected closure eliminated.

12.11 REPORTING AND CERTIFICATION

12.11.1 PRA Report Structure

The PRA report includes: • Executive Summary: Overall health score, critical components, predicted failures, estimated maintenance cost. • Component Predictions: Current age, predicted RUL with confidence intervals, failure mode, recommended action, optimal intervention window, estimated cost. • Maintenance Recommendations: Immediate actions, planned maintenance, long-term investments. • Financial Impact: Baseline vs. predictive annual cost, annual savings, ROI, NPV. • Model Performance: Prediction accuracy, MAE, RMSE, model version, last validation date.

12.11.2 PRA Certification

Aurema Group issues a specific Predictive Risk Assessment Certificate for assets with active PRA. It includes the models used, validated prediction accuracy, prediction horizon, and certification validity (3 months for Level 3, 1 month for Level 4).

12.11.3 Model Audit

An annual mandatory audit reviews: • Prediction accuracy against actual outcomes. • Model drift and retraining logs. • AI governance and ethical compliance. • Financial impact validation. • Corrective actions for underperforming models.

CHAPTER 13: PROP TRUST VERIFIED REPORT STRUCTURE

13.1 PURPOSE AND PHILOSOPHY OF PTVS REPORTING

13.1.1 The Report as a Final Value Product

The **Prop Trust Verified Report** is not merely a technical document. It is the **final value product** that Aurema Group delivers to the market, representing the executable synthesis of all work performed in the previous phases (documentary, physical, technical, financial) and the integration of advanced protocols (CAP, PDBP, SCCIP, CPEP, PRA).
While traditional standards (RICS, IIMS, ISO 17020) produce static reports that lose value over time, the PTVS Report is a **living document** that: • For Levels 1 and 2: Constitutes a certified snapshot of the asset at a given moment. • For Level 3: Updates quarterly via the CAP protocol. • For Level 4: Updates monthly with continuous monitoring and predictive analysis.

13.1.2 Objectives of PTVS Reporting

OBJECTIVE 1: Executive Clarity Provide C-level executives and institutional investors with an immediate understanding of the asset's state, critical risks, and strategic recommendations, without requiring specialized technical training.
OBJECTIVE 2: Technical Depth Offer technical teams, engineers, and auditors the necessary detail to validate findings, understand methodologies, and execute recommendations.
OBJECTIVE 3: Complete Traceability Document every finding, measurement, and decision with cryptographically verifiable evidence (CPEP), enabling independent auditing at any future time.
OBJECTIVE 4: Immediate Actionability Transform findings into concrete, prioritized, and economically quantified recommendations that enable immediate decision-making.
OBJECTIVE 5: Tokenization Integration For tokenized assets, provide the information foundation that feeds the PDBP and SCCIP protocols, enabling automatic execution of compliance rules.
OBJECTIVE 6: Multi-Stakeholder Communication Serve as a single document that satisfies the needs of all stakeholders: owners, investors, token holders, regulators, insurers, and independent auditors.

13.1.3 Guiding Principles

PRINCIPLE 1: INVERTED PYRAMID The most critical information must appear first. The Executive Summary must be self-sufficient for strategic decision-making.
PRINCIPLE 2: PROGRESSIVITY The document must allow drill-down from the executive view to the forensic technical detail, with clear navigation between levels.
PRINCIPLE 3: VISUALIZATION PRIORITY Complex data must be presented through visualizations (dashboards,

heatmaps, charts) that facilitate immediate comprehension.
PRINCIPLE 4: VERIFIABILITY Every assertion must be backed by verifiable evidence, with cryptographic references (CPEP) enabling independent auditing.
PRINCIPLE 5: CONSISTENCY Structure, terminology, and format must be consistent across all PTVS reports, regardless of asset typology or verification level.
PRINCIPLE 6: DYNAMIC UPDATING For Levels 3 and 4, the report must be a living document that updates automatically with CAP protocol data.

13.1.4 Clarification of Terminology:

Consolidated Scores

To avoid ambiguities in reporting, the PTVS establishes the following official nomenclature for scores:

Acronym	Full Denomination	Origin	Chapter
SRD	Documentary Risk Score	Phase 1	Chapter 5
SRF	Physical Risk Score	Phase 2	Chapter 6
SRT	Technical Risk Score	Phase 3	Chapter 7
SRE	Economic-Financial Risk Score	Phase 4	Chapter 8
SRI	Integral Risk Score (Composite)	Consolidation	This Chapter

Formula for SRI (Integral Risk Score): SRI = (0.25 × SRD) + (0.30 × SRF) + (0.25 × SRT) + (0.20 × SRE)
The weights reflect the relative importance of each dimension in the global valuation of the asset.

13.2 PTVS REPORT

ARCHITECTURE

13.2.1 Five-Level Hierarchical Structure

The PTVS Report is organized into **five hierarchical levels** that allow different depths of reading:
LEVEL A: EXECUTIVE SUMMARY (1-3 pages) • Reading time: 5 minutes • Audience: C-Level, Family Office principals, institutional investors • Content: Key conclusions, SRI, strategic recommendations, go/no-go decision
LEVEL B: EXECUTIVE REPORT (10-15 pages) • Reading time: 20-30 minutes • Audience: Investment directors, asset managers, compliance officers • Content: Findings by phase, risk matrix, financial analysis, recommendations
LEVEL C: TECHNICAL REPORT (50-100 pages) • Reading time: 2-4 hours • Audience: Engineers, architects, inspectors, technical auditors • Content: Detailed analysis by systems, technical evidence, models
LEVEL D: TECHNICAL ANNEXES (100-300 pages) • Reading time: As needed • Audience: Specialists, counterparty experts, regulators • Content: Raw data, FEM models, lab analysis, certificates
LEVEL E: CRYPTOGRAPHIC EVIDENCE (Digital) • Access: Via authenticated web portal • Audience: Independent auditors, regulators, token holders • Content: CPEP custody chains, blockchain hashes, DID signatures, ZKPs

13.2.2 Detailed Document Structure

PART 1: COVER AND CERTIFICATION • PTVS Seal with level and AUI • QR code for cryptographic verification •

Date of issuance and validity • Digital signatures of the team
PART 2: LEVEL A - EXECUTIVE SUMMARY • 2.1 General Asset Overview • 2.2 Integral Risk Score (SRI) • 2.3 Critical Findings (Top 5) • 2.4 Strategic Recommendations • 2.5 Recommended Decision (Go/No-Go)
PART 3: LEVEL B - EXECUTIVE REPORT • 3.1 Verification Methodology • 3.2 Phase 1: Documentary Due Diligence (SRD, key findings, legal/regulatory risks, valuation impact) • 3.3 Phase 2: Physical Inspection (SRF, key findings, critical/high defects, estimated repair cost) • 3.4 Phase 3: Technical Evaluation (SRT, key findings, remaining useful life, environmental/ESG risks) • 3.5 Phase 4: Financial Verification (SRE, key findings, market valuation, profitability analysis, stress testing) • 3.6 Consolidated Risk Matrix • 3.7 Alternative Analysis • 3.8 Recommended Action Plan
PART 4: LEVEL C - TECHNICAL REPORT (Levels 3 and 4 only) • 4.1 Active Advanced Protocols (CAP, PDBP, SCCIP, CPEP, PRA) • 4.2 Detailed Structural Analysis • 4.3 Technical Systems Analysis • 4.4 Complete Environmental Analysis • 4.5 Detailed Energy Audit • 4.6 Natural Risks Analysis • 4.7 Predictive Models (PRA) • 4.8 Digital Twin (access)
PART 5: LEVEL D - TECHNICAL ANNEXES • Annex A: Verified legal documentation • Annex B: Plans and 3D models • Annex C: Laboratory reports • Annex D: Drone and LiDAR data • Annex E: Financial models • Annex F: Stress testing scenarios • Annex G: Certificates and licenses • Annex H: Team curriculum vitae
PART 6: LEVEL E - CRYPTOGRAPHIC EVIDENCE (Web Portal) • E.1 CPEP Custody Chains • E.2 Blockchain Registries • E.3 Inspector DID Signatures • E.4 Zero-Knowledge Proofs • E.5 Verification API

13.3 LEVEL A: EXECUTIVE

SUMMARY

13.3.1 Philosophy of the Executive Summary

The Executive Summary is the **most important piece** of the report. It must be completely self-sufficient: a Family Office principal or CEO must be able to make a €50 million investment decision based solely on these 1-3 pages.
GOLDEN RULES:

1. Maximum 3 pages (ideally 2).
2. One key idea per paragraph.
3. Dominant visualizations (60% visual, 40% text).
4. Concrete figures (no approximations).
5. Explicit recommendation (Go/No-Go/Conditional).

13.3.2 Executive Summary Structure

SECTION 1: GENERAL ASSET OVERVIEW • Asset name, typology, location, year of construction. • Owner (redacted if necessary), declared value. • Purpose of verification (e.g., Tokenization Level 4). • Primary photograph of the asset.
SECTION 2: INTEGRAL RISK SCORE (SRI) • Visual gauge of the SRI (e.g., 82/100). • Qualitative level (e.g., "GOOD"). • Breakdown by dimension: SRD, SRF, SRT, SRE with progress bars. • Formula calculation display. • Threshold comparison (e.g., "Level 4 Threshold: ≥90. ⚠ ASSET DOES NOT MEET THRESHOLD"). • Required action to reach threshold.
SECTION 3: TOP 5 CRITICAL FINDINGS • Ranked by severity (Critical, High, Medium). • For each: Description, financial impact, estimated repair cost, time to resolve, risk if unaddressed. • Example: "? #1 - Corrosion in Starboard

Fuel Tank. Impact: MARPOL contamination risk. Cost: €180,000. Time: 3 weeks. Risk: Port detention."
SECTION 4: STRATEGIC RECOMMENDATIONS • Immediate (<30 days): Actions, total cost. • Short-term (3-6 months): Actions, total cost. • Medium-term (6-18 months): Actions, total cost.
SECTION 5: RECOMMENDED DECISION • Clear verdict: GO, CONDITIONAL GO, or NO-GO. • For CONDITIONAL: List specific conditions to be met, adjusted valuation, projected SRI post-correction, maximum resolution deadline. • Digital signature and DID of the Technical Director.

13.4 LEVEL B: EXECUTIVE REPORT

13.4.1 Structure of the Executive Report

The Executive Report develops each of the four verification phases with an intermediate level of detail, providing context, evidence, economic impact, and concrete recommendations.

13.4.2 Phase Summaries (Sections 3.2 to 3.5)

For each phase, the report includes: • The specific Risk Score (SRD, SRF, SRT, or SRE). • A summary of verification activities performed. • A bulleted list of verified items (✓) and findings (□). • For each finding: Description, impact, evidence reference (CPEP hash), recommended action, and estimated cost/time. • A brief conclusion on the phase's overall health.

13.4.3 Consolidated Risk Matrix (Section 3.6)

A visual 3x3 or 4x4 matrix plotting findings by Probability (Low, Medium, High) and Impact (Low, Medium, High). • Each finding is coded (e.g., H1 for Physical High, D2 for

Documentary Medium). • A legend maps codes to specific findings. • This provides an immediate visual representation of the asset's risk profile.

13.4.4 Alternative Analysis and Action Plan (Sections 3.7 - 3.8)

• Presents 3-4 scenarios (e.g., Status Quo, Minimal Repairs, Full Optimization, Major Renovation). • For each scenario: Estimated cost, projected SRI, adjusted valuation, and recommendation. • Concludes with a clear, prioritized roadmap for the owner, including expected ROI for the recommended scenario.

13.5 LEVEL C: TECHNICAL REPORT

13.5.1 Advanced Protocols Active (Section 4.1)

For Levels 3 and 4, this section details the technological infrastructure: • CAP: Number and types of sensors installed, sampling frequencies, configured thresholds, SCCIP integration rules. • PDBP & SCCIP: AUI, physical-digital hash linkage, smart contract address, active compliance rules, current token state. • CPEP: Statistics on signed data points, blockchain anchoring, auditor DIDs involved, public verification URL. • PRA: Active predictive models, primary RUL predictions for critical components, active predictive alerts.

13.5.2 Detailed Technical Analyses (Sections 4.2 to 4.7)

These sections contain the forensic technical detail: • Structural: FEM model results, non-destructive testing results, crack evolution analysis. • Systems: Electrical

single-line diagrams, thermographic analysis reports, oil analysis results, vibration spectra. • Environmental: GHG emissions inventory (Scope 1, 2, 3), waste management, biodiversity impact. • Energy: Complete energy balance, quantified improvement opportunities with NPV/IRR/payback for each. • Natural Risks: Seismic, flood, wind, and fire risk evaluations with specific mitigation measures. • Predictive Models (PRA): Model descriptions, training data validation metrics, detailed RUL forecasts by component. • Digital Twin: Instructions for accessing the interactive 3D model, simulation capabilities, and real-time data integration.

13.6 LEVEL D: TECHNICAL

ANNEXES

13.6.1 Structure of Annexes

Annexes contain the complete documentary evidence supporting the report:
ANNEX A: LEGAL DOCUMENTATION • Copies of all verified certificates, registry notes, licenses, permits, and relevant contracts.
ANNEX B: PLANS AND 3D MODELS • General asset plans, system diagrams, downloadable 3D model files, LiDAR point clouds.
ANNEX C: LABORATORY REPORTS • Oil analysis, fuel analysis, water analysis, material testing, equipment calibration certificates.
ANNEX D: DRONE AND LiDAR DATA • Complete orthophotos, high-resolution 3D models, aerial thermography, underwater (ROV) inspection reports.

ANNEX E: FINANCIAL MODELS • Valuation calculation spreadsheets, cash flow projections, sensitivity analysis, stress testing scenarios.
ANNEX F: CERTIFICATES AND LICENSES • Full class certificates, statutory certificates, crew certifications, safety certificates.
ANNEX G: TEAM CURRICULUM VITAE • CVs of the inspection team, PTVS accreditations, relevant experience, DIDs, and verifiable credentials.

13.7 LEVEL E: CRYPTOGRAPHIC EVIDENCE

13.7.1 Public Verification Portal

For Levels 3 and 4, Aurema Group maintains a public web portal where any third party can verify the report's authenticity: URL:

`https://verify.proptrustverified.com/{AUI}`

FUNCTIONALITIES: • Report Verification: Upload the PDF to verify its hash against the blockchain. • Custody Chain Visualization: Interactive timeline of all events, drill-down to each signed datum, DID signature verification. • Current State Dashboard: Real-time scores (SRD, SRF, SRT, SRE, SRI), token state, active alerts, SCCIP rules. • Public API: Endpoints for programmatic verification of asset state, custody chains, and auditor credentials.

13.7.2 Zero-Knowledge Proofs for Privacy

To protect client confidentiality, the portal offers ZKP verifications. A third party can verify that: • The asset complies with Level 4 (SRI ≥ 90). • The SRF is ≥ 80. •

There are no critical alerts active. • The certificate is valid. WITHOUT revealing: • The owner's identity. • The exact location. • Sensitive technical or financial details.

13.8 FINAL CERTIFICATION AND PTVS SEAL

13.8.1 The Prop Trust Verified Certificate

The certificate is the official document attesting that the asset has been verified according to the PTVS. STRUCTURE: • Certificate Number and Asset AUI. • Owner (redacted or public based on level). • Verification Level (1, 2, 3, or 4). • Scores: SRD, SRF, SRT, SRE, and final SRI. • Final Valuation. • State: GO, CONDITIONAL (with conditions listed), or NO-GO. • Date of issuance, validity period, and next update date. • Issuer: Aurema Group, Accreditation Number. • Technical Director: Name and DID. • Digital Signature, Cryptographic Signature (Ed25519), Blockchain Hash, and QR Code for verification.

13.8.2 The PTVS Seal

The seal is the visual mark awarded to certified assets. SPECIFICATIONS: • Design: Unified hexagonal or shield emblem incorporating the level number (1, 2, 3, 4), the words "PROP TRUST VERIFIED", the SRI score, the AUI, and the year. • Colors: Level 1 (Blue), Level 2 (Green), Level 3 (Gold), Level 4 (Gold/Black Institutional). • Usage Rules: Must be used exactly as provided, without distortion, color changes, or addition of unauthorized elements. Must always be accompanied by the AUI and level.

13.8.3 Issuance and Revocation Process

ISSUANCE FLOW:

1. Complete all phases (1-4).
2. Calculate scores and verify against level thresholds.
3. Determine decision (Go/Conditional/No-Go).
4. Generate report and obtain Technical Director approval.
5. Digitally and cryptographically sign.
6. Register hash on blockchain.
7. Issue certificate and seal.
8. Publish on verification portal and notify stakeholders.

REVOCATION CAUSES: • Discovery of fraud (documentary or physical). • Lack of integrity of the inspection body. • Improper use of the seal. • Significant unreported changes to the asset. • Failure to resolve critical conditions within the stipulated deadline.

13.9 REPORTING BY LEVELS

13.9.1 Differentiation of Deliverables

Deliverable	Level 1	Level 2	Level 3	Level 4
Executive Summary	✓	✓	✓	✓
Executive Report	✓	✓	✓	✓
Technical Report	✗	Summary	✓ Complete	✓Forensic
Technical Annexes	Basic	Standard	Complete	Forensic
Certificate & Seal	✓	✓	✓	✓

Deliverable	Level 1	Level 2	Level 3	Level 4
Verification Portal	✗	Basic	Complete	Advanced + ZKP
Real-Time Dashboard	✗	✗	✓	✓
Digital Twin	✗	✗	✓	✓Advanced
Update Frequency	Annual	Semi-annual	Quarterly	Monthly
Follow-up Audit	✗	✗	Annual	Semi-annual

13.10 INTEGRATION WITH STAKEHOLDERS

13.10.1 Multi-Audience Communication

The PTVS Report serves multiple audiences with different needs: • Asset Owner: Focus on Executive Summary + Recommendations. Frequency: As per level. Confidentiality: Total. • Investors / Token Holders: Focus on Executive Report + Dashboard. Frequency: As per level. Confidentiality: Selective (via ZKP). • Tokenization Platforms: Focus on API + Structured Data. Frequency: Real-time (Levels 3-4). Confidentiality: As per SCCIP rules. • Insurers / Reinsurers: Focus on Technical Report + Annexes. Frequency: Annual or upon significant change. Confidentiality: Limited to risks. • Regulators: Focus on Specific Annexes + CPEP Evidence. Frequency: Upon request. Confidentiality: As per jurisdiction. • Independent Auditors: Focus on Full Access + Cryptographic Evidence. Frequency: Annual. Confidentiality: Under NDA.

13.10.2 Integration APIs

To facilitate integration with third-party systems, Aurema Group provides RESTful APIs: • `GET /api/v1/verify/{aui}`: Current asset state and scores. • `GET /api/v1/monitor/{aui}/realtime`: Real-time CAP data (Levels 3-4). • `GET /api/v1/token/{aui}/state`: Token state and active SCCIP rules. • `POST /api/v1/token/{aui}/zkp`: Generate a Zero-Knowledge Proof for specific compliance statements. • `GET /api/v1/report/{aui}/executive`: Retrieve the executive report.

13.11 PRACTICAL REPORTING CASES

13.11.1 Case 1: Report for a Family Office

Scenario: Family office evaluating a €25M villa in Marbella (Level 3). Deliverable: • Week 1-2: Executive Summary (SRI 76, 2 critical findings: structural dampness, expired license). Recommendation: Conditional Go. • Week 3-4: Executive Report with adjusted valuation (€23.5M) and repair costs (€450k). • Month 2: Technical Report with FEM analysis, energy audit, and environmental analysis. • Month 3: Update after repairs. New SRI: 85. New valuation: €24.8M. Certificate issued. Result: Family office proceeds with purchase at €24.5M, saving €500k vs. initial price.

13.11.2 Case 2: Report for Tokenization

Scenario: €60M megayacht tokenized into 10,000 tokens (Level 4). Deliverable: • Initial: Full 4-phase report, SRI 88. CAP (500 sensors), SCCIP (23 rules), CPEP, PRA configured. • Pre-Tokenization: PTVS Certificate issued, seal granted, verification portal active, API integrated, technical white paper for investors. • Post-Tokenization: Monthly reports to holders, real-time dashboard, valuation updates, automatic SCCIP alerts, ZKPs for privacy. Result: Successful tokenization, 100% subscription in 3 months, stable market price due to transparent, continuous verification.

13.11.3 Case 3: Report for an Insurer

Scenario: Reinsurer evaluating a portfolio of 5 commercial buildings (€200M total) (Level 3). Deliverable: • Portfolio Overview: Average SRI 82. Lowest SRI 74 (requires attention). • Risk Analysis: Seismic (Low-Medium), Flood (Medium for 2 coastal buildings), Fire (Low), Climate (Medium). • Recommendations: Improvements in 2 buildings to reduce premiums, CAP implementation across all, coverage updates. Result: Reinsurer adjusts premiums based on individual building SRI, offers discounts for CAP implementation.

13.12 AUTOMATED REPORT GENERATION

13.12.1 Generation System Architecture

To guarantee consistency and efficiency, the PTVS Report is generated via an automated system:

1.Data Ingestion: Pulls structured data from Phases 1-4 and advanced protocols.
2.Score Calculation: Automatically calculates SRD, SRF, SRT, SRE, and SRI.
3.Template Rendering: Populates pre-approved, level-specific templates (InDesign/PDF generators) with text, charts, and images.
4.Cryptographic Signing: Hashes the final PDF, signs with the Technical Director's DID, and registers the hash on the blockchain.
5.Portal Publishing: Automatically uploads the report, annexes, and cryptographic evidence to the verification portal.

13.12.2 Templates and Customization

The system uses customizable templates by asset typology: • `template_real_estate_residential.ptvs` • `template_real_estate_commercial.ptvs` • `template_yacht_megayacht.ptvs` • `template_industrial_machinery.ptvs` • `template_infrastructure.ptvs` • `template_patrimonial_asset.ptvs`

Customization allows for client logos, corporate colors, sector-specific terminology, and specific annex structures, while maintaining the rigid integrity of the core PTVS structure and cryptographic guarantees.

CHAPTER 14:

CERTIFICATION LEVELS

14.1 PURPOSE AND PHILOSOPHY OF THE LEVELS

14.1.1 The Need for a Tiered System

The market for asset verification is inherently diverse. A €300,000 apartment does not require or justify the same verification rigor as an €80 million megayacht. Traditional standards commit two fundamental errors:

1. Treating all assets equally, applying the same level of rigor regardless of value, complexity, or purpose.
2. Failing to offer scalability, forcing clients to choose between "all or nothing."

The Trust Standard resolves both issues through a **system of four certification levels** that allows the rigor, technology, depth, and cost of verification to be adapted to the specific needs of each asset and use case.

14.1.2 Principles of the Level System

PRINCIPLE 1: PROPORTIONALITY The level of verification MUST be proportional to the asset's value, technical complexity, identified risks, and the purpose of the verification.

PRINCIPLE 2: SCALABILITY The levels MUST allow for upgrades and downgrades as the client's needs evolve, without requiring complete process restarts.
PRINCIPLE 3: ASCENDING COMPATIBILITY Each higher level MUST include all requirements of the lower level, adding additional capabilities.
PRINCIPLE 4: CLARITY OF VALUE Each level MUST offer a clearly identifiable and quantifiable increase in value compared to the previous level.
PRINCIPLE 5: ACCESSIBILITY Level 1 MUST be accessible to the standard residential market, while Level 4 MUST satisfy the requirements of the most demanding Family Offices.

14.1.3 Philosophy of the Four Levels

Level	Philosophy	Analogy
Level 1 - Basic	"Essential verification for informed decisions"	Basic medical check-up
Level 2 - Standard	"Robust due diligence for secure transactions"	Comprehensive medical exam
Level 3 - Premium	"Institutional verification with continuous monitoring"	Hospital monitoring
Level 4 - Institutional	"Trust infrastructure for ultra-high-value assets"	ICU with 24/7 monitoring

14.2 DETAILED DEFINITION OF EACH LEVEL

14.2.1 LEVEL 1: BASIC VERIFICATION

OFFICIAL DEFINITION: Essential verification that provides a general understanding of the asset through visual inspection, basic documentation, and approximate valuation. Designed for rapid decisions and moderate-value transactions.
TYPICAL APPLICATION: • Single-family homes (€150,000 - €500,000) • Small recreational vessels (€50,000 - €200,000) • Medium-value collector vehicles • Preliminary due diligence before higher levels
ADMISSION CRITERIA: • Asset value: < €500,000 • Technical complexity: Low • Purpose: Standard transaction or general information
TECHNICAL REQUIREMENTS: • Phase 1 (Documentary): Title verification, basic registry check, identification of main charges, essential license verification. Minimum SRD: 60/100. • Phase 2 (Physical): Complete visual inspection (4-8 hours), representative digital photography, basic measurements (surfaces, moisture), identification of evident defects. Minimum SRF: 60/100. • Phase 3 (Technical): Qualitative evaluation of main systems, rough useful life estimation, identification of obvious risks. Minimum SRT: 60/100. • Phase 4 (Financial): Valuation by comparison method, basic market analysis, market value estimation. Minimum SRE: 60/100.
DELIVERABLES: • 20-40 page PDF report • 2-page Executive Summary • 15-30 photographs • PTVS Level 1 Certificate and Seal • Validity: 6 months

REQUIRED TEAM: • 1 Level 1 accredited inspector • 1 technical reviewer (validation) • Total time: 5-7 working days
TECHNOLOGIES APPLIED: • Professional digital camera, laser measure, digital hygrometer, basic valuation software. • Basic digital signature of the inspector.
LIMITATIONS: • Not suitable for tokenization or institutional financing. • No continuous monitoring or predictive analysis. • No advanced cryptographic evidence.

14.2.2 LEVEL 2: STANDARD VERIFICATION

OFFICIAL DEFINITION: Robust due diligence combining detailed physical inspection with advanced technology and complete financial analysis. Designed for medium-to-high value transactions and bank financing.
TYPICAL APPLICATION: • Luxury homes (€500,000 - €3M) • Small commercial buildings • Yachts 15-24m (€500,000 - €3M) • High-value industrial machinery • Bank financing requirements
ADMISSION CRITERIA: • Asset value: €500,000 - €5M • Technical complexity: Medium • Purpose: Transaction with financing or due diligence
TECHNICAL REQUIREMENTS: • Phase 1: All Level 1 requirements, plus exhaustive analysis of charges, verification of licenses with issuing authorities, urban planning compliance analysis. Minimum SRD: 70/100. • Phase 2: All Level 1 requirements, plus detailed inspection with drones (roofs, facades), complete infrared thermography, professional moisture analysis, technical systems inspection. Minimum SRF: 70/100. • Phase 3: All Level 1 requirements, plus quantitative systems analysis, basic energy audit, preliminary environmental assessment. Minimum SRT: 70/100. • Phase 4: All Level 1 requirements, plus valuation by three methods

(comparison, cost, income), profitability analysis, basic stress testing. Minimum SRE: 70/100.
DELIVERABLES: • 50-100 page PDF report • 3-page Executive Summary, complete Executive Report • 50-150 professional photographs, thermal and drone images • PTVS Level 2 Certificate and Seal • Validity: 12 months
REQUIRED TEAM: • 2-3 specialized inspectors, 1 certified drone pilot, 1 senior technical reviewer • Total time: 10-15 working days
TECHNOLOGIES APPLIED: • All Level 1, plus 4K drone, professional thermal camera, advanced measurement equipment, professional valuation software. • Basic CPEP (digital signature of key photographs).
LIMITATIONS: • Not suitable for institutional tokenization. • No continuous monitoring or advanced predictive analysis.

14.2.3 LEVEL 3: PREMIUM VERIFICATION

OFFICIAL DEFINITION: Institutional verification with continuous monitoring, predictive analysis, and cryptographic physical-digital linking. Designed for Family Offices, tokenized assets, and high-value assets requiring institutional guarantees.
TYPICAL APPLICATION: • Luxury real estate (€3M - €30M) • Megayachts 24-50m (€3M - €30M) • Commercial and industrial buildings • Tokenized assets (RWA) • Family Offices, large-risk insurance
ADMISSION CRITERIA: • Asset value: €5M - €50M • Technical complexity: High • Purpose: Tokenization, Family Office, large risks
TECHNICAL REQUIREMENTS: • Phase 1: All Level 2, plus exhaustive legal audit, multi-jurisdictional analysis, cryptographic document verification. Minimum SRD: 80/100. • Phase 2: All Level 2, plus complete LiDAR scanning, 3D asset modeling, ultrasonic structural analysis, underwater inspection (naval), IoT sensor

network installation, edge cryptographic signing (CPEP). Minimum SRF: 80/100. • Phase 3: All Level 2, plus FEM structural analysis, complete energy audit, exhaustive environmental assessment, climate risk evaluation, basic predictive models (PRA), digital twin creation. Minimum SRT: 80/100. • Phase 4: All Level 2, plus life cycle cost analysis, 10-year financial projections, advanced stress testing, specific tokenization valuation. Minimum SRE: 80/100.
DELIVERABLES: • 150-300 page PDF report (Executive, Technical, Annexes) • Interactive 3D model, accessible digital twin, monitoring dashboard • PTVS Level 3 Certificate and Seal, verification portal access • Validity: 24 months (with quarterly updates)
REQUIRED TEAM: • 3-5 specialized inspectors, drone pilot, LiDAR operator, IoT technician, financial analyst, technical review team • Total time: 20-30 working days
TECHNOLOGIES APPLIED: • All Level 2, plus terrestrial LiDAR, LiDAR drone, 50-200 IoT sensors, CAP monitoring platform, digital twin, edge cryptographic signing (CPEP), blockchain registry, basic PRA, basic SCCIP integration.

14.2.4 LEVEL 4: INSTITUTIONAL

VERIFICATION

OFFICIAL DEFINITION: Institutional-grade trust infrastructure for ultra-high-value assets. Combines forensic verification, redundant monitoring, advanced predictive analysis, and complete programmable compliance. Designed for ultra-high-net-worth Family Offices, sovereign wealth funds, reinsurers, and large-scale institutional tokenization.
TYPICAL APPLICATION: • Megayachts 50m+ (€30M - €200M+) • Emblematic buildings (€30M - €500M+) • Critical infrastructure, diversified asset portfolios •

Sovereign wealth funds, reinsurers, large-scale RWA tokenization

ADMISSION CRITERIA: • Asset value: > €50M • Technical complexity: Very high • Purpose: Institutional, large-scale tokenization

TECHNICAL REQUIREMENTS: • Phase 1: All Level 3, plus complete forensic legal audit, exhaustive multi-jurisdictional analysis, historical provenance verification, quantitative legal risk analysis. Minimum SRD: 90/100. • Phase 2: All Level 3, plus forensic inspection with multiple techniques, laboratory material analysis, complete underwater inspection (ROV), redundant IoT sensor network, extended baseline monitoring, cryptographic signing of ALL data. Minimum SRF: 90/100. • Phase 3: All Level 3, plus advanced multi-physics structural analysis, complex computational simulations, forensic environmental analysis, advanced climate risk evaluation, advanced predictive models (PRA), high-fidelity digital twin. Minimum SRT: 90/100. • Phase 4: All Level 3, plus detailed life cycle cost analysis, 20+ year financial projections, multi-scenario stress testing, portfolio risk correlation analysis, complete institutional valuation. Minimum SRE: 90/100.

DELIVERABLES: • 300-500+ page PDF report (Forensic Technical and Annexes) • High-fidelity 3D model, advanced digital twin, institutional dashboard, white-label platform (optional) • PTVS Level 4 Certificate and Seal, dedicated Advisory Board • Validity: 36 months (with monthly updates)

REQUIRED TEAM: • 5-8 senior specialized inspectors, laboratory team, computational modeling team, advanced IoT installation team, senior financial analysts, dedicated Technical Advisory Board • Total time: 30-60 working days

TECHNOLOGIES APPLIED: • All Level 3, plus redundant IoT network (200-500+ sensors), advanced laboratory analysis, multi-physics simulations, high-fidelity digital

twin, advanced Zero-Knowledge Proofs (ZKP), multi-sig governance, custom APIs.

14.3 COMPLETE COMPARISON MATRIX

Characteristic	Level 1	Level 2	Level 3
TYPICAL ASSET VALUE	< €500K	€500K - €5M	€5M - €50M
MINIMUM SCORE THRESHOLDS			
SRD / SRF / SRT / SRE / SRI	60	70	80
PHASE 1 - DOCUMENTARY	Basic	Standard	Exhaustive
Cryptographic verification	✗	✗	✓
PHASE 2 - INSPECTION			
Duration	4-8 hours	1-3 days	3-7 days
Drones / Thermography	✗	✓	✓✓
LiDAR / Lab Analysis	✗	✗	✓/ Optional
IoT Sensors	✗	✗	50-200
PHASE 3 - TECHNICAL			
Structural Analysis	Qualitative	Quantitative	FEM
Digital Twin / PRA	✗	✗	✓/ Basic
PHASE 4 - FINANCIAL			

Characteristic	Level 1	Level 2	Level 3
Valuation Methods	1	3	3+ adjustments
Stress Testing	✗	Basic	Advanced
Life Cycle Analysis	✗	✗	✓
DOCUMENT DEPTH	20-40 pages	50-100 pages	150-300 pages
ADVANCED PROTOCOLS			
CAP / PDBP / SCCIP	✗	✗	✓/ ✓/ Basic
CPEP / PRA	Basic	Basic	✓/ Basic
CERTIFICATION			
Validity / Updates	6 mo / ✗	12 mo / ✗	24 mo / Quarterly
Follow-up Audit	✗	✗	Annual
TOKENIZATION READINESS	✗	✗	✓
ERC-3643 Integration	✗	✗	Basic
TEAM & SUPPORT	1 Inspector	2-3 + Pilot	3-5 + Specialists
Support / Portal	Email	Email/Phone	Dedicated

14.4 DETAILED TECHNICAL REQUIREMENTS

14.4.1 Equipment Requirements

LEVEL 1 (Basic): Professional camera (24MP+), laser measure, digital hygrometer/thermometer, moisture detector, PPE, tablet with PTVS software. **LEVEL 2 (Standard):** All Level 1 + 4K GPS drone, thermal camera (≥320×240, NETD <50mK), industrial endoscope, air quality analyzer, multi-gas detector, vibration meter.

LEVEL 3 (Premium): All Level 2 + Terrestrial LiDAR (±2mm), LiDAR drone, 50-200 IoT sensors, edge gateways, Ground Penetrating Radar (GPR), ultrasonic structural analyzer, CAP platform license, digital twin software. **LEVEL 4 (Institutional):** All Level 3 + Redundant sensor networks (200-500+), portable laboratory analysis systems, multi-physics simulation workstations, ROV for underwater inspection, redundant monitoring systems, forensic equipment, dedicated data center/cloud infrastructure.

14.4.2 Personnel Requirements

LEVEL 1: 1 Accredited Inspector (3+ years experience, relevant degree), 1 Technical Reviewer (5+ years). **LEVEL 2:** 1 Lead Inspector (5+ years), 1 Secondary Inspector (3+ years), 1 Certified Drone Pilot, 1 Senior Technical Reviewer (10+ years). **LEVEL 3:** 1 Inspection Director (10+ years), Structural/Systems Inspectors (7+ years, specific certifications like RICS/IIMS), LiDAR Operator, IoT Technician, Financial Analyst (MBA/CFA), Senior Reviewer (15+ years). **LEVEL 4:** 1 Technical Director (PhD or equivalent, 15+ years), Team of Senior Inspectors (10+ years each), Laboratory Specialists, Computational Modelers, Senior Financial Analysts (CFA, CAIA), Dedicated Technical Advisory Board.

14.4.3 Technological Requirements

• **Level 1:** Report management software, comparable database, basic analysis tools. • **Level 2:** All Level 1 + photogrammetry software, thermographic analysis software, advanced valuation software. • **Level 3:** All Level 2 + CAP monitoring platform, LiDAR processing software, digital twin platform, PRA predictive engine, blockchain integration (CPEP), interactive dashboard, integration APIs. • **Level 4:** All Level 3 + Redundant CAP platform, multi-physics simulation software, high-fidelity digital twin,

advanced PRA engine with ML, blockchain integration with ZKP, white-label platform, custom APIs, multi-sig governance, 24/7 infrastructure.

14.5 CERTIFICATION PROCESS BY LEVEL

14.5.1 General Certification Flow

1. **Request & Initial Evaluation:** Reception, admissibility check, optimal level recommendation, economic proposal.
2. **Formalization:** Contract signing, 30% advance payment, team assignment, inspection planning.
3. **Phase 1 (Documentary):** Data collection, registry verification, charge analysis, regulatory compliance, SRD calculation.
4. **Phase 2 (Physical):** In-situ inspection, evidence capture, edge signing (CPEP), lab analysis, SRF calculation.
5. **Phase 3 (Technical):** Structural analysis, systems evaluation, environmental/energy assessment, predictive models (PRA), SRT calculation.
6. **Phase 4 (Financial):** Market valuation, profitability analysis, stress testing, life cycle analysis, SRE calculation.
7. **Consolidation & SRI Calculation:** SRI = (0.25×SRD) + (0.30×SRF) + (0.25×SRT) + (0.20×SRE). Threshold verification. Go/Conditional/No-Go decision.
8. **Report Generation:** Drafting per level, technical review, quality review, final approval.

9.**Certification & Issuance:** Digital/DID signature, blockchain registration, certificate and seal issuance, portal publication.
10.**Delivery & Follow-up:** Client delivery, results presentation, 70% final payment, follow-up plan activation.

14.5.2 Timelines by Level

Phase	Level 1	Level 2	Level 3	Level 4
Request & Evaluation	2 days	3 days	5 days	7 days
Formalization	2 days	3 days	5 days	7 days
Phase 1 - Documentary	3 days	5 days	10 days	15 days
Phase 2 - Inspection	1 day	3 days	7 days	20 days
Phase 3 - Technical	1 day	3 days	7 days	15 days
Phase 4 - Financial	1 day	2 days	5 days	10 days
Consolidation	1 day	1 day	2 days	3 days
Report Generation	2 days	5 days	10 days	15 days
Certification	1 day	1 day	2 days	3 days
Delivery	1 day	1 day	2 days	2 days
TOTAL ESTIMATED	**15 days**	**24 days**	**55 days**	**97 days**

14.5.3 Critical Milestones and Go/No-Go Decisions

• **Milestone 1 (Post-Phase 1):** If SRD < threshold, alert client. If insurmountable defects exist, recommend No-Go.
• **Milestone 2 (Post-Phase 2):** If SRF < threshold, alert client. If critical defects exist, evaluate viability. • **Milestone 3 (Post-Phase 3):** If SRT < threshold, alert client. If risks are unacceptable, recommend No-Go. • **Milestone 4 (Post-Consolidation):** If SRI < threshold: Option A (Downgrade), Option B (Resolve findings and re-evaluate), Option C (Issue "Conditional" certificate). If SRI ≥ threshold: Proceed to certification.

14.6 PRICING STRUCTURE

14.6.1 Pricing Philosophy

The PTVS pricing structure follows three principles:

1. **Transparency:** Clear and predictable pricing.
2. **Proportionality:** Price reflects the value delivered.
3. **Accessibility:** Level 1 is accessible to the mass market.

14.6.2 Base Pricing Structure

LEVEL 1 - BASIC VERIFICATION: • Fixed Fee: €1,500 • Value Fee: 0.05% of asset value (up to €500K) • Travel: €300-€800 • **Typical Total:** €1,800 - €3,500
LEVEL 2 - STANDARD VERIFICATION: • Fixed Fee: €4,500 • Value Fee: 0.08% of asset value (up to €5M) • Drone/Thermography: Included • Travel: €500-€1,500 • **Typical Total:** €6,000 - €15,000
LEVEL 3 - PREMIUM VERIFICATION: • Fixed Fee: €15,000 • Value Fee: 0.12% of asset value (up to €50M) • LiDAR / IoT (50-200 sensors) / Digital Twin / CAP (1 yr): Included (€15k-€30k value) • Travel: €1,500-€5,000 • **Typical Total:** €40,000 - €120,000
LEVEL 4 - INSTITUTIONAL VERIFICATION: • Fixed Fee: €50,000 • Value Fee: 0.15% of asset value (uncapped) • Lab Analysis / IoT (200-500+) / Advanced Digital Twin / CAP (3 yrs) / Advisory Board: Included (€100k+ value) • White-label platform (optional): €30,000 • Travel: €5,000-€20,000 • **Typical Total:** €200,000 - €750,000+

14.6.3 Discounts and Bonuses

• **Volume Discounts:** 5-10 assets (-10%), 11-20 assets (-15%), 21+ assets (-20%). • **Multi-year Contracts:** 2 years (-5%), 3 years (-10%), 5 years (-15%). • **Upgrade**

Bonuses: Level 1→2 (80% of price differential), Level 2→3 (70%), Level 3→4 (60%).

14.6.4 Annual Maintenance Costs

Level	Annual Maintenance Fee	Includes
Level 1	€0	No updates
Level 2	€1,500	Semi-annual update
Level 3	€12,000	3 quarterly updates + CAP
Level 4	€50,000	11 monthly updates + CAP + 24/7 suppo

14.7 MAINTENANCE AUDITS

14.7.1 Philosophy of Maintenance Audits

The PTVS certification is not a one-time event, but a **state maintained over time**. Maintenance audits guarantee that the asset continues to meet the requirements of the certified level.

14.7.2 Frequency by Level

Level	Documentary Audit	Complete Physical Audit	CAP Monit
Level 1	Not applicable	Every 2 years	Not applicable
Level 2	Semi-annual	Annual	Not applicable
Level 3	Quarterly	Annual	Continuous (24/7)
Level 4	Monthly	Semi-annual	Continuous (24/7)

14.7.3 Types of Audits

• **Documentary Audit:** Verification of registry changes, license updates, regulatory compliance review, analysis of new charges, SRD recalculation. • **Basic Physical Audit:** Visual inspection of critical areas, verification of previously identified defects, confirmation of recommendation execution, partial SRF recalculation. • **Complete Physical Audit:** All Phase 2 original activities, 3D model updates,

IoT sensor recalibration, complete SRF recalculation. • **Technical/Financial Audit:** Updates to structural/energy analyses, predictive model updates, valuation updates, stress testing updates, SRT/SRE recalculation.

14.7.4 Decision Matrix Post-Audit

Scenario	Decision	Action
SRI ≥ level threshold	**MAINTAIN**	Renew certification
SRI between (threshold-5) and threshold	**WARNING**	90-day improvement plan
SRI < (threshold-5) but > (threshold-15)	**DOWNGRADE**	New lower-level certificate
SRI < (threshold-15)	**SUSPENSION**	Remediation plan required
Critical finding	**REVOCATION**	Investigation + action

14.8 UPGRADES AND DOWNGRADES

14.8.1 Philosophy of Mobility

The PTVS system is designed to be **dynamic**, allowing assets to move between levels as their circumstances evolve (e.g., increased value, new tokenization requirements, or reduced operational needs).

14.8.2 Upgrade Process

1. **Viability Evaluation:** Gap analysis between current and target level, identification of additional requirements, cost/time estimation.

2.**Economic Proposal:** Differential pricing (with upgrade bonus), execution timeline, specific conditions.
3.**Execution of Complementary Activities:** Only activities not covered in the previous level are performed. Previous data and work are leveraged.
4.**Score Recalculation:** All scores recalculated with new requirements. Threshold verification. New SRI determination.
5.**New Certificate Issuance:** New certificate with higher level, new PTVS seal, portal update, activation of additional protocols (CAP, SCCIP, etc.).

14.8.3 Downgrade Process

1.**Cause Evaluation:** Why the downgrade? (Economic, operational, change of needs). Temporary or permanent?
2.**Economic Proposal:** Adjustment of maintenance fees, deactivation of advanced protocols, new validity period.
3.**Protocol Deactivation:** CAP sensor removal (if applicable), SCCIP rule updates, PRA model deactivation, digital twin simplification.
4.**Score Recalculation:** With lower-level requirements. Threshold verification. New SRI.
5.**New Certificate Issuance:** New certificate with lower level, new PTVS seal, portal update, stakeholder communication.

14.8.4 Typical Mobility Cases

• **Case 1 (Level 1 → Level 3):** Luxury home owner decides to tokenize fractions of the asset. Gap: Phases 2-4 need deepening. Activities: LiDAR, IoT, digital twin, CAP. Bonus applied. Result: Asset ready for tokenization. • **Case 2 (Level 4 → Level 2):** Family office sells megayacht to a private owner who does not require Level

4. Process: Deactivate advanced CAP, PRA, ZKP. Simplify digital twin. New validity: 12 months. Result: Level 2 certification suited to new context. • **Case 3 (Temporary Level 2 → Level 3):** Commercial building owner needs Level 3 certification for refinancing. Process: Upgrade with 24-month validity. Afterward, option to revert to Level 2. Result: Temporary Level 3 certification for specific operation.

14.9 DETAILED USE CASES

14.9.1 Case 1: Family Home Purchase (Level 1)

Scenario: Family evaluating a €380,000 single-family home in Valencia. **Solution:** • Phase 1: Registry note clear, cadastre matches, licenses valid. SRD: 88. • Phase 2: Visual inspection. Minor dampness in main bathroom detected. SRF: 78. • Phase 3 & 4: Systems functional, 25-year useful life. Valuation: €380,000. SRT: 82, SRE: 85. **Result:** SRI = 82. Decision: **GO**. Recommendation: Repair dampness before signing (€1,200). Cost: €2,100. Timeline: 8 days. Value: Family negotiates €5,000 discount, proceeds with informed purchase.

14.9.2 Case 2: Yacht Due Diligence (Level 2)

Scenario: Investor evaluating a 20m yacht for €1.8M. Requires bank financing due diligence. **Solution:** • Phase 1: Naval registry confirmed, class certificate valid (expires in 8 months). SRD: 90. • Phase 2: Detailed inspection, drone for superstructure, thermography for electrical, oil analysis normal. SRF: 84. • Phase 3 & 4: Engines at 8,500h (within specs). Valuation: €1.78M. Operating costs: €180k/yr. SRT: 82, SRE: 80. **Result:** SRI = 84. Decision: **GO**. Recommendation: Schedule dry-dock for underwater inspection in 6 months. Cost: €8,900. Timeline: 18 days.

Value: Bank approves financing, investor negotiates final price to €1.75M.

14.9.3 Case 3: Building Tokenization (Level 3)

Scenario: Family office wants to tokenize a €25M Madrid office building into 5,000 tokens. **Solution:** • Full 4-phase verification. LiDAR scan, 120 IoT sensors installed, digital twin created, PRA models activated. • Protocol Configuration: CAP (120 sensors, 24/7), PDBP (AUI linked), SCCIP (15 compliance rules, auto-pauses), CPEP (edge signing), PRA (RUL predictions). **Result:** SRI = 86. Decision: **GO**. Valuation: €24.5M. Value per token: €4,900. Cost: €78,000. Timeline: 45 days. Value: 100% of tokens subscribed in 6 weeks. Institutional investors trust the PTVS certification.

14.9.4 Case 4: Institutional Megayacht (Level 4)

Scenario: Sovereign wealth fund evaluating an 80m megayacht for €95M as a patrimonial asset. **Solution:** • Forensic legal audit, multi-jurisdictional analysis. • Forensic inspection, lab material analysis, ROV underwater inspection, 420 redundant IoT sensors, 7-day baseline monitoring. • Multi-physics structural analysis, advanced climate risk evaluation, high-fidelity digital twin, advanced PRA. **Result:** SRI = 91. Decision: **GO**. Valuation: €92M. Cost: €420,000. Timeline: 85 days. Value: Fund proceeds with acquisition, negotiates final price to €90M (€5M savings). Continuous monitoring protects the investment.

14.10 REPORTING AND CERTIFICATION BY LEVEL

14.10.1 PTVS Certificate by Level

LEVEL 1: Basic layout. Asset, AUI, SRI, Value. Validity: 6 months. Digital signature, QR verification. **LEVEL 2:** Adds SRD/SRF/SRT/SRE breakdown. Validity: 12 months. Next review date. Basic cryptographic signature. **LEVEL 3:** Adds active protocols list (✓CAP, ✓PDBP, ✓SCCIP, ✓CPEP, ✓PRA). Validity: 24 months. Next update: 3 months. DID signature, Blockchain hash, Portal QR. **LEVEL 4:** Adds advanced protocols list (✓✓CAP, ✓✓PDBP+ZKP, ✓✓SCCIP, ✓✓CPEP+ZKP, ✓✓PRA, ✓Advisory Board). Validity: 36 months. Next update: 1 month. Multi-sig DID signature, Blockchain hash, Portal+ZKP QR.

14.10.2 PTVS Seal by Level

Unified design with color differentiation: • **Level 1:** Blue. "BASIC". • **Level 2:** Green. "STANDARD". • **Level 3:** Gold. "PREMIUM". • **Level 4:** Gold/Black. "INSTITUTIONAL". All seals include the Level number, "PROP TRUST VERIFIED", SRI score, AUI, and Year.

14.10.3 Stakeholder Reporting by Level

Stakeholder	Level 1	Level 2	Level 3	Level 4
Owner	Full Report	Full Report	Full Report	Full Report
	Exec. Summary	Exec. Summary	Exec. Summary	Exec. Summary

Stakeholder	Level 1	Level 2	Level 3	Level 4
	No updates	Annual update	Quarterly update	Monthly update
Investors/Tokens	N/A	N/A	Dashboard	Dashboard (Adv.)
	N/A	N/A	Quarterly reports	Monthly reports
Token Platforms	N/A	N/A	API Access	Custom API
	N/A	N/A	Basic SCCIP	Complete SCCIP
Insurers	Basic risk data	Standard risk data	Complete risk data	Advanced risk data
Regulators	Basic docs	Standard docs	Complete docs	Forensic docs + ZKP

CHAPTER 15: USE OF THE SEAL AND BRAND PROP TRUST VERIFIED

15.1 PURPOSE AND PROTECTION OF THE BRAND

15.1.1 The Seal as a Guarantee of Trust

The **Prop Trust Verified Seal** is not a decorative element. It is the visual manifestation of a complete trust infrastructure that includes: • Rigorous verification according to the PTVS methodologies. • Immutable cryptographic evidence (CPEP). • Continuous monitoring when applicable (CAP). • Programmable compliance

(SCCIP). • Predictive analysis (PRA). • Audit and governance by Aurema Group.
Every use of the seal represents an **institutional guarantee** backed by Aurema Group. Therefore, controlling its use is fundamental to preserving the value and credibility of the entire PTVS ecosystem.

15.1.2 Protected Intellectual Property Assets

REGISTERED TRADEMARKS OF AUREMA GROUP:

Mark	Type	Jurisdictions	Nice Classes
Prop Trust Verified	Denominative + Figurative	EUIPO, USPTO, WIPO	35, 36, 42, 45
The Trust Standard	Denominative	EUIPO, USPTO, WIPO	35, 36, 42, 45
PTVS	Denominative + Figurative	EUIPO, USPTO	35, 36, 42, 45
Prop Trust Marine	Denominative + Figurative	EUIPO, USPTO	35, 36, 42
Aurema Group	Denominative + Figurative	EUIPO, USPTO, WIPO	Multiple

ADDITIONAL PROTECTED ELEMENTS: • Content of the PTVS standard (copyright). • Methodologies SCCIP, CPEP, CAP, PRA, PDBP (potentially patentable). • Design of the seal and its variants. • Corporate color palette and typography. • Verification portal and its interface.

15.1.3 Principles of Brand Use

PRINCIPLE 1: EXCLUSIVITY The PTVS seal may ONLY be used by assets that have been certified by Aurema Group or by an inspection body accredited by Aurema Group.
PRINCIPLE 2: PROPORTIONALITY The use of the seal MUST faithfully reflect the certification level achieved (1, 2,

3, or 4). Using a higher level than actually obtained is strictly prohibited.
PRINCIPLE 3: INTEGRITY The seal MUST NOT be modified, altered, distorted, or combined with other elements that could create confusion about its meaning.
PRINCIPLE 4: VALIDITY The seal may ONLY be used while the certification is valid. It must be removed immediately upon expiration, suspension, or revocation.
PRINCIPLE 5: CONTEXTUALITY The seal MUST be used in contexts that reinforce its meaning of verification and trust, never in contexts that could damage its reputation.
PRINCIPLE 6: ATTRIBUTION Every use of the seal MUST include proper attribution to Aurema Group as the issuing body of the standard.

15.2 GRAPHIC SPECIFICATIONS OF THE SEAL

15.2.1 Master Seal Design

CONSTITUENT ELEMENTS:

1. **Logotype "PROP TRUST VERIFIED"**: Always in this order, using the designated serif typography.
2. **Level Indicator**: Clearly stating the level (1, 2, 3, or 4) and its designation (Basic, Standard, Premium, Institutional).
3. **Integral Risk Score (SRI)**: Displayed in the format XX/100.
4. **AUI Code**: The unique Asset Unique Identifier.
5. **Year of Certification**: e.g., "2026".

15.2.2 Variants by Level

LEVEL 1 - BASIC: • Colors: Primary text in PTVS Blue (#1E3A8A), level indicator in Light Blue (#3B82F6). • Usage: Standard documentation, basic marketing.
LEVEL 2 - STANDARD: • Colors: Primary text in PTVS Green (#065F46), level indicator in Green (#10B981). • Usage: Commercial real estate listings, bank financing documentation.
LEVEL 3 - PREMIUM: • Colors: Primary text in PTVS Gold (#92400E), level indicator in Gold (#F59E0B). • Usage: Family Office reports, tokenization white papers, high-value asset marketing.
LEVEL 4 - INSTITUTIONAL: • Colors: Primary text in Institutional Gold (#78350F), level indicator in Bright Gold (#FBBF24), with a fine gold border. Background: Black or White. • Usage: Sovereign wealth fund reports, institutional tokenization platforms, high-stakes regulatory filings.

15.2.3 Corporate Color Palette

Color	Hex	RGB	Primary Use
PTVS Blue	#1E3A8A	30, 58, 138	Level 1, primary elements
PTVS Green	#065F46	6, 95, 70	Level 2, confirmation
PTVS Gold	#92400E	146, 64, 14	Level 3, premium
Institutional Gold	#78350F	120, 53, 15	Level 4, institutional
PTVS Black	#111827	17, 24, 39	Backgrounds, primary text
Light Gray	#F3F4F6	243, 244, 246	Secondary backgrounds

15.2.4 Institutional Typography

PRIMARY FONT FAMILY: INTER • Logotype: Inter Bold, 24pt (Weight 700) • Level Indicator: Inter Medium, 18pt (Weight 500) • SRI Score: Inter Regular, 14pt (Weight 400) • AUI / Year: Inter SemiBold / Light, 12pt (Weight 600 / 300)
ALTERNATIVE FONTS (when Inter is unavailable): • Primary alternative: Helvetica Neue • Secondary alternative: Arial • Monospaced (for codes/hashes): JetBrains Mono

15.2.5 Dimensions and Clear Space

MINIMUM REPRODUCTION SIZES: • Print: 25mm width minimum (300 DPI) • Screen: 120px width minimum (72-96 DPI) • Favicon/Icon: 32x32px (simplified version) • Social Media: 400x400px minimum
CLEAR SPACE: The clear space around the seal must equal at least 1/4 of the seal's total height. No graphic or textual element may invade this space to ensure maximum legibility and impact.

15.3 PERMITTED USES OF THE SEAL

15.3.1 Authorized Uses for the Asset Owner

PERMITTED USES (Levels 1-4):

1. **Asset Documentation:** Deeds, sale contracts, insurance policies, registry documentation, inventories.

2. **Marketing and Communication:** Brochures, promotional materials, asset websites, commercial presentations, listings on real estate/naval portals.
3. **Physical Signage:** Plaques on the asset (when appropriate), onboard documentation (vessels), signage on properties.
4. **Communications with Third Parties:** Correspondence with advisors, reports to co-owners, documentation for potential investors.

ADDITIONAL USES (Levels 3-4): 5. **Tokenization Platforms:** Token metadata, platform marketplace, emission white paper, investor materials. 6. **Monitoring and Dashboard:** Visualization of asset state, reports to token holders, integration into owner's systems.

15.3.2 Authorized Uses for Aurema Group

INSTITUTIONAL USES:

1. **Standard Marketing:** Case studies (anonymized when applicable), PTVS presentation materials, conferences, specialized publications.
2. **Accreditation of Bodies:** Directory of accredited bodies, training materials, audits, and supervision reports.
3. **Verification Portal:** Listing of certified assets, aggregated statistics, public standard information.

15.3.3 Authorized Uses for Accredited Bodies

Accredited inspection bodies may use the brand in:

1. **Corporate Materials:** Corporate website (with clear attribution), business cards, presentation materials, issued reports.
2. **Accreditation Communication:** "Accredited Body by Aurema Group", "Official Prop Trust Verified Certifier".

RESTRICTION: Accredited bodies CANNOT use the PTVS seal directly on assets. Only Aurema Group issues the final seal upon certification.

15.3.4 Authorized Uses for Tokenization Platforms

Integrated tokenization platforms may:

1. **Display the Seal on Listings:** For effectively verified assets, visual badge in the marketplace, display of level and SRI.
2. **Integrate PTVS Data:** Official APIs, verification dashboards, automated reporting.
3. **Joint Marketing:** Authorized co-branded campaigns, joint events, case studies.

15.4 PROHIBITED USES

15.4.1 Absolute Prohibitions

NEVER PERMITTED:

1. **Use without Certification:** Using the seal on uncertified assets, before obtaining certification, or after expiration/revocation. Falsifying or imitating the seal.
2. **Unauthorized Modifications:** Altering seal colors, changing typography, distorting proportions, adding unauthorized graphic elements, combining with other logos confusingly, applying unauthorized effects (shadows, 3D).
3. **Deceptive Uses:** Indicating a higher level than obtained, displaying a different SRI than certified, using the Aurema Group seal without authorization, suggesting non-existent sponsorship or

endorsement, using in contexts that damage reputation.
4. **Uses on Uncovered Assets:** Purely financial assets (stocks, bonds), intellectual property not linked to a physical asset, intangible services, consumer products.

15.4.2 Context-Specific Prohibitions

• **In Advertising:** Using the seal as a decorative element without context, associating the seal with profitability guarantees, using in unfair comparative advertising. • **On Social Media:** Using the seal as a profile avatar, modifying the seal for memes or informal content, associating the seal with personal opinions or controversial political contexts. • **On Products/Merchandise:** Reproducing the seal on commercial merchandise, clothing, accessories, or corporate gifts without explicit authorization. • **In Legal Documentation:** Using the seal as a substitute for independent legal advice, presenting the seal as an absolute guarantee of investment return, omitting the limitations and conditions of the certification.

15.5 VISUAL STYLE GUIDE FOR APPLICATIONS

15.5.1 Digital Applications

WEBSITES: • Format: SVG (scalable). • Size: 120px width minimum. • Position: Header or dedicated "Certifications" section. • Link: Always linking to the public verification portal. • Alt text: Descriptive and accessible (e.g., "Prop Trust Verified Level 3, SRI 86/100").

EMAILS: • Use only in official asset communications, not in personal email signatures. • Maximum size: 100px width. • Always accompanied by a verification link.
SOCIAL MEDIA: • Authorized hashtags: #PropTrustVerified, #PTVS, #AssetVerification, #VerifiedAsset, #Level3 (or respective level). • Prohibited hashtags: Any suggesting guaranteed profitability, political/controversial tags, or tags confusing the brand with others.

15.5.2 Print Applications

MARKETING MATERIALS: • Resolution: Minimum 300 DPI. • Color Mode: CMYK for print. • Minimum Size: 25mm width. • Paper: Premium quality recommended. • Finish: Matte or gloss according to overall design.
PHYSICAL PLAQUES (Real Estate/Naval): • Material: Engraved stainless steel or premium acrylic. • Size: 20x30 cm (standard). • Location: Main entrance or documentation area. • QR Code: Must link directly to the asset's verification portal. • Note: The PTVS seal does NOT replace mandatory naval or real estate identification markings.

15.6 INTEGRATION WITH TOKENIZATION

15.6.1 Seal Usage in Tokenized Assets

When a PTVS-certified asset is tokenized, the seal acquires additional dimensions:
TOKEN METADATA (ERC-3643 Example):

```
{
```

```
    "name": "Megayacht Aurelia Token",
    "symbol": "AURELIA-PTVS",
    "standard": "ERC-3643",
    "ptvs_certification": {
        "level": 3,
        "sri": 86,
        "aui": "AUI-NAV-ES-VLC001-v3",
        "issue_date": "2026-07-12T16:00:00Z",
        "valid_until": "2027-07-12T16:00:00Z",
        "verification_url":
"https://verify.proptrustverified.com/AUI-NAV-ES-VLC001-
v3",
        "certificate_hash": "0x9h8i7j...",
        "issuer_did": "did:ptvs:es:auditor:0x7f8e9d..."
    },
    "protocols": {
        "cap": true, "pdbp": true, "sccip": true, "cpep": true,
"pra": true
    }
}
```


MARKETPLACE VISUALIZATION: The seal is prominently displayed on the token listing, accompanied by real-time status indicators (e.g., "Verified ✓", "Monitored 24/7 ✓"), linking directly to the certificate and live dashboard.

15.6.2 Dynamic Seal Updating

For Levels 3 and 4, the seal can display real-time updated information: • Current SRI score with a visual progress bar. • Breakdown of SRD, SRF, SRT, SRE. • Current state (e.g., "NORMAL", "ATTENTION"). • Time since last update and number of active sensors.

15.6.3 Seal in Zero-Knowledge Proofs (ZKP)

To protect owner privacy, the seal can be integrated with ZKPs. A third party can verify a "ZKP Seal" confirming that: • The asset is certified at Level 3 or higher. • The SRI is ≥ 80. • There are no critical alerts active. WITHOUT revealing the owner's identity, exact location, or specific financial/technical details.

15.7 BRAND COMPLIANCE AUDITS

15.7.1 Monitoring Program

Aurema Group maintains an active program to monitor brand usage:
DIGITAL MONITORING: • Automated searches (Google Alerts, social media listening). • Analysis of tokenization marketplaces and real estate/naval portals. • Domain name monitoring.
PHYSICAL MONITORING: • Scheduled inspections during maintenance audits. • Presence at industry events and conferences. • Third-party reports (anonymous whistleblowing channels, feedback from accredited bodies).
FREQUENCY: • Level 1-2: Monthly. • Level 3: Weekly. • Level 4: Daily.

15.7.2 Types of Non-Compliance

MINOR NON-COMPLIANCE: • Slightly altered colors, size below minimum recommended, missing verification link, incomplete attribution. • Action: Friendly notification + 30 days to correct.
MODERATE NON-COMPLIANCE: • Using a higher level than obtained, unauthorized design modifications, use after certification expiration (without bad faith). • Action: Formal demand + 15 days to correct + potential fine.

SERIOUS NON-COMPLIANCE: • Use without any certification, fraudulent use, intentional reputational damage, use after revocation. • Action: Immediate legal action + claim for damages.

15.7.3 Audit Process

1. **Detection:** Automated alert, third-party report, or physical inspection.
2. **Classification:** Assess severity (Minor, Moderate, Serious) and intent.
3. **Notification:** Formal communication to the infringer with evidence, correction deadline, and stated consequences.
4. **Follow-up:** Verify correction. If corrected, close case. If not, escalate.
5. **Resolution:** Apply sanctions, update public registries, document lessons learned.

15.8 SANCTIONS FOR MISUSE

15.8.1 Scale of Sanctions

FOR CERTIFIED ASSET OWNERS: • Minor (First time): Warning. • Minor (Repeated): €500 fine + correction plan. • Moderate: €1,000 - €5,000 fine + temporary seal suspension. • Serious: €10,000 - €25,000 fine + certification revocation. • Critical (Fraud): €50,000+ fine + immediate legal action.
FOR ACCREDITED BODIES: • Misuse of seal: €5,000 - €25,000 fine + formal reprimand. • Forgery of certificates: €50,000 fine + immediate revocation + legal action. • Use after suspension: €25,000 fine + revocation.
FOR UNAUTHORIZED THIRD PARTIES: • Use without certification: Cease & desist + €10,000 minimum fine. • Forgery/Imitation: Legal action + maximum damages + €25,000+ fine.

15.8.2 Sanctioning Process

1. **Investigation (5-10 days):** Gather evidence, verify facts, assess damage, identify responsible parties.
2. **Notification (1 day):** Formal communication of non-compliance, presentation of evidence, 10-day period for allegations.
3. **Resolution (15 days):** Analysis of allegations, determination of sanction, issuance of motivated resolution.
4. **Execution (Immediate):** Notification of sanction, 30-day payment deadline, demand to cease use.
5. **Appeal (Optional, 15 days):** Appeal to the Governance Committee, independent review, final decision.

15.9 LEGAL PROTECTION OF THE BRAND

15.9.1 Legal Framework

TRADEMARK PROTECTION: • Spain (OEPM): Applications N0498420 and N0498759 (pending). 10-year renewable terms. • European Union (EUIPO): EU-wide protection. • United States (USPTO): Federal registration. • International (WIPO - Madrid System): Extension to multiple jurisdictions.
COPYRIGHT PROTECTION: • Content of the PTVS standard, graphic design of the seal, texts, and methodologies. • Automatic protection in 181 countries under the Berne Convention.
PATENT PROTECTION (where applicable): • Technical processes of SCCIP, CPEP, CAP, PRA, PDBP. • Protection in key jurisdictions.

15.9.2 Available Legal Actions

1. **Cease & Desist:** Formal communication to cease, 15-30 day deadline. Prerequisite to judicial action.
2. **Opposition to Registration:** Against similar marks in the application phase (3-month window).
3. **Nullity of Trademark:** Against marks registered in bad faith (up to 5 years).
4. **Infringement Lawsuit:** Against unauthorized uses, claiming damages and injunctive relief.
5. **Criminal Complaint:** Against counterfeiting/forgery (penalties of 6 months to 2 years imprisonment in many jurisdictions).

15.9.3 Proactive Protection Strategy

• **Monitoring Services:** Alerts for new trademark registrations, domain names, and social media uses (Annual cost: €3,000-€8,000). • **Renewals and Maintenance:** Strict calendar for trademark renewals (every 10 years) and proof of use to avoid "non-use" cancellation (5 years). • **Active Defense:** Systematic opposition to all similar registrations in all key jurisdictions. Strategic litigation to establish precedents and deter future infringers.

15.9.4 Priority Jurisdictions

Jurisdiction	Importance	Actions	Estimated Annual Budget
Spain	Maximum (HQ)	Registration + active defense	€10,000
EU	High (Single Market)	EUIPO + systematic opposition	€15,000
USA	High	USPTO + strategic	€20,000

Jurisdiction	Importance	Actions	Estimated Annual Budget
	(Institutional Market)	litigation	
UK	Medium-High (Post-Brexit)	UKIPO + mutual agreements	€8,000
Switzerland	Medium (Family Offices)	IPI + specific protection	€5,000
UAE (ADGM)	Medium-High (Tokenization)	Local registration + defense	€7,000
Singapore	Medium (Asian Hub)	IPOS + regional expansion	€6,000
Cayman Islands	Medium (Naval Registry)	Specific protection	€4,000

15.10 USAGE LICENSES

15.10.1 Types of Licenses

TYPE A LICENSE: CERTIFIED ASSET OWNER • Grant: Automatic upon certification, included in certification fee. • Rights: Use on asset documentation, marketing, communications, tokenization platforms (Levels 3-4). • Restrictions: Only for the specific certified asset, non-transferable, non-sublicensable, subject to compliance audits.
TYPE B LICENSE: ACCREDITED INSPECTION BODY • Grant: Upon successful accreditation. Annual fee: €5,000. • Rights: Use of partnership logo, mention of accreditation, use in corporate materials. • Restrictions: Cannot issue PTVS seals directly, cannot use the seal on assets, subject to Aurema Group supervision.
TYPE C LICENSE: TOKENIZATION PLATFORM • Grant: Upon technical integration. Annual fee: €15,000 - €50,000

(based on volume). • Rights: Display PTVS seals on listings, API integration with verification portal, authorized co-branded marketing. • Restrictions: Only for effectively certified assets, subject to technical audit, compliance with quality standards.
TYPE D LICENSE: EDITORIAL OR ACADEMIC USE • Grant: Specific request, generally free for legitimate uses. • Rights: Use in academic publications, market studies, specialized media analysis. • Restrictions: Non-commercial use, mandatory attribution, no modification of the seal, respectful context.

15.10.2 Typical License Agreement Elements

1. **Object:** Grant of non-exclusive license for the "Prop Trust Verified" seal for a specific asset (AUI) and level.
2. **Scope of Use:** Permitted media, territories, and duration (coinciding with certification validity).
3. **Licensee Obligations:** Respect graphic specifications, do not modify the design, include proper attribution, remove the seal immediately upon expiration/revocation, submit to compliance audits, cooperate with the licensor.
4. **Prohibitions:** Explicit list of forbidden uses (as detailed in Section 15.4).
5. **Supervision and Audit:** Right of the licensor to conduct scheduled or surprise audits.
6. **Sanctions:** Graduated penalties for non-compliance (warning, fines, suspension, revocation).
7. **Brand Protection:** Licensee acknowledges licensor's ownership, agrees to cooperate in defense, and waives the right to challenge the mark's validity.

8. **Termination Consequences:** Immediate cessation of use, destruction/return of physical materials, written confirmation of compliance.
9. **Governing Law and Jurisdiction:** Specified jurisdiction, with prior arbitration via AAA Wilmington as established in the PTVS standard.

15.10.3 License Registry

Aurema Group maintains a public registry of active licenses, including: • License ID, Type, Licensee, Asset (AUI), Level, Issuance Date, Expiration Date, and Status (Active, Suspended, Revoked).

15.11 PRACTICAL BRAND MANAGEMENT CASES

15.11.1 Case 1: Launch of a Tokenized Asset

Scenario: A Level 3 certified megayacht is tokenized into 5,000 tokens. **Implementation:** • Pre-tokenization (Month 1): Design materials with PTVS seal, integrate seal into white paper, configure token metadata, prepare verification portal. • Launch (Month 2): Official announcement highlighting the seal, social media campaign (#PropTrustVerified), investor presentations with visible seal, marketplace integration. • Post-launch (Months 3-12): Quarterly updates with dynamic seal, holder reports, continuous marketing highlighting the certification, monitoring of secondary market usage. **Result:** 100% subscription in 6 weeks, stable market price, multiple references to the PTVS seal as a key trust factor, zero incidents of misuse.

15.11.2 Case 2: Detection of Fraudulent Use

Scenario: Monitoring system detects a website using the PTVS Level 4 seal without any certification. **Implementation:** • Detection (Day 1): Automated alert, evidence capture (screenshots, WHOIS), preliminary analysis. • Investigation (Days 2-5): Verify no certification exists, identify responsible party, assess reputational damage, determine applicable jurisdiction. • Action (Days 6-10): Send cease & desist, notify hosting provider, report to search engines (if applicable), prepare legal action if no response. • Resolution (Days 11-30): Website removes fraudulent content, confirmation of cessation, case archived with lessons learned, detection systems updated. **Result:** Fraudulent use eliminated in 15 days, no significant reputational damage, precedent set as a deterrent.

15.11.3 Case 3: International Expansion

Scenario: Aurema Group expands operations to the UAE (ADGM). Needs to protect the brand in the new jurisdiction. **Implementation:** • Analysis (Month 1): Trademark search in UAE, verification of prior registrations, analysis of local uses, identification of risks. • Registration (Months 2-6): Trademark application in UAE, appointment of local agent, process tracking, response to any oppositions. • Implementation (Months 7-12): Adaptation of materials to local market, training of local partners, establishment of monitoring systems, first certifications in the jurisdiction. • Defense (Ongoing): Active market monitoring, opposition to conflicting registrations, market education, reputation building. **Result:** Trademark registered in UAE in 8 months, zero conflicts in the first year, 5 certifications issued, institutional recognition established.

15.12 REPORTING AND CERTIFICATION

15.12.1 Annual Brand Report

Aurema Group publishes an annual report on the state of the brand, including:

1. **Executive Summary:** Key achievements, KPIs, innovation highlights, future perspectives.
2. **State of the PTVS Standard:** Updates made, new protocols implemented, number of accredited bodies, certifications issued.
3. **Compliance Audits:** Audits performed, non-compliance detected, typology of non-compliance, sanctions imposed.
4. **Legal Protection:** Active trademark registrations, oppositions filed, legal actions, precedents established.
5. **Perspectives for the Following Year:** Expansion to new jurisdictions, new authorized uses, improvements in monitoring systems, quantitative goals.

15.12.2 Brand Compliance Certification

For critical institutional uses, Aurema Group can issue a specific **Brand Compliance Certificate**: • Certifies that a specific entity has satisfactorily complied with all requirements for using the "Prop Trust Verified" seal during a specified period. • Includes: Number of audits performed, non-compliance detected (ideally 0), and a qualitative rating (e.g., "EXCELLENT"). • Benefits: Public recognition, differentiated marketing, access to the excellence network, priority in new developments.

CHAPTER 16: REQUIREMENTS FOR INSPECTION BODIES

16.1 PURPOSE AND SCOPE

16.1.1 The Critical Importance of Inspection Bodies

The **Prop Trust Verified Standard (PTVS)** would be merely a theoretical document without the **inspection bodies** that execute it. These bodies are the **operational arm** of the standard: the entities responsible for transforming theoretical protocols into real-world verifications, the guardians of the PTVS seal's credibility, and the first link in the chain of trust that reaches institutional investors and token holders.
The final quality of a PTVS certification depends, ultimately, on the **competence, integrity, and rigor** of the issuing inspection body. A failure in an inspection body not only compromises an individual certification: it contaminates the entire PTVS ecosystem, erodes Family Office confidence, and can trigger the automatic deactivation of tokens via the SCCIP protocol.
Therefore, this chapter establishes the **mandatory requirements** that every inspection body must meet to be accredited by Aurema Group and authorized to issue PTVS certifications.

16.1.2 Objectives of the Chapter

OBJECTIVE 1: Define Universal Minimum Requirements
Establish the baseline requirements that every inspection body must meet, regardless of size, location, or specialization.

OBJECTIVE 2: Establish Level-Specific Requirements Differentiate requirements based on the certification level the body intends to issue (1, 2, 3, or 4).
OBJECTIVE 3: Guarantee Technical Competence Define standards for the qualification, experience, and continuous training of inspection personnel.
OBJECTIVE 4: Ensure Independence and Impartiality Establish mechanisms guaranteeing that technical decisions are not influenced by commercial interests.
OBJECTIVE 5: Integrate with Advanced Protocols Define the technological requirements for implementing CAP, CPEP, PDBP, SCCIP, and PRA.
OBJECTIVE 6: Establish a Rigorous Accreditation System Create an accreditation process that selects only the most qualified bodies.

16.1.3 Guiding Principles

PRINCIPLE 1: TECHNICAL EXCELLENCE Accredited bodies MUST demonstrate a level of technical competence superior to the market standard, validated through independent audits.
PRINCIPLE 2: ABSOLUTE INTEGRITY Bodies MUST operate with total independence, resisting any commercial, political, or other pressure that could compromise the objectivity of their verifications.
PRINCIPLE 3: CONTINUOUS IMPROVEMENT Bodies MUST maintain quality management systems that guarantee the continuous improvement of their processes and results.
PRINCIPLE 4: TRANSPARENCY Bodies MUST operate transparently regarding their methodologies, pricing, and conflicts of interest, facilitating auditing by Aurema Group and third parties.
PRINCIPLE 5: PROPORTIONALITY Requirements MUST be proportional to the certification level issued, becoming more demanding for higher levels.

PRINCIPLE 6: ACCOUNTABILITY Bodies MUST assume full responsibility for the certifications issued, including adequate insurance coverage and claim mechanisms.

16.1.4 Typology of Recognized Bodies

The PTVS recognizes four typologies of inspection bodies: TYPE A: INTEGRATED BODIES • Offer all verification phases (1-4) internally. • Have in-house personnel for all specializations. • Issue complete PTVS certifications. • Require the highest level of accreditation. TYPE B: SPECIALIZED BODIES • Specialize in one or two asset typologies (e.g., naval, real estate, industrial). • Complement with subcontracting for other areas. • Issue PTVS certifications within their scope of specialization. • Require specific sectoral accreditation. TYPE C: SUPPORT BODIES • Provide specific services (laboratory, drones, LiDAR, IoT). • Do not issue PTVS certifications directly. • Work as subcontractors for Type A or B bodies. • Require technical homologation. TYPE D: INTERNATIONAL BODIES • Operate in multiple jurisdictions. • Have offices in several countries. • Issue global PTVS certifications. • Require multi-jurisdictional accreditation.

16.2 ORGANIZATIONAL AND LEGAL REQUIREMENTS

16.2.1 Legal Personality and Structure

REQUIREMENT 16.2.1.1: CONSTITUTED LEGAL ENTITY The body MUST be a validly constituted legal entity with: • Independent legal personality, differentiated from its owners. • Corporate purpose explicitly including inspection, verification, or expert assessment activities. •

Identified and verifiable registered office. • Indefinite duration (not companies with an imminent dissolution date). • Minimum share capital according to jurisdiction and accreditation level.
MINIMUM SHARE CAPITAL BY LEVEL: • Type C (Support): €30,000 (Limited risk) • Type B (Specialized) Level 1-2: €100,000 (Moderate liability) • Type B (Specialized) Level 3: €250,000 (High liability) • Type B (Specialized) Level 4: €500,000 (Very high liability) • Type A (Integrated) Level 1-2: €250,000 (Broad coverage) • Type A (Integrated) Level 3: €750,000 (Institutional coverage) • Type A (Integrated) Level 4: €2,000,000 (Ultra-high coverage) • Type D (International): €5,000,000 (Multi-jurisdiction operation)
REQUIREMENT 16.2.1.2: CORPORATE GOVERNANCE The body MUST have a governance structure including: • Administrative body (Board of Directors, Sole Administrator, etc.). • Technical Director with operational independence. • Quality Manager independent of commercial management. • Ethics Committee (mandatory for Levels 3 and 4). • Impartiality Committee (mandatory for Levels 3 and 4).
REQUIREMENT 16.2.1.3: PERMANENCE AND CONTINUITY The body MUST demonstrate: • Minimum 3 years of antiquity in inspection activities. • Verifiable track record of at least 50 issued certifications (or equivalent). • Documented business continuity plan. • Succession mechanisms for critical positions. • Active civil liability insurance.

16.2.2 Regulatory Compliance

REQUIREMENT 16.2.2.1: MANDATORY CERTIFICATIONS Every accredited body MUST maintain: BASE CERTIFICATION (All levels): • ISO 9001: Quality management systems. • ISO/IEC 17020: Criteria for the

operation of inspection bodies (recommended for Level 1-2, mandatory for Level 3-4).
ADDITIONAL CERTIFICATIONS BY LEVEL: • Level 1-2: ISO 9001 (Mandatory), ISO 17020 (Recommended). • Level 3: ISO 9001, ISO 17020, ISO 14001 (Mandatory), ISO 27001 (Mandatory). • Level 4: ISO 9001, ISO 17020, ISO 14001, ISO 45001, ISO 27001, ISO 31000 (All Mandatory).
REQUIREMENT 16.2.2.2: JURISDICTIONAL COMPLIANCE The body MUST comply with all applicable regulations in the jurisdictions where it operates: • European Union: MiCA Regulation, GDPR, Services Directive, national inspection regulations. • Spain (example): LOE (Building Planning Act), CTE (Technical Building Code), Maritime Navigation Act, Judicial Expert regulations, AML regulations. • United States: USPAP, SEC regulations (for tokenized assets), State-specific licensing. • Middle East (ADGM/DIFC): FSRA regulations, data protection, AML/CFT compliance.
REQUIREMENT 16.2.2.3: PROFESSIONAL LICENSES AND REGISTRIES The body and its personnel MUST hold: • Professional association membership (where mandatory in the jurisdiction). • Current professional authorizations. • Necessary administrative registries. • Specific sectoral accreditations (RICS, IIMS, etc.).

16.2.3 Operational Structure

REQUIREMENT 16.2.3.1: SEPARATION OF FUNCTIONS The body MUST maintain a clear separation between: • Commercial management (separated from technical decisions). • Inspection (independent of commercial). • Technical review (independent of inspection). • Internal accreditation (independent of all the above). • Complaint handling (independent of the team that generated the complaint).

REQUIREMENT 16.2.3.2: MINIMUM ORGANIGRAM For Type A (Integrated) bodies, the structure must include: • Board of Directors. • Commercial Management, Technical Management, Ethics Committee. • Inspection Department, Quality Department, R&D&I Department. • Specialized inspection teams (Real Estate, Naval, Industrial, Patrimonial).
REQUIREMENT 16.2.3.3: MINIMUM FACILITIES • Type C: Equipment storage (optional physical office). • Type B: Physical office, equipment storage, meeting room. • Type A: Physical office, own laboratory (if applicable), equipment storage, meeting room, data center. • Type D: All Type A requirements, plus regional offices and a crisis room.

16.3 PERSONNEL COMPETENCE

16.3.1 Role Structure

The PTVS defines the following roles within an inspection body:
ROLE 1: TECHNICAL DIRECTOR (TD) • Responsibilities: Technical supervision of all verifications, final approval of reports and certificates, technical representation before Aurema Group, management of the inspection team, validation of methodologies. • Requirements (Level 3): Senior engineering/architecture degree, 15+ years total experience, 10+ years in inspection, 2+ relevant professional certifications (e.g., FRICS, IIMS), PTVS Level 3 accreditation, 2 languages (native + English).
ROLE 2: ACCREDITED INSPECTOR (AI) • Responsibilities: Execution of physical inspections, generation of evidence (photographic, technical), cryptographic edge signing of data (CPEP), drafting of technical reports, IoT sensor installation (Levels 3-4). • Requirements: Relevant degree, minimum 3-7 years

experience (depending on specialization), relevant professional certifications, specific PTVS inspector accreditation level.
ROLE 3: TECHNICAL REVIEWER (TR) • Responsibilities: Independent review of reports, validation of technical conclusions, verification of protocol compliance, resolution of technical doubts, joint signing of certificates. • Requirements: Senior degree in the specialty, minimum 10 years experience, PTVS accreditation at least 1 level higher than the inspector being reviewed, independence from the inspection team.
ROLE 4: SPECIALIZED TECHNICIAN (ST) • Responsibilities: Operation of specialized equipment (drones, LiDAR, thermography), laboratory analysis, IoT sensor installation and configuration, advanced data processing. • Requirements: Specific certifications (e.g., EASA/FAA drone pilot, ISO 18436 Level 2 thermographer, ISO 17025 lab analyst), 2-3 years experience.
ROLE 5: FINANCIAL ANALYST (FA) • Responsibilities: Asset valuation, profitability analysis, stress testing, financial modeling, integration with tokenization. • Requirements: Degree in Economics, Finance, or MBA; CFA, RICS Valuation, or equivalent certification; 5+ years valuation experience; knowledge of tokenization and RWA (Levels 3-4).
ROLE 6: QUALITY MANAGER (QM) • Responsibilities: Management of the quality system, internal audits, non-conformity management, personnel training, relationship with accreditation bodies. • Requirements: Senior degree, Lead Auditor certification (ISO 17020/9001), 5+ years quality management experience, independence from commercial management.

16.3.2 Competence Matrix by Asset Typology

Competence	Real Estate Res.	Real Estate Com.	Industrial	Naval Yachts	Naval Com.	Machinery
Construction Pathologies	✓✓✓	✓✓✓	✓✓	✓	✓	✓
Structural Analysis	✓✓	✓✓	✓✓✓	✓✓	✓✓	✓✓
Electrical Installations	✓✓	✓✓✓	✓✓✓	✓✓	✓✓	✓✓✓
HVAC Systems	✓	✓✓✓	✓✓✓	✓✓	✓✓	✓✓
Naval Propulsion	✗	✗	✗	✓✓✓	✓✓✓	✗
Naval Systems	✗	✗	✗	✓✓✓	✓✓✓	✗
Naval Stability	✗	✗	✗	✓✓	✓✓✓	✗
IMO Conventions	✗	✗	✗	✓	✓✓✓	✗
FEM Structural Analysis	✗	✗	✓✓	✓	✓	✓✓
Thermography	✓✓	✓✓	✓✓✓	✓✓	✓✓	✓✓✓
Lab Analysis	✗	✗	✓✓	✓	✓	✓✓
Drones	✓	✓✓	✓✓	✓✓	✓✓	✓
LiDAR	✓	✓✓	✓✓	✓✓	✓	✓
Environme	✓	✓✓	✓✓✓	✓	✓✓	✓✓

Competence	Real Estate Res.	Real Estate Com.	Industrial	Naval Yachts	Naval Com.	Machinery
ntal Analysis						
Real Estate Valuation	✓✓	✓✓✓	✓✓	✗	✗	✗
Naval Valuation	✗	✗	✗	✓✓✓	✓✓✓	✗
Tokenization / RWA	✓	✓	✓	✓	✓	✓
Blockchain / Smart Contracts	✓	✓	✓	✓	✓	✓
Financial Analysis	✓	✓✓	✓✓	✓✓	✓✓	✓✓

(Legend: ✓ = Basic knowledge, ✓✓ = Operational competence, ✓✓✓ = Expert competence)

16.3.3 Continuous Training

ANNUAL TRAINING REQUIREMENTS: • Technical Director: 60 hours minimum (Management, regulations, technology). • Accredited Inspector: 40 hours minimum (Technical, PTVS protocols, technology). • Technical Reviewer: 40 hours minimum (Technical, regulations, auditing). • Specialized Technician: 30 hours minimum (Specific technology). • Financial Analyst: 30 hours minimum (Finance, tokenization, regulations). • Quality Manager: 40 hours minimum (Quality, auditing, regulations).
MANDATORY TRAINING AREAS: • Technical Update (40%): New inspection technologies, regulatory evolution, case studies. • PTVS Protocols (30%): Standard updates, advanced protocols (CAP, CPEP, SCCIP, PRA), PTVS case studies. • Advanced Technology (20%): Blockchain,

IoT, AI/ML, digital twins. • Ethics and Regulation (10%): Professional ethics, AML, GDPR, conflicts of interest. RECOGNIZED TRAINING PROVIDERS: • Aurema Group Academy (PTVS protocols). • RICS / IIMS (Sectoral standards). • Technical Universities (Specializations). • ISO Training Bodies (Management systems). • Blockchain Councils (Blockchain, tokenization).

16.3.4 Performance Evaluation

ANNUAL PERSONNEL EVALUATION: • Technical Competence (30%): Report quality, precision, depth. • Protocol Compliance (25%): Adherence to PTVS, use of technologies. • Client Satisfaction (15%): Feedback, complaints, repeat business. • Professional Development (15%): Training, new certifications. • Ethics and Integrity (15%): Conflicts, confidentiality, impartiality. CONSEQUENCES OF POOR PERFORMANCE: • First failure (<50%): 90-day improvement plan. • Second consecutive failure: 6-month suspension of accreditation. • Third failure or serious breach: Revocation of PTVS accreditation. • Ethical breach: Immediate revocation and communication to Aurema Group.

16.4 REQUIRED TECHNICAL EQUIPMENT

16.4.1 Minimum Inventory by Level

BASIC EQUIPMENT (All bodies): • Professional camera (24MP+, RAW, GPS), laser distance meter (±1.5mm), digital hygrometer/thermometer, moisture detector, complete PPE, tablet with PTVS software, edge cryptographic signing system.

ADVANCED EQUIPMENT (Level 2+): • Professional drone (4K, GPS, 30min flight), professional thermal camera (≥320×240, NETD <50mK), industrial endoscope (Ø≤6mm), Class A electrical network analyzer, anemometer, CO2/VOC meter, photogrammetry and thermography software licenses.
PREMIUM EQUIPMENT (Level 3+): • Terrestrial LiDAR scanner (±2mm, 500k pts/s), LiDAR drone, Ground Penetrating Radar (GPR), phased array ultrasonic tester, 50-200 IoT sensors (stock), IoT gateways, portable weather station, LiDAR processing software, digital twin software license, CAP platform license.
INSTITUTIONAL EQUIPMENT (Level 4): • All previous equipment with redundancy, underwater ROV (HD camera, 100m depth), material analysis system (spectrometer, chromatograph), dedicated FEM simulation workstation, multi-physics software license (COMSOL/ANSYS), 200-500+ advanced redundant IoT sensors, own blockchain node (redundant), dedicated data center, crisis room with redundant connectivity.

16.4.2 Equipment Management

REQUIREMENT 16.4.2.1: EQUIPMENT REGISTRY The body MUST maintain an updated registry including: unique equipment ID, manufacturer, model, serial number, acquisition date, current location, status (operational, calibration, repair, decommissioned), calibration history, maintenance history, and next scheduled calibration.
REQUIREMENT 16.4.2.2: CALIBRATION AND TRACEABILITY • All measuring equipment MUST be calibrated. • Calibration by an ISO 17025 accredited laboratory. • Traceability to national/international standards. • Calibration certificates archived. • Frequencies: Critical equipment (Level 4) semi-annually; standard measuring equipment annually; drones monthly (check) / annually (calibration).

REQUIREMENT 16.4.2.3: PREVENTIVE MAINTENANCE The body MUST implement a preventive maintenance plan according to manufacturer recommendations, including firmware/software updates, cleaning, functional verification, and intervention logging.
REQUIREMENT 16.4.2.4: OBSOLESCENCE MANAGEMENT The body MUST monitor equipment life cycles, plan renewals in advance, maintain compatibility with PTVS standards, and manage the transition to new technologies.

16.5 QUALITY MANAGEMENT SYSTEMS

16.5.1 Normative Framework

The PTVS adopts **ISO/IEC 17020:2012** (Criteria for the operation of various types of bodies performing inspection) as its foundation, adding specific requirements for advanced protocols.
Structure of ISO 17020:

1. General requirements (impartiality, independence).
2. Structural requirements (organization, management).
3. Resource requirements (personnel, facilities, equipment).
4. Process requirements (methods, procedures, reports).
5. Management system requirements (documentation, audits).

16.5.2 Management System Documentation

DOCUMENTATION PYRAMID: • Level 1: Quality Manual (Policy, objectives, structure). • Level 2: Procedures (Cross-cutting processes). • Level 3: Technical Instructions (Specific PTVS protocols). • Level 4: Records and Forms (Evidence of compliance).
MANDATORY DOCUMENTATION (Level 2 Procedures): • PR-001: Document control. • PR-002: Record control. • PR-003: Internal audits. • PR-004: Non-conformity control. • PR-005: Corrective and preventive actions. • PR-006: Complaints and appeals management. • PR-007: Measuring equipment control. • PR-008: Personnel qualification. • PR-009: Subcontracting. • PR-010: Management review. • PR-011: Risk management. • PR-012: Confidentiality and data protection. • PR-013: Conflict of interest management. • PR-014: Business continuity. • PR-015 to PR-024: Specific PTVS protocols (Phases 1-4, CAP, CPEP, SCCIP, PRA, PDBP).

16.5.3 Document Control

REQUIREMENTS: • Approval: All documents must be approved by authorized personnel before issuance. • Review: Periodic review (minimum annually) and update when necessary. • Identification: Unique code, date, version, pagination. • Distribution: Control of copies, withdrawal of obsolete versions. • Legibility: Clear, legible, identified. • External Documents: Identification and control of external documents. • Obsolete Documents: Identification and withdrawal to prevent unintended use.
PTVS VERSION CONTROL: When Aurema Group updates the PTVS standard, bodies MUST:

1. Receive formal notification.
2. Have a transition period (3-6 months).
3. Update internal procedures.
4. Train personnel on changes.

5.Verify implementation via audit.
6.Certify compliance with the new version.

16.5.4 Record Control

REQUIREMENTS: • Identification: Record type, date, responsible person. • Storage: Accessible, protected, organized. • Retention Periods:

•Inspection reports: 10 years.
•Issued certificates: 15 years.
•Calibration records: Equipment life + 5 years.
•Training records: Working life + 5 years.
•Audit records: 5 years.
•CPEP data (blockchain): Permanent.
•CAP data: According to level. • Disposition: Secure destruction at the end of the period. • Protection: Against damage, deterioration, loss.

16.5.5 Internal Audits

OBJECTIVE: Verify compliance with the management system and PTVS protocols.
FREQUENCY: • Level 1-2 bodies: Annual. • Level 3 bodies: Semi-annual. • Level 4 bodies: Quarterly.
SCOPE: All areas and processes, all asset typologies, all certification levels issued, compliance with PTVS protocols.
AUDIT TEAM: Internal qualified auditors (ISO 19011), independent of the audited area, specific knowledge of PTVS.

16.5.6 Management Review

FREQUENCY: Annual (minimum).
INPUTS: Audit results, client feedback, quality objective compliance, status of corrective actions, changes in regulations and PTVS, improvement recommendations.

OUTPUTS: Improvement of the management system, necessary resources, updated quality objectives, strategic decisions.

16.6 CONTINUOUS TRAINING

(Note: Detailed in Section 16.3.3, expanded here with the Aurema Group Academy framework)
The body MUST:

1. Identify training needs for all personnel.
2. Provide training appropriate to responsibilities.
3. Evaluate training effectiveness.
4. Maintain records of all training.
5. Foster continuous professional development.

AUREMA GROUP ACADEMY: Offers official PTVS training, including certified courses for Inspectors (Levels 1-4), Technical Reviewers, Technical Directors, Quality Managers, and Specialists (CAP, CPEP, SCCIP, PRA). Includes theoretical exams, practical cases, and continuous evaluation. Certificates are valid for 2 years.

16.7 INSURANCE AND CIVIL

LIABILITY

16.7.1 Mandatory Coverages

PROFESSIONAL INDEMNITY INSURANCE: Minimum Coverage by Accreditation Level: • Type C: €500,000 (Max deductible: €5,000) • Type B Level 1-2: €2,000,000 (Max deductible: €10,000) • Type B Level 3: €5,000,000 (Max deductible: €25,000) • Type B Level 4: €10,000,000 (Max deductible: €50,000) • Type A Level 1-2: €5,000,000 (Max deductible: €25,000) • Type A Level 3: €15,000,000 (Max deductible: €50,000) • Type A Level 4: €30,000,000 (Max

deductible: €100,000) • Type D: €50,000,000 (Max deductible: €200,000)
PTVS-SPECIFIC COVERAGES MUST EXPLICITLY INCLUDE: • Professional civil liability for errors in verifications. • Liability for issuance of incorrect PTVS certificates. • Liability for erroneous SCCIP activation (e.g., unwarranted token pauses). • Liability for CPEP failures (cryptographic evidence). • Liability for CAP failures (continuous monitoring). • Liability for PRA failures (incorrect predictions). • Liability for breach of confidentiality. • Liability for data protection violations (GDPR). • Legal defense and bail bonds. • Claims in multiple jurisdictions (for Type D).
GENERAL CIVIL LIABILITY INSURANCE: • Minimum coverage: €1,000,000 for all levels.
PERSONAL ACCIDENT INSURANCE: • Field inspectors: €300,000 minimum. • Technical personnel: €150,000 minimum. • Office personnel: €60,000 minimum.

16.7.2 Risk Management

The body MUST maintain a risk matrix identifying, assessing, and mitigating risks such as: inspection errors, technological failures, data breaches, client claims, regulatory sanctions, and reputational damage. A contingency plan with designated response teams, crisis communication protocols, and coordination with Aurema Group is mandatory.

16.7.3 Claim Management

PROCEDURE:

1. Reception: Dedicated channel, immediate logging.
2. Analysis: Initial assessment within 5 days.
3. Investigation: Evidence gathering.

4.Response: Communication to claimant within 30 days.
5.Resolution: Agreement or motivated decision.
6.Follow-up: Verification of satisfaction.
7.Improvement: Lessons learned incorporated.

COOPERATION WITH AUREMA GROUP: For claims affecting the PTVS standard, the body must immediately notify Aurema Group, cooperate in the investigation, accept Aurema Group's decisions, and implement corrective measures.

16.8 STANDARD OPERATING PROCEDURES (SOPs)

16.8.1 SOP Structure

Every procedure MUST include:

1.Object and scope.
2.Normative references.
3.Definitions and abbreviations.
4.Responsibilities.
5.Detailed process description.
6.Flowchart.
7.Generated records.
8.Annexes (forms, checklists).

16.8.2 Specific PTVS SOPs

• SOP-PTVS-001: Request Reception (From request to team assignment). • SOP-PTVS-002: Phase 1 Execution (Documentary due diligence to SRD calculation). • SOP-PTVS-003: Phase 2 Execution (Physical inspection, CPEP edge signing, SRF calculation). • SOP-PTVS-004: Phase 3 Execution (Technical assessment, PRA models, SRT calculation). • SOP-PTVS-005: Phase 4 Execution

(Financial verification, valuation, SRE calculation). • SOP-PTVS-006: Consolidation and Certificate Issuance (SRI calculation, Go/Conditional/No-Go decision, blockchain registration). • SOP-PTVS-007: Advanced Protocols Management (Level 3-4: CAP, CPEP, SCCIP, PRA, PDBP setup and maintenance). • SOP-PTVS-008: Follow-up Audits (Maintenance audits according to level).

16.8.3 Operational Control

KEY PROCESS INDICATORS (KPIs): • Reception response time: < 48 hours. • Phase execution time: According to level standards. • First-time report approval rate: > 95%. • Client satisfaction (NPS): > 8. • Complaint rate: < 2%.

16.9 DOCUMENT AND RECORD CONTROL

(Expanded from 16.5.3 and 16.5.4)
ELECTRONIC MANAGEMENT REQUIREMENTS: • Secure access with authentication. • Automatic version control. • Advanced search by metadata. • Automatic and redundant backups. • Access and modification traceability. • Legally valid electronic signature. • Integration with PTVS systems. • GDPR and applicable regulation compliance.
INFORMATION SECURITY: • Classification: Public, Internal, Confidential, Restricted. • Physical measures: Access control, locked cabinets, secure server rooms. • Logical measures: Firewalls, intrusion detection, data encryption, multi-factor authentication, encrypted backups, access monitoring. • Organizational measures: Documented security policies, personnel training, confidentiality agreements, incident response plans.

16.10 SUBCONTRACTING AND EXPERT NETWORK

16.10.1 Subcontracting Policy

PRINCIPLES:

1. Ultimate responsibility remains with the accredited body.
2. Verified competence of the subcontractor.
3. PTVS compliance by the subcontractor.
4. Guaranteed confidentiality.
5. Transparency with the client.

ALLOWED SUBCONTRACTING: • Laboratory analysis (ISO 17025 accredited). • Drone piloting (EASA/FAA certified). • LiDAR operation (manufacturer certified). • IoT installation (qualified technician). • FEM structural analysis (specialized engineer). • Specialized valuation (recognized expert). • Sworn translation. • Non-destructive testing (certified personnel).
PROHIBITED SUBCONTRACTING: • Technical direction of the verification. • Technical review of reports. • Signing of PTVS certificates. • Accreditation decisions. • Client relationship management.

16.10.2 Aurema Group Expert Network

Aurema Group maintains a global network of independent experts that accredited bodies can mobilize: • Senior Fellows: International recognition (complex cases, arbitration). • Sectoral Experts: Deep specialization (specific inspections). • Technological Experts: Blockchain, IoT, AI (advanced protocols). • Regional Experts: Local knowledge (specific jurisdictions). • Forensic Experts: Specialized investigation (controversial cases).

Requirements for Network Experts: Personal PTVS accreditation, minimum 10 years verified experience, professional recognition, availability for mobilization, demonstrated independence, civil liability insurance, continuous PTVS training.

16.10.3 Subcontractor Control

INITIAL EVALUATION: Technical competence (30%), Experience (25%), Quality systems (20%), Financial capacity (10%), Reputation (10%), Price (5%).
CONTINUOUS MONITORING: Evaluation after each job, periodic (annual) audits, updating of accreditations, verification of insurance, client feedback.
DISMISSAL CRITERIA: Repeated deadline non-compliance, serious technical deficiencies, confidentiality breaches, failure to renew accreditations, unresolved client complaints.

16.11 INDEPENDENCE AND IMPARTIALITY

16.11.1 Impartiality Policy

The body MUST guarantee that its inspection activities are conducted impartially, free from any commercial, financial, or other pressure that could compromise the objectivity of the results.

16.11.2 Impartiality Governance

IMPARTIALITY COMMITTEE (Mandatory for Levels 3-4): • Composition: 3-5 independent members, at least 1 external to the body, stakeholder representation, independent chair. • Functions: Supervise body

impartiality, evaluate impartiality risks, resolve conflicts of interest, approve impartiality policies, review impartiality complaints. • Meetings: Quarterly (minimum), documented minutes, motivated decisions.

16.11.3 Conflict of Interest Management

IDENTIFICATION: The body MUST identify all potential risks to impartiality, including: • Personal conflicts: Family relationships, financial interests, prior employment, significant friendships. • Financial conflicts: Client dependence (>30% revenue), interests in suppliers, referral commissions, result-linked bonuses. • Structural conflicts: Cross-ownership with clients, shared management, incompatible commercial activities, parent/subsidiary pressure. • Activity conflicts: Inspecting and advising the same client, designing and verifying the same asset, competition with clients.
EVALUATION AND MITIGATION: • Client dependence: Diversification. • Personal relationships: Declaration, reassignment. • Commercial pressure: Separation of functions. • Incompatible activities: Prohibition.
MANDATORY DECLARATION: All personnel MUST declare annually: relevant financial interests, client relationships, external activities, gifts and hospitality received.
CONFLICT REGISTRY: The body MUST maintain an updated registry of identified conflicts, evaluations performed, mitigation measures applied, and conflict resolutions.

16.11.4 Pressures and Attempts at Influence

ACTION PROTOCOL: Upon any attempt at undue influence:

1. Immediate rejection of the proposal.
2. Documentation of the attempt.

3.Notification to the Quality Manager.
4.Communication to the Impartiality Committee.
5.Evaluation of the client relationship.
6.Decision on relationship continuity.
7.Logging in the incident history.

PERSONNEL PROTECTION: The body MUST contractually protect personnel, not allow retaliation for maintaining objectivity, provide training on resistance to pressure, and establish confidential reporting channels.

16.12 COMPLAINTS AND APPEALS MANAGEMENT

16.12.1 Complaints Policy

The body MUST have a formal process to receive, evaluate, and make decisions on complaints and appeals. PRINCIPLES: Accessibility, Transparency, Impartiality, Timeliness, Effectiveness.

16.12.2 Complaints Procedure

PHASE 1: RECEPTION (Days 1-3) Reception via dedicated channel, logging in complaint system, assignment of unique code, acknowledgment of receipt to claimant, initial admissibility analysis.
PHASE 2: ANALYSIS (Days 4-15) Information gathering, identification of responsible parties, severity evaluation, assignment of investigator, investigation plan.
PHASE 3: INVESTIGATION (Days 16-45) Documentary review, interviews with involved parties, technical analysis (if applicable), identification of root causes, resolution proposal.

PHASE 4: RESOLUTION (Days 46-60) Motivated decision, communication to claimant, implementation of measures, satisfaction follow-up.
PHASE 5: CLOSURE (Days 61-75) Verification of effectiveness, lessons learned, procedure updates, file archiving.

16.12.3 Appeals to Aurema Group

CASES THAT CAN BE APPEALED: • Decisions on accreditation. • Imposed sanctions. • Internally unresolved conflicts. • Interpretations of the PTVS standard. • Decisions of the Impartiality Committee.
APPEAL PROCEDURE:

1. Formal submission to Aurema Group.
2. Deadline: 30 days from the decision.
3. Supporting documentation.
4. Analysis by the Appeals Committee.
5. Resolution within 60 days.
6. Final decision.

16.13 INTERNAL AND EXTERNAL AUDITS

16.13.1 Internal Audits

OBJECTIVE: Verify compliance with the management system and PTVS protocols.
PROGRAM: • Type C: Annual (1 auditor). • Type B Level 1-2: Annual (2 auditors). • Type B Level 3: Semi-annual (2 auditors). • Type B Level 4: Quarterly (3 auditors). • Type A Level 1-2: Semi-annual (2 auditors). • Type A Level 3: Quarterly (3 auditors). • Type A Level 4: Quarterly + surprise (3 auditors). • Type D: Quarterly, multi-site (4 auditors).

AUDIT TEAM: Internal qualified auditors (ISO 19011), independent of the audited area, specific knowledge of PTVS and advanced protocols.

16.13.2 Aurema Group Audits

ACCREDITATION AUDITS: • Initial: Before accreditation. • Surveillance: Annual (Level 1-2), Semi-annual (Level 3-4). • Renewal: Every 3 years.
SCOPE: All areas of the body, all levels of certification issued, all PTVS protocols, legal and regulatory compliance.
PROCESS: Planning and notification → Opening meeting → Document review → Field audit → Practical case review → Advanced protocol evaluation → Result analysis → Closing meeting → Provisional report → Body's response → Final report → Accreditation decisions → Non-conformity follow-up.

16.13.3 Third-Party Audits

OFFICIAL ACCREDITATION AUDITS: For bodies seeking ISO 17020 accreditation: ENAC (Spain), UKAS (UK), COFRAC (France), DAkkS (Germany), or equivalent in each jurisdiction.
CLIENT AUDITS: Some clients (Family Offices, funds) may require specific audits, compliance verification, capability evaluation, or review of specific cases.

16.14 PTVS ACCREDITATION FOR BODIES

16.14.1 Accreditation Process

PHASE 1: APPLICATION (1-2 months) Required documentation: Application form, statutes and deed, organigram and job descriptions, CVs of key personnel, certifications and accreditations, equipment list, management system procedures, insurance policies, client references, impartiality declaration. Initial analysis by Aurema Group: Verification of minimum requirements, technical solvency evaluation, potential conflict analysis, recommended accreditation level.
PHASE 2: DOCUMENTARY EVALUATION (1-2 months) Review of: Quality management system, PTVS procedures, personnel competence, technical equipment, insurance and guarantees, impartiality policies. Result: Acceptance for field audit, non-conformities to resolve, or rejection.
PHASE 3: FIELD AUDIT (1-2 months) On-site audit: Verification of implementation, competence evaluation, practical case review, advanced protocol testing, facility and equipment evaluation. Result: Accreditation granted, non-conformities to resolve, conditional accreditation, or rejection.
PHASE 4: DECISION AND ISSUANCE (1 month) Accreditation Committee: Review of audit report, analysis of non-conformities, motivated decision, certificate issuance. PTVS Accreditation Certificate includes: Body identification, Type (A, B, C, D), Authorized levels (1, 2, 3, 4), Asset typologies, Jurisdictions, Issue date and validity, Specific conditions.

PHASE 5: SURVEILLANCE (Continuous) Surveillance audits: Annual (Level 1-2), Semi-annual (Level 3-4), Surprise (Level 4). Re-evaluation: Every 3 years, process similar to initial accreditation, certificate renewal.

16.14.2 Accreditation Levels

PTVS Level	Additional Requirements	Accreditation Fee	Annual Fee
Level 1	Basic	€5,000	€2,000
Level 2	Advanced equipment	€10,000	€4,000
Level 3	Advanced protocols	€25,000	€10,000
Level 4	Institutional + ZKP	€50,000	€25,000

Accreditation by typology (Real Estate, Naval, Industrial, etc.) and jurisdiction is also required and evaluated separately.

16.14.3 Maintenance of Accreditation

OBLIGATIONS OF THE ACCREDITED BODY:

1. Maintain accreditation requirements.
2. Communicate relevant changes.
3. Allow surveillance audits.
4. Use the PTVS seal correctly.
5. Cooperate with Aurema Group.
6. Pay maintenance fees.
7. Continuously train personnel.
8. Continuously improve the system.

SANCTIONS FOR NON-COMPLIANCE: • Minor: Warning (30 days to correct). • Moderate: Fine €5,000-€25,000 (15 days to correct). • Serious: 6-month suspension (Immediate). • Critical: Revocation (Immediate).

REVOCATION CAUSES: Fraud or forgery, repeated serious non-compliance, loss of basic requirements, damage to PTVS reputation, voluntary decision. CONSEQUENCES: Certificate withdrawal, seal usage prohibition, client communication, invalidation of issued certificates (case-by-case evaluation).

16.15 PRACTICAL CASES

16.15.1 Case 1: Successful Accreditation of a Naval Specialized Body

Scenario: Spanish company with 15 years of naval inspection experience requests Type B PTVS accreditation for yachts and megayachts, Levels 1-3. Process: • Phase 1: Complete documentation, 8 inspectors with naval experience, advanced equipment (UT, vibrations, ROV), IIMS certification for several inspectors, €5M professional insurance. • Phase 2: ISO 9001 already certified, PTVS procedures adapted, personnel with PTVS Foundation training. Result: Accepted with minor observations. • Phase 3: 3-day field audit, review of 5 practical cases, equipment evaluation, CAP and CPEP protocol tests. 2 minor non-conformities. Result: Accreditation granted. • Phase 4: Certificate issued. Type B - Naval Level 1-3. Scope: Yachts and megayachts. Jurisdiction: Spain, Balearic Islands, Canary Islands. Validity: 3 years. Result: Body issues first PTVS certification at 4 months. 12 certifications issued in the first year. Revenue increased 35%. Recognition in the naval sector.

16.15.2 Case 2: Upgrade from Level 2 to Level 4

Scenario: Type A body with Level 2 accreditation wants to expand to Level 4 to capture the institutional market. Gap Analysis: Share capital (€250K → €2M), Level 4 personnel (0 → 5+), IoT equipment (50 → 500+), Blockchain (No → Own node), Digital Twin (Basic → High-fidelity), ZKP (No → Implemented), Insurance (€5M → €30M). Transformation Plan (18 months): Months 1-6 (Preparation: capital increase, senior hiring, equipment acquisition); Months 7-12 (Implementation: Level 4 training, ZKP development, digital twin implementation, protocol testing); Months 13-18 (Accreditation: complete internal audit, Aurema Group audit, non-conformity resolution, new accreditation issuance). Total Investment: ~€3M. Result: Level 4 accreditation obtained in 20 months. First institutional client at 3 months. Level 4 revenue: €2M in the first year. ROI of investment: 3 years.

16.15.3 Case 3: Sanction for Non-Compliance

Scenario: Level 3 accredited body issues a certificate with a manipulated SRI to meet the threshold. Detection: Aurema Group surveillance audit. Review of a practical case. Discrepancy between data and declared SRI. Deep investigation. Investigation: Analysis of certificates issued (last 12 months). Interviews with personnel. Identification of commercial pressures. Identification of 3 certificates with inflated SRI. Resolution: Immediate revocation of accreditation. Invalidation of affected certificates. €50,000 fine. Communication to affected clients. Notification to tokenization platforms. Activation of SCCIP for affected tokens. Consequences: Body closes in 6 months. Affected clients migrate to other bodies. 3 inspectors lose personal accreditation. Deterrent precedent for the sector. Lessons

Learned: Reinforcement of surveillance audits, implementation of automatic SRI verification, improved whistleblowing channel, reinforced ethics training.

16.16 REPORTING AND

CERTIFICATION

16.16.1 PTVS Accreditation Certificate

STRUCTURE: • Header: AUREMA GROUP - ACCREDITATION CERTIFICATE - PROP TRUST VERIFIED STANDARD. • Body: Certifies that [BODY NAME], [Tax ID], [Address], has been accredited as an Inspection Body according to PTVS. • Details: Type [A/B/C/D], Authorized Levels [1, 2, 3, 4], Asset Typologies, Jurisdictions. • Metadata: Accreditation Number (PTVS-OI-[XXX]), Issue Date, Validity Date (+3 years), Next Audit Date, Status: ACTIVE. • Footer: Signature, Director of Accreditations, Aurema Group, QR Code for verification.

16.16.2 Public Registry of Accredited Bodies

Aurema Group maintains an updated public registry of all accredited bodies. Public Information: Corporate name, Type and authorized levels, Asset typologies, Jurisdictions, Accreditation date and validity, Status (active, suspended, revoked), Contact data. Restricted Information (clients only): Audit history, non-conformities (anonymized), certification statistics, quality rating.

16.16.3 Annual Body Reporting

Each accredited body MUST submit an annual report to Aurema Group including: • Operational Data: Certifications issued by level and typology, jurisdictions of operation,

average execution times, average SRI by level. • Quality Data: Internal non-conformities, complaints received and resolved, audits performed, corrective actions. • Personnel Data: Number of accredited inspectors, training provided, performance evaluations, staff turnover. • Economic Data: Revenue by level, investment in equipment, investment in training, insurance premiums. • Technological Data: Implementation of advanced protocols, technological incidents, improvements implemented, technological roadmap.

CHAPTER 17: ACCREDITATION PROCESS

17.1 PURPOSE AND PHILOSOPHY OF THE PROCESS

17.1.1 Accreditation as a Guarantee of Quality

The **accreditation process** is the mechanism by which Aurema Group validates that an inspection body complies with all requirements of The Trust Standard (PTVS) and is fully capable of issuing PTVS certifications with the highest level of credibility.
Accreditation is not a mere administrative formality. It is a **rigorous validation process** that guarantees: • The technical competence of the inspection body. • The independence and impartiality of its decisions. • The robustness of its management systems. • The capability to implement advanced protocols (CAP, CPEP, SCCIP, PRA, PDBP). • Financial and operational solvency.

17.1.2 Principles of the Accreditation Process

PRINCIPLE 1: RIGOR The process MUST be demanding, selecting only those bodies that demonstrate technical and operational excellence.
PRINCIPLE 2: TRANSPARENCY All stages, criteria, and decisions MUST be transparent and documented.
PRINCIPLE 3: PROPORTIONALITY Requirements and the depth of evaluation MUST be proportional to the requested accreditation level.
PRINCIPLE 4: EFFICIENCY The process MUST be efficient in time and resources, avoiding unnecessary bureaucracy.
PRINCIPLE 5: CONTINUOUS IMPROVEMENT The process MUST evolve based on lessons learned and industry best practices.
PRINCIPLE 6: CONFIDENTIALITY Sensitive information from the applicant body MUST be treated with the highest level of confidentiality.

17.1.3 Overview of the Process

The complete accreditation process is structured into five phases, with an estimated total duration of 6 to 12 months:

1. **PHASE 1: PRE-ACCREDITATION (1-2 months)** • Initial information and consultation. • Body self-assessment. • Formal application.
2. **PHASE 2: DOCUMENTARY EVALUATION (2-3 months)** • Document review. • Compliance analysis. • Documentary evaluation report.
3. **PHASE 3: FIELD AUDIT (1-2 months)** • Opening meeting. • Systems audit. • Technical audit (practical cases). • Advanced protocols audit. • Closing meeting.
4. **PHASE 4: DECISION AND ISSUANCE (1 month)** • Accreditation Committee analysis. •

Motivated decision. • Resolution of non-conformities (if applicable). • Certificate issuance.

5. **PHASE 5: SURVEILLANCE AND RENEWAL (Continuous)** • Surveillance audits. • Annual reporting. • Renewal every 3 years.

17.2 PHASE 1: PRE-

ACCREDITATION

17.2.1 Initial Information and Consultation

OBJECTIVE: Provide the interested body with all necessary information to evaluate its viability and prepare an informed application.

ACTIVITIES:

1. **Initial briefing session:** Presentation of the PTVS, explanation of requirements by level, estimation of timelines and costs, Q&A.
2. **Preliminary viability assessment:** Initial gap analysis, identification of potential barriers, preparation recommendations.
3. **Assignment of an accreditation manager:** Single point of contact, guidance throughout the process, internal coordination at Aurema Group.

INITIAL DOCUMENTATION PROVIDED: • PTVS Accreditation Guide (Interactive PDF). • Self-Assessment Checklist (Excel). • Fee Schedule (PDF). • Typical Timeline (PDF). • Accreditation FAQ (PDF).

17.2.2 Body Self-Assessment

OBJECTIVE: Allow the body to evaluate its own compliance before formal application, identifying gaps and areas for improvement.

The self-assessment covers seven key areas:

1.Organizational Requirements (15 points)
2.Regulatory Compliance (10 points)
3.Personnel Competence (20 points)
4.Technical Equipment (15 points)
5.Management Systems (15 points)
6.PTVS Protocols (15 points)
7.Independence and Impartiality (10 points)

SCORING INTERPRETATION: • 90-100%: Excellent. Proceed with application. • 75-89%: Good. Proceed after correcting minor gaps. • 60-74%: Acceptable. 3-6 month improvement plan required. • 40-59%: Deficient. 6-12 month improvement plan required. • <40%: Critical. Reconsider viability.

17.2.3 Formal Application

ADMISSIBILITY REQUIREMENTS: For an application to be admitted, the body MUST:

1.Meet minimum requirements (minimum share capital, 3+ years antiquity, minimum qualified personnel, active insurance).
2.Not be subject to exclusion causes (serious sanctions in the last 5 years, fraud proceedings, previous PTVS revocations, insurmountable conflicts of interest).
3.Submit complete documentation (signed, sealed, in digital and physical format).

APPLICATION FORM ELEMENTS: • Body data (Name, Tax ID, Address, Registration, Capital, Object). • Contact person details. • Requested accreditation type (A, B, C, D) and levels (1, 2, 3, 4). • Requested asset typologies and jurisdictions. • Economic data (Revenue, employees, inspectors, certifications issued). • Existing certifications and accreditations (ISO 9001, 17020, etc.). • Mandatory declarations (truthfulness, no exclusion causes, acceptance of PTVS ethics code, authorization for

verification). • Attached documentation checklist. • Proof of application fee payment.
APPLICATION FEES (Non-refundable): • Level 1: Type A (€2,000), Type B (€1,000), Type C (€500), Type D (€5,000). • Level 2: Type A (€4,000), Type B (€2,000), Type C (€1,000), Type D (€8,000). • Level 3: Type A (€8,000), Type B (€5,000), Type D (€15,000). • Level 4: Type A (€15,000), Type B (€10,000), Type D (€25,000).

17.3 PHASE 2: DOCUMENTARY EVALUATION

17.3.1 Objective and Scope

OBJECTIVE: Verify that the applicant body meets all documentary requirements of the PTVS before proceeding to the field audit.
SCOPE: • Review of all submitted documentation. • Verification of requirement compliance. • Identification of documentary non-conformities. • Decision on the appropriateness of a field audit.
EVALUATION TEAM: • Evaluation Leader: Coordination, final decision (PTVS Lead Auditor). • Technical Evaluator: Technical review (Expert in requested typology). • Quality Evaluator: Management systems (ISO 17020 Auditor). • Protocols Evaluator: Advanced protocols (CAP/CPEP/SCCIP Expert).

17.3.2 Evaluation Process

STEP 1: RECEPTION AND REGISTRATION (Days 1-3)
Verification of document completeness, registration in the accreditation system, assignment of file code, assignment of evaluation team, communication to the applicant.

STEP 2: DOCUMENTARY ANALYSIS (Days 4-30) Review by area: • Organizational/Legal: Section 16.2 requirements (10 days). • Regulatory/Certifications: Section 16.2.2 requirements (5 days). • Personnel Competence: Section 16.3 requirements (10 days). • Equipment: Section 16.4 requirements (5 days). • Management Systems: Section 16.5 requirements (10 days). • PTVS Protocols: Section 16.8 requirements (10 days). • Impartiality: Section 16.11 requirements (5 days).
STEP 3: CONSOLIDATION (Days 31-40) Consolidation of findings, classification of non-conformities, drafting of the evaluation report, decision on proceeding.
STEP 4: COMMUNICATION (Days 41-45) Sending the report to the applicant, communication of the decision, 15-day period for allegations.

17.3.3 Evaluation Criteria

CLASSIFICATION OF FINDINGS:
CONFORMITY (C): The requirement is fully met with adequate evidence.
MAJOR NON-CONFORMITY (MaNC): Total absence or fundamental failure in the implementation of a requirement. Prevents accreditation until resolved. Examples: Lack of mandatory ISO 9001 certification, Technical Director without required qualifications, absence of a procedure for a PTVS phase, inactive professional liability insurance.
MINOR NON-CONFORMITY (MiNC): Partial non-compliance or weakness in implementation that does not compromise the system's overall capability. Examples: Outdated procedure, incomplete but existing record, missing secondary equipment, unrecorded training.
OBSERVATION (OBS): Opportunity for improvement that does not constitute non-compliance.
DECISION CRITERIA: • 0 MaNC + 0-2 MiNC: Proceed to field audit (resolve MiNC during audit). • 0 MaNC + 3-5

MiNC: Proceed conditionally (resolve MiNC before audit). • 0 MaNC + >5 MiNC: Pause (3-month improvement plan). • 1-2 MaNC: Pause (Resolve MaNC in 3 months). • >2 MaNC: Rejection (New process after 6 months).

17.3.4 Documentary Evaluation Report

The report includes:

1. Executive Summary (Result, total findings, recommendation).
2. Evaluation by Area (Conformities, MaNCs, MiNCs, OBS, Conclusion per area).
3. Detail of Findings (Description, requirement reference, evidence, impact, required action, deadline).
4. Conclusion and Recommendation (Motivated recommendation, next steps).
5. Annexes (Complete evaluation checklist, reviewed documentation, evidence of findings).

17.3.5 Applicant's Response

DEADLINE: 15 business days from receipt of the report.
CONTENT:

1. Acceptance of findings: Agreement with each finding, corrective action plan, proposed deadlines.
2. Allegations (if applicable): Motivated disagreement, additional evidence, proposal for reclassification.
3. Complementary documentation: Missing documents, requested updates.

EVALUATION OF RESPONSE: Analysis of proposed corrective actions, verification of additional evidence, final decision on proceeding, communication to the applicant.

17.4 PHASE 3: FIELD AUDIT

17.4.1 Audit Planning

OBJECTIVE: Verify in practice the compliance with all PTVS requirements, evaluating the body's actual capability to issue quality certifications.
DURATION BY LEVEL: • Level 1: 2 days minimum (1 auditor) + 1 day per additional typology. • Level 2: 3 days minimum (2 auditors) + 1 day per additional typology. • Level 3: 5 days minimum (2-3 auditors) + 2 days per additional typology. • Level 4: 7 days minimum (3-4 auditors) + 3 days per additional typology.
AUDIT TEAM: • Audit Leader: PTVS Lead Auditor + ISO 17020 (Coordination, final report). • Technical Auditor: Expert in the specific asset typology (Technical evaluation). • Protocols Auditor: CAP/CPEP/SCCIP/PRA Expert (Advanced protocols evaluation). • Quality Auditor: ISO 17020 Auditor (Management systems).
TYPICAL AGENDA (Level 3, 5 days): • Day 1: Opening meeting, management system audit, documentation review, impartiality review. • Day 2: Personnel competence (file review, interviews, competence tests). • Day 3: Equipment and practical cases (facility inspection, equipment verification, calibration review, simulation cases). • Day 4: Advanced protocols (CAP, CPEP, SCCIP, PRA, PDBP evaluation). • Day 5: Consolidation, internal team meeting, closing meeting.

17.4.2 Opening Meeting

PARTICIPANTS: Full audit team, body management, Technical Director, Quality Manager, key personnel.
AGENDA:

1. Introductions (15 min).

2.Plan confirmation (15 min): Agenda review, logistics confirmation.
3.Process explanation (30 min): Audit methodology, evaluation criteria, confidentiality, findings management.
4.Impartiality declaration (15 min): Confirmation of no conflicts, signature by auditors.
5.Logistics and communication (15 min): Communication channels, document access, contact persons.

17.4.3 Management Systems Audit

Areas evaluated: • Quality Manual and Documentation: Current version, approved, document/record control. • Internal Audits: Annual program, execution according to plan, qualified auditors, follow-up of findings. • Non-Conformities and Corrective Actions: Registration system, root cause analysis, corrective actions, effectiveness verification. • Management Review: Annual execution, complete inputs, documented outputs, follow-up. • Complaints Management: Documented procedure, complaint registry, investigation and response, trend analysis.

17.4.4 Personnel Competence Evaluation

METHODS OF EVALUATION:

1.File Review: Original or certified degrees, detailed CV + references, valid certifications, training and performance records, signed annual interest declarations, signed ethics code, completed PTVS training.
2.Technical Interviews: • Real Estate: "How do you evaluate a structural pathology?" "What equipment do you use for moisture detection?" "How do you calculate the SRF in a complex case?" • Naval: "How do you evaluate hull osmosis?" "What

parameters do you analyze in engine oil?" "How do you verify MARPOL compliance?" • Advanced Protocols: "How do you implement edge signing (CPEP)?" "How do you configure thresholds in CAP?" "How does it integrate with SCCIP?" 3.Practical Tests: • Case 1: Simulated Inspection. The inspector performs a complete inspection of a simulated asset, identifies defects, calculates the SRF, and generates a partial report. Evaluated on methodology, equipment use, defect identification, score calculation, report quality, and time. • Case 2: Technical Analysis. Analysis of provided sensor data from a yacht, identifying patterns, applying PRA, and generating recommendations. • Case 3: Advanced Protocols. The team configures CAP, implements CPEP, defines SCCIP rules, and creates a basic digital twin for a Level 3 asset.

17.4.5 Equipment Evaluation

PHYSICAL VERIFICATION: • Measuring equipment: Valid calibration (<1 year). • Drones: Operational and certified (test flight). • Thermography: Functional and calibrated (field test). • LiDAR: Operational and calibrated (test scan). • IoT Sensors: In stock and functional (random sampling). • Software: Valid licenses. • CAP Platform: Operational (live demo). • Digital Twin: Functional (live demo).
OPERATIONAL TESTS: • Drone Test: 15-minute flight, 4K image capture, orthophoto generation. • Thermography Test: Electrical panel inspection, hot spot identification, report generation. • LiDAR Test: Room scanning, point cloud generation, quality evaluation. • IoT Test: Installation of 5 sensors, platform configuration, data verification.

17.4.6 Advanced Protocols Evaluation

• CAP: Platform operational (live demo), sensors configured (real-time data), thresholds defined, alerts

functioning (simulation), digital twin integration, automatic reporting. • CPEP: Body and inspector DIDs verified, edge signing operational (real test), custody chain verified, blockchain integration verified, public verification portal functional. • SCCIP: Smart contracts deployed, rules configured, oracles operational (test), automatic pauses simulated, multi-sig governance verified, ERC-3643 integration tested. • PRA: Models implemented, training data verified, model validation metrics reviewed, operational predictions demonstrated, CAP integration demonstrated. • PDBP: AUI generated, physical-digital linking verified, ZKP implemented (Level 4), dynamic updating demonstrated, SCCIP integration verified.

17.4.7 Closing Meeting

PARTICIPANTS: Full audit team, body management, key personnel.
AGENDA:

1. Acknowledgments (5 min): Thanks for collaboration, recognition of effort.
2. Summary of Findings (30 min): Highlighted conformities, major non-conformities, minor non-conformities, observations.
3. Discussion of Findings (30 min): Clarifications, additional context, factual corrections.
4. Preliminary Classification (15 min): Audit team recommendation, next steps, deadlines.
5. Signing of Closing Report (10 min): Signatures of auditors and body representatives, notation of any disagreements.

17.5 PHASE 4: DECISION AND ISSUANCE

17.5.1 Accreditation Committee

COMPOSITION: • Director of Accreditations (Aurema Group) - Chair (Voting). • PTVS Technical Director (Aurema Group) - Member (Voting). • External Lead Auditor - Member (Voting). • Sectoral Expert (based on typology) - Member (Voting). • Representative of Accredited Bodies - Member (Non-voting in conflicts). **QUORUM:** Minimum 4 members present, Chair mandatory, at least 1 external member. **DECISION:** Simple majority. Tie-breaker: Chair's casting vote. Motivated and documented decisions.

17.5.2 File Analysis

DOCUMENTS REVIEWED:

1. Application and initial documentation.
2. Self-assessment report.
3. Documentary evaluation report.
4. Field audit report.
5. Body's response to findings.
6. Evidence of corrective actions.
7. Audit team's recommendation.

DECISION CRITERIA: • No MaNC + 0-2 MiNC: **ACCREDITATION GRANTED** (Resolve MiNC in 3 months). • No MaNC + 3-5 MiNC: **CONDITIONAL ACCREDITATION** (Resolve MiNC in 2 months + follow-up audit). • No MaNC + >5 MiNC: **PAUSE** (3-month improvement plan + re-audit). • 1-2 MaNC: **PAUSE** (Resolve MaNC in 3 months + re-audit). • >2 MaNC:

REJECTION (New process after 6 months). • Fraud detected: **REJECTION + SANCTIONS** (5-year exclusion).

17.5.3 Communication of Decision

ACCREDITATION GRANTED: Formal letter detailing: Accreditation number, type, levels, typologies, jurisdictions, issue date, validity date, next surveillance date, conditions (if any), and next steps (contract signing, fee payment, portal onboarding, welcome kit, supervised first certification).
CONDITIONAL ACCREDITATION: Accreditation granted subject to resolving MiNCs within 2 months and a follow-up audit at 3 months.
PAUSE: Process suspended until an improvement plan is presented, corrective actions are implemented, and a re-audit is scheduled.
REJECTION: Accreditation denied with detailed reasons. Possibility of a new process after the specified period. Right to appeal within 30 days.

17.5.4 Issuance of Certificate and Contract

ACCREDITATION CONTRACT: A formal agreement between Aurema Group (Licensor) and the Body (Licensee) covering: Object (accredation details), Duration (3 years, renewable), Body Obligations (maintain requirements, allow audits, communicate changes, pay fees, use seal correctly, cooperate, train staff, improve continuously), Licensor Obligations (maintain standard, conduct audits, provide support, maintain registry, protect confidentiality), Fees, Suspension/Revocation conditions, Liability, Brand Protection, Confidentiality, and Governing Law/Arbitration (AAA Wilmington).
PTVS ACCREDITATION CERTIFICATE: A formal document stating the body's name, tax ID, address, accreditation as a PTVS Inspection Body, Accreditation Number (PTVS-OI-[XXX]), Type, Authorized Levels, Asset

Typologies, Jurisdictions, Issue Date, Validity Date (+3 years), Next Surveillance Date, Status (ACTIVE), Digital Signature, Cryptographic Signature, Blockchain Hash, and QR Code for public verification.

17.5.5 Welcome Kit

Upon signing, the body receives: • Physical and digital Accreditation Certificate. • Signed Accreditation Contract. • PTVS Seal Usage Guide and Brand Manual. • PTVS Report Templates. • Access to the Accredited Bodies Portal. • Access to the Training Platform (Aurema Group Academy). • Access to the Accreditation Management System. • Integration APIs. • Monitoring Dashboard Access. • Physical Accreditation Plaque, seals, marketing materials. • Assigned Accreditation Manager, priority support channel, onboarding meeting, and a "First 100 Days" plan.

17.6 PHASE 5: SURVEILLANCE AND RENEWAL

17.6.1 Surveillance Audits

OBJECTIVE: Verify the continuous maintenance of accreditation requirements and the quality of issued certifications.
FREQUENCY: • Level 1: Annual (Documentary + Field, 1 day). • Level 2: Annual (Complete, 2 days). • Level 3: Semi-annual (Complete, 3 days). • Level 4: Quarterly + Surprise (Complete, 3-5 days).
SCOPE: • Review of a sample of issued certifications. • Verification of maintained requirements. • Evaluation of complaints and non-conformities. • Review of continuous

training. • Equipment verification. • Advanced protocols evaluation.

17.6.2 Annual Body Reports

Each accredited body MUST submit an annual report to Aurema Group including: • Operational Data: Certifications issued by level/typology, jurisdictions, average execution times, average SRI by level. • Quality Data: Internal non-conformities, complaints received/resolved, audits performed, corrective actions. • Personnel Data: Number of accredited inspectors, training hours, performance evaluations, turnover. • Economic Data: Revenue by level, equipment investment, training investment, insurance premiums. • Technological Data: Advanced protocol implementation, technological incidents, improvements, roadmap.

17.6.3 Renewal of Accreditation

FREQUENCY: Every 3 years.
PROCESS:

1.Renewal Request (6 months before expiration): Automatic notification, confirmation of intent, documentation update.
2.Renewal Audit (3-4 months before): Similar to initial audit, complete evaluation of all requirements, review of 3-year history, continuous improvement evaluation.
3.Decision (2 months before): Accreditation Committee analysis. Decision: Renewal / Conditional / Non-renewal.
4.Issuance (1 month before): New 3-year certificate, registry update, public communication.

RENEWAL CRITERIA: • Excellent compliance: Direct renewal. • Minor non-critical MiNCs: Renewal with conditions. • Minor MaNCs: Conditional renewal + follow-

up. • Serious MaNCs: Non-renewal + improvement plan. • Systematic non-compliance: Non-renewal + possible revocation.

17.6.4 Special Cases

LEVEL UPGRADE: Scenario: Level 2 body wants to expand to Level 3. Process: Simplified. Gap analysis (documentary), specific audit (only new requirements), decision, new certificate issuance. Timeline: 3-4 months. Fee: 60% of the new accreditation fee.
TYPOLOGY EXTENSION: Scenario: Real estate accredited body wants to add naval. Process: Simplified. Evaluation of technical competence in the new typology, specific audit (new typology only), decision, certificate update. Timeline: 2-3 months. Fee: 40% of the new accreditation fee.
JURISDICTIONAL EXTENSION: Scenario: Spanish body wants to operate in Portugal. Process: Request, evaluation of competence in the new jurisdiction, verification of local regulatory compliance, specific audit (if necessary), decision, update. Timeline: 2-4 months. Fee: 30% of the new accreditation fee.
SIGNIFICANT CHANGES: Mandatory communication within 30 days for: Change of corporate name, change of address, change of Technical Director, >30% change in capital, merger/acquisition, change in controlling shareholders.
VOLUNTARY SUSPENSION: Scenario: Body needs to temporarily pause operations. Process: Motivated request, agreement on conditions, suspension of certification issuance, maintenance of basic requirements, reactivation via specific audit. Maximum suspension period: 12 months.

17.7 NON-CONFORMITY MANAGEMENT

17.7.1 Classification of Non-Conformities

DURING ACCREDITATION: • Major Non-Conformity (MaNC): Total absence or fundamental failure. Direct impact on certification quality or PTVS seal integrity. • Minor Non-Conformity (MiNC): Partial non-compliance or weakness. Does not compromise overall capability.
DURING SURVEILLANCE: In addition to MaNC and MiNC: • Observation (OBS): Opportunity for improvement, not a non-compliance. • Area for Improvement (AFI): Weakness identified that may evolve into a MiNC if not addressed.

17.7.2 Management Process

1.IDENTIFICATION AND REGISTRATION: Detection, logging in the system, classification, assignment of responsible party.
2.ROOT CAUSE ANALYSIS: Investigation of origin, identification of root cause, impact analysis, documentation of findings.
3.CORRECTIVE ACTION PLAN: Description of the NC, root cause, corrective action, responsible person, deadline, required resources, verification method.
4.IMPLEMENTATION: Execution of the plan, progress logging, communication to the auditor, adjustments if necessary.
5.EFFECTIVENESS VERIFICATION: Auditor verifies implementation, evaluates effectiveness, closes the non-conformity, logs in history.

6.FOLLOW-UP: Verification of maintenance, trend analysis, lessons learned, system improvement.

17.7.3 Resolution Deadlines

Type	Maximum Deadline	Consequences
MaNC during accreditation	3 months	Process paused
MiNC during accreditation	1 month	Conditions accreditation
MaNC during surveillance	1 month	Suspension if unresolved
MiNC during surveillance	3 months	Follow-up in next audit
OBS	6 months	Recommendation, not mandatory
AFI	3 months	Improvement plan required

17.7.4 Escalation of Non-Compliance

• First Non-Compliance: Formal notification, corrective action plan, deadline according to type. • Recurrence: Increased severity, reduced deadline, potential financial penalty. • Persistent Non-Compliance: Accreditation suspension, extraordinary audit, supervised remediation plan. • Critical Non-Compliance: Accreditation revocation, invalidation of certificates, legal action if applicable.

17.8 SANCTIONS AND DISCIPLINARY REGIME

17.8.1 Typology of Sanctions

REPRIMAND: Formal communication of non-compliance, logged in the file. No economic impact. For first-time minor non-compliances.
ECONOMIC PENALTY: • Minor repeated: €1,000 - €5,000. • Moderate: €5,000 - €15,000. • Serious: €15,000 - €50,000. • Very serious: €50,000 - €100,000.
TEMPORARY SUSPENSION: • 1-3 months: Repeated MiNCs. Consequence: No certification issuance. • 3-6 months: Unresolved MaNCs. Consequence: No issuance + extraordinary audit. • 6-12 months: Serious non-compliance. Consequence: Supervised remediation plan.
REVOCATION: Definitive loss of accreditation. Invalidation of certificates (case-by-case evaluation). Public communication. Possible legal action.
EXCLUSION: Revocation + prohibition of a new process. Period: 5 years for fraud, 2 years for other serious cases. Communication to other bodies and clients.

17.8.2 Sanctioning Process

1.INITIATION (5 days): Identification of non-compliance, notification to the body, opening of file, precautionary measures if necessary.
2.INSTRUCTION (30 days): Evidence gathering, hearing of the body (15 days), resolution proposal, instructor's report.
3.RESOLUTION (15 days): Analysis by the Disciplinary Committee, motivated decision, notification to the body, execution of the sanction.

4.APPEAL (30 days): Possibility of appeal, review by the Appeals Committee, final decision.

17.8.3 Mitigating and Aggravating Factors

MITIGATING FACTORS: • Self-reporting of the non-compliance. • Collaboration in the investigation. • Immediate corrective actions. • Absence of harm to third parties. • Good prior history.
AGGRAVATING FACTORS: • Intentionality. • Concealment of the non-compliance. • Harm to third parties. • Recurrence. • Illicit economic benefit.

17.9 PRACTICAL CASES

17.9.1 Case 1: Successful Accreditation of an International Body

Scenario: UK-based company with 20 years of naval inspection experience requests Type D, Level 4 accreditation to operate in multiple jurisdictions. **Process:** • Phase 1: Briefing in London. Self-assessment: 92% compliance. Formal application. Fees paid: €25,000. • Phase 2: Extensive documentation (500+ pages). 15 MiNCs identified, 0 MaNCs. MiNCs resolved in 2 months. Result: Proceed to audit. • Phase 3: Field audit (7 days). Team: 4 auditors (UK, Spain, UAE, Singapore). Audit in 3 locations (London, Gibraltar, Dubai). Advanced protocols evaluation: Excellent. Practical cases: 5 yachts of different sizes. 1 MiNC (documentation in a new jurisdiction). Result: Accreditation granted. • Phase 4: Committee: Unanimous approval. Certificate issued: PTVS-OI-UK-001. Type D, Level 4. Jurisdictions: UK, Spain, UAE, Singapore, Cayman Islands. • Phase 5: First surveillance audit at 6 months. Result: No findings. Certifications issued in year 1: 12 megayachts. PTVS Revenue: €2.5M.

Lessons Learned: Importance of multi-jurisdictional coordination, value of the body's prior experience, need for auditors with local knowledge.

17.9.2 Case 2: Conditional Accreditation with Follow-up

Scenario: Spanish body specialized in real estate requests Type B, Level 3 accreditation. **Process:** • Phase 1: Self-assessment: 78% compliance. Formal application. Fees: €5,000. • Phase 2: 3 MaNCs identified: (1) Missing PRA procedure, (2) Personnel without advanced PTVS training, (3) CAP platform not operational. 8 MiNCs. MaNCs resolved in 3 months. Result: Conditional. • Phase 3: Field audit (5 days). Verification of MaNC resolution: Satisfactory. Protocols evaluation: Acceptable. 2 additional MiNCs. Result: Conditional accreditation. • Phase 4: Accreditation granted with conditions: Resolve MiNCs in 2 months, follow-up audit at 4 months, supervision of the first 5 certifications. • Phase 5: Follow-up: MiNCs resolved satisfactorily. Follow-up audit: No findings. First 5 supervised certifications: Correct. Full accreditation granted. **Lessons Learned:** Importance of prior preparation, value of initial supervision, continuous improvement post-accreditation.

17.9.3 Case 3: Rejection for Serious Non-Compliance

Scenario: New body (2 years old) requests Level 4 accreditation without prior experience in advanced protocols. **Process:** • Phase 1: Self-assessment: 45% (biased). Formal application. Initial warning about viability. • Phase 2: 5 MaNCs identified: (1) Personnel without Level 4 experience, (2) Lack of institutional equipment, (3)

Advanced protocols not implemented, (4) Deficient management systems, (5) Insufficient insurance. Result: Rejection. • Decision: Rejection of accreditation. Recommendation: Start with Level 2. New process possible after 6 months. • Body's Reaction: Appeal filed. Appeals Committee review. Rejection confirmed. Acceptance of recommendation. • Final Result: Body requests Level 2 after 6 months. Level 2 accreditation granted after 8 months. Plan to evolve to Level 3-4 in 3 years. **Lessons Learned:** Importance of realism in self-assessment, value of starting at appropriate levels, fair and transparent appeal processes.

17.9.4 Case 4: Suspension and Revocation

Scenario: Level 3 accredited body issues certificates with manipulated scores after 2 years of operation. **Detection:** Surprise surveillance audit by Aurema Group. Statistical analysis of issued SRIs. Discrepancy between data and scores. Deep investigation. **Investigation:** Review of 50 issued certificates. Identification of 12 with manipulated SRI. Personnel interviews. Identification of commercial pressure as the cause. **Sanctioning Process:** File initiation. Precautionary suspension of accreditation. Instruction: 30 days. Resolution: Revocation + €75,000 fine. **Consequences:** Accreditation revoked. 12 certificates invalidated. Notification to affected clients. Activation of SCCIP for affected tokens. Public communication (anonymized). 5-year exclusion. **Lessons Learned:** Importance of surprise audits, value of statistical analysis, need for whistleblowing channels, reinforcement of quality controls.

17.10 REPORTING AND CERTIFICATION

17.10.1 Accreditation Statistics

Aurema Group publishes a quarterly Accreditation Report including: • Applications received. • Accreditations granted. • Grant rate (%). • Average process time (months). • Active bodies. • Distribution by type (A, B, C, D). • Distribution by level (1, 2, 3, 4). • Distribution by typology. • Surveillance audits performed. • Sanctions imposed.

17.10.2 Public Registry

A public directory of accredited bodies is maintained at:

`https://accreditation.proptrustverified.com`

Public Information: • Corporate name. • Accreditation number. • Type and authorized levels. • Authorized typologies. • Jurisdictions. • Accreditation date and validity. • Status (active, suspended). • Contact data.
Verifiable Information: • QR code for certificate verification. • Blockchain hash of the certificate. • Audit history (summarized).

17.10.3 Process Certification

For each certification issued by an accredited body, a certificate is generated that includes: • Asset description and AUI. • Owner (redacted or public based on level). • Issuing body name and accreditation number (PTVS-OI-[XXX]). • Verification level (1, 2, 3, or 4). • Scores: SRD, SRF, SRT, SRE, SRI. • Value. • Issue date and validity. •

Inspector name and DID. • Digital signature, cryptographic signature, blockchain hash, and QR code for verification at https://verify.proptrustverified.com/[AUI].

CHAPTER 18: MAINTENANCE AND CONTINUOUS IMPROVEMENT

18.1 PURPOSE AND PHILOSOPHY

18.1.1 Continuous Improvement as a Pillar of the Standard

The **Prop Trust Verified Standard (PTVS)** is not a static document. It is a **living organism** that must evolve at the pace of: • Technological advances (blockchain, AI, IoT, quantum computing). • Regulatory evolution (MiCA, national regulations, ISO standards). • Market needs (new asset types, new use cases). • Lessons learned from daily practice. • Feedback from all stakeholders.
While previous chapters have established **what** must be done and **how**, this chapter establishes **how to maintain and improve** the system over time. It is the guarantee that the PTVS is not only the best standard today, but will continue to be so tomorrow.

18.1.2 Principles of Continuous Improvement

PRINCIPLE 1: PDCA CYCLE (Plan-Do-Check-Act) Every improvement MUST follow the Deming cycle: Plan the change, implement it on a controlled scale, verify results against objectives, and standardize if successful or adjust if not.
PRINCIPLE 2: DATA-DRIVEN Every improvement decision MUST be based on objective data, not intuition.

PRINCIPLE 3: UNIVERSAL PARTICIPATION Continuous improvement MUST involve all levels: management, inspectors, support staff, clients, and stakeholders.
PRINCIPLE 4: CLIENT FOCUS Every improvement MUST be oriented toward increasing the perceived value by the end client.
PRINCIPLE 5: SYSTEMIC THINKING Improvements MUST consider the impact on the entire system, not just isolated areas.
PRINCIPLE 6: EVIDENCE-BASED DECISION MAKING Decisions MUST be based on the analysis of data and information, not on assumptions.

18.1.3 Scope of Continuous Improvement

The PTVS identifies **seven areas** where continuous improvement must be applied systematically:

1. Verification Processes (Phases 1-4, advanced protocols).
2. Technology and Infrastructure (CAP, CPEP, SCCIP, PRA, PDBP, digital twin).
3. Personnel and Competencies (Training, accreditation, professional development).
4. Management Systems (Quality, security, environment, information).
5. Client Relations (Satisfaction, loyalty, new services).
6. Market Positioning (Competitiveness, differentiation, expansion).
7. Sustainability and Impact (ESG, carbon footprint, social impact).

18.2 CERTIFICATION

MAINTENANCE AUDITS

18.2.1 Differentiation from Accreditation

Audits

It is fundamental to distinguish between two types of audits: • **Accreditation Audit (Chapter 17):** Focuses on the inspection body. Goal: Grant or maintain the body's authorization to issue PTVS certifications. • **Maintenance Audit (Chapter 18):** Focuses on the certified asset. Goal: Maintain the validity of the asset's specific certification over time.

18.2.2 Frequency by Certification Level

Level	Documentary Audit	Complete Physical Audit	CAP Monitoring
Level 1	Annual	Every 2 years	Not applicable
Level 2	Semi-annual	Annual	Not applicable
Level 3	Quarterly	Annual	Continuous (24/7)
Level 4	Monthly	Semi-annual	Continuous (24/7) redundant

18.2.3 Maintenance Audit Process

PHASE 1: PLANNING (30 days prior) • Review of previous certification. • Analysis of CAP data (Levels 3-4). • Identification of risk areas. • Scheduling with the owner. • Assignment of audit team.

PHASE 2: EXECUTION • **Documentary:** Verification of registry changes, license updates, regulatory compliance review, analysis of new charges, SRD recalculation. • **Physical:** Inspection according to defined scope, verification of previous defects, confirmation of recommendation execution, evidence update, SRF recalculation. • **Technical:** Update of analyses, system review, predictive model update, SRT recalculation. • **Financial:** Valuation update, market condition review, stress testing update, SRE recalculation.
PHASE 3: EVALUATION • Comparison with previous certification. • Calculation of new SRI. • Identification of significant changes. • Determination of decision.
PHASE 4: DECISION MATRIX • SRI ≥ level threshold: **MAINTAIN** (Renew certification). • SRI between (threshold - 5) and threshold: **WARNING** (90-day improvement plan). • SRI < (threshold - 5) but > (threshold - 15): **DOWNGRADE** (New lower-level certificate). • SRI < (threshold - 15): **SUSPENSION** (Remediation plan required). • Critical finding: **REVOCATION** (Investigation + action).
PHASE 5: COMMUNICATION • Maintenance report to the owner. • Certificate update. • Public portal update. • Stakeholder notification. • SCCIP update (if applicable).

18.3 COMPETENCE UPDATE

18.3.1 Continuing Professional Development (CPD) Framework

DEFINITION: CPD is the process by which personnel maintain, improve, and expand their competencies throughout their professional lives.
OBJECTIVES:

1. Maintain current technical competence.

2.Adapt to new standards and regulations.
3.Incorporate new technologies.
4.Develop transversal skills.
5.Prepare personnel for higher levels.

18.3.2 Individual Development Plan (IDP)

Every staff member MUST have an annually updated IDP, including: • Current competence analysis (gap analysis against target level). • Specific, measurable development objectives. • Training plan by quarter (courses, activities, providers, budget). • Quarterly progress reviews and adjustments. • Final annual evaluation (objectives achieved, training hours, new certifications).

18.3.3 PTVS Training Catalog

MANDATORY TRAINING (All levels): • PTVS Professional Ethics (8h, Annual, Online). • Regulatory Updates (16h, Annual, Hybrid). • Data Protection (GDPR) (8h, Annual, Online). • Anti-Money Laundering (8h, Annual, Online). • Information Security (8h, Annual, Online).
ROLE-SPECIFIC TRAINING: • Inspectors: PTVS Inspector Advanced (40h), New Inspection Technologies (24h), Complex Practical Cases (16h). • Technical Directors: Technical Leadership (24h), Advanced Quality Management (32h), Innovation in Verification (16h). • Quality Managers: Advanced Internal Auditing (40h), ISO 17020 Updates (16h), Risk Management (24h). • Specialized Technicians: Advanced Drone Operation (24h), Professional LiDAR (32h), Level 2 Thermography (40h), IoT for Inspectors (24h), Applied Blockchain (32h).

18.3.4 Mentorship Program

OBJECTIVE: Accelerate the development of junior inspectors through the guidance of senior inspectors.
STRUCTURE: Senior Level 4 Inspector mentors Level 2

Inspector for 12 months (Goal: Reach Level 3). Experienced DT mentors new DT for 6 months. **REQUIREMENTS:** Mentor must have 10+ years experience, proven communication skills, and 4 hours/month availability. Mentee must show commitment to development. **FOLLOW-UP:** Monthly mentor-mentee meetings, quarterly progress evaluation, final development report, mentor recognition.

18.3.5 360-Degree Competence Evaluation

SOURCES OF EVALUATION: • Self-evaluation (10%) • Supervisor evaluation (30%) • Peer evaluation (20%) • Client feedback (25%) • Objective results (15%)
COMPETENCIES EVALUATED: • Technical (50%): Sector knowledge, PTVS protocol mastery, advanced technology use, report quality, valuation precision. • Transversal (30%): Written/oral communication, teamwork, problem-solving, time management, adaptability. • Ethical (20%): Professional integrity, impartiality, confidentiality, responsibility, transparency.
SCALE: 1 (Basic) to 5 (Master/International Recognition).
CONSEQUENCES: >85% (Promotion, bonus, recognition), 70-85% (Maintenance, development plan), 50-69% (Intensive 90-day improvement plan), <50% (Reassignment or exit).

18.4 SYSTEM CONTINUOUS IMPROVEMENT

18.4.1 Continuous Improvement Methodology

The PTVS continuous improvement cycle follows a structured approach:

1. **IDENTIFY OPPORTUNITY:** Detect the problem or area for improvement.
2. **ANALYZE CAUSES:** Find the root cause, not just the symptom.
3. **PLAN SOLUTION:** Define the action, responsible party, deadline, and resources.
4. **IMPLEMENT PILOT:** Test the solution on a small scale.
5. **VERIFY RESULTS:** Measure outcomes against objectives.
6. **STANDARDIZE:** If successful, implement globally and update procedures.

18.4.2 Sources of Improvement Opportunities

INTERNAL SOURCES: • Internal and external audit findings. • Operational data analysis (execution times, error rates, costs). • Staff feedback (climate surveys, suggestion boxes, improvement groups). • Management review.
EXTERNAL SOURCES: • Client feedback (satisfaction surveys, complaints, NPS). • Market evolution (new competitors, demand changes, technological trends). • Regulatory changes (new regulations, standard updates). • Benchmarking (industry best practices, international standards).

18.4.3 Improvement Tools

1. 5 WHYS (Root Cause Analysis): Asking "why" five times to reach the root cause. Example: Reports are late → Inspectors lack information → Lab data arrives late → No SLA with the lab → No subcontractor management procedure. Root Cause: Lack of subcontractor management procedure.
2. ISHIKAWA DIAGRAM (Fishbone): Analyzes causes of a problem from multiple perspectives: Personnel,

Methods, Machinery, Material, Environment, Measurement.
3. PRIORITIZATION MATRIX: Scores improvements based on Impact (1-5), Effort (1-5), and Urgency (1-5). Score = Impact + Urgency - Effort. Prioritize high-score items.

18.4.4 Improvement Projects

Significant improvements are managed as formal projects, including: • Title, sponsor, and project manager. • Context and justification (data supporting the need). • SMART objectives and success KPIs. • Scope (inclusions and exclusions). • Work plan (Gantt chart, milestones, deliverables). • Budget and resource allocation. • Risk management plan. • Governance and communication plan. • Closure criteria and sustainability plan.

18.4.5 Improvement Registry

A centralized database tracks all improvement initiatives: • ID, Title, Status (Identified, Planned, In Progress, Completed, Cancelled). • Responsible party, Start/End dates. • Measured Impact (time reduction %, cost savings, quality improvement). • Calculated ROI.

18.5 CHANGE MANAGEMENT

18.5.1 Change Management Framework

Change management is the structured approach to preparing, supporting, and helping individuals, teams, and organizations make organizational transitions. The PTVS applies the **ADKAR** model:

1. **Awareness:** Communicate why the change is necessary.

2. **Desire:** Generate motivation to support the change.
3. **Knowledge:** Train on how to change.
4. **Ability:** Develop the capability to implement the change.
5. **Reinforcement:** Sustain the change over time.

18.5.2 Types of Change in PTVS

• **Technological Change:** Implementation of new technologies, protocol updates (e.g., CPEP v2.0), platform migrations. • **Organizational Change:** Team restructuring, new roles, process changes, mergers. • **Regulatory Change:** New regulations (MiCA, GDPR updates), ISO standard updates, new PTVS requirements. • **Market Change:** New asset types, new use cases, competitive shifts.

18.5.3 Change Management Plan

For any significant change, a plan must be developed:

1. **Impact Analysis:** Who is affected? What changes are required? What resistance is expected?
2. **Stakeholder Analysis:** Map stakeholders by interest and influence to define communication strategies.
3. **Communication Plan:** Kick-off, monthly newsletters, updated FAQs, Q&A channels.
4. **Training Plan:** Mandatory online course, in-person workshop, individual mentoring, reference materials.
5. **Pilot Phase:** Select a small group (e.g., 5 inspectors), implement, gather feedback, adjust.
6. **Rollout:** Progressive deployment, intensive support, adoption monitoring.

7. **Consolidation:** Recognize early adopters, share best practices, celebrate successes, adjust final details, measure adoption KPIs.

18.5.4 Managing Resistance

Resistance is natural. Strategies to manage it include: • **Education and Communication:** Explain the "why" clearly. • **Participation and Involvement:** Involve affected parties in the design of the change. • **Facilitation and Support:** Provide resources, training, and time to adapt. • **Negotiation and Agreement:** Seek compromises where feasible. • **Coercion (Last Resort):** Explicit or implicit enforcement, only in extreme cases.

18.5.5 Measuring Change Success

KPIs for change management: • Adoption rate: Active users / Total users (Target: >90%). • Time to competence: Days until full proficiency (Target: <60 days). • Usage rate: Actual uses / Expected uses (Target: >85%). • Change satisfaction: Internal NPS (Target: >7). • Change ROI: Benefits / Costs (Target: >200%). • Change retention: % maintaining the change at 6 months (Target: >95%).

18.6 TECHNOLOGICAL INNOVATION

18.6.1 PTVS Technology Horizon

The PTVS actively monitors emerging technologies across three horizons:
SHORT-TERM HORIZON (1-2 years): • Generative AI: Assistance in report drafting (Emerging, HIGH priority). • Computer Vision: Automatic defect detection in images (Mature, HIGH priority). • Private 5G: Real-time monitoring

in remote areas (Deployment, MEDIUM priority). • Advanced Digital Twins: Multi-physics simulation (Mature, HIGH priority). • Layer 2 Blockchain: Reduction of transaction costs (Mature, MEDIUM priority).
MEDIUM-TERM HORIZON (3-5 years): • Quantum Computing: Post-quantum cryptography (Emerging, MEDIUM priority). • Autonomous Robotics: Automated inspections (Emerging, MEDIUM priority). • Augmented Reality (AR): Remote guided inspections (Mature, HIGH priority). • Edge AI: Advanced local processing (Mature, HIGH priority). • Advanced ZKP: Enhanced privacy protocols (Emerging, HIGH priority).
LONG-TERM HORIZON (5-10 years): • AGI (Artificial General Intelligence): Advanced predictive analysis (Emerging, LOW priority). • Nanosensors: Nano-scale monitoring (Emerging, MEDIUM priority). • Advanced Biometrics: Seamless inspector authentication (Emerging, MEDIUM priority).

18.6.2 Technological Innovation Committee

COMPOSITION: • Director of Innovation (Chair, Aurema Group CTO). • Blockchain Expert. • AI/Data Science Senior Expert. • IoT/Telecommunications Engineer. • CISO (Cybersecurity). • Representative of Accredited Bodies (Senior Inspector). • External Academic (University Professor).
FUNCTIONS: • Technology scouting and monitoring (Gartner Hype Cycle analysis). • Evaluation of applicability to PTVS. • Recommendations for adoption or rejection. • Supervision of pilot projects. • Personnel training on new technologies. • Contribution to industry thought leadership (publications, conferences).

18.6.3 Technology Adoption Process

1. **Identification (1-2 months):** Detect emerging technology, assess maturity, evaluate PTVS relevance, preliminary feasibility analysis.
2. **Evaluation (2-3 months):** Detailed technical analysis, vendor evaluation, Proof of Concept (PoC), cost-benefit analysis, risk assessment.
3. **Pilot (3-6 months):** Select use case, controlled implementation, measure results, gather feedback, optimize.
4. **Decision (1 month):** Analyze pilot results, evaluate ROI, decide on adoption or discard, create deployment plan.
5. **Deployment (6-12 months):** Implementation plan, personnel training, progressive rollout, adoption monitoring, continuous optimization.

18.6.4 Strategic Collaborations

UNIVERSITIES AND RESEARCH CENTERS: • Polytechnic University of Valencia: Advanced FEM models. • MIT Media Lab: ZKP applied to RWA. • Imperial College London: Computer Vision for inspection. • ETH Zurich: Post-quantum cryptography.
TECHNOLOGY COMPANIES: • Chainlink: Oracle integration. • Microsoft Azure: PTVS cloud infrastructure. • NVIDIA: Computer Vision and AI. • Siemens: Digital twin technology. • DJI: Aerial inspection solutions.
CONSORTIA AND STANDARDS: • Digital Twin Consortium: Digital twin standards. • Token Alliance: Tokenization standards. • RWA Working Groups: Real World Asset standards. • ISO/TC 59: Construction standards participation. • W3C DID Working Group: Decentralized identity contribution.

18.7 BENCHMARKING AND BEST PRACTICES

18.7.1 Strategic Benchmarking

DEFINITION: The process of comparing the processes and performance metrics of the organization with those of other leading organizations to identify opportunities for improvement.
TYPES APPLICABLE TO PTVS: • **Internal:** Comparison between accredited bodies to share best practices. • **Competitive:** Comparison with competing standards (RICS, IIMS, ISO) to identify competitive advantages and gaps. • **Functional:** Comparison with organizations in other sectors to learn best practices (e.g., aviation safety protocols applied to asset inspection). • **Generic:** Comparison of universal processes (e.g., customer service, data management) against global excellence standards.

18.7.2 Benchmarking Metrics

OPERATIONAL METRICS: • Average inspection time: PTVS (3-7 days) vs. RICS (5-10 days) vs. IIMS (7-14 days). Target: <3 days. • Report issuance time: PTVS (7-14 days) vs. Competitors (10-30 days). Target: <7 days. • Error rate: PTVS (<1%) vs. Competitors (2-5%). Target: <0.5%. • Client satisfaction (NPS): PTVS (>8) vs. Competitors (5-7). Target: >9.
TECHNOLOGICAL METRICS: • % inspections with drones: PTVS (80% for L3-4) vs. Competitors (30-40%). Target: 95%. • % cryptographically signed data: PTVS (100% for L3-4) vs. Competitors (0%). Target: 100%. • %

assets with continuous monitoring: PTVS (100% for L3-4) vs. Competitors (5-10%). Target: 100%.

18.7.3 Identified Best Practices

PRACTICE 1: Autonomous Drone Inspection. Programmed drones for repetitive inspections with minimal human intervention. Benefits: 60% time reduction, enhanced safety (no working at heights), consistent data capture, cost reduction. PTVS Implementation: Pilot in 2025, full rollout in 2026, CPEP integration.
PRACTICE 2: Predictive Digital Twin. A digital twin that not only reflects the current state but predicts future states. Benefits: Problem anticipation, maintenance optimization, better decision-making, higher client value. PTVS Implementation: Development with Siemens, pilot in Q4 2026, commercial launch in 2027.
PRACTICE 3: Remote Verification with Augmented Reality. Inspections conducted remotely with AR support. Benefits: Reduced travel, access to global experts, cost reduction, faster response. PTVS Implementation: Initial research, Microsoft HoloLens collaboration, pilot planned for 2027.

18.7.4 PTVS Best Practices Network

OBJECTIVE: Create a community of practice among accredited bodies to share experiences and improve collectively.
STRUCTURE: • Real Estate Group (Quarterly). • Naval Group (Quarterly). • Technology Group (Monthly, CTOs). • Quality Group (Semi-annual, Quality Managers). • Leadership Group (Annual, CEOs).
ACTIVITIES: • Success case presentations. • Body visits and peer reviews. • Training webinars. • White paper publications. • Development of common standards.

Best Practices Repository: A database accessible to accredited bodies containing documented cases with metrics, templates, tools, and lessons learned.

18.7.5 PTVS Excellence Awards

An annual program to recognize outstanding performance:
• **Best Body of the Year:** Overall excellence, technological innovation, client satisfaction, contribution to the standard. • **Technological Innovation of the Year:** Implementation of disruptive technology with measurable impact. • **Best Verification Case:** Technical complexity, report quality, value added to the client, use of advanced protocols. • **Inspector of the Year:** Technical excellence, contribution to standard development, mentorship, client recognition. • **Sustainability of the Year:** Positive environmental impact, ESG innovation, contribution to climate goals.
Selection Process: Nominations by bodies and clients, evaluation by an independent committee, objective and measurable criteria, annual awards ceremony at the PTVS conference.

18.8 LESSONS LEARNED

18.8.1 Knowledge Management System

DEFINITION: The systematic process of identifying, capturing, organizing, sharing, and applying organizational knowledge to improve performance.
TYPES OF KNOWLEDGE IN PTVS: • **Explicit Knowledge:** Documented in procedures, manuals, reports. Easy to share and transfer. • **Tacit Knowledge:** In people's minds (experience, intuition, skills). Difficult to document and transfer. • **Embedded Knowledge:** In processes, systems, technologies. Coded in procedures and tools.

18.8.2 Lessons Learned Capture Process

1. **Identification:** At the end of each project/certification, after incidents or problems, during periodic reviews, or via staff suggestions.
2. **Documentation:** Using a standardized template: • Title, Date, Project/Case Reference, Author, Category (Success/Failure/Improvement). • Context: Initial situation, objectives, scope. • What Happened: Objective description of facts. • What Went Well: Positive aspects. • What Went Wrong: Negative aspects. • Root Causes: Underlying reasons. • Recommendations: Specific, actionable recommendations. • Applicability: Asset typologies, verification levels, similar situations. • Expected Impact: Time reduction %, cost savings, quality improvement.
3. **Validation:** Peer review, technical validation, committee approval, publication in the repository.
4. **Dissemination:** Lessons learned newsletter, sharing sessions, integration into training, procedure updates.
5. **Application:** Incorporation into new projects, impact monitoring, metric updates, feedback to the system.

18.8.3 Lessons Learned Repository

Structured by: • **Category:** Physical inspection, documentary analysis, technical evaluation, financial verification, advanced protocols, project management, client relations, technological innovation. • **Asset Typology:** Real Estate (Res/Com/Ind), Naval (Yacht/Commercial), Machinery, Infrastructure, Patrimonial. • **Verification Level:** Level 1, 2, 3, 4. • **Lesson Type:** Successes (what to do), Failures (what to

avoid), Improvements (how to optimize), Innovations (new ways).

18.8.4 Practical Cases

CASE 1: Megayacht Inspection with Access Problems. Category: Physical Inspection / Naval. Level: 4. Type: Improvement. Context: 80m megayacht inspection in a shipyard with tight deadlines. What Happened: Limited access to certain areas due to parallel work. Impossible underwater inspection. 5-day deadline vs. 7-day standard. What Went Well: Prior coordination with the shipyard, use of drones for hard-to-reach areas, constant communication with the client. What Went Wrong: Underestimation of required time, lack of Plan B for underwater inspection, team fatigue due to tight deadlines. Root Causes: Insufficient planning, lack of flexibility in deadlines, absence of contingency protocols. Recommendations: Always include a 20% time buffer in planning, define a Plan B for critical inspections, implement team rotation for long inspections, develop a specific shipyard checklist. Expected Impact: 40% reduction in team stress, 15% improvement in report quality, 10% increase in client satisfaction.
CASE 2: CPEP Protocol Implementation in a New Body. Category: Advanced Protocols / CPEP. Level: 3. Type: Success. Context: Newly accredited body implementing CPEP for the first time. What Happened: Successful implementation in 4 months, 100% staff adoption, zero security incidents. What Went Well: Intensive initial training (40h per inspector), mentorship from an experienced body, technical support from Aurema Group, clear communication of benefits. What Went Wrong: Initial resistance from senior inspectors, slower learning curve than expected, minor technical issues with devices. Root Causes of Success: Committed leadership, gradual approach (pilot → rollout), recognition of early

adopters. Recommendations: Involve senior inspectors from the beginning, plan for 6 months for full adoption, establish 24/7 technical support channel, celebrate milestones and successes.

18.8.5 Learning Culture

CHARACTERISTICS OF A STRONG LEARNING CULTURE:

1. **Psychological Safety:** People feel safe admitting mistakes.
2. **Transparency:** Information flows freely throughout the organization.
3. **Curiosity:** The questions "why?" and "how can we improve?" are encouraged.
4. **Experimentation:** Trying new ways of doing things is permitted.
5. **Recognition:** Learning is valued, not just success.
6. **Collaboration:** Knowledge is shared across teams and individuals.
7. **Reflection:** Time is dedicated to analyzing what happened.

INDICATORS OF A STRONG LEARNING CULTURE: • Lessons documented per project: >2. • % of lessons applied in new projects: >70%. • Participation in sharing sessions: >80% of staff. • Improvement suggestions per employee/year: >2. • Time dedicated to learning: >40h/year per employee. • Internal NPS (work climate): >8.

18.9 STRATEGIC IMPROVEMENT PLAN

18.9.1 PTVS Strategic Vision 2026-2030

VISION: "To be the global reference standard for the verification of physical assets in the digital era, recognized for its technical excellence, technological innovation, and institutional integrity."
MISSION: "To provide the world's most robust trust infrastructure for the verification of physical assets, enabling the secure tokenization of Real World Assets and protecting the wealth of institutional investors and Family Offices."
VALUES: • **Integrity:** Absolute honesty in all verifications. • **Excellence:** Continuous pursuit of the highest quality. • **Innovation:** Pioneering adoption of disruptive technologies. • **Transparency:** Clear and verifiable communication. • **Collaboration:** Joint work with all stakeholders. • **Sustainability:** Commitment to positive impact.

18.9.2 Strategic Objectives 2026-2030

OBJECTIVE 1: MARKET LEADERSHIP • Goal: 50% market share in RWA tokenization verification. • KPIs: Number of certified assets, tokenized volume. • Initiatives: Geographical expansion, strategic alliances.
OBJECTIVE 2: OPERATIONAL EXCELLENCE • Goal: Reduce verification times by 40%. • KPIs: Average time, error rate, client satisfaction. • Initiatives: Automation, AI, process optimization.
OBJECTIVE 3: TECHNOLOGICAL INNOVATION • Goal: Implement 3 disruptive technologies. • KPIs: New

technologies adopted, patents, publications. • Initiatives: R&D, collaborations, pilot projects.
OBJECTIVE 4: GLOBAL EXPANSION • Goal: Presence in 30 jurisdictions. • KPIs: Countries with accredited bodies, international certifications. • Initiatives: Multi-jurisdictional accreditations, partnerships.
OBJECTIVE 5: SUSTAINABILITY AND IMPACT • Goal: 100% of certifications include ESG analysis. • KPIs: Carbon footprint, social impact, governance. • Initiatives: ESG standards, compensation, reporting.
OBJECTIVE 6: TALENT DEVELOPMENT • Goal: 1,000 PTVS accredited inspectors. • KPIs: Number of accredited personnel, satisfaction, retention. • Initiatives: Aurema Group Academy, mentorship, development.

18.9.3 Implementation Roadmap

YEAR 1 (2026): CONSOLIDATION • Q1-Q2: Official launch of the complete PTVS standard, accreditation of 20 initial bodies, first 100 certifications issued, establishment of the body network. • Q3-Q4: Expansion to 5 additional jurisdictions, implementation of advanced protocols in all bodies, first annual impact report, 500 certifications issued.
YEAR 2 (2027): GROWTH • Q1-Q2: Expansion to 15 jurisdictions, 50 accredited bodies, launch of generative AI for reports, 2,000 certifications issued. • Q3-Q4: First tokenized assets with PTVS, expansion to 20 jurisdictions, implementation of advanced ZKP, 5,000 certifications issued.
YEAR 3 (2028): MATURITY • Q1-Q2: Internationally recognized standard, proposal of PTVS as an ISO norm, 100 accredited bodies, 15,000 certifications issued. • Q3-Q4: Expansion to 25 jurisdictions, implementation of quantum computing (cryptography), 10,000 tokenized assets with PTVS, 30,000 certifications issued.

YEARS 4-5 (2029-2030): LEADERSHIP • 30 jurisdictions, 200 accredited bodies, 100,000 certifications issued, de facto standard for tokenized RWAs, recognition as an ISO standard.

18.9.4 Investment Plan

TOTAL INVESTMENT 2026-2030: €50M

Area	Investment	%	Expected ROI
Technology and R&D	€18M	36%	400%
Geographical Expansion	€12M	24%	300%
Talent Development	€8M	16%	250%
Marketing and Communication	€6M	12%	350%
Operations	€4M	8%	200%
Sustainability	€2M	4%	150%

FUNDING SOURCES: • Internal resources: 40% • Strategic investors: 35% • Technology partners: 15% • Grants and subsidies: 10%

18.9.5 Balanced Scorecard

FINANCIAL PERSPECTIVE: • Revenue growth: Target €100M (Current: €5M). • Profitability: EBITDA margin >30% (Current: 28%). • Efficiency: Cost per certification -30% (Current: Baseline). • Shareholder value: ROI >25% (Current: 22%).
CLIENT PERSPECTIVE: • Satisfaction: NPS >9 (Current: 8.2). • Retention: Retention rate >95% (Current: 92%). • Perceived value: Premium over competition >50% (Current: 30%). • Reputation: Brand awareness Top 3 (Current: Top 10).
INTERNAL PROCESS PERSPECTIVE: • Operational excellence: Average verification time -40% (Current: Baseline). • Quality: Error rate <0.1% (Current: 0.8%). •

Innovation: New services/year >5 (Current: 2). • Efficiency: Automation >70% (Current: 30%).
LEARNING AND GROWTH PERSPECTIVE: • Talent: Accredited inspectors: 1,000 (Current: 50). • Training: Training hours/employee >60h (Current: 40h). • Innovation: Registered patents >20 (Current: 3). • Work climate: Internal NPS >8 (Current: 7.5).

18.9.6 Strategic Review

REVIEW FREQUENCY: • Operational: Monthly (Management + middle management) → KPI tracking. • Tactical: Quarterly (Management + committee) → Tactical adjustments. • Strategic: Annual (Board + committee) → Strategy review. • Transformation: Biennial (Board + stakeholders) → Long-term vision.
ANNUAL STRATEGIC REVIEW PROCESS:

1. **Environment Analysis:** Market evolution, regulatory changes, technological advances, competitor movements.
2. **Performance Evaluation:** Objective compliance, deviation analysis, lessons learned, benchmarking.
3. **Strategy Review:** Validity of vision and mission, adequacy of objectives, priorities for the next year, resource allocation.
4. **Action Plan:** Annual objectives, key initiatives, budgets, responsible parties and deadlines.
5. **Communication:** Report to the board, communication to staff, reporting to stakeholders, annual publication.

18.10 REPORTING AND CERTIFICATION

18.10.1 Annual Continuous Improvement Report

Aurema Group publishes an annual Continuous Improvement Report including:

1. **Executive Summary:** Main achievements of the year, key KPIs, innovation highlights, perspectives for the next year.
2. **State of the PTVS:** Updates made, new protocols implemented, accredited bodies, certifications issued.
3. **Process Improvement:** Completed improvement projects, measured impact (time, cost, quality), lessons learned, upcoming projects.
4. **Technological Innovation:** Technologies implemented, ongoing R&D projects, strategic collaborations, patents and publications.
5. **Talent Development:** Training provided, new personnel accreditations, mentorship programs, recognitions.
6. **Client Relations:** Measured satisfaction, complaints and claims, new services launched, success cases.
7. **Sustainability and Impact:** Carbon footprint, ESG initiatives, social impact, future objectives.
8. **Perspectives for the Following Year:** Strategic objectives, key initiatives, planned investments, quantitative goals.

18.10.2 Seal of Continuous Improvement

Aurema Group awards a specific **Seal of Continuous Improvement** to bodies that demonstrate excellence in this area.
CRITERIA FOR OBTAINING: • >5 improvement projects completed per year. • >200% average ROI of improvements. • >10 lessons learned documented per year. • >50h training/employee per year. • Client NPS >8. • >1 new technology implemented.
BENEFITS: • Public recognition. • Differentiated marketing. • Access to the excellence network. • Priority in new developments. • Discounts on training and accreditation fees.

18.10.3 Continuous Improvement Audit

SPECIFIC AUDIT SCOPE: • Continuous improvement management system. • Implemented projects and results. • Learning culture. • Technological innovation. • Talent development.
FREQUENCY: • Annual for Level 1-2 bodies. • Semi-annual for Level 3-4 bodies.
EVALUATION CRITERIA: • Existence and effectiveness of the system. • Measurable results of improvements. • Staff participation. • Integration with strategy.

ANNEX I: GLOBAL IMPLEMENTATION STRATEGY AND OVERCOMING MARKET BARRIERS

I.1. Purpose of the Annex

The Prop Trust Verified Standard (PTVS) is not merely a set of technical protocols; it is a governance framework designed for mass adoption in mature and highly regulated markets. This annex details the operational, legal, and economic strategies designed to overcome traditional barriers to entry and consolidate PTVS as the definitive trust operating system for Real World Assets (RWA).

I.2. "Forced Symbiosis" Strategy and Liability Segmentation

To avoid friction with traditional institutions (e.g., RICS, Lloyd's Register, professional engineering associations), PTVS positions itself as a **layer of digital integrity verification** that validates and enhances traditional reports, rather than as a direct competitor.
To protect the standard's reputation and manage the risk of "false attribution" (mediocre reports cryptographically signed by third parties), the standard establishes a **Strict Segmentation of Liability**:

- **Levels 1 and 2 (Integrity Certification):** The PTVS seal at these levels exclusively certifies the **authorship, immutability, traceability (CPEP), and absence of data manipulation** presented by the expert. It does not certify the technical suitability or market value of the asset. The hash metadata and the resulting PDF will explicitly state: "This certification validates the cryptographic integrity of the data capture process, not the technical

valuation of the asset, which remains the exclusive responsibility of the original signatory."
This segmentation allows legacy appraisers to instantly bridge their traditional practice into the Web3 ecosystem without altering their existing liability insurance structures.

- **Levels 3 and 4 (Technical Suitability Certification):** At these levels, PTVS assumes responsibility for technical validation through AI coherence auditing, Blind Peer Review, and continuous monitoring (CAP). Only at these levels does the standard endorse the technical and financial health of the asset.

I.3. The Jurisprudential Accelerator: International Commercial Arbitration

Leveraging the active Judicial Expert status of Aurema Group's Technical Director, PTVS seeks to establish the legal incontestability of its chain of custody (CPEP). Recognizing that ordinary civil courts may delay this process, the strategy relies on **International Commercial Arbitration**.

- **The PTVS Clause:** Aurema Group and its strategic partners will insert an arbitration submission clause in their verification and tokenization contracts (e.g., ICC International Court of Arbitration in Paris, or DIFC-LCIA in Dubai).
- **The Mechanism:** The clause will stipulate that "The parties accept reports generated under the Prop Trust Verified Standard, along with their

cryptographic hashes and blockchain-registered chains of custody, as definitive, immutable, and binding expert evidence for the resolution of any dispute arising from the asset."

Leveraging the 1958 New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards, these decisions will be internationally enforceable across more than 160 jurisdictions, bypassing local judicial delays.

•**The Result:** A favorable arbitral award sets a fast, enforceable legal precedent (in weeks or months, not years), validating the standard before Family Offices and global regulators much faster than ordinary civil jurisprudence.

I.4. The PropTrust Certified Expert (PCE) Network and Scalability

To resolve the challenge of global scalability and the high cost of advanced hardware without sacrificing quality control, PTVS implements a licensed, decentralized network model.

•**Enablement, Not Employment:** Instead of deploying proprietary staff globally, Aurema Group certifies and licenses independent local experts (naval engineers, architects, appraisers) under the **PropTrust Certified Expert (PCE)** program.

•**The "PTVS Toolkit":** Each PCE expert receives access to the PTVS SaaS platform, a decentralized auditor identity (DID), and access to verified hardware providers at corporate rates.

•**Market Stratification:** PCE experts can offer Level 1 and 2 verifications using their own basic

equipment (low cost, high volume), while Level 3 and 4 verifications are executed in joint venture with Aurema Group's elite teams and advanced hardware partners.

I.5. The Algorithmic Guardian and the Economic Incentive of Peer Review

To eradicate the "Garbage In, Garbage Out" problem on the blockchain, PTVS implements a two-layer validation system prior to the minting of the certificate for Levels 3 and 4:

1. **AI Coherence Audit (Pre-Signature):** Before the expert can cryptographically sign the report, an AI algorithm mandatorily cross-references the data. If it detects anomalies (e.g., thermography shows a critical hot spot not mentioned in the text, or the photo's geolocation does not match the asset), the system blocks the signature and demands justification or correction.
2. **Blind Peer Review with Smart Escrow:**

- The client pays the certification fee, which is locked in a smart contract (Smart Escrow).
- The system randomly assigns the report to a second PCE expert in a different jurisdiction, under mutual anonymity.
- The reviewer completes a digital methodological coherence checklist.
- **The assignment carries a strict time-lock (e.g., 72 hours). If the reviewer fails to complete the checklist within the window, the smart contract**

automatically reassigns the report to another peer, penalizing the delayed expert's internal rating.

- **Economic Incentive:** Once the AI and the reviewer approve the report, the smart contract automatically releases the funds: a predefined fraction is transferred as a fee to the reviewer for their work, and the remainder is released to the original expert. If the report is rejected, it is returned to the original expert for correction at no additional cost to the client.

I.6. Strategic Conclusion

By combining symbiosis with traditional institutions, accelerated legal validation via arbitration, scalability through the PCE network, and technical infallibility guaranteed by AI and crypto-economic incentives, the Prop Trust Verified Standard does not merely overcome adoption barriers; it turns them into its definitive competitive advantages. PTVS is designed to be, and remain, the de facto standard of the digital physical economy.

ADDENDUM TO ANNEX I: TACTICAL EXECUTION AND LOW-CASH-BURN BULWARK OPTIMIZATIONS

I.7. Low-Cost Distributed Governance (Swiss Verein Structure)

To project immediate neutrality and institutional solidity without incurring the operational costs of early physical

infrastructure, the standard adopts the structure of a **Swiss Association (Verein) based in Zug**.

- **The Custody Structure:** The Verein will act as the umbrella non-profit entity exclusively responsible for safeguarding, updating, and auditing the intellectual property of PTVS, guaranteeing the independence of the standard from the commercial activities of Aurema Group.
- **The Geopolitical Advisory Council (GAC):** Instead of deploying physical offices in key markets, a virtual council of three high-level strategic advisors is established in the following areas:
  1. Common Law and Maritime Litigation (London).
  2. Financial Asset Regulation and Tokenization (ADGM/Dubai).
  3. Regulatory Compliance and Cryptographic Trust Standards (Switzerland). These advisors will be compensated through Advisory Equity / Token Warrants of the commercial entity, aligning their interests with the global success of the standard from day one.

I.8. The Trusted Intermediary Trojan Horse (B2B2C Strategy)

Recognizing the high sensitivity and natural paranoia of Ultra-High-Net-Worth Individuals (UHNWI) regarding their assets, PTVS will not direct its Level 4 commercial prospecting efforts directly at the final owner, but rather at the **administrators of their risk**:

- **Insurance (Lloyd's Syndicates) and Elite Law Firms Channels:** The pilot Level 4 diagnosis (funded and absorbed by the PTVS development fund) will be presented as a loss mitigation tool for insurance underwriters or as definitive expert

evidence for law firms handling high-complexity litigation.

- **The Value Exchange:** By reducing the insurer's risk premium or securing the judicial success of the law firm, PTVS obtains in return the institutional validation of the intermediary and the right to publish the case study (duly anonymized), leveraging the reputational trust these giants already possess with the end client.

I.9. The Council of the Twelve Cardinals (Elite Governance)

The recruitment of the world's top experts (PCE) will not be based solely on transactional economic incentives, but on the **cession of power, status, and governance**:

- **The Technical Advisory Board:** A surgical selection of the 12 most influential experts in the international landscape (key representatives in naval engineering, structural pathology, and judicial valuation) will be made to form the hard core of governance.
- **Ownership Incentives:** These 12 original members will be granted voting rights on future updates to the standard and a stake in the governance of the Verein through deferred financial instruments. The exclusivity of this council will generate a reputation-driven attraction effect (FOMO) over the remaining 38 experts needed to complete the first phase of the global network.

I.10. Co-authorship of ERC Standards (The Invisible Infrastructure)

To avoid the risk of being an easily replaceable API provider, the technological development of PTVS will focus on **standardization at the protocol level**:

- **The Asset Identity Extension:** Aurema Group's development team will work in collaboration with leading RWA consortia to propose a specific extension within the ERC-3643 security standard (proposed under the technical nomenclature of physical verification extension).
- **The Highway, Not the Toll:** By making the verification of the PTVS immutable hash a native requirement within the official issuance templates for Real World Assets, the standard transitions from being an optional auditing tool to becoming the mandatory infrastructure for technical compliance and property transfer on the blockchain.

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